

**NI 43-101 Technical Report
for the
Island Gold Mine,
Dubreuilville, Ontario, Canada**

Prepared for



ALAMOS GOLD INC.

**181 Bay Street, Suite 3910
Toronto, ON M5J 2T3, Canada**

Prepared by

**Nathan Bourgeault – P.Eng.
Raynald Vincent – P.Eng., M.G.P.
Colin Webster – P.Eng.
Neil Lincoln – P.Eng**

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For a more detailed discussion of such risks and other factors that may affect Alamos' ability to achieve the expectations set forth in the forward-looking statements contained in this report, see Alamos' latest 40-F/Annual Information Form and Management's Discussion and Analysis, each under the heading "Risk Factors" available on the SEDAR website at www.sedar.com or on EDGAR at www.sec.gov. The foregoing should be reviewed in conjunction with the information and risk factors and assumptions found in this report.

Alamos disclaims any intention or obligation to update or revise any forward-looking statements whether as a result of new information, future events or otherwise, except as required by applicable law.

Cautionary Note to U.S. Investors

Alamos prepares its disclosure in accordance with the requirements of securities laws in effect in Canada. Unless otherwise indicated, all Mineral Resource and Mineral Reserve estimates included in this document have been prepared in accordance with National Instrument 43-101 - Standards of Disclosure for Mineral Projects ("NI 43-101") and the Canadian Institute of Mining, Metallurgy and Petroleum (the "CIM") - CIM Definition Standards on Mineral Resources and Mineral Reserves, adopted by the CIM Council, as amended (the "CIM Standards"). NI 43-101 is a rule developed by the Canadian Securities Administrators, which established standards for all public disclosure an issuer makes of scientific and technical information concerning mineral projects. Mining disclosure in the United States was previously required to comply with SEC Industry Guide 7 ("SEC Industry Guide 7") under the United States Securities Exchange Act of 1934, as amended. The U.S. Securities and Exchange Commission (the "SEC") has adopted final rules, to replace SEC Industry Guide 7 with new mining disclosure rules under sub-part 1300 of Regulation S-K of the U.S. Securities Act ("Regulation S-K 1300") which became mandatory for U.S. reporting companies beginning with the first fiscal year commencing on or after January 1, 2021. Under Regulation S-K 1300, the SEC now recognizes estimates of "Measured Mineral Resources", "Indicated Mineral Resources" and "Inferred Mineral Resources". In addition, the SEC has amended its definitions of "Proven Mineral Reserves" and "Probable Mineral Reserves" to be substantially similar to international standards.

Investors are cautioned that while the above terms are "substantially similar" to CIM Definitions, there are differences in the definitions under Regulation S-K 1300 and the CIM Standards. Accordingly, there is no assurance any mineral reserves or mineral resources that Alamos may report as "proven mineral reserves", "probable mineral reserves", "measured mineral resources", "indicated mineral resources" and "inferred mineral resources" under NI 43-101 would be the same had Alamos prepared the mineral reserve or mineral resource estimates under the standards adopted under Regulation S-K 1300. U.S. investors are also cautioned that while the SEC recognizes "measured mineral resources", "indicated mineral resources" and "inferred mineral resources" under Regulation S-K 1300, investors should not assume that any part or all of the mineralization in these categories will ever be converted into a higher category of mineral resources or into mineral reserves. Mineralization described using these terms has a greater degree of uncertainty as to its existence and feasibility than mineralization that has been characterized as reserves. Accordingly, investors are cautioned not to assume that any measured mineral resources, indicated mineral resources, or inferred mineral resources that Alamos reports are or will be economically or legally mineable.

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In addition to disclosing results determined in accordance with generally accepted accounting principles (GAAP), Alamos may also disclose certain non-GAAP financial measures, which are presented in accordance with International Financial Reporting Standards (IFRS), including the following: (1) total mine-site free cash flow; (2) total cash cost per ounce of gold sold; and (3) all-in sustaining cost per ounce of gold sold. The Company believes that these measures, together with measures determined in accordance with IFRS, provide investors with an improved ability to evaluate the underlying performance of the Company.

Non-GAAP financial measures do not have any standardized meaning prescribed under IFRS, and therefore they may not be comparable to similar measures employed by other companies. The data is intended to provide additional information and should not be considered in isolation or as a substitute for measures of performance prepared in accordance with IFRS. Management's determination of the components of non-GAAP and additional measures are evaluated on a periodic basis influenced by new items and transactions, a review of investor uses and new regulations as applicable. Any changes to the measures are fully noted and retrospectively applied as applicable. A reconciliation of historical non-GAAP and additional GAAP measures are available in the Company's latest Management's Discussion and Analysis available online at www.alamosgold.com and on the SEDAR website www.sedar.com or on EDGAR at www.sec.gov.

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1 SUMMARY

1.1 Introduction

In a press release dated June 28, 2022, Alamos Gold Inc. (“Alamos” or “Alamos Gold” or the “Company”) announced the results of its Phase 3+ Expansion Study (“Phase 3+”) completed on the Island Gold Mine (“Island Gold”), located in Northern Ontario, Canada. Having successfully completed expansions to 1,100 tpd and 1,200 tpd at Island Gold and having announced in July 2020 an expansion to 2,000 tpd, the objective of Phase 3+ was to consider if further expansion beyond 2,000 tpd to increase annual production and mine the current Mineral Reserves, and a portion of the Mineral Resources located to a depth of 1,500 metres, was warranted. This report outlines the results of that study and conforms to National Instrument 43-101 Standards of Disclosure of Mineral Projects (“NI 43-101”).

The study involved the evaluation of several scenarios, which demonstrated that the Shaft Expansion at a new production profile of 2,400 tpd was the most economic, efficient, and productive alternative. This also best positions Island Gold to capitalize on further growth in Mineral Reserve and Resources. The 2,400 tpd Shaft Expansion will result in the construction of a new shaft amongst other infrastructure upgrades and additions.

Island Gold utilized the services of several consulting firms to design and cost the components making up the various options. They included: Hatch Ltd., Redpath Mining Inc., Golder Associates Ltd., Halyard Inc., DRA Global Ltd, and Paterson and Cooke.

All costs are in Q2 2022 Canadian dollars unless otherwise stated.

All units of measurement are in metric, unless otherwise stated.

1.2 Property Description

The Island Gold Mine and its surrounding project lands (collectively, the “Island Gold Property”) is situated 43 km northeast of Wawa, Ontario within the Sault Ste. Marie Mining Division. The town of Dubreuilville, a forestry center, is 10 km to the northwest of the mine site. The Island Gold Property, which is divided into ten (10) property areas, is comprised of patented fee simple and/or patented leasehold mining rights and surface rights claims, mining licences of occupation and unpatented cell claims covering approximately 15,524 hectares. Alamos holds 100% of all mining titles related to the Island Gold Property.

Collectively, the Island Gold Property is subject to different obligations and royalties. Based on the currently defined Mineral Reserves and Mineral Resources, the only royalties to apply are:

- The Lochalsh property is subject to a 3% net smelter returns (NSR) royalty payable to Osisko Gold Royalties Ltd. (“Osisko”). The Island Main and Lochalsh zones, as well as a part of the Island Gold Mineral Resources below the 400 m Level, are located on this property;
- The Goudreau Lake property is subject to a 2% NSR royalty payable to Osisko as to a 69% interest and to Franco Nevada Corporation as to a 31% interest; and
- The Goudreau property is subject to a 2% NSR royalty payable to Osisko.

1.3 Accessibility, Climate, Local Resources, Infrastructure and Physiography

Access to the area is provided by the Trans Canada Highway (Highway 17), which continues north from Wawa for 35 km, and Highway 519 to Dubreuilville which is 10 km to the northwest of the mine site. The Goudreau Road, an all-weather road, extends from east of Dubreuilville for 17 km to the mine site.

The Island Gold Property is contained within the Lake Superior Regional climatic zone and is described as "modified continental". The mean annual temperature is about 10°C, with extremes of -51°C and 38°C being recorded. Precipitation is in the range of 980 mm per year, with about 600 mm as rainfall and evaporation at 517 mm/year principally during the summer months.

Wawa has a population of approximately 3,500 inhabitants and, Dubreuilville, originally a forestry community, has a population of approximately 635 permanent residents and contains accommodations for mine personnel. The Island Gold Mine is also within a few kilometres of railway lines operated by Canadian National and Algoma Central Railways. A hydro-electric power substation, water supply, gravel roads, and living accommodations are all available within the general mine area. Power is connected to the provincial power grid and is supplied by Algoma Power Inc. (API).

The Island Gold Mine infrastructure includes a primary tailings pond, a secondary settling pond, the Kremzar mill (the "mill"), the Lochalsh ramp and portal, a mine access road, power lines, and an electrical substation. Offices, core logging and storage facilities, a fire hall and separate mine dries for men and women are also located at the mill site. The mill currently has a capacity of 1,200 tpd and the fully permitted tailings area is located at Miller Lake, west of the historic Kremzar Mine.

The Island Gold Property lies in the Superior Province of the Canadian Shield. Topography within the mine area varies from a high of 488 metres above sea level (masl) in the vicinity of Miller and Maskinonge Lakes to a low of 381 masl at Goudreau Creek. Periods of intense glacial activity have contributed to the hummocky, rock knelled and largely bedrock-controlled topography, characteristic of the region. Glacial advance from the north deposited a thin mantle of stony sand till over a scoured rock surface.

1.4 History

The Goudreau – Lochalsh Gold Camp area has been the subject of interest dating back to the early 1900's and has attracted prospectors and mining companies in search of iron ore, gold, and base metal deposits. The Wawa – Michipicoten area has been recognized for its long history of iron exploration which has resulted in the development and production of several iron ore mining operations. Gold exploration followed shortly thereafter, resulting in several gold discoveries which were subsequently developed and brought into commercial production in the area which would later become the Island Gold property.

The initial discovery of gold was made by a group of prospectors at Emily Bay on Dog Lake in Riggs Township in 1900. Up to 1944, prospecting, geological mapping, trenching, shaft sinking, and 1,732 m of diamond drilling were completed to explore various gold prospects. Ultimately this period is marked principally by various exploration efforts by several companies carrying out surface trenching and diamond drilling on several gold prospects.

After an extended period of relatively little interest and activity in the area, exploration was resumed by Amax Inc. and its Canadian division, Canamax Resources Inc. ("Canamax") in 1974. In 1985, drilling approximately two kilometers south of the Kremzar mine intersected a series of sub-parallel lenses containing gold mineralization within deformed rocks of the Goudreau Lake

Deformation Zone (GLDZ). In December 1988, the Canamax Kremzar project began commercial production. From 1988 to 1990, production from the Kremzar mine was 306,000 tonnes grading 4.80 g/t Au. Over 1989 and 1990, underground access was established into the Island Gold deposit with an adit from the north shore of Goudreau Lake. A 4,167 tonne bulk sample was extracted and processed at the Kremzar Mill. At the end of 1990, Canamax suspended all operations at both the Kremzar and Island Gold projects.

In 1996 the Island Gold property was acquired from Canada Tungsten Inc. by Patricia Mining Corp. ("Patricia"). From 1996 to 2002, various exploration activities on the property included prospecting, surface trenching, geological and geophysical surveys, and diamond drilling was carried out to explore for both Island Gold and Kremzar styles of gold bearing prospects and zones. In 2003, Patricia and Richmont Mines Inc. ("Richmont"), entered into a joint venture agreement. Work completed during the joint venture included 72,984 m of surface and underground diamond drilling to test the various zones. On January 1, 2005, Richmont became the operator of the project.

Commercial production at Island Gold began on October 1, 2007. Richmont acquired Patricia's 45% interest in December 2008, becoming 100% owner of the property and operations. Exploration activities ramped up in 2009 with a minimum of 30,000 metres of drilling completed in each of the next several years, increasing sharply to more than 80,000 metres in 2012. This included drilling below the 400 metre Level as part of the Island Gold deep exploration program, which was successful in extending the main C Zone at depth with an initial Inferred Mineral Resource being calculated on the high-grade deep C Zone in January 2013.

A large exploration program commenced at the end of 2015 to explore beneath the Island Gold Mine. Directional diamond drilling was used to reach targets at depth, allowing greater accuracy than conventional drilling techniques. As a result of this program, Mineral Resources were added in the C zone at depth and to the east in the E1E zone in the Extension 2 area. A total of 226,824 m of directional drilling was completed between 2015 and 2021.

1.5 Geological Setting and Mineralization

The Island Gold Property is in the Michipicoten Greenstone Belt (MGB) which is part of the Wawa Subprovince within the Archaean Superior Province. The MGB is approximately 140 km long and up to 45 km wide. The metamorphic grade of the subprovince is greenschist but amphibolite facies can be seen locally or proximal to intrusions. A major regional deformation zone called the Goudreau Lake Deformation Zone (GLDZ) is situated throughout the area. It is a north-easterly trending structure which has been traced along strike for 30 km with a width of 4.5 km and is believed to be the main control of gold mineralization for the Project area. It is a high angle oblique-slip fault zone with an overall dextral movement cutting stratigraphy at a shallow angle. There are three main splays to the GLDZ in the area, the southernmost of which hosts the Island Gold Mine structure which contains a stacked sequence of east-northeast striking, steeply dipping, and subparallel zones of gold mineralization.

Lithologies appear to form a conformable homoclinal volcano-stratigraphic sequence, facing and younging to the north in the project area. Tight to isoclinal folds and local attenuation or boudinage of units along fold limbs appear to occur regionally. Fold axes are subparallel to the regional foliation at N070°E to N095°E.

The Island Gold Mine is stratigraphically positioned in the upper portion of the Wawa Assemblage, on the northern limb of the Goudreau Anticline. This assemblage is mostly composed of felsic volcanic rocks of various facies of tuffs and lavas.

Quartz veins commonly bear visible gold in the form of aggregates, disseminated fine grains or along chlorite-sericite slickensides within the veins. The degree of veining appears to change at

depth, transitioning from a stringer style quartz-carbonate vein on scales between millimeter to larger scale veins which can be over 4 m in width.

The Island Gold deposit is composed of multiple, stacked, south dipping lenses. The mineralized corridor expands from 50 m wide in the upper levels to over 150 m wide at depth. The zone's dip varies from sub-vertical to vertical from -50° to -90° south. Locally, north dip reversals occur but are not common. Rare instances of offset or folding have been seen. Around the 400 m elevation there is a shallow dipping southern inflection of the mineralized zones. It is not yet clear if this inflection is related to a fault, a shear zone, or a fold. This inflection point is the division of what is locally referred to as the Upper Island Gold Mine and the Lower Island Gold Mine.

1.6 Deposit Types

The Island Gold Mine is an Archean orogenic lode gold deposit. It is a structurally hosted quartz-carbonate vein system situated within the GLDZ, a major regional brittle-ductile structure. The host terrane is a sequence of felsic to intermediate volcanic rocks of the Wawa Assemblage which are in the greenschist metamorphic range as is common for this type of deposit. High strain zones associated with the GLDZ have the tendency to develop at variable scales along lithologic unit contacts where complex geology and related competency contrasts can control stress patterns and facilitate shearing and the consequent development of dilatancy zones and concomitant quartz carbonate vein formation. It is generally accepted that these Archean orogenic lode gold deposits are related to compressional and transpressional tectonics and the associated metamorphic dewatering and devolatilization of magma processes from which the gold bearing fluids are derived.

1.7 Exploration

The deep exploration diamond drilling program started in October 2015 and, at the end of 2021, a total of 274,345 metres of surface drilling were completed. Directional diamond drilling was used to reach targets at depth allowing greater accuracy than conventional drilling techniques. A total of 226,824 m of directional drilling was completed between 2015-2021. During the same period, 179,759 metres of underground diamond drilling were completed.

Since 2015, the underground and surface exploration programs have added close to 3.7 million ounces of gold to mineral inventory, net of depletion. Measured and Indicated Mineral Resources increased by 214,000 ounces of gold, Inferred Mineral Resources increased by 2,686,000 ounces of gold and Mineral Reserves increase by 776,000 ounces of gold, in the seven year period. This includes the addition of nearly 1.4 million ounces of Mineral Reserves and Resources since the release of the July 2020 Phase III study. The discovery cost has averaged approximately CAD \$18 per ounce during this period.

The results of these exploration programs up to December 31, 2021, were used in the December 31, 2021, Mineral Resource estimate and have been incorporated into the Phase 3+ Expansion Study.

1.8 Drilling

An optimal drilling pattern of 20 m by 20 m hole spacing is sought during the planning of the delineation-definition drilling. A 50 m to 100 m spacing pattern is used for the first phase of exploration drilling in new sectors. Island Gold employees use a Leica Global Positioning System to survey surface collar locations. Surveying of underground drill holes collars is performed using a Leica Total Station. Single shot Reflex down-hole survey measurements start at 15 m from the collar and are carried out at every 30 m thereafter along the hole. Surface exploration holes have

used Reflex or a gyro to survey the hole and measurements are taken every 30 m in surface holes. Diamond drill holes are grouted at the collar once they are completed or abandoned.

A total of 813 holes representing 447,187 m have been drilled from surface and 6,094 holes totaling 918,210 m have been drilled from underground within the Island Gold Mine area.

Under the direct supervision of Qualified Persons, geologists prepare a detailed description of the drill core. A computerized log is entered for each drill hole with the following information: collar location, down hole surveys, rock quality designation (RQD), primary and secondary geological units, texture, structure, mineralization, alteration, mineralogy, thickness, sample type, location, and core photos are taken.

1.9 Sample Preparation, Analyses and Security

Until April 2021, most drill core samples were prepared and assayed by Laboratoire Expert Inc (LabExpert) in Rouyn-Noranda, Quebec, which has been operating a fire assay laboratory for over 20 years. In April of 2021, Island Gold began shipping most drill core samples for preparation and assay to AGAT Laboratories Ltd (AGAT) in Thunder Bay, Timmins, and Mississauga. AGAT is recognized as an accredited testing laboratory as defined by the Standards Council of Canada (SCC), that conforms with the requirements of ISO/IEC 17025.

Gold is analyzed by a 50 gram fire assay with an Atomic Absorption (AA) finish. Samples greater than 6.5 g/t Au are re-analyzed using gravimetric finish methods. AGAT has internal quality control (QC) programs that include insertion of reagent blanks and reference materials, and analysis of pulp duplicates.

A small portion of the definition drill core as well as all underground production samples were assayed at the Wesdome laboratory, in Wawa, Ontario. This laboratory also processes assay samples from Wesdome's Eagle River Mine.

Alamos maintains an internal QA-QC program at the Island Gold Mine which is used to validate core and production chip assay analyses. Certified Reference Materials (CRMs) are purchased from Rocklabs (New Zealand) or from Ore Research & Exploration ("Oreas") (Australia) and inserted with diamond drill core samples. Geologists randomly place 2 standards for every 50 samples. At the same rate, in-house blank material is inserted in the core sample stream. The blanks consist of washed and cleaned diabase dyke core which is drilled from the Island Gold property. Cross check assays are completed on a regular basis in a secondary accredited laboratory.

Island Gold Mine's QA-QC procedures were audited in 2019 by ASL Canada and they concluded that Island Gold's assay quality control program meets or exceeds industry standards. In addition, ASL audited the LabExpert and Wesdome laboratories in 2019 and made recommendations for improvements which have been completed.

1.10 Data Verification

The Qualified Person considers that the Island Gold Mine database is suitable for use in the Mineral Reserve and Resource estimation. The SQL database is adequate and acceptable for supporting Mineral Resource estimation. This database contains all the information related to drill holes, drift sampling, assay results and the laboratory certificates. Some verification of the original data was performed, and modifications were completed if needed prior to the calculation of any estimates. The verification of, and corrections to, the Island Gold database were done prior to the Mineral Resource and Mineral Reserve estimates of December 31, 2021.

1.11 Metallurgical Test Work

The process plant has been treating ore since 2008 and the metallurgy is well understood. Since 2016, the ore feed has been almost exclusively from the Lower Island Gold Domain. Historical data shows that the Lower Zone ore behaves similarly to ore from the other zones and therefore has no significant effect on mill gold recoveries. The process plant has consistently achieved recoveries of greater than 96%.

1.12 Mineral Resource Estimates

The December 31, 2021, Mineral Resource and Mineral Reserve Estimation was carried out by the Island Gold Mine Technical Services department's staff under the supervision of Raynald Vincent, P.Eng., M.G.P., Exploration Superintendent and Nathan Bourgeault, P.Eng., Chief Engineer of the Island Gold Mine. Both are considered Qualified Persons within the meaning of Canadian Securities Administrators' National Instrument 43-101.

The Mineral Resource evaluation methodology involved the following procedures:

- Database compilation and validation;
- Construction of wireframe models for the boundaries of the gold mineralization;
- Geostatistical analysis and variography;
- Block modelling and grade interpolation;
- Definition of Mineral Resource domains;
- Assessment of "reasonable prospects for economic extraction" and selection of appropriate cut-off grades;
- Preparation of the Mineral Resource Statement.

Mineral Resources as of December 31, 2021, are found in Table 1-1.

Table 1-1 Island Gold Mineral Resources as of Dec 31, 2021

Mineral Resource	Tonnes	Grade (g/t Au)	Ounces
Measured	20,350	4.92	3,200
Indicated	1,075,950	8.18	283,100
Total Measured and Indicated	1,096,300	8.12	286,300
Inferred	7,905,600	13.59	3,453,800

Notes:

- CIM definitions of Mineral Resources were followed.
- Mineral Resources are estimated at an undiluted cut-off grade of 3.74 g/t Au.
- Depending on the zones, the capping value for the high-grade samples varies from 45 g/t Au to 225 g/t Au.
- Mineral Resources are estimated using a long-term gold price of \$1,400 per ounce.
- A minimum mining width of 2.00 m was used.
- Mineral Resources, which are not Mineral Reserves, do not have demonstrated economic viability.
- Totals may not match due to rounding

1.13 Mineral Reserve Estimate

The global Mineral Resource was reviewed by the Island Gold engineering department, with assistance from the geological staff, to define the Mineral Reserve blocks that could be economically extracted with a mining plan. The conversion of Mineral Resources into Mineral Reserves is based on the economic parameters detailed in Table 1-2. Only Mineral Resources that are classified as Measured or Indicated Mineral Resource categories were used in the economic calculations to estimate Mineral Reserves as of December 31, 2019.

Table 1-2 Mineral Reserve Estimation Parameters

Mineral Reserves Parameter	Value
Gold Price (USD)	\$1,250
Exchange Rate (USD/CAD)	0.75
Stope Cut-off Grade (g/t Au)	3.74
Development/Marginal Cut-off Grade (g/t Au)	3.06
Stope Dilution ¹ (%)	15%-40%
Development ¹ Dilution (%)	20%-30%
Dilution Grade (g/t Au)	0.50
Mining Recovery ¹ (%)	67%-95%
Process Recovery (%)	96.5%
Ore Specific Gravity ² (t/m ³)	2.78
Minimum Mining Width (m)	2.0
Mining, Processing and G&A Cost (CAD \$/t)	193

Notes:

1. Dependant on sector and mining method.
2. 2.82 t/m³ for Upper Mine.

Mineral Reserves as of December 31, 2021, are presented in Table 1-3.

Table 1-3 Island Gold – Mineral Reserve Estimate as of Dec 31, 2021

Mineral Reserve	Tonnes	Grade (g/t Au)	Ounces
Proven	834,100	9.33	250,150
Probable	3,277,750	10.33	1,088,300
Total Proven and Probable	4,111,850	10.12	1,338,450

Notes:

- CIM definitions of Mineral Reserves were followed.
- Mineral Reserves are estimated at cut-off grade of 3.06 g/t Au for developed areas and 3.74 g/t Au for undeveloped areas
- Mineral Reserves are estimated using a long-term gold price of \$1,250 per ounce.
- A minimum mining width of 2.00 m was used.
- Totals may not match due to rounding.

1.14 Mining Production Plan

The Island Gold deposit is accessed via a single decline from surface down to the 425 Level, at which point multiple ramps are utilized to access the main IG, IG West, Extension and East zones. These ramps are also connected at numerous points throughout the mine allowing for easy travel between mining zones.

The life of mine plan (LOM) includes the addition of a mine shaft which will be constructed between 2022 and 2026. Once commissioned, the shaft will be utilized to hoist ore and waste from the 1350 Level to surface. Additionally, the shaft will be used to transport personnel and materials to any of the three shaft stations. From the shaft collar location ore and waste will be trucked to either the mill or the surface waste stockpile.

A total of 124 km of lateral and vertical development are planned as part of the life of mine plan. Of this total approximately 28% is operating development, 66% is capital development and 6% is planned to support exploration activities.

Presently, level accesses are designed towards the center of the ore vein and stopes are mined longitudinally from sill extremities towards the level intersection. As mining progresses deeper level accesses are designed to access the extents of the deposit with stopes being mined from the center towards the extremities to support improved mining stress management.

The mining method for a particular stope is selected based on a variety of factors such as overall geometry of the mineralization, width of the ore zone, local stresses, mapping and geotechnical data, spatial location of the stope, and existing nearby development and infrastructure. Other factors considered include equipment size and limitations as well as available fill type. Presently, stoping is undertaken with longitudinal open stoping (modified Avoca) and transverse open stoping. A limited amount of Alimak stoping will be undertaken, beginning in 2023.

Island Gold presently uses unconsolidated rockfill for most of the longitudinal stoping and cemented rockfill for transverse stoping. With the Phase 3+ expansion, a paste fill plant will be constructed and paste fill underground will be implemented.

An internal scoping study, in 2019, determined that expanding the mining rate beyond 1,200 tpd, to take advantage of the growing Mineral Resource, was viable. Furthermore, the scoping study concluded that two material transport options, continued truck haulage and a shaft for ore and waste, should be advanced to a pre-feasibility level of design and engineering.

In 2020, an NI 43-101 technical report on a potential Phase III expansion concluded that constructing a shaft and hoisting facility and expanding capacity to 2,000 tpd was the most viable option of five scenarios studied.

In undertaking the Phase 3+ Expansion Study, it was assumed that the Mineral Reserves and a significant portion of the Mineral Resources would ultimately be available for mining. Total Mineral Reserves and Mineral Resources included within the mine plans for the Study are 13.5 Mt at a gold grade of 10.59 g/t.

Three different mine capacity rates were examined, 1,200 tpd with ramp haulage (R1200), 2,000 tpd with a shaft (S2000), and 2,400 tpd with a shaft (S2400). The current mill capacity of Island Gold is 1,200 tpd and this case was deemed the Basecase for the Study.

Detailed capital and operating costing models were developed for each of the scenarios studied. Combining these models with the physicals, cash flow models were constructed that allowed for sensitivity analysis of costing and productivity input parameters.

Capital costs increase moving from the R1200 scenario to the S2000 scenario to the S2400 scenario. The increase in capital from the R1200 scenario to S2000 scenario is attributable to construction of the shaft and hoisting facility and the capital required to expand the mill to 2,000 tpd, partially offset by less capital required for trucks, ventilation infrastructure and capital development. The increase in capital from the S2000 to the S2400 is attributable to higher capacities required in the process and paste fill plant, as well as the accelerated mine development required to prepare additional mining horizons to allow for higher mining rates.

Productivity levels dramatically increase with the use of the shaft to transport personnel. Using 2022 as a baseline, effective time at the face increases by 24% with the shaft. With additional time at the face, development metres per employee and stoping tonnes per employee increase. Increased productivity results in less personnel required to attain the same tonnage or alternatively the same number of personnel to achieve higher tonnage rates.

With the ramp scenario, mining costs increase over the mine life due to increased haulage cycle times from lower mine horizons. Mine unit operating costs decrease moving from the R1200 scenario to the S2000 scenario to the S2400 scenario. Costs are reduced because of economies of scale, operating less trucks and reduced ventilation costs.

- In expanding the Island Gold Mine, the shaft at a 2,400 tpd mining rate (S2400) scenario should be adopted given that it:
 - o Provides the strongest economics (free cash flow, after-tax NPV, and IRR) of all the scenarios studied;
 - o Provides the highest annual gold production;
 - o Has the lowest operating costs, as well as the lowest cash costs and mine-site all-in sustaining costs per ounce;
 - o Has the lowest combined operating and capital costs;
 - o Has the lowest capital intensity per ounce produced;
 - o Provides access to higher grade stopes sooner in the mine life
 - o De-risks the lower mine operation;
 - o Provides for reduced congestion and reduced personnel and mobile fleet requirements; and
 - o Provides a significantly reduced carbon footprint and reduced exposure to diesel price and carbon tax increases.

Based upon these conclusions Island Gold is proceeding with the permitting and construction of the paste plant and the shaft complex and expanding the milling and mining rates to 2,400 tpd.

1.15 Processing

The existing process plant at Island Gold treats ore via a conventional cyanidation process. Ore is processed using a conventional two-stage crushing circuit followed by a two-stage ball mill grinding circuit, cyanide leach, partial carbon-in-leach (CIL), carbon-in-pulp (CIP) circuit and associated gold recovery and carbon handling circuits to produce gold doré.

To match the expanded mine plan, the mill throughput will be increased from 1,200 t/d to 2,400 t/d. The existing process plant will be expanded as follows:

- Replace the existing crushing plant with a new primary and secondary crushing circuit to process the higher ore tonnage;
- Install a new crushed ore storage day bin with a new associated reclaim system;
- Install a new primary ball mill and convert the existing grinding circuit to a twin train grinding circuit to produce a primary grind size of P80 of 65 µm;
- Upgrade the feed well, pumps and pipes at the existing pre-leach thickener;

- Convert the existing five leach/CIL tanks to CIL and add three new larger CIL tanks providing approximately 20 hours residence time;
- Decommission the existing CIP circuit;
- Replace the existing 1.7 tonne ADR plant with a new packaged 5 tonne ADR plant;
- Replace the existing gold room with a new gold room for the increased doré production;
- Install a cyanide destruction circuit; and,
- Upgrade the tailings pumping system for the increased throughput and pumping to a new paste backfill plant.

1.16 Infrastructure

1.16.1 Surface Site Infrastructure

The Island Gold Mine is accessed via a singular portal and decline. The ore stockpile pad and waste storage are located directly adjacent to the mine portal. The mill feed is hauled approximately 1 km from the stockpile to the mill complex. The maintenance facility, surface office and dry complex are located adjacent to the mill. The Kremzar Portal, accessing old mine workings, is detached from the current underground mine, and is utilized to access the bottom of the surface ore bins, which are used to feed the crusher section of the mill.

The primary tailings pond is located approximately 500 m west of the mill and the secondary pond is located just west of the portal area. Mine ventilation is via two surface fresh air fans and raises located adjacent to Goudreau Road, approximately 2.5 km away from the mill site along with one exhaust raise located across from the fresh air fans.

1.16.2 Tailings Management

The tailings management facility (TMF) represents the main water retention structures on the mine site. It consists of two ponds, the Primary Pond and the Secondary Pond which acts as a polishing pond in addition to water transfer system via a siphon system. The Primary Pond (built in the former Miller Lake basin) occupies an area of 109 ha. The Secondary Pond has an area of 22 ha.

Tailings slurry is conveyed by a pressurized pipeline from the mill and spigotted around the inside perimeter of the Primary Pond. The surface of the tailings forms a sloped beach allowing for a pond to form at the lowest part. Water is reclaimed (pumped) from the Primary Pond to the mill. Both the primary tailings and reclaim pipes are placed in an engineered ditch, with drainage to an emergency catchment section (with an area of 0.8 ha) at its lowest points and reinforced by construction of earthen berms. The TMF also includes seepage collection and pump back systems at dykes Nos. 1 and 2 at the Primary Pond; these were built to prevent any migration of seepage to Maskinonge Lake.

Water treatment is managed through natural degradation in the Primary Pond and Secondary Pond. Natural degradation is primarily active during ice-free periods, and batch discharge cycles are timed to accommodate the natural processes. Cycle duration is typically 40 days:

- 10 days transfer from the Primary Pond to the Secondary Pond;
- 20 days of final polishing at the Secondary Pond; and

- 10 days discharge of treated water from the Secondary Pond to the receiving environment in a series of streams, wetlands, and ponds, eventually discharging into the central part of Goudreau Lake.

Water quality is routinely monitored in the Primary Pond and Secondary Pond, and in Goudreau Lake at the discharge point and downstream. A comprehensive water monitoring program has been implemented for the site, and includes twelve compliance sampling locations, and effluent limits as mandated by the Ministry of Environment and Conservation and Parks (MECP).

The current dam footprint is sufficient for the construction of additional lifts that will accommodate the minable resource incorporated in Phase 3+. Should additional Mineral Reserves and Resources be delineated and brought into a mine plan, these can be accommodated as well.

1.16.3 Phase 3+ Infrastructure Expansion

Island Gold recently undertook engineering and economics studies on a possible Phase 3+ Expansion to increase underground tonnage rates and implement associated infrastructure upgrades as required. The study demonstrated that the Shaft Expansion at a new production profile of 2,400 tpd of ore was the most economic, most efficient, and productive alternative. This also best positions Island Gold to capitalize on further growth in Mineral Reserves and Mineral Resources. The Phase 3+ Expansion will result in the construction of a new shaft amongst other infrastructure upgrades as listed below:

- Development of a new production/service shaft down to 1,380 m in depth (initial depth);
- Development/implementation of a new ore and waste handling system underground;
- Upgrade to the main site power supply;
- Construction of a paste plant and underground distribution system;
- Upgrade of the existing mill from 1,200 tpd to 2,400 tpd; and
- Upgrade of the mine water treatment system.

1.17 Environmental Studies, Permitting and Social or Community Impact

From exploration to operations to closure, one of the goals at the Island Gold Mine is to safeguard the environment, educate its employees and the communities about the mine's environmental programs and commitments, and apply best management practices to prevent or mitigate any potential environmental impacts. The operations at Island Gold use a range of materials and consumables that includes explosives, chemicals, and fuels.

The Island Gold Mine is located within the Maskinonge Lake and Goudreau Lake sub-watersheds (total area of 48.2 km²), approximately 40 km south of the Arctic drainage divide. Both sub-watersheds are part of the Michipicoten-Magpie watershed and Lake Superior Drainage Basin. Surface water drainage at the site is bedrock-controlled, generally flowing from northeast to southwest within the valleys between the elongated hills and ridges.

A comprehensive environmental monitoring program is in place at the Island Gold Mine. It includes inspections, sampling schedules, data management and reporting. Also included in the program are sampling frequency, various parameters of concern (for field and laboratory analyses) and QA-QC procedures. Key performance indicators are tracked, and any deviations from targets are addressed and corrected.

Tailings, water management, and final effluent monitoring and quality requirements are regulated under an amended Environmental Compliance Approval (ECA) (No. 544-BNPL46) which was issued in April 2020. This ECA also allows for a mill production rate of up to 38,480 tonnes per month. Final treated water from the mine flows into the upper portion of Goudreau Lake via Goudreau Creek, which flows into the Michipicoten River system, entering Lake Superior near Wawa.

Additional monthly surface water quality monitoring is conducted by Island Gold Mine at two locations in Goudreau Lake (the receiving water body), one on Maskinonge Lake and one on Pine Lake. Both Maskinonge Lake and the upper basins of Goudreau Lake would be characterized as meeting provincial objectives. The majority of metal concentrations were below their respective Provincial Water Quality Objectives (PWQO). Annual results have been comparable from 2007 to 2022.

Air and noise discharges are regulated under an amended ECA, No. 1821 BAWLAC which was issued in May 2019, to the Island Gold Mine, allowing for an annual ore processing rate of 461,760 tonnes per year.

The ECA requires that the Island Gold Mine be in compliance with Ontario Regulation 419/05, applicable MECP Guidelines for Air and Noise, and other performance requirements as specified as permit conditions. It allows modifications such as process changes, de-bottlenecking, or addition of new equipment subject to limits on operational flexibility.

Alamos Gold's strategy is to reduce consumption, reuse any waste generated, and dispose final waste in a safe and responsible manner. A Waste Management Procedure (WMP) has been developed and implemented for the site; it provides guidance to site and non-site personnel on the handling, processing, and disposal of waste, including hazardous waste and domestic materials generated during the normal operations of the facility.

Excess underground waste rock is transported to the surface and stockpiled for use as future backfill and/or maintaining site roads and future dam raise projects. In 2019, WSP Golder was subcontracted to conduct an assessment on the geochemistry of Island Gold tailings and waste rock. Historical documentation, from Wood PLC, (formerly AMEC) was also reviewed. WSP Golder determined that the waste rock did not generate any acid-rock drainage (ARD) nor metal leaching and recommended reducing the sample analyses to monthly. Weekly analyses were conducted for tailings and waste for metal leaching and ARD and kinetic testing of these materials is underway. Results to date have shown that geochemistry of the tailings and waste rock are not a significant issue.

Anticipated permitting activities identify and address the various municipal, provincial, and federal regulatory requirements applicable to the Island Gold Mine. Relevant agencies for Phase 3+ permitting needs include Department of Fisheries and Oceans (DFO), MECP, Ministry of Natural Resources and Forestry (MNRF), and Ministry of Mines (Mines).

Island Gold Mine's philosophy is to maximize local hiring of employees from the labour pool in the surrounding communities. This has increased the economic stability of the local communities of Dubreuilville, Wawa and White River who have been hit hard by the downturn of the forestry industry.

The following Indigenous groups have been identified as having varying degrees of interest around the Island Gold Mine: Michipicoten First Nation (MFN), Missanabie Cree First Nation (MCFN), Batchewana First Nation (BFN), and the Garden River First Nation (GRFN). A Community Benefits Agreement (CBA) was signed with the Missanabie Cree First Nation in March 2017 and is valid until March 2024. Alamos signed a Community Consultation and Benefit

Agreement with Michipicoten First Nations in April 2022. Alamos is actively working with the other First Nations towards future agreements.

The Island Gold Closure Plan Amendment (CPA) details the decommissioning strategy for the Island Gold Mine. It reflects the current and expected site conditions and defines a program which ensures the long-term chemical and physical stability of the site. The goal of the CPA is to ensure that chemical and physical impacts to the site are minimized during operations and that the site is returned as closely as possible to pre-development conditions at close-out. The CPA has been developed using data collected during physical, chemical, and biological studies of the site (treated effluent, surface water, ground water, ore/waste rock) and the surrounding environment during advanced exploration and production phases.

1.18 Capital and Operating Costs

1.18.1 Capital Expenditures

As this report covers the planned expansion of Island Gold to 2,400 tpd, capital expenditures are divided into two distinct categories: growth capital and sustaining capital.

Growth capital is defined as expenditures that allow the mine to expand from the current throughput of 1,200 tpd to 2,400 tpd. Once the Shaft Expansion is completed and production has sustainably reached the targeted tonnage for a period of three months, all further capital expenditures are classified as sustaining capital.

Sustaining capital is defined expenditures related to sustaining the existing production and operating plan and allow Island Gold to mine its current Mineral Reserves, and a portion of its Mineral Resources, during, and post, the project period. The project period is defined as 2022 to 2026 and the post project period is defined as 2027 to the end of the life of the mine. Table 1-4 summarizes the total capital costs.

Table 1-4 Total Capital Costs

Total Capital Cost	LOM C\$M	LOM US\$M
Sustaining Capital	\$995	\$777
Growth Capital	\$966	\$756
Total Capital Costs	\$1,961	\$1,533

The sustaining capital requirements reported in the Phase 3+ study for the S2400 Shaft Expansion total \$995 million and include \$15 million of reclamation costs, and \$26 million of delineation drilling. In the Shaft Expansion mine development will require the largest sustaining capital investment, totaling \$559 million from 2022 to 2039. Table 1-5 presents the life of mine sustaining capital expenditures for the S2400 Shaft Expansion scenario.

Table 1-5 Sustaining Capital Costs

Sustaining Capital Cost	LOM C\$M	LOM US\$M
Tailings Storage Facility	\$41	\$32
Underground Infrastructure	\$197	\$154
Mobile Equipment	\$143	\$112
Delineation Drilling	\$41	\$32
Capital Development	\$559	\$436
Total Sustaining Capital	\$980	\$765
Reclamation	\$15	\$12
Total Sustaining Capital (including Reclamation)	\$995	\$777

The Island Gold growth capital expenditures for the Shaft Expansion are estimated to be \$966 million, including; \$507 million of direct costs, \$82 million of indirect costs (contractor indirects and owner's costs), \$70 million of contingency, and \$207 million of capital development (Table 1-6). This is expected to be spent between 2022 and 2026, until the completion of the shaft and mill expansion in 2026 with the bulk of this spending occurring between 2022 and 2025.

Within the \$966 million of growth capital there are \$307 million of expenditures that are accelerated spending in that they are expenses that would have been incurred without the Shaft Expansion, albeit at a later point in time. These accelerated growth capital expenditures are required to be spent during the project period to both advance the ramp system to the bottom of the shaft and to prepare mining areas in advance of expanding the mining rate to 2,400 tpd. These expenditures consist of lateral capital development and the establishment of underground infrastructure including ventilation and electrical expansions.

Table 1-6 Growth Capital Costs

Growth Capital Cost	LOM C\$M	LOM US\$M
Shaft Surface Complex	\$75	\$59
Shaft	\$217	\$170
Mill	\$97	\$76
Paste Plant	\$66	\$52
Power Upgrade	\$30	\$24
Effluent Treatment Plant	\$21	\$16
General Indirect Costs	\$82	\$64
Contingency	\$70	\$55
Total Growth Capital	\$659	\$516
Underground Equipment & Infrastructure (mostly accelerated)	\$101	\$79
Accelerated Capital Development	\$207	\$162
Total Growth Capital (including Accelerated Spend)	\$966	\$756

1.18.2 Operating Costs

Operating expenses were calculated using Island Gold's 2022 budget as a reference point where applicable and were developed from first principles when budgetary items were not available. The 2022 budget costs were adjusted to reflect increases related to mining at greater depths, increased operational efficiencies associated with shaft access, higher underground throughput, and recent inflationary impacts on inputs. Fixed and variable components of cost centers were considered. Costs were adjusted to reflect the total volume of material moved (waste and ore) per year. Costs were also adjusted to reflect the reduction in labour related to the completion of capital development activities.

Total life of mine operating costs, as presented in Table 1-7, amount to \$2,412M (US \$1,882M), including silver by-product credits, royalties and refining and transportation charges. This translates into an average cost of \$178/t of mill feed processed over the life of mine.

Table 1-7 Summary of Operating Costs

Operating Cost	C\$/t Processed	LOM C\$M	US\$/t Processed	LOM US\$M
Mining	\$101	\$1,373	\$79	\$1,071
Processing	\$35	\$470	\$27	\$367
G&A	\$37	\$507	\$29	\$395
Subtotal	\$173	\$2,350	\$135	\$1,834
Silver Credit	-\$1	-\$18	-\$1	-\$14
Royalties	\$6	\$80	\$5	\$62
TOTAL Operating Costs	\$178	\$2,412	\$139	\$1,882

1.19 Economic Analysis

1.19.1 Assumptions

All costs and economic results are reported as CAD\$, unless otherwise noted. Table 22-1 outlines the planned life of mine tonnage and grade estimates.

Table 1-8 Life of Mine Plan Summary

Parameters	Unit	Value
Mine Life ¹	Years	18
Total Mill Feed	Kt	13,550
Processing Rate ²	tpd	2,400
Average Au Head Grade	g/t	10.59
Total Au Production over Life of Mine	koz	4,460
Au Production (Years 2022 to 2039)	Average koz/a	255
Au Production (Years 2026 to 2039) ³	Average koz/a	287

Notes:

1. 2022 to 2039

2. 1,200 tpd from 2022 until shaft completion in 2026, after which the mill ramps up to 2,400 tpd by 2027
3. Post-shaft completion in 2026

Other economic factors and assumptions used in the economic analysis include the following:

- US \$1,800/oz gold, US \$22.00/oz silver and a \$0.79 USD/CAD were used in the cash flow model for 2022;
- US \$1,650/oz gold, US \$18.00/oz silver and a \$0.78 USD/CAD were used in the cash flow model for 2023 onwards;
- Discount rate of 5%;
- Closure cost of \$15.5M (US \$12.1M);
- No salvage assumed at the end of mine life; and
- Exclusion of all costs prior to 2022.

1.19.2 Taxes

Island Gold will be subject to provincial, federal, and mining taxes as follows:

- Ontario Mining Tax: 10%;
- Ontario Provincial Income Tax: 10%; and
- Federal Income Tax: 15%.

The rates above are current as of the date of this report and are subject to change in the future. Based on these rates and the financial assumptions used in this report, Island Gold is expected to have payable income and mining taxes of \$1,301M (US\$1,015M) over its 18-year life. Alamos has various Canadian tax pools that could be applied against future income from its Canadian operations, and 60% of the tax pools as of December 31, 2021, were used in this study to reduce taxes payable at Island Gold in the economic analysis.

1.19.3 Royalties

Production from Island Gold is subject to third-party royalties that average approximately 2.5% over the life of mine based on ounces produced. However, approximately 90% of the royalties at Island Gold are paid in-kind (ounces) to a third-party. The accounting treatment requires that in-kind royalties be recorded at production cost, which lowers the royalty expense with an offsetting reduction in revenue given in-kind ounces transferred to royalty holders do not meet the definition of sales. As a result, the average royalty included in the economic analysis is approximately 0.9% of revenue over the life of mine. There is no net impact on gross margin from the accounting for in-kind royalties compared to cash-paid royalties given the lower royalty expense is offset by a reduction in ounces sold. Total royalties included in this report are \$80M (US\$62M).

1.19.4 Economic Analysis

The expansion option is economically viable with an after-tax internal rate of return (IRR) of 23% IRR which is calculated on the differential after-tax cash flow between the expansion scenario and the scenario of continuing to mine at 1,200 tpd with ramp-only access. The after-tax net present value at 5% (NPV^{5%}) is \$2,093M (US \$1,632M).

Table 1-9 shows the detailed results of this evaluation.

Table 1-9 Summary of Economic Results

Category	Unit	Value (C\$)	Value (US\$)
Net Revenues	\$M	\$9,247	\$7,216
Operating Costs ¹	\$M	\$2,412	\$1,882
After-Tax Cash Flow from Operations ²	\$M	\$5,533	\$4,318
Total Capital & Closure Costs	\$M	\$1,961	\$1,533
Total Cash Cost (2022-2039)	\$/oz	\$553	\$432
Mine Site All-In Sustaining Cost (2022-2039)	\$/oz	\$781	\$610
Total Cash Cost (2026-2039) ³	\$/oz	\$545	\$425
Mine Site All-In Sustaining Cost (2026-2039) ³	\$/oz	\$739	\$576
Net After-Tax Cash Flow	\$M	\$3,572	\$2,786
After-Tax NPV5%	\$M	\$2,093	\$1,632
After-Tax IRR	%	23%	

Notes:

1. Operating Costs include mining, processing, G&A, royalties, transport & refining costs, and silver credit.
2. Cash Flow from Operations includes payable taxes.
3. Post-shaft completion in 2026.
4. IRR is calculated on the differential after-tax cash flow between the expansion scenario and the scenario of continuing to mine at 1,200 tpd with ramp-only access.

A sensitivity analysis was performed to test value drivers on Island Gold's after-tax NPV using a 5% discount rate. The results of this analysis are demonstrated in Table 1-10. Island Gold proved to be most sensitive to changes in metal price followed by foreign exchange, capital costs and operating costs.

Table 1-10 After-Tax NPV^{5%} Sensitivity Results

(\$M of CAD)	-10%	-5%	100%	5%	10%
Gold Price	\$1,698	\$1,896	\$2,093	\$2,290	\$2,486
Canadian Dollar	\$2,531	\$2,300	\$2,093	\$1,905	\$1,733
Capital Costs	\$2,209	\$2,151	\$2,093	\$2,035	\$1,976
Operating Costs	\$2,200	\$2,147	\$2,093	\$2,039	\$1,986

(\$M of USD)	-10%	-5%	100%	5%	10%
Gold Price	\$1,324	\$1,479	\$1,632	\$1,785	\$1,939
Canadian Dollar	\$1,772	\$1,702	\$1,632	\$1,562	\$1,491
Capital Costs	\$1,723	\$1,678	\$1,632	\$1,587	\$1,541
Operating Costs	\$1,716	\$1,674	\$1,632	\$1,590	\$1,548

1.20 Interpretations and Conclusions

Alamos personnel reviewed and audited the historical exploration data available for the Island Gold Mine as well as the exploration methodologies adopted to generate the data. Exploration work is professionally managed, and procedures are adopted that meet accepted industry best practices. The Qualified Person is of the opinion that the exploration data is sufficiently reliable to interpret with confidence the boundaries of the gold mineralization and support evaluation and classification of Mineral Resources in accordance with generally accepted CIM Estimation of Mineral Resource and Mineral Reserve Best Practices Guidelines and CIM Definition Standards for Mineral Resources and Mineral Reserves.

The drilling database includes information from 6,907 drill holes (813 from surface and 6,094 from underground) comprising 1,365,397 m of drilling.

The mining methods used at Island Gold include longitudinal open stoping and transverse open stoping and are deemed suitable considering the geometry of the orebody.

Island Gold performs regular reconciliations between production and the reserve block model and results have generally been within industry acceptable ranges. Island Gold uses the reconciliation process to validate its Mineral Resource estimation parameters and procedures.

Island Gold has undertaken detailed engineering and economic studies of a possible expansion scenarios to mine the Mineral Reserves and a portion of the Mineral Resources. Island Gold has concluded that constructing a shaft to a depth of 1,380 m and expanding the mining and milling capacity to 2,400 tpd is the best way to proceed.

The current 1,200 tpd mill consistently achieves recoveries of greater than 96%. In undertaking the Shaft Expansion, the mill will be expanded to 2,400 tpd with the addition of an additional primary ball mill, replacement of the crushing circuit and other upgrades, additions, and expansions within the circuit.

As part of the Shaft Expansion, and to support sustainable development of mine going forward, the following infrastructure upgrades will be undertaken:

- Development of a new production/service shaft down to 1,380 m in depth (initial depth);
- Development/implementation of a new ore and waste handling system underground;
- Upgrade to the main site power supply;
- Construction of a paste plant and underground distribution system;
- Upgrade of the existing mill from 1,200 tpd to 2,400 tpd; and
- Upgrade of the mine water treatment system.

The Island Gold Mine is operating within environmental compliance.

Several operational permits will need to be amended to allow for the expansion up to 2,400 tpd and are expected to take between 18 and 24 months to acquire. Island Gold is in receipt of a Closure Plan Amendment allowing it to proceed with shaft sinking and construction of the associated surface infrastructure.

The Island Gold Mine has been and will continue to be major contributor to the local economy. Alamos will continue to engage and work with area Indigenous communities and other communities of interest.

The 2,400 tpd Shaft Expansion option is economically viable with an after-tax IRR of 23%, which is calculated on the differential after-tax cash flow between the 2,400 tpd Shaft Expansion scenario and the base case scenario of continuing to mine at 1,200 tpd with ramp only access. The after-tax net present value at 5% (NPV^{5%}) is \$2,093M (US \$1,632M).

1.21 Recommendations

At the conclusion of the Phase 3+ Expansion Study the following recommendations are being made:

- Continue to invest in the surface exploration drilling program to potentially add to the Mineral Resource base;
- Continue with the underground delineation drilling program to convert Inferred Mineral Resources to Indicated Mineral Resources;
- Utilize the geology model to aid in identifying additional targets on the Island Gold Property;
- Continue with the production to Mineral Reserve reconciliations to further refine Mineral Resource estimation parameters and methodologies;
- Proceed with the paste fill plant construction;
- Proceed with defining the detailed scope of the process plant expansion through a feasibility study and then commence engineering and construction;
- Continue with the 2,400 tpd Shaft Expansion project;
- Continue with the environmental baseline program to support the permitting program; and
- Continue with the process of amending existing operational permits and acquire a series of new permits and/or authorizations for both future operational requirements and Phase 3+ construction related activities.

2 INTRODUCTION

In a press release dated July 14, 2020, Alamos announced the results of its Phase III Expansion Study (“Phase III”) completed on the Island Gold Mine, located in Northern Ontario, Canada. Having successfully completed expansions to 1,100 tpd and 1,200 tpd at Island Gold, the objective of Phase III was to consider the most cost and capital effective strategy to increase annual production and mine the current Mineral Reserves, and the Mineral Resources located to a depth of 1,500 metres. The results of that study indicated that the sinking of a production shaft and the expansion of the mining and milling infrastructure to 2,000 tpd was the most attractive option.

With the continued exploration success at Island Gold and 37% increase in Mineral Reserve and Resource gold ounces in the following two years, Alamos investigated the possibility of expanding the mining and milling capacity beyond the 2,000 tpd envisaged in the Phase III Expansion Study. In a press release on June 28, 2022, Alamos announced the results of the Phase 3+ study that determined that an expansion to 2,400 tpd was technically and economically warranted. This report outlines the results of that study and conforms to National Instrument 43-101 Standards of Disclosure of Mineral Projects (“NI 43-101”).

The study involved the evaluation of several scenarios, which demonstrated that the Shaft Expansion at a new production profile of 2,400 tpd was the most economic, most efficient, and productive alternative. This also best positions Island Gold to capitalize on further growth in Mineral Reserve and Resources. The Phase 3+ Expansion will result in the construction of a new shaft amongst and other infrastructure upgrades.

Island Gold utilized the services of a several consulting firms to design and cost the components making up the various options. They included:

- Hatch Ltd (“Hatch”) was commissioned to undertake the design and engineering of the overall surface infrastructure as well as the underground ore and waste handling system;
- Redpath Mining Inc. (“Redpath”) was contracted to do the shaft design, engineering and shaft sinking;
- Golder Associates Ltd. (“Golder”) was commissioned to evaluate the life of mine tailings requirements and water treatment options and develop the design and engineering around them. Golder is currently undertaking the environmental baseline monitoring and permitting for the site;
- Paterson and Cooke undertook the paste testwork and paste plant design and engineering.
- Halyard Inc. (“Halyard”) and DRA Global Ltd. (“DRA”) developed various flowsheet options to expand the existing milling facility from 1,200 tpd to 2,400 tpd and undertook the design and engineering for these options;
- All the consultants were engaged to provide input and contributed to the development of the Operating Cost (OPEX) and Growth and Sustaining Capital Expenditures (CAPEX); and
- Island Gold reviewed and developed those elements of the Project relating to the geological setting and mineralization, Mineral Resources, mine plan, market studies and contracts and economic analysis. Island Gold and Alamos compiled the overall Technical Report.

2.1 Terms of Reference

All costs are in Q2 2022 Canadian dollars unless otherwise stated.

All units of measurement are in metric, unless otherwise stated.

2.2 List of Qualified Persons

Table 2-1 sets out the Qualified Persons (QPs) responsible for each section of this Technical Report.

Table 2-1 Section Qualified Persons

Section	Description	Qualified Person	Company
1	Summary	All in part	
2	Introduction	Nathan Bourgeault & Raynald Vincent	Island Gold
3	Reliance on Other Experts	Nathan Bourgeault & Raynald Vincent	Island Gold
4	Property Description and Location	Raynald Vincent	Island Gold
5	Accessibility, Climate, Local Resources, Infrastructure and Physiography	Raynald Vincent	Island Gold
6	History	Raynald Vincent	Island Gold
7	Geological Setting and Mineralization	Raynald Vincent	Island Gold
8	Deposit Types	Raynald Vincent	Island Gold
9	Exploration	Raynald Vincent	Island Gold
10	Drilling	Raynald Vincent	Island Gold
11	Sample Preparation, Analyses and Security	Raynald Vincent	Island Gold
12	Data Verification	Raynald Vincent	Island Gold
13	Mineral Processing and Metallurgical Testing	Neil Lincoln	LMI
14	Mineral Resource Estimates	Raynald Vincent	Island Gold
15	Mineral Reserve Estimates	Raynald Vincent & Nathan Bourgeault	Island Gold
16	Mining Methods	Nathan Bourgeault	Island Gold
17	Recovery Methods	Neil Lincoln	LMI
18 except:	Project Infrastructure	Nathan Bourgeault	Island Gold
18.3.3.1 - 18.3.3.3	Paste Backfill	Neil Lincoln	LMI
19	Market Studies and Contracts	Nathan Bourgeault	Island Gold
20	Environmental Studies, Permitting and Social or Community Impact	Colin Webster	Island Gold
21	Capital and Operating Costs	Nathan Bourgeault	Island Gold
22	Economic Analysis	Nathan Bourgeault	Island Gold
23	Adjacent Properties	Raynald Vincent	Island Gold
24	Other Relevant Data and Information	n/a	
25	Interpretations and Conclusions	All in part	
26	Recommendations	All in part	
27	References	All in part	

2.3 Site Visits

The following QPs visited the Island Gold Site as indicated below:

- Raynald Vincent, P.Eng., M.G.P., Exploration Superintendent, Island Gold Mine is employed at the site;
- Nathan Bourgeault, P.Eng., Chief Engineer, Island Gold Mine is employed at the site;
- Colin Webster, P.Eng., Vice President – Sustainability and External Affairs, Alamos Gold Inc, has visited the site on numerous occasions during the previous year with his last site visit occurring April 11th, 2022; and
- Neil Lincoln, P.Eng., Independent Metallurgical Consultant, Lincoln Metallurgical Inc. (LMI), has visited the site several times during the previous year with his last site visit occurring August 9th, 2022.

3 RELIANCE ON OTHER EXPERTS

The Qualified Persons have relied on the input from Alamos and qualified independent consulting companies in preparing this report. This report was prepared using the reports and documents noted in Section 27 “References”.

The Qualified Persons responsibilities were to assure that this NI 43-101 Technical Report met the stipulated guidelines and standards considering that the designs, engineering, and costing in this Report were contributed by Hatch, Redpath, Golder, Paterson and Cooke, Halyard, DRA, Alamos, or other Alamos consultants.

The information, conclusions, opinions, and estimates contained herein are also based on data, reports, and other information supplied by Alamos and the other party sources.

4 PROPERTY LOCATION AND MINING TITLES

4.1 Location

The Island Gold Property is situated 43 km northeast of Wawa, Ontario within the Ontario Ministry of Northern Development, Mines, Natural Resources and Forestry (MNDMNR) Sault Ste. Marie Mining Division. The town of Dubreuilville, a forestry center, is 10 km to the northwest of the mine site. Access to the area is provided by the TransCanada Highway (Highway 17), which continues north from Wawa for 35 km then following Highway 519 to Dubreuilville. The Goudreau Road, an all-weather road, extends east from Dubreuilville for 17 km to the mine site (Figure 4-1).

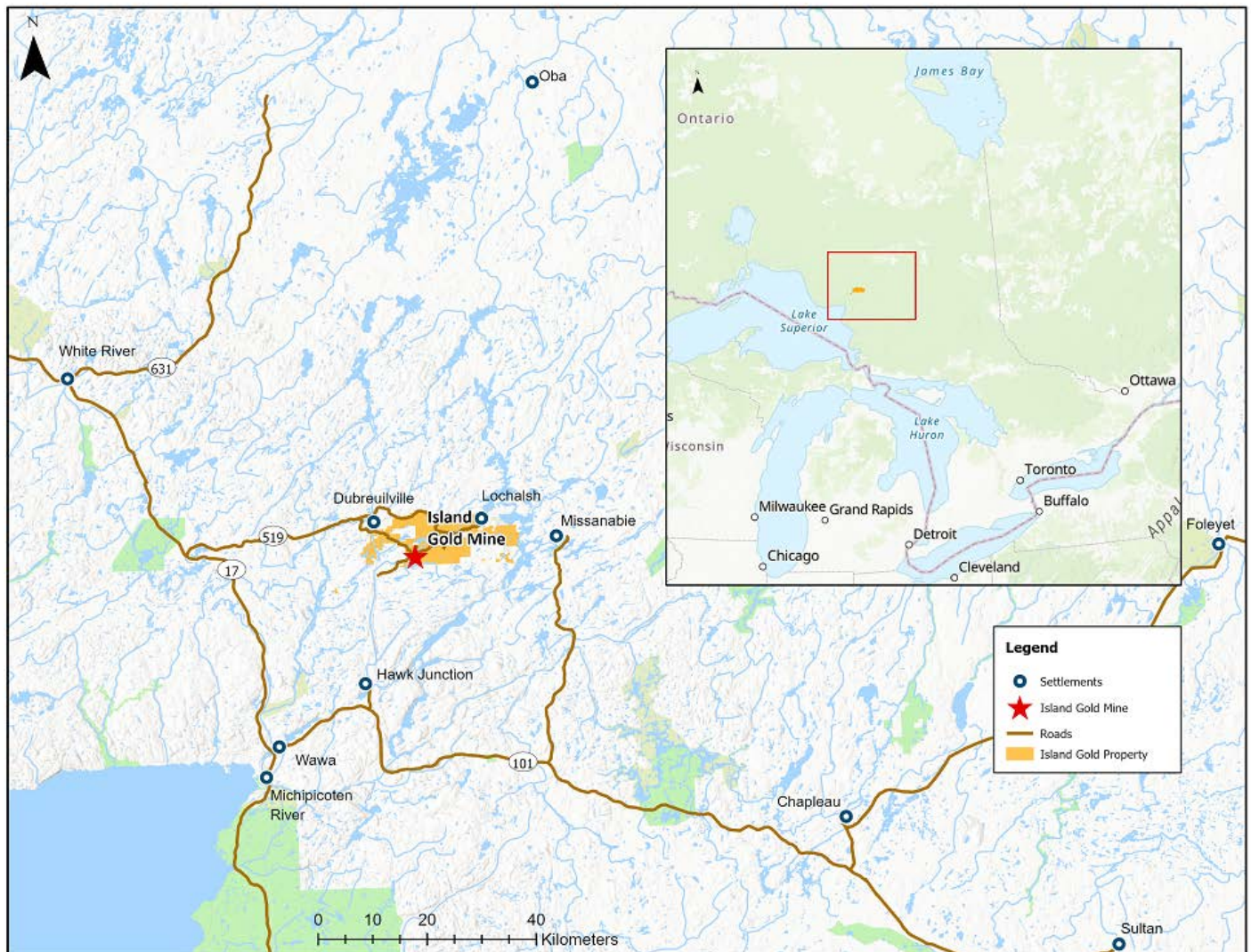


Figure 4-1 Island Gold Mine Location
(Alamos 2022)

4.2 Description of Mining Titles and Recorded Interests

The Island Gold Property (Figure 4-2), which is divided into ten (10) property areas, namely: Argonaut, Edwards, Ego, Goudreau, Goudreau Lake, Island Gold, Kremzar, Lochalsh, Salo, and Trillium, consisting of patented fee simple and/or patented Crown leasehold mining rights and surface rights claims (collectively, the “Patented Claims”), mining licences of occupation (the “MLOs”) and unpatented cell claims (the “Cell Claims”) covering approximately 15,524 hectares. Alamos, itself or through its wholly owned subsidiaries, Trillium Mining Corp. (“Trillium”), and/or 2663200 Ontario Inc. (“2663200 Ontario”), holds 100% of the title and/or interest in the afore mentioned tenures, except for:

- Part of one (1) mining lease, for which it holds 100% below 100 m in elevation, on the Lochalsh property;
- Six (6) patented fee simple claims, for which it owns 100% below 400 m in elevation, and part of one (1) patented fee simple claim for which it owns 100% below 100 m in elevation, both situated on the Goudreau property;
- Four (4) patented fee simple claims, for which it owns 100% below 400 m, situated within the Kremzar property; and,
- Three (3) patented fee simple claims, for which it owns 100% below 400 m in elevation, on the Argonaut property.

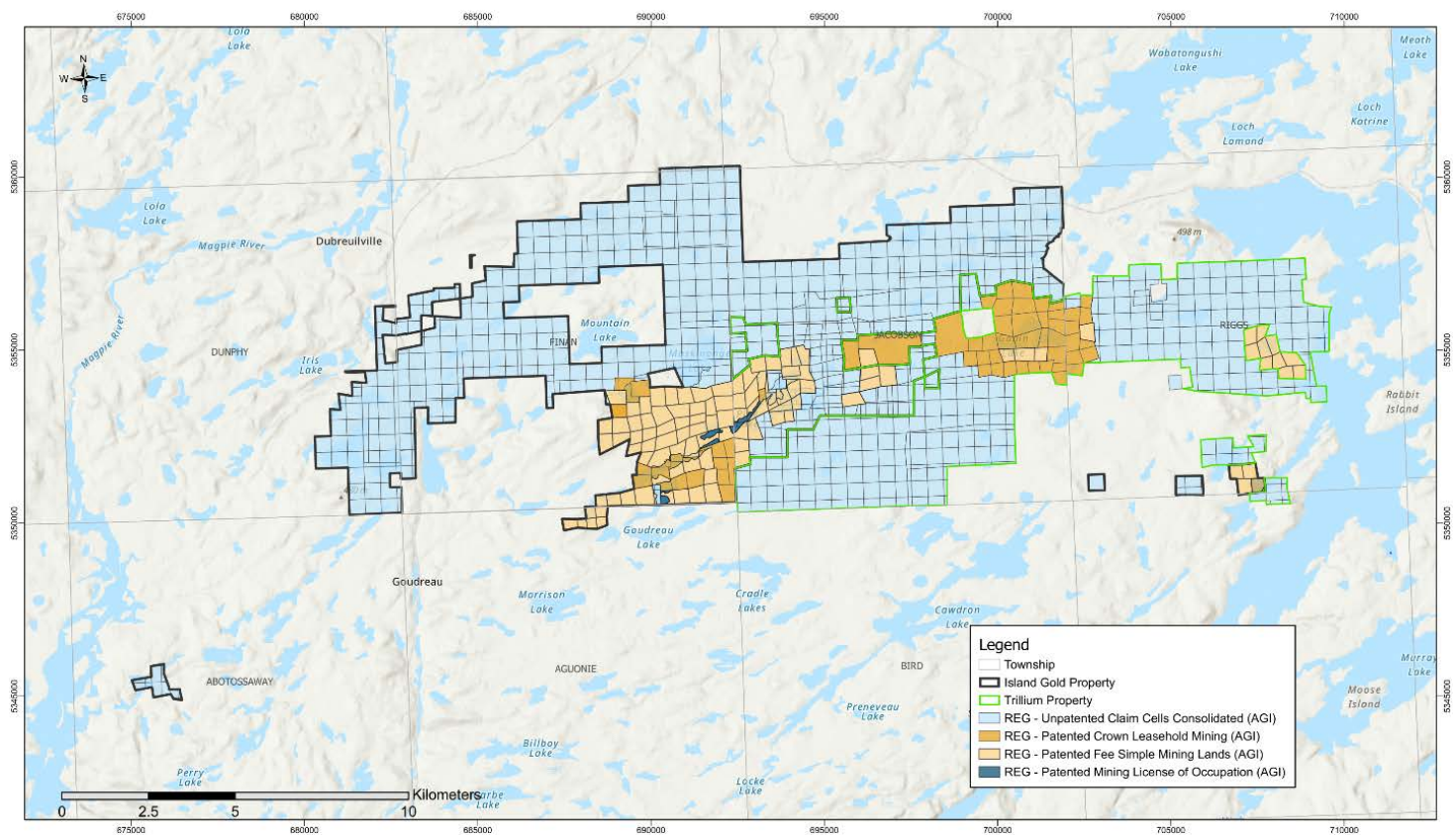


Figure 4-2 Mining Titles Map – Island Gold Property
(Alamos 2022)

Alamos is in receipt of an updated title opinion (Dentons, 2022) for the Island Gold Property dated August 25, 2022, prepared by Dentons LLP. Notwithstanding the foregoing, while Alamos has carried out reviews of the registered title and recorded interests to its Patented Claims, MLOs and Cell Claims, this should not be construed as a guarantee that such title and/or such interests will not be challenged or impugned. Said Patented Claims, MLOs and Cell Claims may be subject to prior unregistered agreements or transfers or native land claims, and therefore title and/or interests may be affected by undetected defects.

Each of the Patented Claims and MLOs are surveyed and do not have annual assessment work obligations. Taxes covering provincial land tax levies and MNDMNRF mining land taxes, along with Crown rents and fees are paid annually to the provincial government to keep the Patented Claims and MLOs in good standing. While Cell Claims are not surveyed but are staked, said Cell Claims do require that minimum assessment work be completed annually either by conducting exploration work or by distributing banked assessment work credit available from contiguous Patented Claims and/or Cell Claims, to facilitate renewal of the expiring Cell Claim on or before its anniversary date.

Alamos has a dedicated land and tenure manager, along with internal procedures and measures to ensure compliance, validity, and good standing of its Patented Claims, MLOs, and Cell Claims.

The mineralized zones, including those containing the Mineral Resources and Mineral Reserves, are located on patented Crown leasehold claims SSM724370, SSM725372, SSM825287, SSM825288, SSM825289, SSM825290, SSM837118, SSM991852, SSM991853, SSM991854, SSM991855, SSM991856, and SSM991857, and on patented fee simple claims SSM2264, SSM2490, SSM2491, SSM2666, SSM2667, and SSM3817. The ramp and waste pad are on patented fee simple claims SSM1776 and SSM1710.

4.3 Ownership of Mineral Rights

All registered titles and recorded interests pertaining to the Patented Claims, MLOs, and Cell Claims relating to the Island Gold Property are owned and/or held by Alamos, Trillium, and/or 2663200 Ontario.

4.4 Mining Royalties

Collectively, the Island Gold Property is subject to different obligations and royalties. Based on the currently defined Mineral Reserves and Mineral Resources, the only royalties to apply are:

- The Lochalsh property is subject to a 3% net smelter returns (NSR) royalty payable to Osisko Gold Royalties Ltd. ("Osisko"). The Island Main and Lochalsh zones, as well as a part of the Island Gold Mineral Resources below the 400 m Level, are located on this property;
- The Goudreau Lake property is subject to a 2% NSR royalty payable to Osisko as to a 69% interest and to Franco Nevada Corporation as to a 31% interest; and
- The Goudreau property is subject to a 2% NSR royalty payable to Osisko.

5 ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE AND PHYSIOGRAPHY

5.1 Access

Access to the area is provided by the TransCanada Highway (Highway 17), which continues north from Wawa for 35 km then following Highway 519 to Dubreuilville which is 10 km to the northwest of the mine site. The Goudreau Road, an all-weather road, extends from east Dubreuilville for 17 km to the mine site.

5.2 Climate

The Island Gold Property is contained within the Lake Superior Regional climatic zone. This area borders the north shore of Lake Superior from Sault Ste. Marie to Thunder Bay and extends inland approximately 40 to 80 km. The climate is described as "modified continental", the modification being due to impacts of Lake Superior. Climatologic records for temperature, precipitation and wind obtained from the Wawa weather station are representative of the actual conditions at the Island Gold Mine site.

The mean annual temperature is about 10°C, with extremes of –51°C and 38°C being recorded. January is the coldest month and July the warmest.

Precipitation is in the range of 980 mm per year, with about 600 mm as rainfall and evaporation at 517 mm/year principally during the summer months. Peak months for rainfall are August and September, with over 100 mm typically in September. Snow cover generally persists from late October to early May, with 50 to 60 mm (water equivalent) occurring monthly.

Approximately 45% of the annual precipitation is lost as runoff, with 50 to 60% of the total annual runoff occurring in April and May in association with spring melt and spring rains.

Average annual wind speeds are in the range of 7 to 15 km/hr. Winds from the northwest through north are most prevalent during the winter, while winds from the southwest through west dominate in the summer months. East winds are infrequent in all months. The percentage of calm is high at 21 to 36%.

5.3 Local Resources

Wawa has a population of approximately 3,500 inhabitants and, Dubreuilville, originally a forestry community, has a population of approximately 635 permanent residents and contains accommodations for mine personnel. The Island Gold Mine is also within a few kilometres of railway lines operated by Canadian National and Algoma Central Railways. Sidings for each of these railway lines are in the villages of Goudreau and Lochalsh.

A hydro-electric power substation, water supply, gravel roads, offices, maintenance buildings, and living accommodations are all available within the mine's general area. Power is connected to the provincial power grid and is supplied by Algoma Power Inc. (API).

Island Gold offers temporary living accommodations and flexible schedules to its non-local employees. Training is offered to maintain a local qualified workforce.

5.4 Surface Infrastructure

The Island Gold Mine infrastructure includes a primary tailings pond, a secondary settling pond, the mill, the Lochalsh ramp and portal, a mine access road, power lines, and an electrical substation. Offices, core logging and storage facilities, a fire hall, and separate mine dries for men and women are also located on the mine site.

The mill, which was originally built in 1988, was designed to process 650 tpd. Since then, the milling capacity has been increased to 850 tpd in 2010, to 900 tpd in 2015, to 1,100 tpd in 2018, and to 1,200 tpd in 2020. The fully permitted tailings area is located at Miller Lake, west of the Kremzar Mine. The tailings and waste rock have been tested for acid mine drainage and are not acid generating. All permits for the existing mining and milling operations are in good standing.

5.5 Physiography

The Project lies in the Superior Province of the Canadian Shield. Topography within the mine area varies from a high of 488 metres above sea level (masl) in the vicinity of Miller and Maskinonge Lakes to a low of 381 masl at Goudreau Creek. Land topographic variation is most strongly developed west and north of Miller Lake in an area of rock knob-controlled terrain. Extreme slopes in the area measure 30-49%. Elsewhere, particularly the at the south, end east of Maskinonge Lake, the terrain is relatively flat.

Periods of intense glacial activity have contributed to the hummocky, rock knelled and largely bedrock-controlled topography, characteristic of the region. Glacial advance from the north deposited a thin mantle of stony sand till over a scoured rock surface (Boissonneau, 1966). The till is generally less than 1 metre thick on the crests of hills but can exceed 5 metres on some slopes and valleys (Gartner and McQuay, 1979).

Water depths in Goudreau Lake vary substantially. The deepest areas, up to 13 metres, occur in the northern portion of the lake, upstream of the first narrows. Downstream of the narrows, lake depths are shallow, generally being less than 2-3 metres. Considerable areas of marginal swamp are associated with the lower portions of Goudreau Lake.

6 HISTORY

The Goudreau – Lochalsh Gold Camp area has been the subject of interest dating back to the early 1900's and has attracted prospectors and mining companies in search of iron ore, gold, and base metal deposits. The Wawa – Michipicoten area has been recognized for its long history of iron exploration which has resulted in the development and production of several iron ore mining operations.

Gold exploration followed shortly thereafter, resulting in several gold discoveries which were subsequently developed and brought into commercial production in the area which would later become the Island Gold Property.

6.1 Work History

Five distinct periods of exploration (Table 6-1) have been identified:

- Period 1: 1901 to 1954;
- Period 2: 1974 to 1991;
- Period 3: 1996 to 2002;
- Period 4: 2003 to 2014; and
- Period 5: 2015 to the present

6.2 Period 1: 1901 to 1954

The initial discovery of gold was made by a group of prospectors at Emily Bay on Dog Lake in Riggs Township in 1901. Up to approximately 1944, prospecting, geological mapping, trenching, shaft sinking, and 1,732 m of diamond drilling were completed to explore various gold prospects.

From 1916 to 1954, Algoma Ore Properties Limited carried out extensive exploration work on the Morrison No. 1 iron sulphide property in Finan Township, defining a sizable iron-bearing deposit. In the later years of this period the deposit was further explored for gold to define 491,000 tonnes grading 1.59 g/t Au.

Ultimately this period is marked principally by various exploration efforts by several companies carrying out surface trenching and diamond drilling on several gold prospects. A total of 4,917 m of drilling was carried out to explore the various prospects. In the overall area during this period, 158 holes totaling 12,066 m of diamond drilling were carried out, exploring for gold and iron.

6.3 Period 2: 1974 to 1991

After an extended period of relatively little interest and activity in the area, exploration was resumed by Amax Inc. and its Canadian division, Canamax Resources Inc. ("Canamax"). Canamax carried out assorted exploration efforts in Finan and Jacobson Townships, consisting of various types of geophysical and geological surveys followed up with diamond drilling.

In 1985, drilling approximately two kilometers south of the Kremzar Mine intersected a series of sub-parallel lenses containing gold mineralization within deformed rocks of the Goudreau Lake Deformation Zone (GLDZ). These lenses are known as the Lochalsh, Island Gold, Shore, and Goudreau Lake Zones.

In December 1988, Canamax's Kremzar project began commercial production. From 1988 to 1990, production from the Kremzar mine was 306,000 tonnes grading 4.80 g/t Au.

Over 1989 and 1990, underground access was established into the Island Gold deposit with an adit from the north shore of Goudreau Lake. A 4,167 tonne bulk sample grading 6.50 g/t Au was extracted and processed at the Kremzar Mill. At the end of 1990, Canamax suspended all operations at both the Kremzar and Island Gold Projects.

During this period, a total of 66,661 m of diamond drilling was completed in 336 holes on various parts of the different Canamax properties.

6.4 Period 3: 1996 to 2002

The Island Gold Property was acquired from Canada Tungsten Inc. by Patricia Mining Corp. ("Patricia"). From 1996 to 2002, various exploration activities on the property including prospecting, surface trenching, geological and geophysical surveys, and diamond drilling were carried out to explore for both Island Gold and Kremzar styles of gold bearing prospects and zones.

During this period of exploration, 20,237 m of diamond drilling was completed in 115 holes at various locations on various properties.

6.5 Period 4: 2003 to 2014

In 2003, Patricia and Richmond Mines Inc. ("Richmont"), entered into a joint venture agreement. Work completed during the joint venture included 72,984 m of surface and underground diamond drilling to test the various zones. On January 1, 2005, Richmont became the operator of the project.

Commercial production at Island Gold began on October 1, 2007. Richmont acquired Patricia 45% interest in December 2008, becoming 100% owner of the property and operations.

Exploration activities ramped up in 2009 with a minimum of 30,000 metres of drilling completed in each of the next several years, increasing sharply to more than 80,000 metres in 2012. This included drilling below the 400 m elevation as part of the Island Gold Deep exploration program which was successful in extending the main C Zone at depth with initial Inferred Mineral Resource being calculated on the high-grade deep C Zone in January 2013.

Over the next year, drilling in the Island Gold Deep sectors from the west, below Lochalsh, to the east and below Extension 1 confirmed the presence and continuity of the deep C Zone and some parallel zones. This drove a substantial increase in Inferred Mineral Resources to 3.6 million tonnes grading 9.07 g/t for 1.04 million ounces of gold as of April 2014. This represented an increase of nearly one million ounces at a 46% higher grade from the end of 2012.

During this period of exploration, Richmont completed a number of ground and airborne geophysical surveys, extensive surface and underground exploration diamond drilling programs as well as underground delineation, and definition diamond drilling for a total of 495,155 m of diamond drilling completed in 2,578 holes on different properties.

6.6 Period 5: 2015 to Present

A large exploration program commenced at the end of 2015 to explore beneath the Island Gold Mine. Directional diamond drilling was used to reach targets at depth which allowed greater

accuracy than conventional drilling techniques. As a result of this program, Mineral Resources were added in the C zone at depth and to the east in the E1E zone in the Extension 2 area. A total of 274,345 m of directional drilling was completed between 2015-2021.

Between 2016 and 2018 a drill program totalling 9,669 m was conducted to explore the Kremzar Mine. During this period further exploration was carried out along the GLDZ east and west of the Island Gold Mine along strike to test the extent of mineralization. In addition to this exploration, in 2017 a 3,302 m condemnation program was completed beneath the claims held by Argonaut Gold at the Magino Mine to the west of the Island Gold Mine to facilitate a claim trade.

Two Master's theses on Island Gold geology were completed in 2019 from the University of Waterloo focussing on structural geology (Jellicoe, 2019) and alteration (Cuifo, 2019).

Other exploration activities undertaken during this period include stripping a 145 m long trench along the GLDZ and 185 soil and gas hydrocarbon samples over the Island Gold property in 3 transects. Geophysical surveys and remote sensing such as LIDAR and VLF surveys were conducted as well.

In 2018, a 2,200 line-kilometre high-sensitivity aeromagnetic and HeliFALCON Airborne Gravity Gradiometer survey was completed over the Island Gold Property by CGG Canada Services Ltd.

A geological modelling project of the Island Gold Mine was completed between 2018 and 2021. The project comprised targeted re-logging of drill core, collection and analysis of geochemical, spectral, and structural data sets, and 3D modelling. The results of the project include an updated understanding of the geological framework of the Island Gold deposit and a mine-scale to property wide targeting framework for ongoing exploration. These concepts are being actively applied within current exploration programs.

In 2019 a regional mapping program on Alamos property surrounding the mine mapped approximately 35 km². During mapping 883 samples were collected of which 141 were sent for lithogeochemical analysis, 402 were sent for gold and metal assaying and 41 were sent for thin sections analysis.

Table 6-1 Summary of Work

	Year	Company	Area	Type of work	# Holes	Metres	Comment
Period 1	1916	Algoma	Morrison #1	Trenching / Diamond drilling	19	1,524	277,000 tons pyrite 35% sulphur
	1925	Patrice Kremzar	Kremzar Property	Prospecting, staking			Staking of part of present mine property, first gold discovery
	1925-1930	Algoma	Kremzar Property	Diamond drilling	6	203	#2, #7, #8 zone work
	1930	M.J. O'Brien Lmt.	Kremzar Property	Diamond drilling	10	1,476	#1, #2 zones, Tent vein
	1935-1936	Cockshutt and Hopkins	Kremzar Property	Diamond drilling	12	611	Local high grade Tent vein result
	1940	O'Brien Gold Mines	Kremzar Property	Diamond drilling	41	2,628	#2, #7, and Tent veins, discovery of Kremzar deposit
	1944	Algoma	Emily Bay	Diamond drilling	38	1,732	Gold bearing iron formation
	1953-1954	Algoma	Morrison #1	Diamond drilling	33	3,896	Up to 3.8 g/t over 30 meters intersected, BP series
	TOTAL			139	159	12,070 m	
Period 2	1976-1979	Amax	Finan, Jacobson, Riggs	Diamond drilling	21	2408	Edward and Island claim
	1983	Canamax	Algoma property	Joint ventures Algoma-Canamax			117 patent claims
	1983-1986	Canamax	Goudreau Project	Diamond drilling	75	10,186	Island zone discovery, Defined Inferred resources of 1.1M tons @ .235 opt Au.
	1985-1986	Canamax	Kremzar Property	Ramp driven and mill constructed			Ramp to 240L
	1987	Canamax	Goudreau Lake	Diamond drilling	77	23,860	Drilling on several gold showings: Breccia zone, #2, #8, Tent Vein, Pine Zone, Morrison #1, Portage Showing, Portal Zone
	1988	Canamax	Goudreau Lake	Diamond drilling	65	18,400	Surface exploration Goudreau project
	1989	Canamax	Goudreau Lake	Diamond drilling	29	5,295	Surface exploration Goudreau project
	1989	Canamax	Kremzar Mine	Beginning of production			Beginning of production 4th quarter 1988
	1989-1990	Canamax	Island Gold Zone	Development			2,062 m of development (ramp, 125 and 140 Levels)
	1990	Canamax	Kremzar Mine	Production			Mine shutdown 1990, 46,798 oz of gold were produced.
	1990	Canamax	Goudreau Lake	Diamond drilling	12	1,528	Surface exploration Goudreau project
	1990	Canamax	Island Gold Zone	Bulk sample			4,167 tonnes of ore grading 6.5 g/t
	1991	Canamax	Island Gold Zone	Diamond drilling	57	4,984	Definition drilling on Island deposit
	TOTAL			180 DH's	336	66,661	
Period 3	1996	Patricia	Island Property	Acquisition of property from Canamax	-	-	Patricia acquired Kremzar, Lochalsh and Goudreau claim groups
	1996-1997	Patricia	Island, Lochalsh zones	Diamond drilling	43	15,610	Surface exploration holes, PL series
	1997	Patricia	Kremzar property	Trenching	-	-	Trenching in 15 areas to expand historic showings
	1997	Patricia	Island Gold project	Technical report	-	-	Technical report on the Island Gold deposit by RPA
	1998	Patricia	Island, Lochalsh zones	Diamond drilling	58	2,206	Surface exploration drilling, north of Island
	2000	Patricia	Island Gold project	Technical report	-	-	Technical report on the Island Gold deposit by RPA
	2000	Patricia	Island Gold Project	Mapping, Geophysics, Diamond drilling	2	289	Exploration drilling work focused on Pine Zone
	2001	Patricia	North Shear	Diamond drilling	5	1,027	Surface exploration holes, PL series
	2002	Patricia	North Shear	Diamond drilling	7	1,105	Surface exploration holes, PL series
	2002	Patricia	Island Gold Project	Relogging			re-logging of 24 diamond drill holes
	2002	Patricia	Island Gold Project	Technical report			Addendum to 2000 technical report by RPA
	TOTAL				115	20,237	

Table 6-1 Summary of Work (continued)

	Year	Company	Area	Type of work	# Holes	Metres	Comment
Period 4	2003	Richmont / Patricia	Island Gold Project	Joint venture agreement			Richmont acquired 55% interest of Island Gold
	2004	Richmont / Patricia	Island Gold Project	Surface and U/G Diamond drilling / Development			Surface 10 holes on North Shear zone,
	2005	Richmont / Patricia	Island Gold Mine	U/G drilling	72	18,330	E1E and CD zone work
	2006	Richmont / Patricia	Island Gold Mine	Surface and U/G Diamond drilling	81	7,906	Surface 26 holes-10903 m
	2007	Richmont / Patricia	Island Gold Mine	U/G drilling / Production	215	38,696	Beginning of commercial production
	2008	Richmont	Island Gold Mine	U/G drilling	134	13,060	Acquisition of Island Gold Mine from Patricia
	2009	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	87	16,126	Delineation and exploration, one hole on Edward
	2010	Richmont	Island Gold Mine	U/G drilling	219	33,107	Delineation and definition of zones
	2011	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	272	66,710	Island: Deep drilling, delineation, and definition drilling, 4 holes on Edward property
	2012	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	323	64,981	Island: Deep drilling, delineation, and definition drilling, 30 holes on Kremzar and other targets
	2013	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	440	96,067	Deep drilling Island zone, delineation, and definition drilling
	2014	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	384	98,175	Deep drilling Island zone, delineation, and definition drilling
	TOTAL				2,578	495,155	
Period 5	2015	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	563	102,997	Deep drilling Island zone, delineation, and definition drilling
	2016	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	722	149,585	Deep drilling Island zone, delineation, and definition drilling
	2017	Richmont	Island Gold Mine	Surface and U/G Diamond drilling	835	139,243	Deep drilling Island zone, delineation, and definition drilling
	2017	Richmont	Island Gold Mine	Trenching / Lidar survey			145 m trench / Lidar survey of the mine and surrounding area ~270 km2
	2017	Alamos Gold	Island Gold/Kremzar	Acquisition			Alamos Gold acquired Richmont Mines in November 2017
	2018	Alamos Gold	Island Gold Mine	VLF survey / Airborne Survey			2,200 km of HeliFALCON Airborne Gravity Gradiometer of IGM aera, VLF on 4 transects over IGM
	2018	Alamos Gold	Island Gold Mine	Surface and U/G Diamond drilling	715	136,565	Deep drilling Island zone, delineation, and definition drilling
	2018	Alamos Gold	Island Gold Mine	Geologic modelling project			3D geological modelling of IGM, completed in 2021
	2019	Alamos Gold	Island Gold Mine	Soil and gas hydrocarbon sampling			3 transects of samples on the IGM property, 185 samples
	2019	Alamos Gold	Island Gold Mine	Surface and U/G Diamond drilling	502	119,333	Deep drilling Island zone, delineation, and definition drilling
	2019	Alamos Gold	Island Gold Mine	2 Masters theses			University of Waterloo thesis: IGM geology and alteration
	2019	Alamos Gold	Island Gold Mine	Mapping			Regional mapping surrounding IGM and GLDZ, ~35 km2
	2020	Alamos Gold	Island Gold/Regional	Acquisition			Alamos Gold acquires Trillium land package of 54.2 m2
	2020	Alamos Gold	Island Gold Mine	Surface and U/G Diamond drilling	294	73,647	Deep drilling Island zone, delineation and definition drilling regional drilling 21 Zone, Eastern Extension and North Zone
	2020-2021	Alamos Gold	Island Gold Mine	Geochemical Sampling Program			Litho-geochemical characterization of IGM, PCA analysis, 12,500 samples
	2020	Alamos Gold	Island Gold Property	Gold in Till Sampling			Island Gold Property, 102 samples
	2020	Alamos Gold	Island Gold Mine	Airborne Mag Survey			Drone Magnetic survey on Original Island Property. 1,515 km of lines covering 65 km2
	2021	Alamos Gold	Island Gold Mine	Surface and U/G Diamond drilling	469	97,243	Deep drilling Island zone, delineation and definition drilling, North Zone
	2021	Alamos Gold	Island Gold/Trillium	Airborne Mag Survey			Drone Magnetic survey SE part Island Property. 3,106 km of lines covering 90 km2
	2021	Alamos Gold	Island Gold/Trillium	Gold in Till Sampling			Till sampling, 340 samples,
	2021	Alamos Gold	Island Gold/Kremzar	Mapping			Regional mapping surrounding IGM, Pine Breccia, Kremzar. 439 Geochem samples, ~55 km2
	2021	Alamos Gold	Island Gold/Pine-Breccia	2 Masters theses			University of Waterloo thesis
	TOTAL				4,100	818,613	

6.7 Historical and Existing Island Gold Mine Mineral Resource and Reserves Estimates

Several Mineral Resource and Mineral Reserve estimates were prepared over the years within the Island Gold Mine area. They are summarized in Table 6-2.

These “Mineral Resources” are historical in nature and should not be relied upon. Additionally, assumptions used to determine cut-off grades are likely to have changed since they were done. Consequently, these “Mineral Resources” cannot be considered as current. They are included in this section for illustrative purposes only and should not be disclosed out of context.

6.8 Historical production from the Island Gold Mine

Since beginning of the underground mining in 2006, and up to December 31, 2021, the Island Gold Mine produced 1,092,384 ounces of gold. Commercial production started in October 2007 and since then approximately 1,068,150 ounces of gold were produced. Details per year are given in Table 6-3.

Table 6-2 Historical and Existing Island Gold Mine Mineral Resources and Reserves Estimates

Year End	Company	Type of Mineral Resources and Reserves	Tonnage (t)	Grade (g/t)	Gold (oz)
2001	Kallio	Inferred	20,610,000	2.35	1,557,173
		Inferred	4,210,000	6.00	812,128
		Inferred	2,034,000	8.30	542,775
2003	RPA/Hubacheck Consulting Geologists, (HCG)	Measured and Indicated	72,000	12.30	108,000
2006	Genivar/HCG	Proven and Probable	1,013,854	8.55	278,711
		Measured and Indicated	454,705	10.26	149,972
		Inferred	610,728	9.96	195,549
2007	Genivar	Proven and Probable	1,058,880	8.39	285,536
		Measured and Indicated	590,167	10.14	192,422
		Inferred	613,635	9.8	193,350
2008	Genivar	Proven and Probable	1,031,187	8.72	289,069
		Measured and Indicated	422,197	10.77	146,254
		Inferred	676,608	9.65	209,985
2009	Richmont	Proven and Probable	927,143	8.86	264,085
		Measured and Indicated	456,353	10.55	154,813
		Inferred	640,614	9.69	199,569
2010	Richmont	Proven and Probable	818,066	6.13	161,197
		Measured and Indicated	796,475	7.36	188,511
		Inferred	604,729	7.14	138,732
2011	Richmont	Proven and Probable	959,523	5.57	171,814
		Measured and Indicated	679,359	7.05	153,920
		Inferred	344,382	6.07	67,238
2012	Richmont	Proven and Probable	785,221	5.60	141,456
		Measured and Indicated	502,910	6.86	110,958
		Inferred	279,569	6.20	55,744
2013	Richmont	Proven and Probable	733,347	6.09	143,506
		Measured and Indicated	739,700	9.81	233,330
		Inferred	3,558,972	9.07	1,037,327
2014	Richmont	Proven and Probable	894,859	6.39	183,726
		Measured and Indicated	733,295	9.29	219,047
		Inferred	3,547,693	8.79	1,002,761
2015	Richmont	Proven and Probable	2,115,634	8.26	561,704
		Measured and Indicated	348,232	6.40	71,685
		Inferred	2,814,742	8.49	768,031
2016	Richmont	Proven and Probable	2,551,006	9.17	752,209
		Measured and Indicated	478,811	5.94	91,427
		Inferred	3,041,836	10.18	995,717
2017	Alamos Gold	Proven and Probable	2,702,858	10.20	886,773
		Measured and Indicated	590,596	5.86	111,253
		Inferred	2,958,204	9.55	908,478
July 2018	Alamos Gold	Proven and Probable	2,789,731	10.69	958,842
		Measured and Indicated	841,244	8.18	221,156
		Inferred	3,673,426	9.99	1,179,810
2018	Alamos Gold	Proven and Probable	3,047,249	10.28	1,007,274
		Measured and Indicated	696,233	8.77	196,213
		Inferred	4,178,079	11.71	1,573,133
2019	Alamos Gold	Proven and Probable	3,642,933	10.37	1,215,135
		Measured and Indicated	878,639	6.51	183,993
		Inferred	5,392,311	13.26	2,297,994
2020	Alamos Gold	Proven and Probable	4,197,369	9.71	1,309,991
		Measured and Indicated	718,044	7.18	165,854
		Inferred	6,915,259	14.43	3,208,279

Table 6-3 Island Gold Mine Production by Year

Year	Tonnes	Head grade (g/t Au)	Gold Recovery (%)	Ounces Produced
2006	41,521	4.80	93.45	3,255
2007	159,493	6.02	94.36	29,281
2008	162,158	7.74	95.83	39,224
2009	223,345	5.85	94.52	39,794
2010	246,712	6.03	95.49	43,762
2011	255,103	6.05	95.91	49,443
2012	247,833	5.47	96.45	41,952
2013	239,766	4.57	96.06	34,691
2014	230,828	5.91	96.26	42,042
2015	242,137	7.31	96.75	52,835
2016	297,757	9.02	96.52	81,799
2017	338,603	9.36	96.82	97,932
2018	369,767	9.20	96.16	105,823
2019	401,276	11.85	97.05	150,355
2020	386,591	11.62	96.75	138,987
2021	435,297	10.35	96.01	141,209
Total	4,278,187	8.26	96.29	1,092,384

Notes:

- Bulk sample in 2006
- 20,983 ounces produced prior to commercial production (October 2007).

7 GEOLOGICAL SETTING

7.1 Regional Geology

The Island Gold Project is in the Michipicoten Greenstone Belt (MGB) which is part of the Wawa Subprovince within the Archaean Superior Province (Figure 7-1). The MGB is approximately 140 km long and up to 45 km wide. The metamorphic grade of the subprovince is dominantly greenschist with amphibolite facies seen locally or proximal to intrusions.

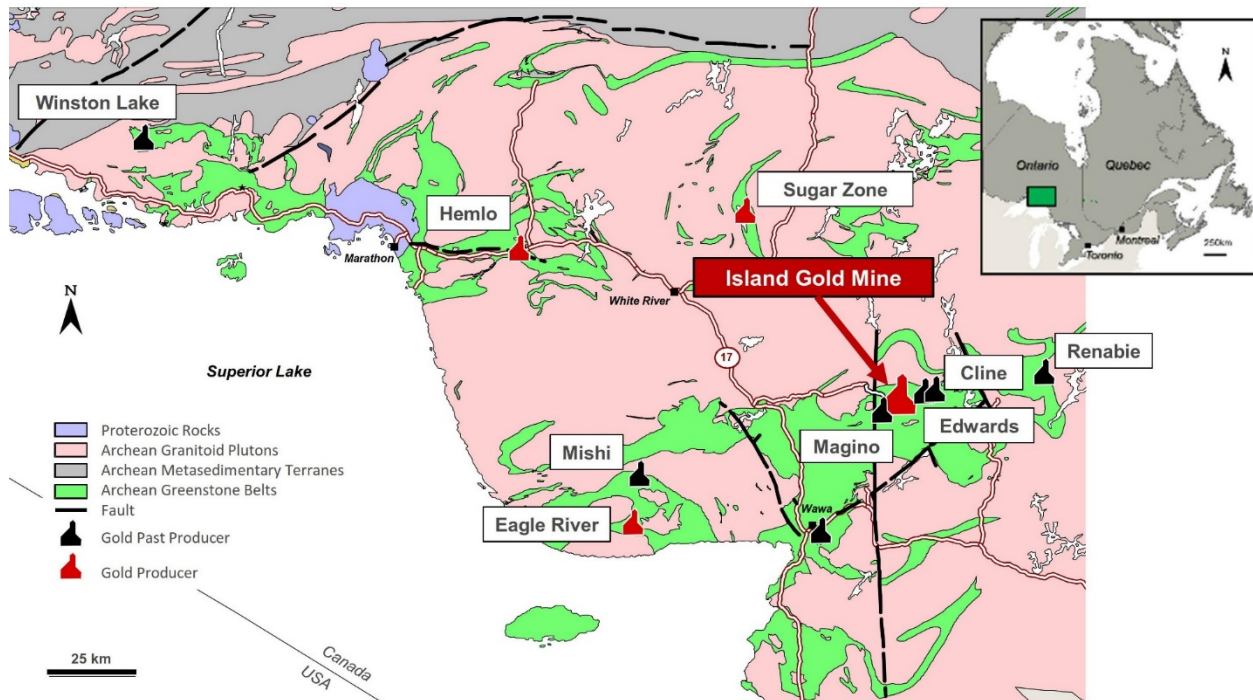


Figure 7-1 Geological Map of the Western Part of the Wawa Subprovince
(Alamos 2022)

The MGB comprises three bimodal (rhyolite-basalt) volcanic cycles capped by iron formations. Rocks vary in age from 2,889 Ma for the Hawk Assemblage (Cycle 1) to 2,750 Ma for the Wawa Assemblage (Cycle 2), and to 2,700 Ma for the Catfish Assemblage (Cycle 3). Shearing along contacts has obscured the original relationship between the cycles.

A regional deformation zone called the Goudreau Lake Deformation Zone (GLDZ) is situated throughout the area at the interface of the Wawa and Catfish Assemblage cycles. It is a north-easterly trending structure which has been traced along strike for 30 km with a width of 4.5 km and is believed to be the main control of gold mineralization for the Project area. It is a high angle oblique-slip fault zone with an overall dextral movement cutting stratigraphy at a shallow angle.

The Island Gold Mine occurs as a sequence of stacked east-northeast striking, steeply dipping, and subparallel zones of gold mineralization within the GLDZ.

7.2 Geology of the Island Gold Mine Area

The lithological setting of the Island Gold Mine area is characterized by a mafic to felsic volcano-sedimentary sequence intruded by multiple intrusions. The mine sequence of rocks is folded in an upright orientation with sub-vertical axial planes trending east-northeast and fold axes plunging very shallow to the east-northeast. On a regional-scale, the property is cut by brittle-ductile east-northeast, northeast, and southeast faults, and an array of north-northwest diabase dykes and sub-parallel brittle faults.

The Island Gold Mine is stratigraphically positioned in the upper portion of the Wawa Assemblage, on the northern limb of the regional-scale Goudreau Anticline. The hinge is south of the area displayed in Figure 7-2. This assemblage is dominantly composed of felsic volcanic rocks of various facies of tuffs and lavas.

Around the periphery of the mine area are felsic intrusions ranging in size from one to several kilometres across. The Webb Lake Stock lies to the north of the deposit (Figure 7-2). Narrow quartz-feldspar porphyry dykes and minor mafic volcanic rocks are present within the dominantly felsic volcanic sequence. North-trending diabase dykes crosscut all stratigraphy (in purple in Figure 7-2). The volcanic units generally strike at N70°E to N90°E strike with a subvertical to steep dip.

The past producing Kremzar mine is hosted in the lower portion of the Catfish Assemblage (Figure 7-2), within a sequence of massive and pillowed magnesium and iron-rich tholeiitic flows. The mafic flows of the Catfish Assemblage face north and are cut by the Herman Lake nepheline-syenite intrusive complex in orange and the Maskinonge Lake granite in pink in Figure 7-2.

7.3 Alteration and Mineralization

Alteration within the Island Gold deposit is characterized by the presence of silica, sulfides, white mica, biotite, and carbonate. (Cuifo, 2019). Pyrite content ranges from 1% to 11% in disseminated form and less commonly as millimeter scale discontinuous stringers. Pyrrhotite and chalcopyrite can be present but are uncommon. Alteration minerals which are not consistently associated with the ore body may include tourmaline, apatite, epidote, chloritoid and local garnets in the deeper levels. The alteration occurs primarily in linear south dipping envelopes which can pinch, swell, and vary in thickness between decimeter scale to over 15 m in thickness. There appears to be a gradual change in alteration with depth as silicification becomes more dominant. The alteration envelopes are termed Alteration Package Island in mine geology nomenclature.

The alteration envelopes generally possess a strong degree of deformation. Structural elements such as a deposit-wide strong penetrative foliation, shear fabric, boudinaged structures, crenulation cleavage, mylonitization and smaller scale thrust faults are noted to be present throughout the Island Gold Mine.

Quartz veins commonly bear visible gold in the form of aggregates, disseminated fine grains or along chlorite-sericite slickensides within the veins (Jellicoe, 2019). Metallurgical studies indicate that free gold flakes are typically less than 25 microns in diameter. The quartz veins host most of the gold, however, the surrounding altered rock within the zone can host gold mineralization. The degree of veining appears to change at depth, transitioning from a stringer style quartz-carbonate vein on millimeter scales to larger scale veins which can be over 4 metres in width.

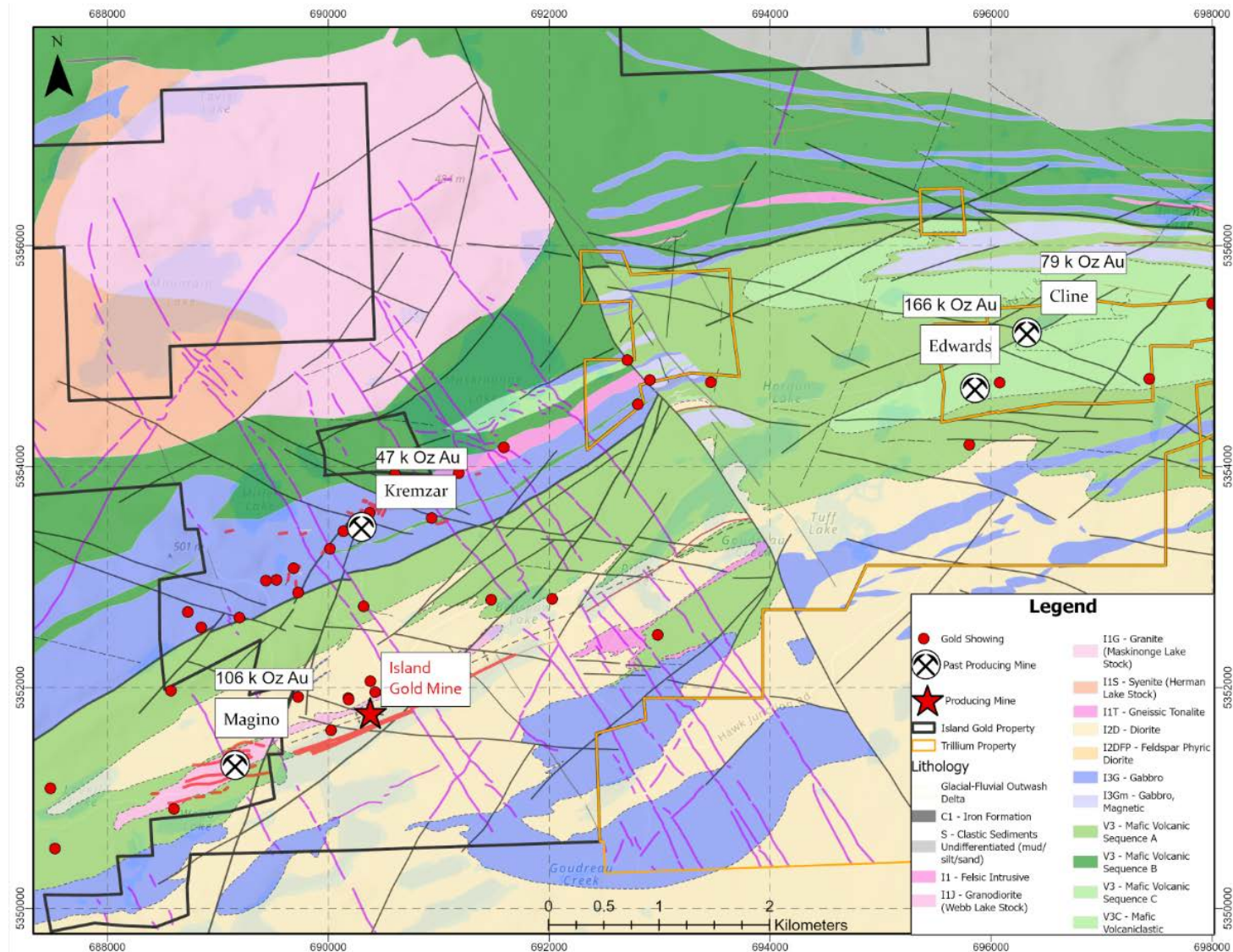


Figure 7-2 Geological Map of the Island Gold Mine Area, Including Former Gold Mines and Past Production
(Alamos 2022)

Five main types of quartz veining have been observed (Jellicoe, 2019) and are listed below in order of timing from oldest to youngest.

- V_{GD} veins occur exclusively in the Goudreau Domain and their timing relationship to the V1 - V4 veins is unknown. They are subhorizontal extensional veins composed of quartz-carbonate with minor sericite and frequently have tourmaline cores. The veins are often extremely folded and vary in thickness from 1 cm to 40 cm. Visible gold is common and is in the form of nuggets. The gold grade can be discontinuous and varies dramatically throughout the vein. The very high grades produced in the Goudreau Domain are derived from the fold noses where the gold is concentrated. These veins have been found to crosscut all units in the Island Gold deposit.
- V1 veins are parallel to foliation and are the dominant auriferous veins. They are smoky grey to milky white in colour and are most often laminated but can be massive. Wall rock, strongly altered by sericite, silica, and chlorite, defines the lamination within the quartz veins. The V1 veins are the primary host of visible gold. These veins can be folded, boudinaged or brittlely deformed.
- V2 veins are found with the altered wall rock and range from millimeters to centimeters in thickness. They are discontinuous crack-seal veinlets closely spaced together. Most often they are parallel to the foliation of the ore body and host minor gold associated with pyrite.
- V3 veins occur as boudin necks, extensional veins that crosscut the ore zones and as en echelon veins. They are most often noted within the V1 veins forming a “ladder appearance” within the vein but they may also extend past the vein boundary. The V3 veins are composed of white quartz, tourmaline, and carbonate. Typically, they are unmineralized but in some rare cases have produced low grade gold values.
- V4 veins crosscut all other vein sets and established fabrics. They are primarily composed of euhedral tourmaline needles with white quartz. These veins are always barren of gold.

7.4 Island Gold Deposit

The Island Gold deposit is composed of multiple, stacked, south dipping lenses. The mineralized corridor expands from 50 m wide in the upper levels to over 150 m wide at depth. The zone's dip varies from -50° to -90° south. Locally, north dip reversals occur but are not common. Instances of offset or folding are also observed. Around the 400 metre elevation there is a shallow dipping southern inflection of the mineralized zones. It is not yet clear if this inflection is related to a fault, a shear zone, or a fold. This inflection point is the division of what is locally referred to as the Upper Island Gold Mine and the Lower Island Gold Mine (Figure 7-3 and Figure 7-4).

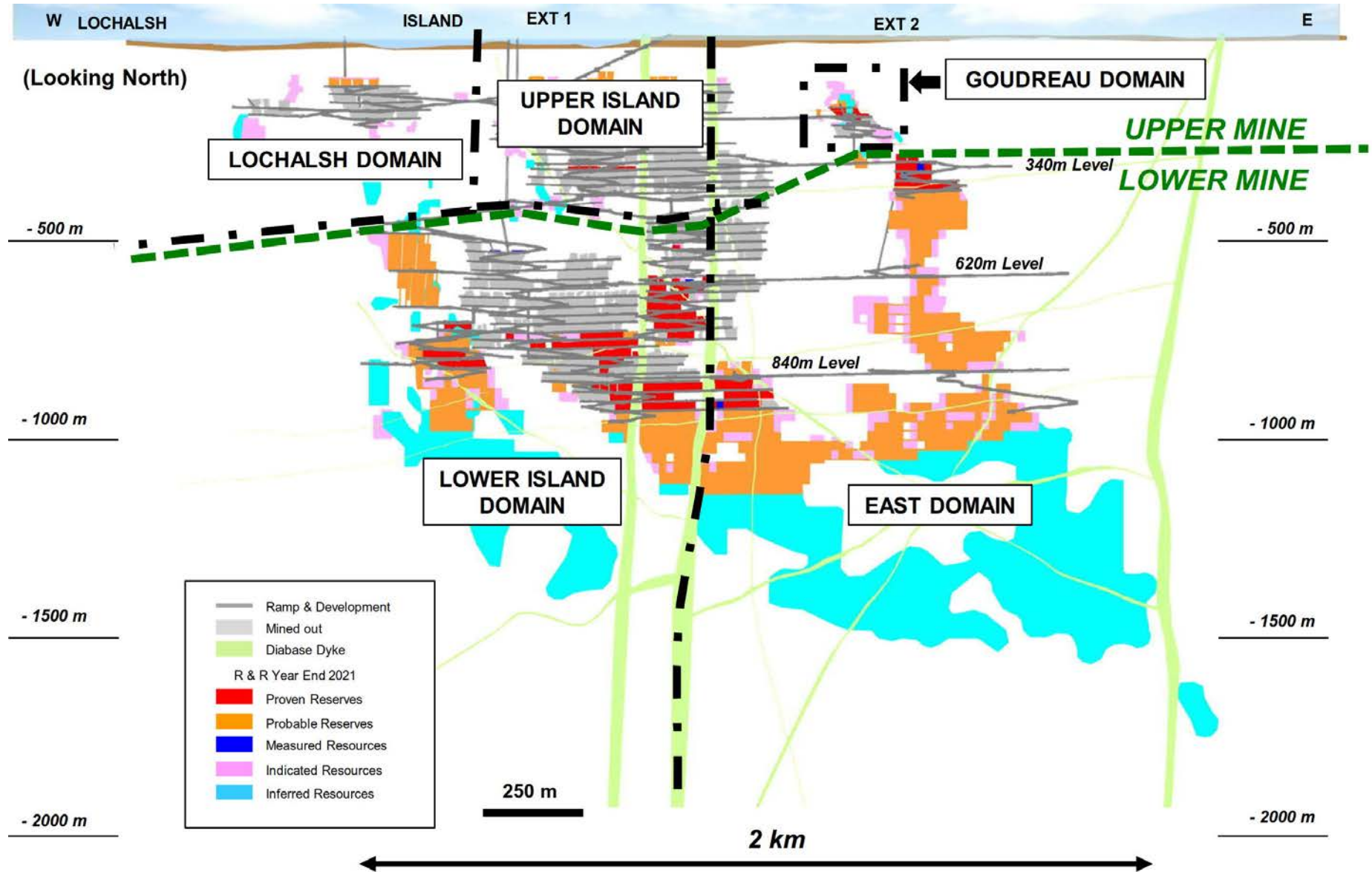


Figure 7-3 Longitudinal View of Domains and Mineral Resources and Reserves with Existing Infrastructure

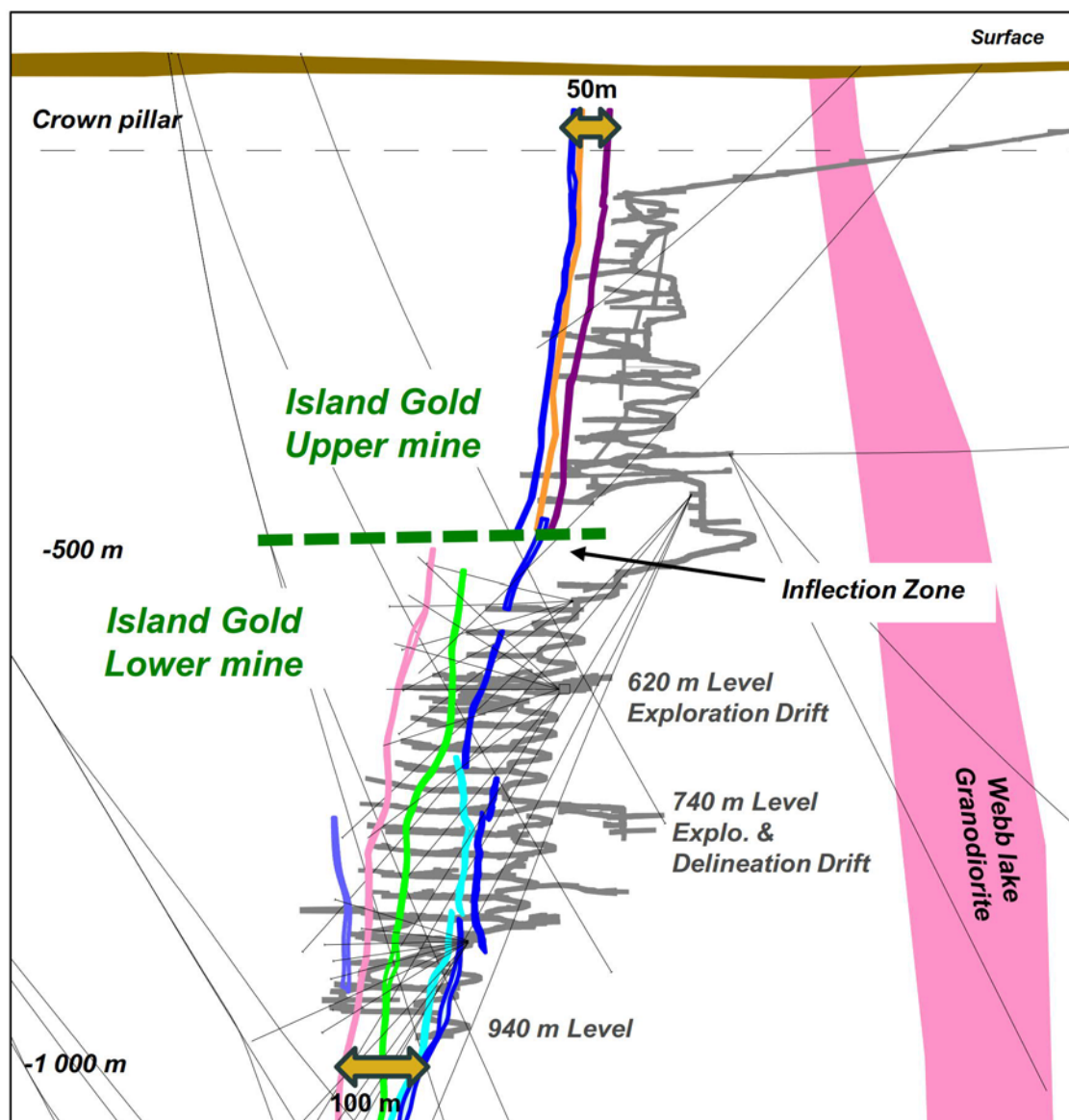


Figure 7-4 Cross Section Showing the Inflection to the South of Mineralized Zones

7.4.1 Upper Island Gold Mine

Three main domains are found in the Upper Island Gold Mine, the Upper Island Domain, the Lochalsh Domain, and the Goudreau Domain.

7.4.1.1 Upper Island Domain

Five mineralized zones have been recognized in the Upper Island Domain which include, from north to south, E2, E1E, D1, D, and C. The relationship between the different zones in their respective domains can be complex, they pinch and swell, merge, and anastomose. The complexities of the zones are well documented by sill development and drilling. Most of the Upper

Island Domain has been mined out. This domain also includes part of what is often referred to as Extension 1. A diabase dyke crosscuts the mineralized zones however, there is no offset therefore both sides of the dyke are grouped into the same domain.

7.4.1.2 Lochalsh Domain

The Lochalsh Domain has a 450 m strike length between depths of 100 m to 450 m below surface. The geology, mineralization, and alteration of the Lochalsh Domain are similar to the Upper Island Domain. Four mineralized zones have been recognized in the Lochalsh Domain which include, from north to south, E2, E1E, D, and C.

Drilling between the Upper Island Domain and Lochalsh Domain has shown that the zones reduce in thickness and in grade toward the west of the Island Gold deposit. Drilling at the same easting, lower in elevation, indicates economic mineralization in the area but at deeper levels that fall within the Lower Island Domain.

7.4.1.3 Goudreau Domain

The Goudreau Domain is situated between 15,600E and 15,900E approximately 200 m north of the main mine structure which hosts the other domains. There are seven zones in the Goudreau Domain, GD2, GD3, GD6, GD7, and GD9 are vertical and GP2 and GP5 are horizontal. The vertical zones are stacked, steeply-dipping to vertical mineralized lenses consisting of quartz-sericite-carbonate-pyrite alteration envelopes and contain high-grade gold in quartz veining.

GP2 and GP5 are interpreted to be very folded flat lying gently east dipping zones which may link the vertical zones. The majority of the gold comes from decametric gold-bearing quartz veins. Alteration surrounding these flat lying zones is limited to a maximum of 0.5 m surrounding the quartz vein or is altogether absent. These veins are hosted within volcanics or a quartz diorite intrusion which they crosscut. The horizontal nature of GP2 and GP5 have been confirmed by mining development.

7.4.2 Lower Island Gold Mine

The majority of the Mineral Resources and Mineral Reserves are within the Lower Island Gold Mine in the Lower Island Domain and East Domain.

7.4.2.1 Lower Island Domain

Nine mineralized zones have been recognized in the Lower Island Domain, they are, from north to south, E1E, D, D1, C, B, G, GNW, G1, and STH. The Lower Island Domain also includes the lower part of what was often referred to as Extension 1. A diabase dyke crosscuts the mineralized zones however, there is no offset, therefore both sides of the dyke are grouped into the same domain. It is bounded by a north-south trending vertical diabase dyke in the east and is open on the west.

The C zone is the most laterally and vertically continuous of the nine different zones of the Lower Island Domain. It contains by far the most Mineral Resources and Mineral Reserves in this domain and has been successfully mined since 2014. Other zones have been mined with some success but demonstrate significantly less lateral and vertical continuity. Only small portions of the other eight zones form part of the Mineral Reserves, with the remainder categorized as Inferred and Indicated Mineral Resources. The C Zone Mineral Reserve currently being mined below the 450 metre Level has a larger average width and a higher average grade than what was previously mined in the upper part of the mine. From the 760 metre Level to 900 metre Level in the main ramp the zone dramatically increases in width and grade. The width reaches a maximum of approximately 15 m with smoky grey quartz veins as large as approximately 5 m in width.

Within the domain is a thick assemblage of intermediate volcanic and/or intrusive rocks (diorite/crystal tuff, lapilli tuff, ash tuff and possible flows) and often mineralized zones follow the contacts between these units. Mafic dykes/plutons emplaced pre-mineralization are found throughout the mine and can be mineralized but it is rare. Several later mafic dykes of a different composition are present, usually between 0.1 m and 1 m wide and can conform to the stratigraphy and proliferate within the mineralized corridor. A dyke of uncertain composition ranging from 0.3 m to 2 m in width is observed in several locations in the mine and crosscuts the zone but does not seem to offset it.

7.4.2.2 East Domain

East Domain is composed of two defined zones, the E1D to the north and the E1E to the south. Other potential zones have been noted in this domain but have yet to demonstrate the continuity laterally or in elevation to be able to confirm their presence. The E1E zone is open at depth and reaches as high as the 100 metre Level. On the western limit of this domain is a north-south trending vertical diabase dyke however, it is still open to the east. East domain is considered a different domain because the zone is offset approximately 10 to 30 m north from the Lower Island Domain across the diabase dyke. The zone also has differing geological characteristics, such as zone width and grade, from the Island C zone, which affirms the domain division. The E1E zone in East Domain host the most Mineral Reserves and Mineral Resources at the Island Gold Mine and it is where most of the underground and surface diamond drilling is focused.

The extent of the E1E is confirmed for 1.2 km laterally. Above the 840 metre Level the economically viable areas of the E1E are concentrated into narrow corridors that extend vertically but are rather short laterally. Below the 840 metre Level the zone opens to become more laterally extensive and continuous as can be seen by the Mineral Reserve and Mineral Resource block shapes (Figure 7-3).

Alteration is similar between the Lower Island Domain and East Domain however, in general there are fewer large veins. Veining above the 840 metre Level remains at less than 1 m and more frequently centimeter to decimeter width. Below the 840 metre Level in the E1E zone, the mineralized corridor increases in width and increases in the amount of visible gold, grade and quartz veining are observed.

7.5 Other Gold Zones

Several other gold zones and showings occur throughout the property area, mainly north of the Island Gold mine where multiple mineralized zones have been identified north of the Webb Lake Granodiorite. These zones are north dipping shear zones that strike parallel to the mine trend. They are interpreted to be within the north limb of a deposit scale antiform, which accounts for the change in dip. Exploration in this northern domain was dominant during Patricia era of exploration (1997-2004) and has received new attention as Mine Exploration and Regional exploration projects at Island move further from Island Main Zone. The Northern Domain was an area of focus for regional exploration drilling in 2021 and 2022.

7.5.1 North Shear

The North Shear Zone is generally located along the northern contact of the Webb Lake Granodiorite Stock and has been traced over a 1 km strike length between sections 14,200E and 15,300E. The North Shear structure is marked by a broad brittle to ductile deformation zone which dips from -75° to -80° north. The zone contains quartz-tourmaline stringer veining, stockwork quartz which occurs within highly strained and crenulated felsic volcanic and Webb Lake granodiorite host rocks. Alteration is characterized by strong silicification, sericitization and pyritization. The gold mineralization is hosted in chlorite-quartz-tourmaline stringer and stockwork veining containing visible gold, minor pyrite, and trace chalcopyrite. Mineralized zones can be up

to 50 m wide in places, grading 0.5-1.0g/t Au. In 2005, the 140 Level vent drift development cut through the North Shear. The shear is observed dipping 65° to 70° north and is believed to follow the contact between the Webb Lake Granodiorite Sill and a massive feldspar porphyry unit. Recent drilling has confirmed the presence of the wide low grade mineralized zone and extended it to ~450 m below surface. Higher grade structures exist near both the north and southern contact of the wider zone, but drilling has not established significant their continuity.

In 2004, the Kallio Resource estimate of the North Shear zone is sited as 229,000 tonnes at 6.57 g/t Au totaling 48,429 ounces of gold (5 g/t Au cut-off) in the NI 43-101 RPA technical report on the Island Deposit (RPA, 2004).

These “Mineral Resources” are historical in nature and should not be relied upon. Additionally, assumptions used to determine cut-off grades are likely to have changed since the estimate was done. Consequently, these “Mineral Resources” cannot be considered as current. They are included in this section for illustrative purposes only and should not be disclosed out of context.

7.5.2 Zone 21

In April 1997, Patricia intersected a series of gold-bearing, quartz veins in drill hole PL-21 at a vertical depth of 250 m. The intersection averaged 52.2 g/t Au over 19 m with erratically distributed values and visible gold within a strongly deformed feldspar porphyry. Drilling in 2021 and Q1 2022 determined the mineralized zone comprised a single narrow 10-20 cm smoky quartz vein, with frequent VG has been identified ~20 m into the hanging wall of the Northern Shear. Multiple intercepts returned more than 100 g/t Au over 0.3 m intercepts. The vein has demonstrated along strike continuity between 14,425E to 14,525E (approx. ~100 m). Vertically, it shows 60 m continuity.

7.5.3 Kremzar Mine Property

The Kremzar Deposit occurs 1,200 m to the north of the GLDZ on a northwest-trending fault structure at 120° azimuth that dips at -75° to the southwest in what has previously been termed the northern splay of the GLDZ on the mine property. The Kremzar Mine, which was in production from 1988 to 1990, produced 306,000 tonnes at a grade of 4.8 g/t totaling 47,000 ounces of gold. A historic Mineral Reserve at the time of mine closing (Derry, et. al.) estimated a Proven and Probable Mineral Reserve of 181,944 tonnes at a grade of 6.27 g/t Au (non NI 43-101 compliant).

These “Mineral Resources” are historical in nature and should not be relied upon. Additionally, assumptions used to determine cut-off grades are likely to have changed since the estimate was done. Consequently, these “Mineral Resources” cannot be considered as current. They are included in this section for illustrative purposes only and should not be disclosed out of context.

The alteration style of the Kremzar zones is characterized by strong envelopes of biotite, carbonate, and silica in widths from 1 m to 3 m. Mineralization within the envelopes ranges generally from 2% to 5% disseminated pyrite/pyrrhotite. Quartz carbonate vein development is present as grey blueish siliceous bands, sinuous lenses and broad pervasive silicification. The siliceous bands can also extend into the footwall and hanging wall. Fourteen other historic mineralized zones occur along this trend developing at or near the southern contact of an east-west trending gabbroic sill. Drilling in 2016 indicated that gold extends at deeper levels below the Kremzar Mine as a zone grading 9.71 g/t Au over 8 m was intersected in drill hole KZ-16-02 approximately 600 m below surface.

7.5.4 Portal Zone

Near surface, Island Gold ramp intersects a series of east-west-striking quartz-ankerite veins called the Portal Zone. Locally, these veins assayed up to 20 g/t Au over 1 m and averaged 4.0

g/t Au over 11 m in ramp wall samples. A series of four short holes totaling 1,227 m were drilled along strike to the east and west of the ramp portal without extending the zone.

7.5.5 Portage Showing

The Portage showing occurs along the Bearpaw Lake portage and consists of a series of quartz veins which occur within a deformed feldspar porphyry. Quartz veins were located along the stream bed as well as in nearby trenches dating back to the 1920's and 1930's. Grab samples have averaged 2.3 g/t Au in past sampling. Limited shallow drilling to the north and northeast of the zone under Pine Lake encountered only weak alteration structures and negligible gold values. Canamax drilling encountered gold in the immediate area. Drill hole 061-02-23 east of north Bearpaw Lake intersected 95.9 g/t Au over 1.4 m, drill hole 061-03-24 intersected 9.9 g/t Au over 0.6 m north of Pine Lake in Jacobson Township and drill hole 061-02-66 intersected 1.7 g/t Au over 0.7 m.

7.5.6 Pine Zone and Breccia Zone

The Pine Zone and the Breccia Zone are located north-east of Bearpaw Lake in Jacobson Township and adjacent and east of the Maskinonge Lake Fault (MLF). The MLF a major structure trending north to northwest at 320° azimuth with a geophysical inferred sinistral strike slip movement of over a kilometer.

The Pine Zone is a folded sulphide-oxide iron formation which is part of the Goudreau Iron Range. It contains gold proximal to the MLF. Surface trenching and a 21 drill hole program have defined a small tonnage of sub-economic gold mineralization.

The Breccia Zone is a silicic fault breccia to quartz stockwork zone which crosscuts the stratigraphy at right angles and can be traced over several kilometres along the Maskinonge lake Fault. Limited drilling on the showing has encountered 6.0 g/t Au over 1.0 m. 2021 and 2022 field programs, consisting of mapping and channel sampling are working to identify the near-surface potential at Pine-Breccia. A Masters thesis, investigating the kinematics, timing, and relationship to gold mineralization of the Maskinonge Lake Fault is currently ongoing.

7.5.7 Morrison Number One Zone

The Morrison #1 iron deposit is an oxide-sulphide-carbonate iron formation which is part of the larger Goudreau Iron Range. In 1985, Canamax drilled a short four hole program totaling 375 m with the best gold value being 18.7 g/t Au over 1 m. In 1954, Algoma drilled a hole that returned 2.7 g/t Au over 30.5 m. Gold mineralization is related to quartz-carbonate fracture fillings within the carbonate facies of the iron formation in a hinge zone of a tight Z fold.

8 DEPOSIT TYPES

The Island Gold Mine is an Archean orogenic lode gold deposit. It is a structurally hosted quartz-carbonate vein system situated within the Goudreau Lake Deformation Zone (GLDZ), a major regional brittle-ductile structure. The host terrane is a sequence of felsic to intermediate volcanic and intrusive rocks of the Wawa Assemblage which are in the greenschist to amphibolite metamorphic range as is common for this type of deposit.

High strain zones associated with the GLDZ have the tendency to develop at variable scales along lithologic unit contacts where complex geology and related competency contrasts can control stress patterns and facilitate shearing and the consequent development of dilatancy zones and concomitant quartz carbonate vein formation. It is generally accepted that these Archean orogenic lode gold deposits are related to compressional and transpressional tectonics and the associated metamorphic dewatering and devolatilization of magma processes from which the gold bearing fluids are derived.

Gold mineralization in the Goudreau-Lochalsh area is not restricted to any rock type with the general exception of the late intruding north-west trending Matachewan diabase dykes which show no evidence of mineralization. Deposits may be hosted by one or several rock types, with past-producing mines and numerous other gold occurrences in the area exhibiting a close spatial association with felsic, intermediate, and even mafic intrusive rocks. East of the Island Gold Mine, in Jacobson Township, the past producing Edwards and Cline Lake gold mines are associated with felsic intrusive complexes and dykes. The past producing Magino Mine to the immediate west of the Island Gold property is hosted by the Webb Lake stock, a trondhjemite intrusive. The past producing Kremzar Mine, located on the Island Gold property, is hosted by a regional gabbroic sill.

Mineralization in the Goudreau Camp occurs along a 30 kilometre strike length of the GLDZ which transects the Island Gold property area in a roughly east-west direction. The GLDZ is a major regional deformation structure, and it is believed to be the main control on gold mineralization for the area. The GLDZ and subsidiary splays have been subdivided into four structural domains (Southern, Northern, Eastern and Western) based on the style of deformation, lineation patterns, and the orientation and sense of shear displacement on sets of shear zones. The Island Gold mine mineralized zones are within the Southern domain of the GLDZ (Heather and Arias 1992). Most mineralization in this domain is hosted by quartz veining and/or shear zones with an orientation of 075°. The zones with this orientation are roughly parallel with the overall deformation zone and are considered to have formed along shear planes related to the dextral oblique slip movement of the GLDZ and are potentially localized near fold hinges and intersecting oblique structures.

Typical alteration mineralogy associated with gold deposits of the Goudreau Camp includes variable amounts of carbonatization (Fe-carbonate $\pm$ calcite), silicification, sulphidization, biotitization, sericitization, feldspathization, and chloritization. Deposits and gold occurrences with a felsic rock association are generally associated with a quartz-sericite-pyrite $\pm$ pyrrhotite alteration package. Deposits and occurrences hosted by mafic host rocks, such as the Kremzar Mine and the historic showings along this trend are generally altered to biotite, Fe-carbonate, pyrrhotite $\pm$ pyrite, quartz, and minor K-feldspar. Chloritization is common throughout the belt. Gold presence in the Goudreau-Lochalsh area is primarily associated quartz stringers, fracture fillings and veins. Gold can be associated with pyrite disseminated in alteration envelopes but generally only in low grade levels.

9 EXPLORATION

9.1 Introduction

The deep exploration diamond drilling program started in October 2015, and, at the end of 2021, a total of 179,759 m was completed from underground diamond drilling and a total of 274,345 m was completed from surface diamond drilling. (Table 9-1).

The underground and surface exploration programs have added, since December 31, 2015, close to 3.7 million ounces of gold to the mineral inventory, net of mining depletion. Measured and Indicated Mineral Resources increased by 215,000 ounces of gold, Inferred Mineral Resources increased by 2,686,000 ounces of gold and Mineral Reserves increased by 777,000 ounces of gold in the seven-year period. The discovery cost has been approximately CAD \$18 per ounce during this period.

The results of these programs up to December 31st, 2021, were used in the December 31, 2021, Mineral Resources estimate and have been incorporated into the Phase 3+ Expansion Study. Drilling results have also been published on a regular basis since 2016 in press releases available on SEDAR.

Table 9-1 2015 – 2021 Summary of Diamond Drilling

Type of diamond drilling	2015 (m)	2016 (m)	2017 (m)	2018 (m)	2019 (m)	2020 (m)	2021 (m)	Total (m)
U/G Exploration	23,552	28,712	30,813	25,378	24,462	23,522	23,320	179,759
Surface Exploration	14,306	54,399	51,152	57,725	47,608	21,415	27,740	274,345
TOTAL	37,858	83,111	81,965	83,103	72,070	44,937	51,060	454,104

9.2 2021-2022 Exploration Drilling program

After a hiatus, field programs resumed in 2021. Mapping stepped further away from the mine and an additional ~50 km² of property geology maps were traversed and updated. In the summer of 2021, a total of 439 samples were collected and added to the Geochem database. During this period, two areas under cover and one historic gold occurrence were stripped and exposed to facilitate detailed mapping and channel sampling.

Since 2020 the focus of the regional program shifted to systematic data collection at the property scale. Included in this effort were high resolution drone magnetic surveys, collected in three-phases from 2020-2021. A total of 4,621 kilometers of line were flown at low altitude, often ~5 m above treetops and at 25 m line spacing to cover an area of approximately 150 km². Additionally, a property wide gold-in-till survey was completed across the combined Island Gold and Trillium properties during the same period (2020-2021). The objective was to test glacial till for anomalous gold grain populations that would provide insight on the prospectivity of areas with limited outcrop exposure and covered in glacial till. In 2020 185 till samples were collected and an additional 340 were collected in 2021.

Exploration activities at Island Gold remain focused on continuing to define new near mine Mineral Resources. The underground diamond drilling consists of 2-4 underground diamond drill rigs in operation including some underground directional drilling. Since 2020, the surface directional diamond drilling program has consisted of 3-5 diamond drill rigs.

In 2022, a total of 17,988 m of surface diamond drilling and 11,837 m of underground exploration diamond drilling, from the 360, 620, 790 and 840 Level exploration drifts, has been completed as of June 28th, 2022.

Highlights from the 2020-2021 exploration drilling program (Figure 9-1 and Table 9-2) include:

- 15.86 g/t Au (10.74 cut) over 16.04 m (MH25-10);
- 11.90 g/t Au (11.90g/t cut) over 4.60 m (MH27-02);
- 9.07 g/t Au (9.07 g/t cut) over 2.12 m (MH27-01);
- 12.03 g/t Au (12.03 g/t cut) over 8.26 m (840-602-23).

This continues to confirm the E1E Zone extends vertically over 1.2 km, between a depth of 300 m and 1,500 m in the eastern part of the Island Gold deposit.

The results of the 2022 diamond drilling exploration program were not included to the Mineral Resources used for the Phase 3+ Expansion study.

Table 9-2 Diamond Drilling Highlights Results – 2020-2021 Exploration Programs

Hole ID	Zone	Domain	From (m)	To (m)	Core Length (m)	True Width (m)	Au Uncut (g/t)	Au Cut (g/t)	Vertical Depth (m)
MH22-04	C	Lower Island	1531.50	1546.40	14.90	5.58	25.41	23.07	1377
MH21-04	D1	Lower Island	1299.00	1311.50	12.50	11.26	4.03	3.10	1120
MH25-01	D1	Lower Island	1570.20	1576.50	6.30	6.08	3.82	3.82	1296
MH20-04	E1E	Lower Island	1174.70	1180.72	6.02	4.45	3.33	3.33	1111
MH20-05	E1E	Extension 2	1141.80	1145.80	4.00	3.73	4.22	4.22	1056
MH21-04	E1E	Lower Island	1288.70	1291.20	2.50	2.25	44.30	44.30	1108
MH21-05	E1E	Lower Island	1280.60	1283.85	3.25	3.19	3.90	3.90	1079
MH23-01	E1E	Extension 2	1077.00	1079.40	2.40	2.09	10.98	10.98	1000
620-610-26	E1E	Extension 2	318.60	333.70	15.10	6.30	8.41	8.41	808
620-616-01	E1E	Extension 2	143.00	148.00	5.00	2.94	6.28	6.28	641
620-616-02	E1E	Extension 2	267.00	273.40	6.40	2.24	21.30	15.52	751
620-616-04	E1E	Extension 2	79.06	83.10	4.04	3.83	8.15	8.15	561
620-616-07	E1E	Extension 2	79.40	83.32	3.92	3.64	18.72	16.44	532
620-MH2-01	E1E	Extension 2	806.20	812.20	6.00	4.86	29.05	26.67	1177
840-566-01	E1E	Lower Island	291.75	310.20	18.45	10.31	52.10	22.54	1041
840-566-02	E1E	Extension 2	260.35	264.75	4.40	2.36	13.53	13.53	1002
840-566-05	E1E	Extension 2	176.10	185.50	9.40	7.96	4.48	4.48	880
840-566-06	E1E	Lower Island	154.20	159.90	5.70	4.26	21.01	21.01	842
840-566-07	E1E	Lower Island	283.50	292.60	9.10	3.73	6.90	6.90	1028
840-572-02	E1E	Lower Island	293.30	297.65	4.35	2.22	31.19	31.19	1032
620-610-26	Unknown	Extension 2	101.60	110.20	8.60	?	3.90	3.90	664
620-MH2-01	Unknown	Extension 2	759.90	762.65	2.75	?	40.34	40.34	1164
840-572-03	Unknown	Lower Island	200.00	202.20	2.20	?	6.33	6.33	915
MH26-03	E1E	East	1551.70	1558.4	6.70	4.78	6.52	6.52	1478
MH26-05	E1E	East	1580.80	1593.2	12.40	10.65	2.80	2.80	1532
MH25-11	E1E	East	1642.07	1654	11.93	10.82	15.76	15.76	1541
MH25-10	E1E	East	1622.60	1641.20	18.60	16.04	15.86	10.74	1505
MH25-08	E1E	East	1589.80	1614.9	25.10	21.33	71.21	39.24	1464
MH25-07	E1E	East	1562.90	1569.30	6.40	5.98	34.87	34.87	1382
MH27-01	E1E	East	1764.90	1767.60	2.70	2.12	9.07	9.07	1649
MH27-02	E1E	East	1824.45	1829.90	5.45	4.60	11.9	11.9	1725
340-580-15	E1E	East	185.70	190.30	4.60	2.34	5.20	5.20	441
340-580-13	E1E	East	175.35	180.45	5.10	2.38	3.15	3.15	431
620-623-11	E1E	East	98.95	101.40	2.45	2.11	3.21	3.21	511
620-623-12	E1E	East	55.50	59.00	3.50	2.07	4.33	4.33	570
840-584-04	E1E	East	151.40	156.10	4.70	3.37	5.77	5.77	892
840-584-03	E1E	East	175.73	187.40	11.67	8.45	8.88	8.88	932
840-584-05	E1E	East	208.20	214.50	6.30	3.82	4.62	4.62	970
840-590-06	E1E	East	134.4	137.50	3.10	2.63	3.20	3.20	880
840-590-10	E1E	East	173.90	177.40	3.50	2.64	16.75	16.75	939

Table 9-2 Diamond Drilling Highlights Results – 2020-2021 Exploration Programs (Continued)

Hole ID	Zone	Domain	From (m)	To (m)	Core Length (m)	True Width (m)	Au Uncut (g/t)	Au Cut (g/t)	Vertical Depth (m)
840-590-08	E1E	East	136.00	138.50	2.500	2.04	4.94	4.94	880
840-590-07	E1E	East	174.55	177.70	3.15	2.20	12.01	12.01	937
840-590-03	E1E	East	227.90	232.60	4.70	2.31	7.52	7.52	998
840-602-23	E1E	East	146.10	157.80	11.7	8.26	12.03	12.03	911
840-602-32	E1E	East	190.80	194.10	3.30	2.14	8.65	8.65	960
840-602-29	E1E	East	153.00	161.80	8.80	3.77	14.24	13.88	928
840-605-28	E1E	East	184.30	1890	4.70	3.48	4.67	4.67	968
840-618-01	E1E	East	150.10	152.60	2.50	2.50	11.39	11.39	810

Notes:

- *Composite intervals greater than 3 g/t Au weighted average, capping values:*
 - *Lower Island Domain @ 225 g/t Au;*
 - *East Domain @ 185 g/t Au.*
- *Unknown zone corresponds to gold intercepts outside known ore zones and for which continuity is not yet established and therefore true width has not been calculated.*

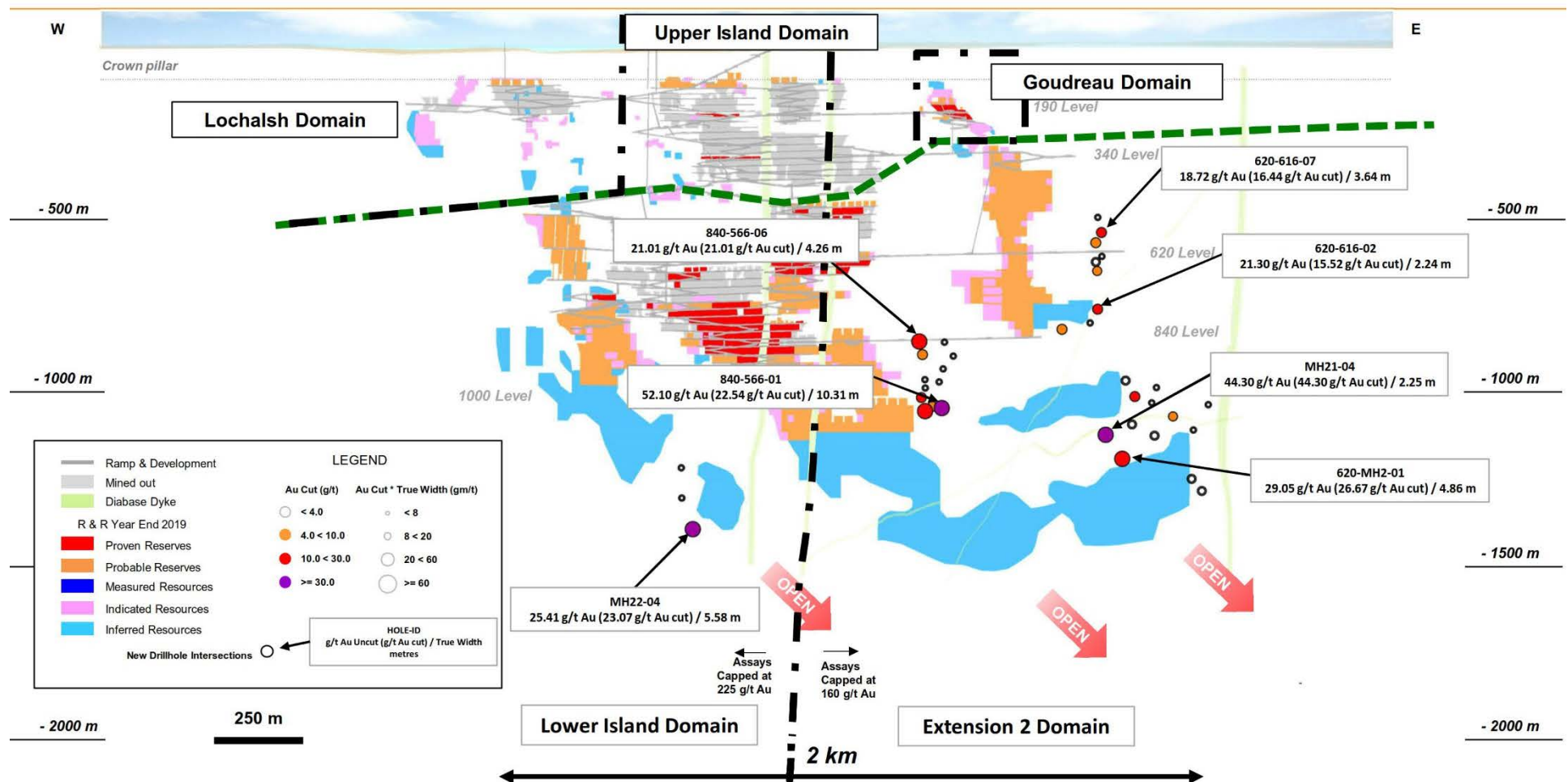


Figure 9-1 Diamond Drilling Exploration Highlights – 2020/2021

10 DRILLING

10.1 Methodology and Planning

Drill hole planning begins by marking targets on longitudinal sections of the targeted zone. Cross sections are used to confirm the positioning of the drill targets and distance of planned drill holes to any existing infrastructure.

An optimal drilling pattern of 20 m by 20 m hole spacing is sought during the planning of the delineation and definition drilling. The pattern can be decreased to 10 m by 15 m in geologically complex areas. At surface, a 50 m to 100 m spacing pattern is used for the first phase of exploration drilling in new sectors. A 40 m by 40 m drill pattern is used for underground exploration drilling.

Diamond drill holes are planned to intersect all the known zones. Holes are stopped after the last mineralized zone is crossed depending on the confidence of the interpreted zone intersection, 20-50 m for definition and delineation drilling and at least 100 m for exploration drilling.

10.2 Drill hole Alignment

Beginning in August of 2020, underground drill holes were aligned to drifts using the DeviAligner rig alignment system. The DeviAligner is a north-seeking gyro alignment system that records azimuth, inclination, and roll angle measurements. The DeviAligner tool is attached directly to the drill rod and provides the measurements to the driller. Prior to the use of the DeviAligner tool, surveyors would mark a back and foresight on drift walls with the use of a Leica Total Station surveying instrument.

Some drill holes that occur along drifts are marked up by members of the production geology team or surveyors. Drill holes that are planned within drill bays are not required to be marked up. Surface drill hole locations are found and marked by using GPS. The drill is aligned by using a compass or the Leica GPS. Alignment of the holes were confirmed using a gyro system.

10.3 Collar Surveying

Island Gold employees use a Leica Global Positioning System to survey surface collar locations. Drill hole coordinates are recorded in the local grid system of the Island Gold Mine which is rotated 22° to the west from the geographic north.

In the underground, drill hole locations are tagged, identified, and recorded by drilling contractors. This information is provided to the Island Gold surveyors after drill holes have been completed. The underground diamond drill hole collars are then surveyed by Island Gold employees with a Leica Total Station. The collar information is recorded by geologists into acQuire software.

10.4 Down-Hole Surveying

For underground diamond drill holes Single shot Reflex down-hole survey measurements start at 15 m from the collar and are carried out at every 30 m thereafter along the hole and at the end of the hole. If the azimuth and dip are outside of a $\pm 1.5^\circ$ tolerance of the requested values at the first reflex measurement, the hole is attempted again. Reflex or gyroscopic multi-shot surveys that are taken after the hole has been drilled are only completed if there are concerns regarding the single shot surveys, such as if the magnetic readings are anomalous. Single-shot gyroscopic surveys are also completed if there are concerns drilling near areas of anomalous magnetism

such as the vertical diabase dykes that intersect the island gold deposit. All measurements are converted to the Island Gold Mine grid for entry into the SQL database.

Surface exploration holes have used Reflex or a gyro to survey the hole. Surface directional drilling holes use a north seeking gyro and a PeeWee survey tool when drilling the cut. Measurements are taken every 30 m in surface holes and multi shot Gyro surveys are taken in each hole.

10.5 Cementing of Drill Holes

Underground diamond drill holes are grouted at the collar once they are completed or abandoned. Cementing of drill holes is recorded in the acQuire database. Generally, fifteen sticks of cement are used to cement a diamond drill hole collar, or to 9 meters depth. If the hole has intersected water, then a grout pump is used to pump cement into the hole until the water stops.

Surface drill holes for the exploration directional drilling program are cemented from the bottom of the hole until 100 m above the expected zone. Many surface exploration holes which are not immediately adjacent to the mine were left uncemented to provide an option for re-entry if desired or to bore-hole geophysical surveys.

10.6 Drill Core Logging

Core logging is carried out by professional geologists or geologists in training under the supervision of the Qualified Person. Island Gold geologists enter core logging data into an acQuire database. The user interface of the acQuire system has been customized for Island Gold geologists' use. The data entries follow a pre-established structure with consistent lithology codes and structural descriptions, creating uniformity in the geological description. With the acQuire logging program mandatory data such as lithology, sample lengths etc. are required fields that if not completed will alert the user and halt any further progress until the mandatory information is entered.

10.7 Geology and Analysis

Under the direct supervision of Qualified Persons, geologists prepare a detailed description of the drill core. A computerized log is entered for each drill hole with the following basic information:

- collar location;
- down hole surveys;
- rock quality designation (RQD);
- main and secondary geological units;
- texture and structure;
- mineralization and alteration: mineralogy, thickness, type;
- sample location; and
- core photos.

The length and limits of core samples are defined by the geology: (*i.e.*, geological unit, alteration package, mineralized zone and deformation zone). In the case of exploration holes, the sampled intervals are sawed in two along the core length to keep half of the core on site as a reference sample. Since 2016, several definition and delineation holes were drilled from the same drill bays,

one in five holes of these holes have a portion kept for reference. For the four other holes, the entire section of the core that is to be assayed, is sent to the laboratory, and the remaining core is discarded.

Assay results are plotted on sections and level plans at the appropriate scale. Nomenclature and symbols for the geological units follow an in-house legend, which is modified from the legend prepared by the *Ministère de l'Énergie et des Ressources naturelles du Québec* (MERN). Horizontal thickness and true thickness of drill hole intercepts (composites) are computed in the Datamine software, based on a modeled representation of the ore zone. True thickness and grade of composites are plotted on the vertical longitudinal sections.

11 SAMPLE PREPARATION, ANALYSES AND SECURITY

The Mineral Resource and Mineral Reserve estimates as of December 31, 2021, are supported by both diamond drill hole samples and underground chip sampling. Other sample types, including muck samples (rock fragments collected after an underground blast) and test holes (sludge from jack-leg drilling), helped guide production decisions but are not used in Mineral Resource estimates. The assays for these sample types are not discussed further in this report as the results do not impact Mineral Resource estimates.

Until April 2021, diamond drilling samples were sent to Laboratoire Expert Inc. ("LabExpert") located in Rouyn-Noranda, Quebec, and a small portion to the Wesdome Gold Mines Inc. laboratory ("Wesdome") in Wawa, Ontario. Starting in April 2021, diamond drilling samples are primarily sent to AGAT Laboratories Ltd. ("AGAT") laboratories in Thunder Bay, Timmins, and Mississauga, Ontario. All production samples (chip, mucks, and test-holes) were sent exclusively to the Wesdome laboratory. Sample volumes for 2021 are summarized in Table 11-1. Activation Laboratories Ltd. ("Actlabs") is only utilized for QA-QC pulp check assays.

Table 11-1 Island Gold Mine – 2021 Sample Volumes

Sample Type	LabExpert	Actlabs (<i>Pulps</i>)	Wesdome	AGAT
Drill core	7,168	337	4,768	23,322
Underground chips	-	92	3,437	-
Mucks	-	-	13,446	-
Test hole sludges	-	-	1,388	-

Notes:

- Assays that were added since the last Mineral Resource estimation Dec 1st, 2020, to Dec 1st, 2021
- Includes pulps and rejects submitted for check assays

11.1 Core Sampling and Collection

11.1.1 Drill Core Sampling

Intervals of core to be sampled for analysis are marked by a geologist. Sampling is completed over the mineralized section along regular intervals. The sample lengths vary from 0.3 m to 1.5 m. When present, lithological boundaries such as geological units or alteration limit the sample intervals. Sample positions are identified on the core by the geologist while logging and sample tags are placed under the core in the core boxes at the end of each sample. Sample intervals, sample numbers, standards and blanks are manually entered into the database by Island Gold personnel.

Core recovery is considered excellent and is generally close to 100%. Very minor isolated centimetre-scale fault gouge and blocky core can be identified but do not impact the reliability of the analytical results of drill core samples.

The core is cut into two halves by the core shack technician using an electric core saw equipped with a diamond tipped blade. For all surface drilling, underground directional drilling, and approximately 20% of other underground drilling, one half of the core is placed into a plastic sample bag for assaying, and the remaining half of the core is returned to the core box for future reference. For approximately 80% of the combined unidirectional underground drill holes, the

entire core is placed in the plastic sample bag. Island Gold routinely inserts quality control samples and tracks sample shipments to the commercial laboratories.

11.1.2 Core Size

The majority of the underground diamond drill holes were carried out using either BQ size coring with a small proportion being NQ or AQTK size. NQ size core are used for all surface drilling except when steering the directional drill holes. The core size during steering of directional holes is AQTK.

11.1.3 Core Storage

The drill core is stored outdoors in covered racks or as separate cross-piles on the mine site. The reference portions of the drill core are stored and catalogued for future reference in the core library located at the Island Gold Mine site.

11.2 Chip Sample Collection

The chip sampling method consists of taking horizontal representative samples of the exposed ore zone either from the drift face or from the adjacent walls. The geological technician or the geologist takes a 1.5 to 2 kg sample. Walls are chipped with a hammer horizontally across geological units. Minimum and maximum width of sample intervals taken from the rock walls are 0.3 m to 1.0 m respectively. The sampler notes the location and the lithology of each chip sample. Assays are then entered into a Promine module on the AutoCAD Software, and then, since 2020, transferred into Studio RM software (previously into the Gemcom software).

11.3 Laboratory Procedures

Until April 2021 most drill core samples were prepared and assayed by LabExpert in Rouyn-Noranda, Quebec, which has been operating a fire assay laboratory for over 20 years. In April of 2021 Island Gold began shipping most drill core samples for preparation and assay to AGAT in Thunder Bay, Timmins, and Mississauga, Ontario.

AGAT is recognized as an accredited testing laboratory as defined by the Standards Council of Canada (SCC) and conforms with the requirements of ISO/IEC 17025.

A small portion of the definition drill core, as well as all underground production samples, are assayed at the Wesdome laboratory, in Wawa, Ontario. This laboratory also processes assay samples from Wesdome's Eagle River Mine.

Pulps are sent to Actlabs in Thunder Bay, Ontario. Actlabs is headquartered in Ancaster, Ontario. Actlabs is ISO/IEC 17025 accredited and has maintained its Certificate of Laboratory Proficiency (PTPMAL) for over 10 years.

The laboratories use industry-standard sample preparation and assay methods that are summarized in Table 11-2.

All the laboratories mentioned have internal quality control (QC) programs that include insertion of reagent blanks, reference materials, and pulp duplicates. Actlabs and AGAT also routinely participate in international round robins, monitor preparation of duplicates, and maintain measurement systems as required by ISO 17025 accreditation. As typical of mining laboratories, Wesdome does not include QC materials with muck and test hole samples, and in general their QC protocols are less rigorous than those of commercial laboratories.

Table 11-2 Summary of Preparation and Assay Methods

Analyses	LabExpert	Actlabs	Wesdome	AGAT
Crushing	80% - 2 mm	80% -2 mm	Approx. 1/4 inch	80% -2 mm
Splitting	300 g	350 g	150 to 300 g	300 g
Pulverizing	90% -200 mesh	95% -105 microns	~90% at 200 mesh (not measured)	90% at 200 mesh
Gold by Fire Assay Sample Weight	1 assay-ton (29.16 g)	30 g	30 g	50 g
AA Finish	AA to 10 g/t; Detection limit: 5 ppb	AA to 3.75 g/t; Detection limit: 5 ppb	-	AA to 10g/t; Detection limit: 5 ppb
Gravimetric Finish	Gravimetric finish > 6.8 g/t Au*	Gravimetric finish >3.75 g/t Au	Gravimetric finish; detection limit 0.04 g/t Au	Gravimetric finish > 6.8 g/t Au
Internal quality control	Rigorous (>10%)	Rigorous (>10%; to ISO standards)	5 to 10% (reliance on duplicates)	Rigorous (>10%; to ISO standards)

Notes:

- *LabExpert prepared a second pulp of high-grade samples and assayed by fire assay with a gravimetric finish.*

11.4 Security

Security personnel control access to the mine site at all times. Individual sample bags are sealed with zip ties. The samples are placed in large Fabrene bags identified and sealed before being placed on pallets. The core samples are picked-up and delivered to laboratories via transportation companies (typically Manitoulin Transport or Gardewine). Beginning in late 2021, muck samples sent to Wesdome Lab are sealed with red security tags. Extra red security tags are also provided to the laboratory assaying mucks for returning any reject materials. Tag numbers are tracked in a spreadsheet. The assay laboratories provide a letter upon reception of the samples detailing the shipment they received.

11.5 Database Security

The database, which contains all diamond drilling assays, logging and surveys, is stored on the Alamos Gold private network which can only be accessed by employees. In 2020 data entry into the structured query language database (SQL) was changed from Microsoft Access to acQuire. Additional restrictions were put in place to limit the number of employees who have access to the database. Security groups are used to limit individuals to the parts of the database that is needed for their work. Access must be granted by a supervisor in the Geology Department.

Sample intervals, sample numbers, standards and blanks are manually entered into the database by Island Gold Mine geologists. Once the assays are completed at the various laboratories, they are sent via email to a list of Island Gold Mine personnel. They are received in an excel sheet directly from the lab and are uploaded electronically into the database by an in-house program. The assays automatically are matched to the sample numbers in the database, with no manual entry required. In general, the acceptable threshold for blanks is 10x the lower detection limit, and for standards, is the three times the standard deviation of the expected mean assay. If an assay exceeds these thresholds the geologist then decides how to proceed with the batch of assays in one of several ways:

- Request a partial or full certificate re-assay of the batch (pulp or rejects);

- Defer as pending investigation, or;
- Accept the batch.

All actions taken are recorded in the database.

11.6 2021 Island Gold Internal Quality Assurance and Quality Control Program (QA-QC)

Alamos maintains an internal QA-QC program at the Island Gold Mine which is used to validate core and production chip assay analyses. Certified Reference Materials (CRMs) are purchased from Rocklabs (New Zealand) or from Ore Research & Exploration (“Oreas”) (Australia) and inserted with diamond drill core samples. Geologists randomly place 2 standards for every 50 samples. At the same rate, in-house blank material is inserted in the core sample stream. The blanks consist of washed and cleaned diabase dyke core which is drilled from the Island Gold property.

Most visible gold diamond drill core samples are shipped separately from the regular samples. A minimum of one high grade standard and one blank are added to the shipment with the samples containing visible gold. This is undertaken with the intent of minimizing the contamination by the visible gold samples to the other samples if they were kept in the same sample stream. All samples containing visible gold are identified with flagging tape. Implemented in November 2021, the visible gold samples are listed on chain of custody (COC) documents. The COC documents list all the samples and assay instructions to the various laboratories that assay Island Gold samples. Identifying and separating a visible gold batch indicates to the lab to take extra care to clean the crushers, pulverisers, and splitters diligently between samples.

In 2021, fourteen CRMs with different gold grades were used for the QA-QC program. The CRMs' grades range from 1.58 g/t Au to 42.96 g/t Au and reflect expected gold grades within the mineralization found at the Island Gold deposit. The CRMs used by Alamos Gold from Rocklabs consist of feldspar, basalt and iron pyrite mixed with fine gold. The CRMs from Oreas are from orogenic lode gold deposits with primary gold and a greenstone matrix. These CRMs have a similar composition to Island Gold mineralized zones.

For CRMs sent to AGAT and LabExpert, the criteria used by Island Gold to consider whether results are acceptable are based on a 3x standard deviation from the expected value. Any value greater or equal to 3x standard deviation of the mean expected assay will be investigated further by the geologist reviewing the certificate. Often this means requesting a re-assay or partial re-assay of a certificate. For samples sent to the Wesdome laboratory criteria for CRMs is not to exceed 10% of the expected assay.

11.6.1 Laboratory Performance Based on Certified Reference Materials (2021)

During 2021, a total of approximately 305 CRMs were analyzed by LabExpert. Assays demonstrated good accuracy with the average of the observed values of the 9 different CRM values falling between 99.0% and 100.1% of the expected value, with a weighted average of 99.7% (Table 11-3). A small number of QC failures triggered requests for re-assaying of the affected sample batch. Based on the insertion of CRMs as part of the QA-QC program, LabExpert's accuracy provided acceptable results for the Mineral Resource and Mineral Reserve estimates prepared for 2021.

A total of 501 CRMs from 13 different CRMs were assayed by Wesdome in 2021 (Table 11-4 and Table 11-5). The average of the observed values was between 95.7% and 108.8% of the expected value, with a weighted average of 102.3%. The Wesdome Lab demonstrated a significant improvement from the previous year's performance. Only 3.0% of all the standards

assayed at the Wesdome Laboratory exceeded 3x standard deviation of the expected assay result, 86.2% of the assay results were within 1x standard deviation of the mean.

In 2021, a total of 1,095 CRMs were analyzed by AGAT (Table 11-6). This represents approximately 4.4% of the total samples analyzed by AGAT. For the 14 different CRMs assayed at AGAT, the weighted average value of the assays was within 98.6%. The range of these values were between 95.7% and 100.2%. Many of the assay results from AGAT that exceeded Island Gold's QC thresholds were amongst the first batches sent for assaying. Each result that exceeded the threshold was requested to be re-assayed along with 5 samples above and below the list of assay results within the same certificate. No significant bias has been identified within the assay results AGAT has provided.

In 2021, 6 different CRMs for a total of 27 samples were assayed at Actlabs (Table 11-7) representing slightly more than 1% of the total number of CRMs submitted for assay. CRM performance at Actlabs indicates a good degree of accuracy with all the observed assayed results being between 91.5% and 106.5% of the expected value. The weighted average assay results from Actlabs were 100.7% relative to the expected assay values.

Actlabs has been used as a secondary laboratory for pulp check assays and has provided reliable assay data for the rechecking program.

Table 11-3 LabExpert Reference Material Results 2021 - from DDH

Standard	Number of Samples	Expected Values			Observed Values for the year					
		Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Within 1 SD	Within 2 SD	Failure or ≥3 SD
SQ87	41	30.87	0.717	2.3%	30.57	0.252	0.8%	39	2	0
OREAS239	37	3.55	0.086	2.4%	3.55	0.030	0.9%	37	0	0
OREAS229b	32	11.95	0.288	2.4%	11.93	0.125	1.1%	31	1	0
OREAS228b	39	8.57	0.199	2.3%	8.57	0.058	0.7%	39	0	0
OREAS226	44	5.45	0.126	2.3%	5.46	0.028	0.5%	44	0	0
OREAS216b	41	6.66	0.158	2.4%	6.60	0.040	0.6%	38	1	2
OREAS209	44	1.58	0.044	2.8%	1.57	0.028	1.8%	39	5	0
OREAS245	27	25.73	0.546	2.1%	25.67	0.268	1.0%	24	1	2
Blanks	321	-	-	-	-	-	-	-	-	4

Table 11-4 Wesdome Reference Material Results 2021 - from Chips and Test Holes

Standard	Number of Samples	Expected Values			Observed Values for the year					
		Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Within 1 SD	Within 2 SD	Failure or ≥3 SD
OREAS229b	89	11.95	0.288	2.4%	12.17	0.22	1.8%	82	7	0
OREAS228b	41	8.57	0.199	2.3%	8.75	0.16	1.8%	41	0	0
OREAS216b	98	6.66	0.158	2.4%	6.87	0.32	4.6%	78	14	6
OREAS226	45	5.45	0.126	2.3%	5.63	0.18	3.1%	36	8	1
OREAS243	11	12.39	0.306	2.5%	11.75	1.79	15.2%	9	1	1
OREAS240	9	5.51	0.139	2.5%	5.65	0.31	5.4%	7	1	1
Blanks	147	-	-	-	-	-	-	-	-	9

Table 11-5 Wesdome Reference Material Results 2021 - from DDH

Standard	Number of Samples	Expected Values			Observed Values for the year					
		Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Within 1 SD	Within 2 SD	Failure or ≥3 SD
SQ87	4	30.87	0.717	2.3%	30.12	0.82	2.7%	3	1	0
OREAS229b	68	11.95	0.288	2.4%	12.15	0.18	1.5%	65	3	0
OREAS228b	12	8.57	0.199	2.3%	9.07	0.96	10.6%	8	3	1
OREAS216b	64	6.66	0.158	2.4%	6.81	0.13	2.0%	61	3	0
OREAS226	3	5.45	0.126	2.3%	5.58	0.10	1.8%	3	0	0
OREAS239	20	3.55	0.086	2.4%	3.66	0.16	4.5%	14	4	2
OREAS245	4	25.73	0.546	2.1%	25.77	0.67	2.6%	4	0	0
OREAS237	13	2.21	0.054	2.4%	2.35	0.13	5.7%	1	9	3
OREAS240	11	5.51	0.139	2.5%	5.56	0.12	2.2%	11	0	0
OREAS242	2	8.67	0.215	2.5%	8.43	0.42	5.0%	2	0	0
OREAS243	4	12.39	0.306	2.5%	12.17	0.21	1.7%	4	0	0
OREAS235	1	1.59	0.038	2.4%	-	-	-	1	0	0
OREAS247	2	42.96	0.9	2.1%	44.00	0.00	0.0%	2	0	0
Blanks	245	-	-	-	-	-	-	-	-	3

Table 11-6 AGAT Reference Material Results 2021 - from DDH

Standard	Number of Samples	Expected Values			Observed Values for the year					
		Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Within 1 SD	Within 2 SD	Failure or ≥3 SD
SQ87	33	30.87	0.717	2.3%	30.64	1.10	3.6%	26	6	1
OREAS229b	148	11.95	0.288	2.4%	11.86	0.30	2.5%	99	48	1
OREAS228b	109	8.57	0.199	2.3%	8.53	0.50	5.8%	77	24	8
OREAS216b	152	6.66	0.158	2.4%	6.45	0.51	7.9%	68	71	13
OREAS226	167	5.45	0.126	2.3%	5.37	0.28	5.2%	55	90	22
OREAS209	40	1.58	0.044	2.8%	1.52	0.05	3.0%	12	25	3
OREAS239	209	3.55	0.086	2.4%	3.52	0.10	2.9%	141	63	5
OREAS245	21	25.73	0.546	2.1%	25.57	1.38	5.4%	9	11	1
OREAS237	89	2.21	0.054	2.4%	2.21	0.07	3.1%	71	16	2
OREAS240	37	5.51	0.139	2.5%	5.43	0.12	2.3%	19	18	0
OREAS242	24	8.67	0.215	2.5%	8.57	0.10	1.2%	23	1	0
OREAS243	18	12.39	0.306	2.5%	12.15	0.23	1.9%	13	5	0
OREAS247	9	42.96	0.9	2.1%	41.10	4.76	11.6%	3	3	3
OREAS235	39	1.59	0.038	2.4%	1.54	0.03	1.7%	14	25	0
Blanks	1044				-	-	-	-	-	11

Table 11-7 Actlabs Reference Material Results 2021 – Pulp Duplicates

Standard	Number of Samples	Expected Values			Observed Values for the year					
		Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Mean (g/t Au)	Standard Deviation (g/t Au)	1 Relative Standard Dev.	Within 1 SD	Within 2 SD	Failure or ≥3 SD
OREAS229b	8	11.95	0.288	2.4%	11.90	0.42	3.5%	5	2	1
OREAS228b	4	8.57	0.199	2.3%	8.55	0.47	5.5%	1	2	1
OREAS216b	6	6.66	0.158	2.4%	6.76	0.13	1.9%	4	2	0
OREAS226	5	5.45	0.126	2.3%	5.51	0.16	2.9%	4	0	1
OREAS209	1	1.58	0.044	2.8%	1.60	-	-	1	0	0
OREAS239	3	3.55	0.086	2.4%	3.64	0.12	3.3%	2	0	1

11.6.2 Blanks

Field blanks were included in the 2021 Island Gold drill program to monitor possible contamination during the sample preparation and analytical processes. The field blank material used by Island Gold is made with drill core that has transected the Matachewan diabase dyke located on the mine property. Testing of the diabase dyke has indicated that the material is suitable as blank material. The blank is carefully selected to not include any veins, mineralization,

or inclusions. Each submitted blank of cut core is washed and bagged before use. When new batches of blanks, are prepared several samples are taken for assay to ensure the material is still acceptable. A blank is inserted at a rate of 2 in 50 samples. Blanks are also inserted along with groups or individual samples containing visible gold. Assays higher than 10% of the laboratory's detection limit are considered as failures. This limit is 0.5 g/t for Wesdome and 0.05 g/t for AGAT and LabExpert. Any blank that exceeds those thresholds are requested, along with some or all the rest of the assays in the certificate, to be re-assayed. The results, acceptance or otherwise, is recorded in the acQuire software along with the reasoning for the action taken.

In 2019 the Island Gold chip assay program was improved by inserting diabase dyke as field blanks amongst channel samples. These blanks are chip samples taken from underground exposures of the diabase dyke. Blank assays with a value between 0.5 ppm and 1 ppm are evaluated before being accepted or re-assayed. Every blank that has an assay result greater than 1 ppm is considered as a failure and the rejects are automatically re-assayed along with the rest of the samples listed in the certificate.

LabExpert tested 321 blanks and had a 1.2% failure rate. During LabExpert's pulverization process, the ring-and-puck crusher is cleaned with compressed air and undergoes a cycle of sand cleaning. Only 4 blank samples assayed exceed 10x the lower detection (0.05 g/t). It is possible there was contamination of the samples during the crushing process or potentially the diabase blank may have contained anomalous gold values. These samples were re-assayed at the laboratory and mostly showed similar results and some passing the QC threshold upon re-assay.

Wesdome tested 392 blanks and had 12 assays that exceeded Island Gold's QC threshold or a 3.1% failure rate. All failures were requested to have their rejects re-assayed along with the rest of the assays in the certificate. Depending on the results of the re-assay certificate, usually either the original or re-assay results were accepted into the database. Wesdome has seen an increased performance in acceptance of blanks with respect to previous years.

In April 2021, approximately 1,000 samples were forwarded from LabExpert to AGAT for 'rush' assay analysis. The rush samples were assayed with a gravimetric finish. This is significant because the lower detection limit of the gravimetric finish is 0.5 g/t, which exceeds the IG typical QC threshold for any blanks amongst these samples. For the select few blank samples amongst these 'rush' samples it was decided to accept the assay results unless the assay exceeded 0.5 g/t. Other than the blanks included with the 'rush' samples, all blanks are expected to perform at <10x of the detection limit of the AA assay method (0.05 g/t). AGAT laboratories analysed 1,044 blanks in 2021 with a failure rate of 1.0%. The drill core blanks submitted to each laboratory are summarized in Table 11-8.

Table 11-8 Summary of 2021 Blank Performance

Description	LabExpert	Wesdome	AGAT
Number of blank insertions	321	392	1044
Maximum grade for failure	0.05 g/t Au	0.5 g/t Au	0.05 g/t Au
Number of failures	4	12	11
Percent of failures	1.2%	3.1%	1.0%
Maximum value	0.12 g/t Au	1.03 g/t Au	306 g/t

11.6.3 Long Term QA-QC

Over the course of Island Gold's drilling programs from 2020 to 2021, a total of 14 different CRMs were used. The grades of the CRMs ranged from 1.58 g/t Au to 42.96 g/t Au and reflected expected assay grades within the mineralization of the Island Gold deposit. During those years, a total of approximately 1,928 pulp samples using these CRMs were analyzed by four different laboratories and retained for statistics (mislabels are excluded).

The Wesdome laboratory showed a distinct improvement with a decreased rate of quality control samples exceeding Island Gold's QC assay thresholds, with 3.0% of CRMs and 3.06% of blanks failing the QC test. The Wesdome laboratory is mostly used by Island Gold for underground production samples (mucks, chips, test holes), although in 2021 they were used to assay some definition drilling core during the transition of the primary assay laboratory from LabExpert to AGAT. An external third-party audit of the Wesdome lab was completed in June 2019. The replacement of a pulveriser in July 2019, and increased communication between the Wesdome mine laboratory personnel and the Island Gold employees in charge of the QA/QC were some of the recommended solutions implemented after the ASL Consulting audit in 2020. There has been an improvement in the quality of the assays as indicated by the reduction of blanks that exceed QC thresholds and comparison with pulp assays at Actlabs.

11.6.4 Pulp Duplicate Analysis (Original vs Duplicate Values)

LabExpert, AGAT and Wesdome are routinely verified by fire assaying second 30-g (50-g for AGAT) pulp aliquots from the original pulp samples. Such re-assays are carried out at a rate of approximately 6% for LabExpert, 8% for AGAT and approximately 7% for Wesdome. At Wesdome any samples identified with VG have the pulps assayed twice. Results are presented in Table 11-9 and Figure 11-1 and Figure 11-2.

The mean grades of the re-assays are very similar to the original assays. There is an acceptable correlation coefficient for each laboratory. These results demonstrate the ability of the labs to reproduce the global average of the first assays despite discrepancies in the individual assays.

Table 11-9 2021 Statistics of Duplicate Assays (Internal QA-QC)

	LabExpert		Wesdome	
	Primary Assay	Check Assay	Primary Assay	Check Assay
Number of pairs	560		183	
Minimum (g/t)	0.0025	0.0025	0.002	0.002
Maximum (g/t)	10.56	9.98	16.23	9.97
Mean (g/t)	0.689	0.686	0.843	0.794
Median (g/t)	0.044	0.040	0.07	0.1
Standard deviation	1.580	1.563	1.931	1.651
Coeff. Correlation	1.00		0.95	
Avg Abs Value of relative difference	9.3%		104.1%	

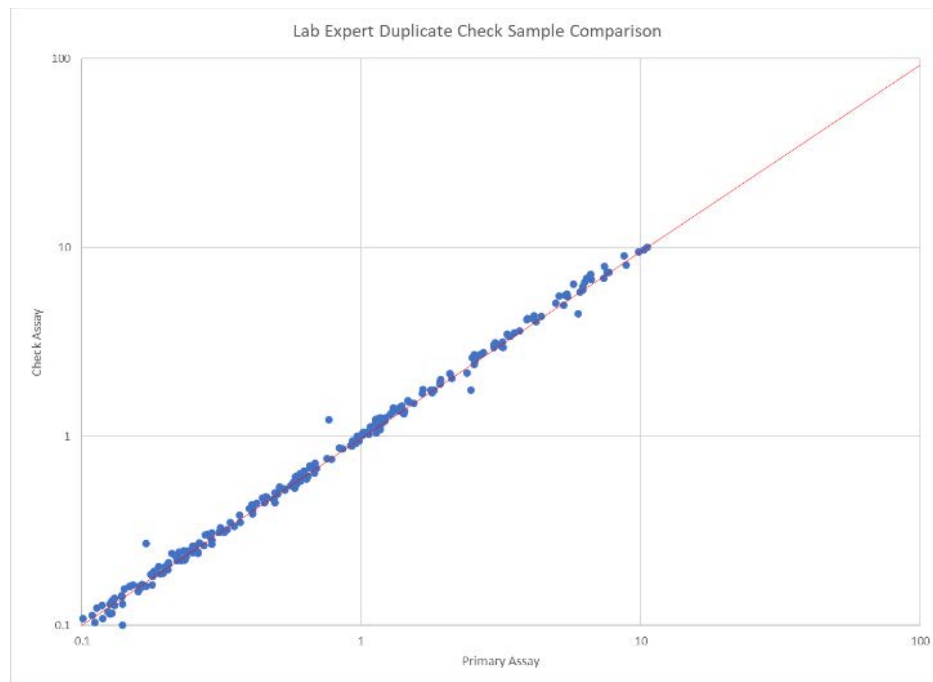


Figure 11-1 2021 Scatterplot LabExpert Duplicate Assay Comparison (Internal QA-QC)

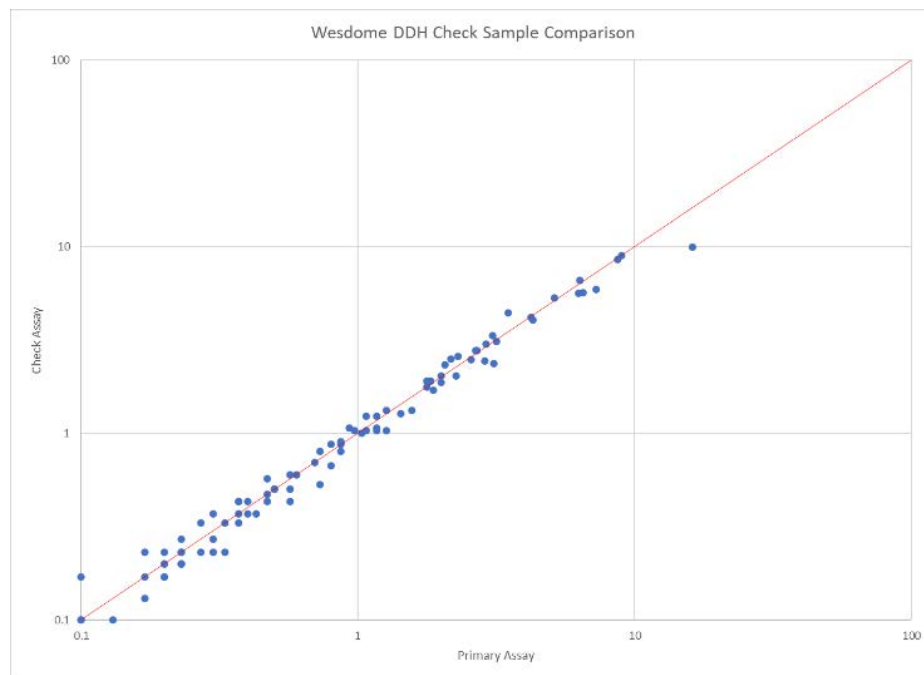


Figure 11-2 2021 Scatterplot Wesdome Duplicate Assay Comparison (Internal QA-QC)

11.7 Laboratory Cross Check Sampling (Pulp Samples)

Additional testing for accuracy is performed at a secondary or umpire laboratory. The CRMs monitor analytical accuracy but are not blind to the laboratory. To ensure that the laboratory gold assaying technique presents no problem or bias, a series of pulps from the primary laboratory are resubmitted to another laboratory for re-assay.

In 2021, 127 pulps assayed at Wesdome, 138 pulps assayed at LabExpert, and 165 pulps assayed at AGAT were sent to Actlabs to be re-assayed (Table 11-10). Scatterplots comparing the labs' duplicate pulp sample assays versus the original assays are shown in Figure 11-3, Figure 11-4, and Figure 11-5. No outliers have been removed and the graphs have been adjusted to a logarithmic scale for easier visual interpretation.

Results are reasonably distributed on both sides of the line of best fit. Globally, the statistics presented in the Table 11-6 show no evidence of strong bias between laboratories. Wesdome sample assay results demonstrate a tendency for over estimation compared to Actlabs, likely influenced by the singular use of gravimetric finish (Figure 11-4).

Sixty five percent of the samples analyzed by LabExpert are slightly lower than the values reported by Actlabs, indicating there is no risk of overestimating the Mineral Resource grade. Wesdome shows the opposite trend with 65% of the assays returning higher assay values in comparison to Actlabs.

The AGAT and ActLabs data show significantly similar results across the extent of the data. 76% of the assay results from AGAT were lower than Actlabs. The strong correlation and minor variation indicate there is no significant discernable bias of assay result between the labs.

Table 11-10 2021 Statistics of Duplicate Assays (External QA-QC)

	LabExpert VS Actlabs		Wesdome VS Actlabs		AGAT VS Actlabs	
	LabExpert	Actlabs	Wesdome	Actlabs	AGAT	Actlabs
Number of pairs	138		126		165	
Minimum (g/t)	0.14	0.03	0.002	0.005	0.003	0.006
Maximum (g/t)	521.11	586.00	2034.53	2560.00	177.00	180.00
Mean (g/t)	52.24	55.9	25.05	28.29	2.07	2.21
Median (g/t)	22.34	24.15	1.45	1.39	0.10	0.09
Standard Deviation	207.10	215.89	182.42	228.66	14.30	14.56
Coefficient of Correlation	0.99		1.00		1.00	

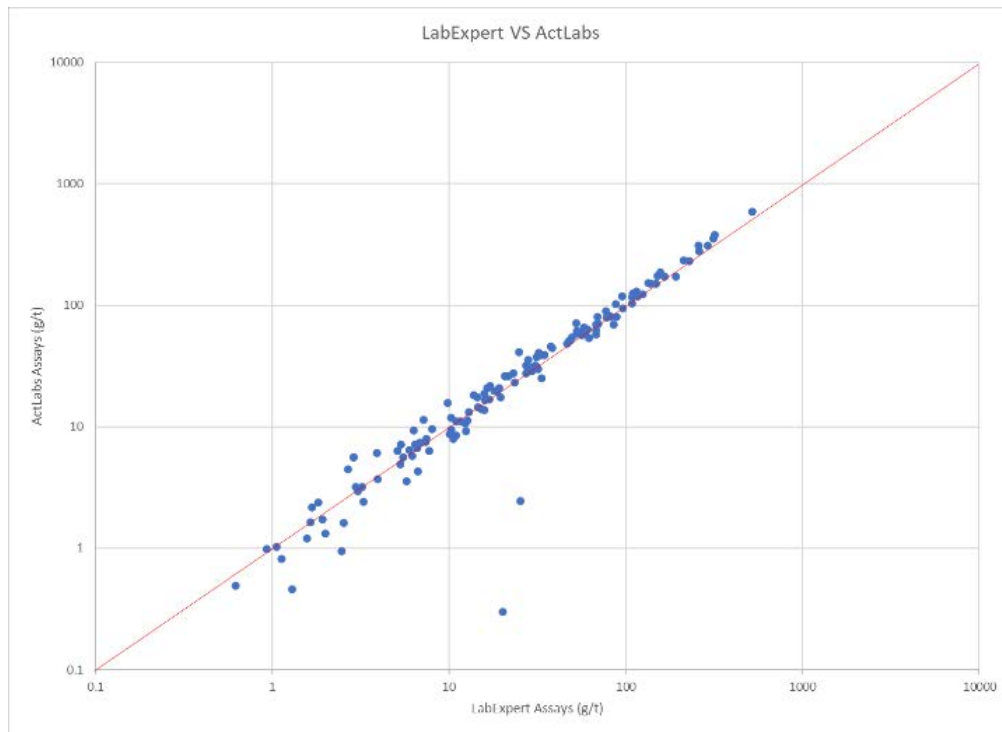


Figure 11-3 Scatterplot of Pulp Re-assays (Actlabs) vs Original Assays (LabExpert)

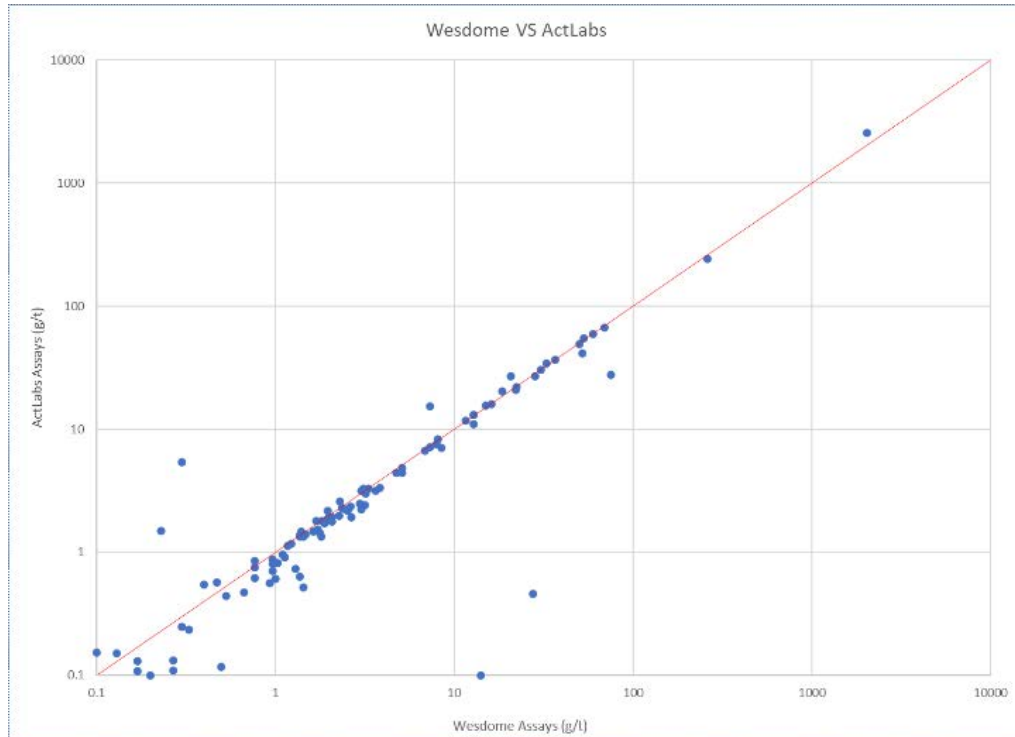


Figure 11-4 Scatterplot of Pulp Re-assays (Actlabs) vs Original Assays (Wesdome Lab)

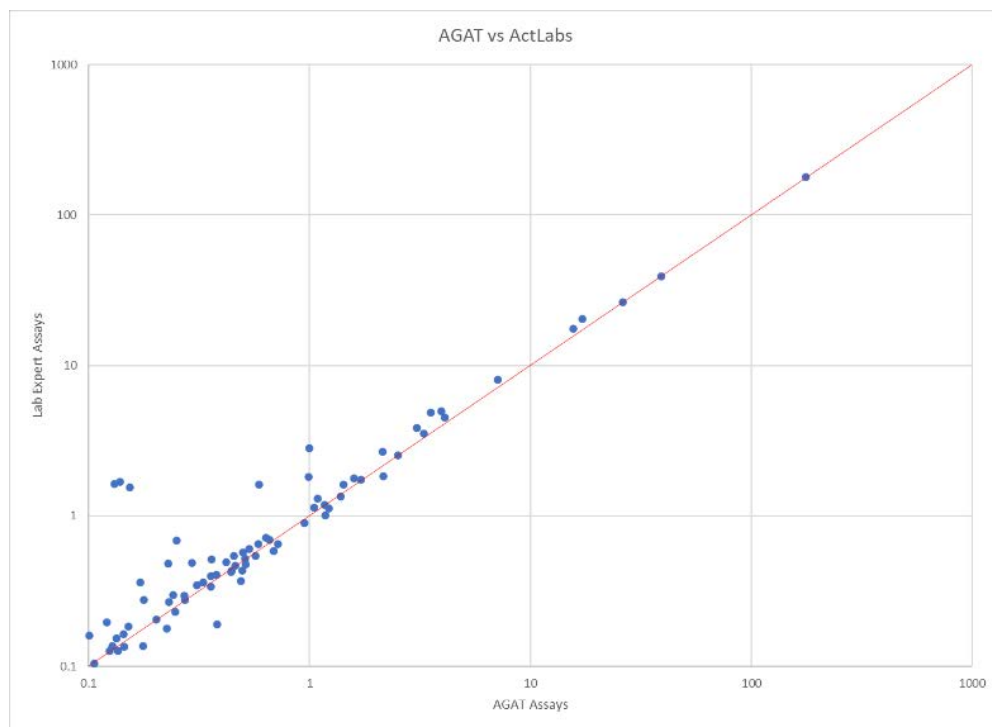


Figure 11-5 Scatter Plot of Pulp Re-assays (Actlabs) vs Original Assays (AGAT)

11.8 Core Duplicates

Until 2015, drill core duplicates were sent to LabExpert and Actlabs to be assayed. The audit conducted by ASL in November 2015 (Bloom, 2015), demonstrated that the core duplicates exhibit poorer reproducibility than the pulp duplicates. This data can not be used to improve or monitor the sampling procedures or assay quality. The precision for core duplicates is within the expected range for the deposit style. Following the recommendation from the audit, Island Gold reviewed the assay quality control program and decided to no longer assay core duplicates.

11.9 Underground Muck Tracking

Internal tracking of broken rock material underground, either ore or waste, is completed using machine monitoring systems installed on haul trucks, coordinated by an underground dispatcher. This system ensures that all daily underground muck movement is accounted for. Daily assays and sample tags received, are entered into the database and support production decisions.

11.10 Summary and Comments

Island Gold received over 75,000 gold assay results in 2021. In switching primary laboratories from LabExpert to the SCC accredited laboratory AGAT, additional confidence in accuracy, precision, and turn around time of assay results were expected. In addition, there has been a distinct improvement by Wesdome laboratory in 2021, as evidenced by a reduction in proportion of blanks and standards that exceed Island Gold's quality standards. Quality control testing at

Island Gold has shown the various contracted laboratories have similar results between primary and umpire laboratories, and reliable results for quality control assays (CRMs and blanks).

Island Gold Mine's QA-QC procedures were audited in 2019 by ASL Canada. The consulting service concluded that Alamos Gold's assay quality control program meets or exceeds industry standards, and that the gold assays are reliable for the purpose of Mineral Resource estimates.

12 DATA VERIFICATION

The Qualified Person considers that the Island Gold Mine database is suitable for use in the Mineral Reserve and Mineral Resource estimation. The SQL database is adequate and acceptable for supporting Mineral Resource estimation. This database contains all the information related to drill holes, drift sampling, assay results and the laboratory certificates. Some verification of the original data was performed, and modifications were completed, if needed, prior to the calculation of any estimates. The verification of, and corrections to, the Island Gold database were accomplished prior to the Mineral Resource and Mineral Reserve estimates of December 31, 2021.

An external Mineral Reserve and Mineral Resource audit (RPA, 2016) was completed in December 2016 by Roscoe Postle Associates Inc. (RPA) and based on a desktop review of the Island Gold Mine Mineral Resources, RPA was of the opinion that the Mineral Resource estimates were reasonable, had been adequately prepared using standard industry practices, and conformed to the Canadian Institute of Mining, Metallurgy and Petroleum (CIM) Definition Standards for Mineral Resources and Mineral Reserves ("CIM definitions") as incorporated into National Instrument 43-101 ("NI 43-101").

Apart from the transition from Gemcom software to Datamine Studio software, most of the methodologies and procedures utilized in the 2021 Mineral Resources and Mineral Reserves estimate are identical to those used in the 2016 estimate therefore, it is presumed that the methodologies used in 2021 are still adequate and acceptable.

In September 2019, the LabExpert laboratory and the Wesdome laboratory were audited by Analytical Solutions LTD. (ASL). The audits were a thorough review of lab procedures, policies and methods which included on site observations and discussions of the processes involved in handling and processing the Island Gold Mine samples. ASL's audit of the Wesdome Lab (Bloom 2019b) demonstrated some deficiencies in their assaying of values lower than 2 g/t however, most samples sent to Wesdome are used for daily production decisions, not reporting of Mineral Resources and Reserves Mineral Resource is 3.74 g/t Au and Mineral Reserve cut off grade is 3.06 g/t Au for developed areas and 3.74 g/t Au for undeveloped areas, therefore any inaccuracy below 2.0 g/t Au will not have an impact. The suggestions of ASL for both laboratories have been implemented.

13 MINERAL PROCESSING AND METALLURGICAL TESTING

13.1 Overview

The Island Gold process plant has been treating ore since 2008 and the metallurgy is well understood. Since 2016, the ore feed has been almost exclusively from the Lower Island Gold Domain. Historical data shows that the Lower Zone ore behaves similarly to ore from the other zones and therefore has no significant effect on mill gold recoveries.

The process plant has consistently achieved recoveries of greater than 96%. Figure 13-1 shows the gold recovery from the last six years.

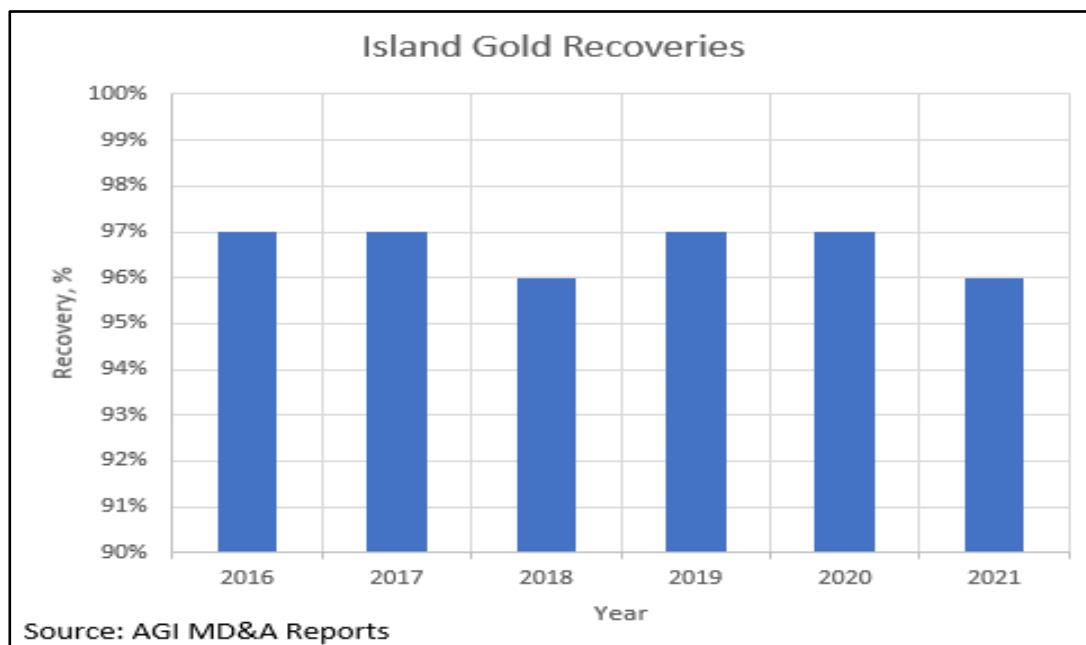


Figure 13-1 Mill Gold Recovery of Island Ore

Historical metallurgical test work has been completed on ore samples from the mine and has been documented in previous technical reports. The most recent metallurgical test work was completed in 2016 and pertinent information is summarized in this section.

A confirmatory metallurgical test work program has been planned and will commence in Q3 2022.

13.2 Ore Mineralogy

The Island Gold ore contains the presence of strong silicification, sericitization and carbonatization. Alteration minerals associated with auriferous quartz veining include biotite, Ca-Mg-Fe carbonates, chlorite (ripidolite), plagioclase, quartz, sulphides (pyrite, pyrrhotite and chalcopryrite) and white mica (muscovite +/- phengite). A small percentage of ore exists (called “cooked ore”) that has stronger silicification and increased hematite staining proximal to the dykes, and occasional enrichment in regions very close to the contact. The three main rock types are: dacitic volcanics (T2), gabbro (I3G) and quartz-dabase dykes (I3DD).

The tables below provide a summary of the chemical and mineral content of samples tested during the 2016 metallurgical test work program at ALS Global (“ALS”).

Table 13-1 Chemical Content Summary

Sample	Assay				
	% Cu	% Fe	g/t Au	g/t Ag	% As
Upper Part	0.005	3	8.99	2	0.006
Lower Part	0.004	3.2	14.3	2	0.007
Concentrator Feed	0.007	3.3	13.6	2	0.020

Source: ALS 2016

Table 13-2 Mineral Content Summary

Mineral	Mineral Content %		
	Concentrator Feed	Lower Part	Upper Part
Chalcopyrite	<0.1	<0.1	<0.1
Pyrite	2.9	2.1	2.5
Quartz	48.4	40.3	39.1
Micas	15.2	18.3	18.0
Feldspars	18.9	26.8	27.5
Kaolinite	0.5	0.4	0.3
Chlorite	5.4	5.3	5.8
Carbonates	5.3	3.3	3.4
Others	3.4	3.5	3.5
Total	100	100	100

Source: ALS 2016

Mineralogical analysis of these samples was by QEMSCAN BMA protocols and measured most of the samples as silicate minerals such as quartz, micas, feldspars, or chlorite; some carbonate minerals were also measured. Some sulphide content was present, primarily as pyrite. Some copper was detected within chalcopyrite; chalcopyrite would exhibit limited cyanide solubility and at that time sodium cyanide consumption did not increase, as expected, in the existing cyanidation leach process.

13.3 Comminution Tests

Historical bench scale comminution tests were completed at ALS during 2016 and at SGS during 2017. A total of 19 samples were submitted for the Bond suite of tests (AI, CWI, BWI) and SMC tests. A bulk rock sample was submitted for JK Drop Weight test (DWT). All SMC tests were completed on the -31.5+26.5 mm size fraction, except for two samples that were done on the -22.4+19 mm size fraction. All samples for Bond ball grindability tests, except for one, were conducted at a closed side setting of 106 µm. Ore hardness is considered soft to moderately soft in terms of ball milling.

Table 13-3 Comminution Summary

	Relative SG (SMC)	Axb	BWI (kWh/t)	RWI (kWh/t)	CWI (kWh/t)	AI (g)
Average	2.76	37.3	12.2	14.9	16.2	0.117

Source: Orway 2021

13.4 Cyanidation Tests

During the 2016 ALS test work program, cyanidation bottle roll tests were completed on three composites at three primary grind sizes along with a gravity recoverable test.

Cyanidation tests were conducted at primary grind sizes of 130, 100, and 65 µm K80. The leach tests were conducted at a sodium cyanide concentration of 300 ppm at a pH of 11 over a 48-hour period. The pH of the slurry was modulated with lime.

Overall gold extractions were high and ranged from about 94 to 98% of the feed gold. Extractions of about 98% of feed gold were achieved at a nominal primary grind sizing of 65 µm K80. Gold extractions ranged from 94 to 96% at a nominal 130 µm K80 grind size, silver extractions ranged from about 82 to 90%. This increased to between 89 and 93% at a nominal 65 µm K80 grind size. Gold and silver leach kinetics were generally rapid, with most of the leach extraction taking place over the first 24 hours. The test results are summarized in the Table 13-4.

Table 13-4 Leach Tests Summary

Sample	Grind µm K80	48h Extraction Au%	Feed Grade g/t Au	Residue Grade g/t Au	NaCN kg/t	Lime kg/t
Upper Part	129	95.5	11.0	0.49	0.2	0.7
	105	95.9	9.74	0.4	0.1	0.8
	63	97.7	12.4	0.29	0.4	0.8
Lower Part	129	93.7	14.0	0.89	0.1	0.4
	105	95.3	13.7	0.64	0.2	0.4
	63	97.7	13.9	0.32	0.4	0.4
Concentrator Feed	129	95.3	11.9	0.56	0.1	0.4
	105	96.2	12.4	0.47	0.1	0.4
	63	97.7	12.4	0.29	0.3	0.4

Source: ALS 2016

13.5 Gravity Recoverable Gold Tests

A gravity recoverable gold test was completed using 30 kg of the Concentrate Feed sample from the 2016 ALS program. The test was completed by running the sample through a Knelson

concentrator using a 100 g cone at sequentially finer grind sizes. The feed was first ground to 743 µm K80, run through the gravity concentrator, then gravity tailings were ground finer to 336 µm K80 and the gravity concentration was repeated. The gravity tailings were ground finally to 121 µm K80 and introduced once more through the Knelson Concentrator.

In the first concentration stage at the coarsest grind sizing, about 30% of the gold was recovered to the gravity concentrate. About 21% of the gold was recovered in the intermediate stage, and at the finest stage, a further 19% of the gold was recovered. The overall gold recovery was about 70% of the feed gold to a combined gravity concentrate grading about 646 g/t. The Concentrate Feed sample appeared to be amenable to gravity separation. Currently there is no gravity recovery circuit at Island Gold.

13.6 Future Metallurgical Test Work

A metallurgical test work program has been planned and will commence Q3 2022 for the following objectives:

- Confirm the metallurgical response of ore from future underground mining at Island Gold; and
- Generate sufficient metallurgical data to support engineering design for a plant expansion.

The metallurgical test work program will be performed on existing core samples. Four composites, representing four future periods of mining, and 17 variability samples will be tested. The scope of the metallurgical test work program includes:

- Sample preparation of all samples to generate sufficient mass and the required crush sizes for the various tests;
- All samples will be subject to BWI and RWI. The four composites will be subject to SMC tests. Select samples will be subject to BWI at a finer closing screen size, and AI;
- Duplicate head assay samples will be assayed for Au, Ag, C, TOC, S, and Fe. An ICP and WRA multi-element scan will also be performed on one of the head samples;
- A separate head sample will be assessed for quantitative mineralogy using the BMA protocol on QEMSCAN for each master composite and the select variability samples that will be evaluated for leaching characteristics;
- A grind calibration will be performed on each composite to establish the grind time to achieve a target grind sizes between 53 and 75 µm K80;
- The master composites will each be subjected to a three-stage gravity recoverable gold test (E-GRG);
- A primary grind leach series, with and without gravity, at 53, 63 and 75 µm K80 will be conducted on each master composites, and the select variability samples;
- At the selected primary grind size, pre-oxidation, and air/oxygen sparging will be evaluated;
- Carbon loading tests (Sequential triple contact, Equilibrium isotherms) will be conducted on each of the master composites;

- Oxygen uptake rate determinations will be conducted on each of the master composites;
- Cyanide destruction (SO₂/Air) will be evaluated for each of the master composites;
- Dewatering testing, by way of static and dynamic thickening, together with viscosity determinations will be tested on each of the master composites and the select variability samples.
- All leach tests will be bottle roll tests with 2, 6, 24, 36, and 48 hour measurement intervals. Leach solutions and tailings will be assayed for Au and Ag; and
- Select tailings will be evaluated for ARD and shake flask extraction.

14 MINERAL RESOURCE ESTIMATES

14.1 Introduction

The December 31, 2021, Mineral Resource and Mineral Reserve Estimation was carried out by the Island Gold Mine Technical Services department's staff under the supervision of Raynald Vincent, P.Eng., M.G.P., Chief Geologist and Nathan Bourgeault, P.Eng., Chief Engineer. Both are considered Qualified Persons within the meaning of Canadian Securities Administrators' National Instrument 43-101.

This section describes the Mineral Resource estimation methodology and summarizes the key assumptions considered by Island Gold personnel. In the opinion of the authors, the Mineral Resource evaluation reported herein is a reasonable representation of the gold Mineral Resources contained within the Island Gold property at the current level of sampling. The Mineral Resources have been estimated in conformity with generally accepted CIM Estimation of Mineral Resource and Mineral Reserves Best Practices Guidelines and are reported in accordance with the Canadian Securities Administrators' National Instrument 43-101. Mineral Resources are not Mineral Reserves and do not have demonstrated economic viability. There is no certainty that all or any part of the Mineral Resources will be converted into Mineral Reserves.

Island Gold's technical staff carried out geological interpretation of the mineralized zones according to local conventions used at the Island Gold Mine as follows:

- C, D, D1, E2, and E1E for the Upper Island Domain (IG) and Lochalsh Domain (LC);
- GD2, GP2, GD3, GP5, GD6, GD7, and GD9 for the Goudreau Domain;
- E1E, B, D, D1, C, G, G1, GNW, and STH for the Lower Island Domain (EX); and
- E1E and E1D for the East Domain (X2).

The zones are clearly defined on longitudinal sections (Figure 14-1 and Figure 14-2).

Prior to outlining Mineral Resources on vertical longitudinal sections, a global Mineral Resource model was carried out to identify mineralized zones that meet defined technical parameters. The Mineral Resources presented here are exclusive of Mineral Reserves.

14.2 Mineral Resources Classifications, Categories and Definitions

- Mineral Resources were classified according to the Canadian Institute of Mining, Metallurgy and Petroleum (CIM) Definition Standards published on May 10, 2014 and adopted by CIM Council on May 19, 2014. A Mineral Resource is a concentration or occurrence of solid material of economic interest in or on the Earth's crust in such form, grade or quality and quantity that there are reasonable prospects for eventual economic extraction. The location, quantity, grade or quality, continuity and other geological characteristics of a Mineral Resource are known, estimated, or interpreted from specific geological evidence and knowledge, including sampling.
- An Inferred Mineral Resource is that part of a Mineral Resource for which quantity and grade or quality are estimated based on limited geological evidence and sampling. Geological evidence is sufficient to imply but not verify geological and grade or quality continuity. An Inferred Mineral Resource has a lower level of confidence than that applying to an Indicated Mineral Resource and must not be converted to a Mineral Reserve. It is reasonably expected that most of the Inferred Mineral Resources could be upgraded to Indicated Mineral Resources with continued exploration.

- An Indicated Mineral Resource is that part of a Mineral Resource for which quantity, grade or quality, densities, shape, and physical characteristics are estimated with sufficient confidence to allow the application of Modifying Factors in sufficient detail to support mine planning and evaluation of the economic viability of the deposit. Geological evidence is derived from adequately detailed and reliable exploration, sampling and testing and is sufficient to assume geological and grade or quality continuity between points of observation. An Indicated Mineral Resource has a lower level of confidence than that applying to a Measured Mineral Resource and may only be converted to a Probable Mineral Reserve.
- A Measured Mineral Resource is that part of a Mineral Resource for which quantity, grade or quality, densities, shape, and physical characteristics are estimated with confidence sufficient to allow the application of Modifying Factors to support detailed mine planning and final evaluation of the economic viability of the deposit. Geological evidence is derived from detailed and reliable exploration, sampling and testing and is sufficient to confirm geological and grade or quality continuity between points of observation. A Measured Mineral Resource has a higher level of confidence than that applying to either an Indicated Mineral Resource or an Inferred Mineral Resource. It may be converted to a Proven Mineral Reserve or to a Probable Mineral Reserve.
- Modifying Factors are considerations used to convert Mineral Resources to Mineral Reserves. These include, but are not restricted to, mining, processing, metallurgical, infrastructure, economic, marketing, legal, environmental, social, and governmental factors.

14.3 Methodology

The Mineral Resource evaluation methodology involves the following procedures:

- Database compilation and validation for the diamond drillholes used in the mineral resource estimate;
- Construction of wireframe models for the boundaries of the gold mineralization;
- Generation of drillhole interzones for each mineralized zone;
- Geostatistical analysis and variography;
- Block modelling and grade interpolation;
- Definition of Mineral Resource domains;
- Assessment of “reasonable prospects for economic extraction” and selection of appropriate cut-off grades; and
- Preparation of the Mineral Resource Statement.

The Qualified Person believes that the current drilling information is sufficiently reliable to interpret the boundaries for gold mineralization with confidence and that the assay data is sufficiently reliable to support Mineral Resource estimates. Datamine Studio RM version 1.10.10.0 (“Studio RM”) software was used to prepare assay data for geostatistical analysis, construct the block model, estimate metal grades, tabulate Mineral Resources, and construct all geological solids except the Goudreau Domain. The Goudreau Domain has not been drilled since 2019, so resource estimation and geological solids were finalized and created with Gemcom software for this domain.

The Island Gold geology department determines mineralized intersections for each drill hole and underground development face. The diamond drill hole intersections are determined from the

interpretation of vertical cross-sections and horizontal plans, while the development intersections are interpreted using face mapping and assay results of each development face. Each mineralized intersection is coded in the database according to its respective zone name.

14.4 Databases

Two different databases are used for Mineral Resource and Mineral Reserve estimates, one for surface and underground drill holes and a second one for channel sampling, both imported to Studio RM from acQuire.

The drill hole database contains all the holes drilled within the mine area, more specifically between mine grid 13,800E to 18,910E and 3,000N to 5,500N (mine grid). The drill hole database is managed using acQuire GIM Suite 4.0.3. As of December 31, 2021, the database contained 6,907 holes totaling 1,365,397 m of surface and underground diamond drilling (Table 14-1). Several surface holes were drilled prior to 1985 in programs coordinated by Canamax and previous owners. The remainder were drilled in later programs managed by Patricia, Richmond, and lastly by Alamos.

A total of 813 holes representing 447,187 m have been drilled from surface and 6,094 holes totaling 918,210 m have been drilled from underground at the Island Gold Mine. Underground drill holes are drilled from north to south because the mine infrastructure is in the footwall of the mineralized zone. Surface drill holes are drilled from south to north to drill perpendicular to the south dipping zones.

Surface holes are usually oriented perpendicular to the main trend of mineralization (Figure 14 1). Conversely, the underground drill holes have highly variable azimuths and dips, as they were drilled in a series of fans from individual drill stations, which creates a multitude of angles between drill holes and mineralization (Figure 14 2). Some drill holes may end up being parallel to the mineralization and not represent true mineralized width. However, the modelling technique mitigates this impact by limiting all intercepts within interpreted domains to those that comply with the true mineralized widths.

Table 14-1 Island Gold Drill Hole Database Summary (Dec 31, 2021)

Location	Number of Holes	Length (metres)
Surface	813	447,187
Underground	6,094	918,210
Total	6,907	1,365,397

In addition to the drill holes, underground channel samples of the Island Gold mine are also used in Mineral Resource estimation. The channel samples are taken from the 125 Level (upper mine) to the 920 Level (Lower Zone). Table 14-2 summarizes the channel sampling.

Table 14-2 Island Gold Channel Sample Database Summary (Dec 31, 2021)

Sample type	Number of Channels	Length (metres)
Channel sampling	9,931	41,984

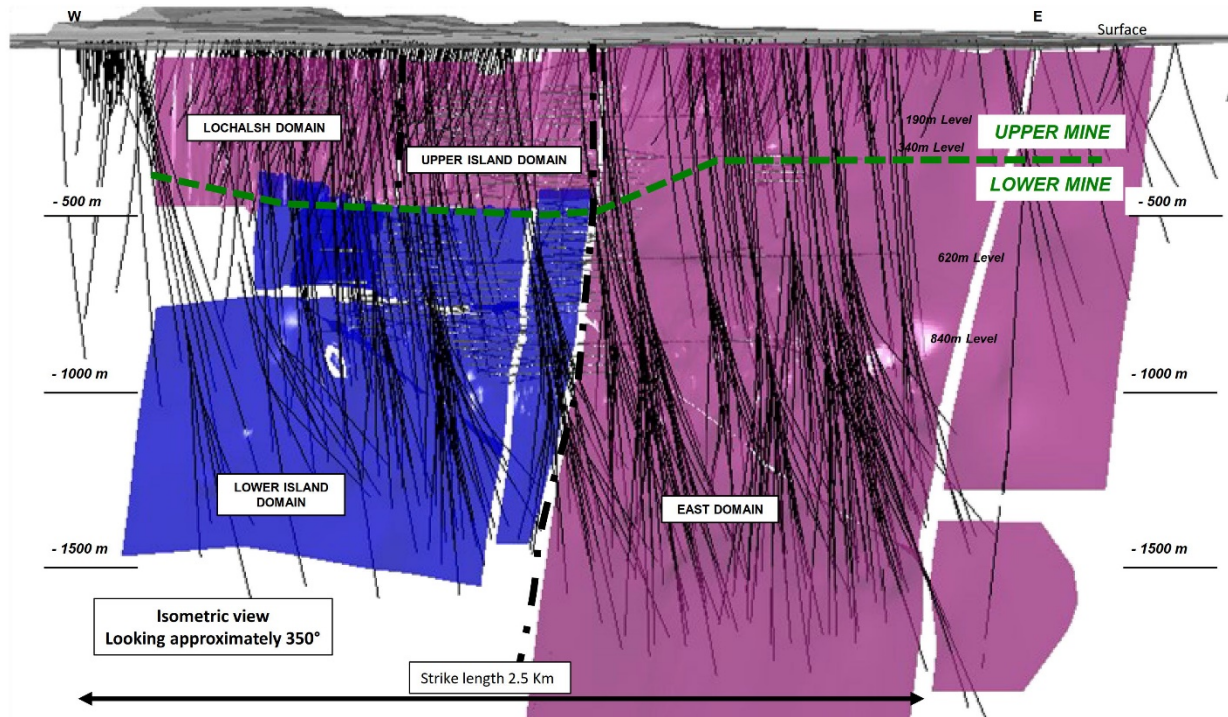


Figure 14-1 Surface Diamond Drilling – Island Gold Mine

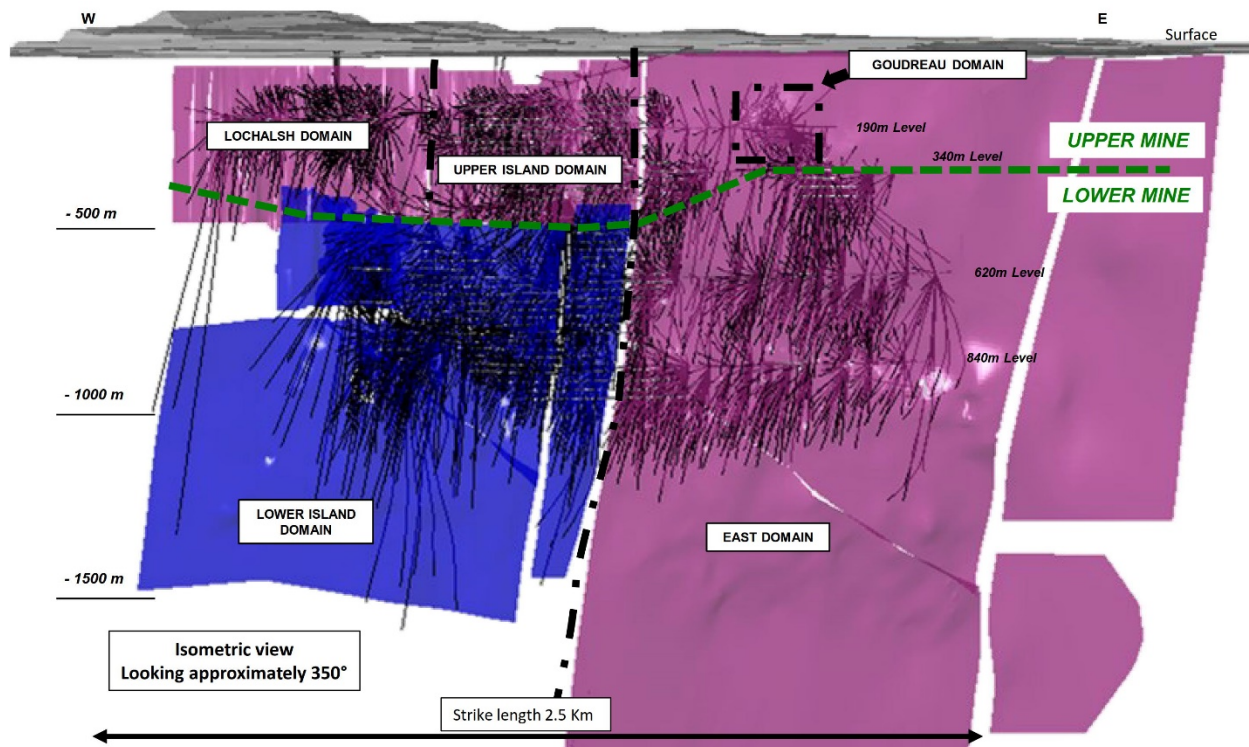


Figure 14-2 Underground Diamond Drilling – Island Gold Mine

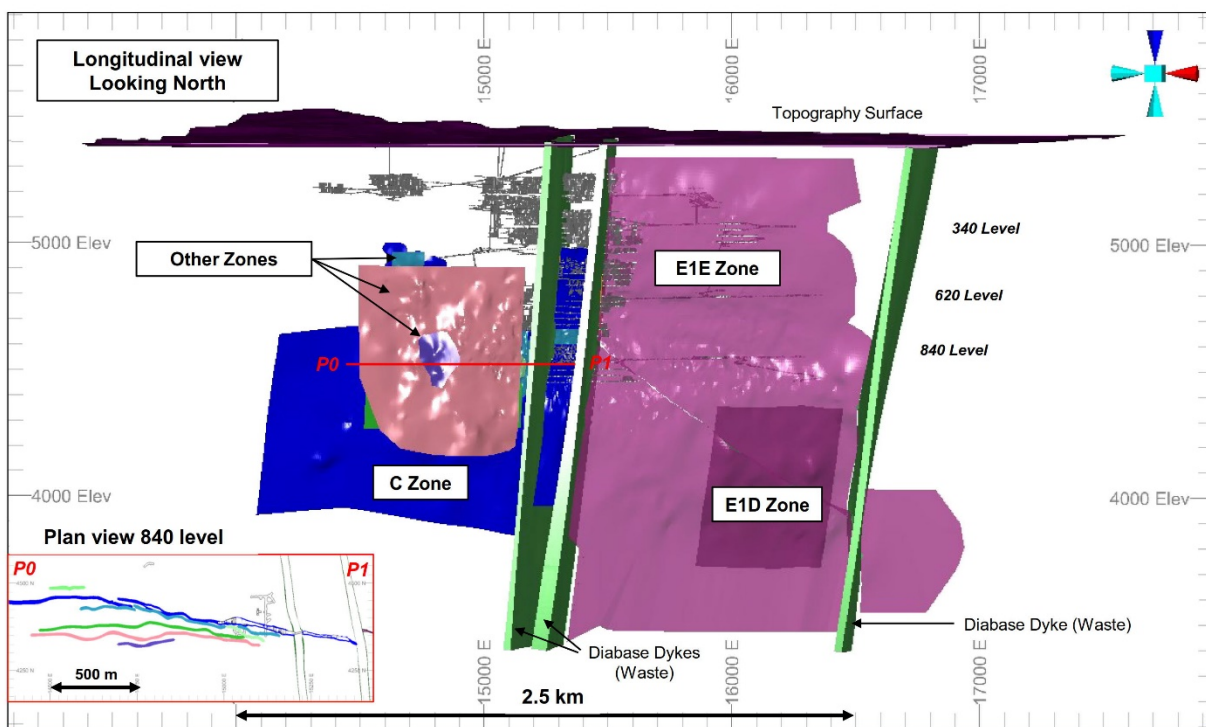


Figure 14-4 3D Mineralized Solid Island Gold Lower Zones

14.6 Statistical Analysis and Grade Capping

Drill hole assay intervals that intersect interpreted domains were coded in the database and were used to (a) analyze sample lengths, (b) generate statistics, and (c) compositing and variography. Table 14-3 and Table 14-4 summarizes some statistics drawn from the original (raw) assay data with the corresponding proposed grade capping value and the number of samples capped for the Island Gold Lower Zones. The statistics show that about 1.54% of the DDH samples and 3.70% of the chip samples were capped. All zone capping grades are displayed except for GP2, GP5, GD7 and GD9, which were not updated; therefore, historic capping grades of 75g/t for DDH and faces were applied.

Figure 14-5 and Figure 14-6 show an example of the histogram and the probability plot used for the grade capping study of Zone EX-C (Island Lower C Zone – code 730).

Table 14-3 Summary Statistics of Original Assay Samples from Diamond Drill Holes

Zone	Rock Code	Number Samples	Uncapped Mean	C.V.	Capping Value	Number Capped	Capped Mean	Capped C.V.	Apparent Loss	% Capped
EX-C	730	18,911	19.96	6.55	225	276	15.22	2.56	23.7%	1.5%
EX-D1	725	2,313	10.12	6.01	45	66	4.23	2.39	58.2%	2.9%
EX-D	720	2,708	6.29	4.74	50	55	3.87	2.61	38.5%	2.0%
EX-G	740	3,393	10.1	11.4	70	47	4.53	2.65	55.1%	1.4%
EX-G1	745	1,773	7.87	7.76	45	33	3.48	2.48	55.8%	1.9%
EX-GNW	742	644	7.91	2.8	70	12	6.86	2.1	13.3%	1.9%
EX-B	760	6,263	11.06	9.6	90	122	5.92	2.67	46.5%	1.9%
EX-STH	750	103	8.14	3.13	50	2	5.85	1.76	28.1%	1.9%
X2-E1E	310	13,917	14.4	4.92	185	173	11.05	2.66	23.3%	1.2%
X2-E1D	305	442	9.7	4.87	80	8	5.65	2.88	41.8%	1.8%
LC-E1E	410	3,203	8.61	4.33	75	46	6.09	2.3	29.3%	1.4%
LC-D	420	1,660	2.75	4.61	45	12	2.16	2.91	21.5%	0.7%
LC-C	430	1,481	6.93	7.87	60	21	3.61	2.79	47.9%	1.4%
LC-E2	450	1,490	5	5.22	55	13	3.41	2.54	31.8%	0.9%
IG-E1E	110	4,173	9.2	3.3	100	52	7.71	2.35	16.2%	1.2%
IG-D	120	2,611	8.74	5.26	75	43	5.71	2.39	34.7%	1.6%
IG-C	130	2,506	8.95	5.68	100	28	6.14	2.65	31.4%	1.1%
IG-E2	150	572	13.88	6.67	75	11	6.13	2.36	55.8%	1.9%
IG-D1	160	1,505	3.96	3.43	75	8	3.6	2.62	9.1%	0.5%
X1-E1E	210	1,307	8.33	2.57	75	16	7.31	1.9	12.2%	1.2%
GD-G2	620	488	16.72	3.55	100	22	9.96	2.43	40.4%	4.5%
GD-G3	630	618	12.9	6.38	75	17	5.38	2.76	58.3%	2.8%
GD-G6	660	434	29.95	2.88	100	34	16.66	1.79	44.4%	7.8%
All zones	DDH	72,515	12.87			1,117	9.07		29.5%	1.54%

Table 14-4 Summary Statistics of Original Assay Samples from Faces

Zone	Rock Code	Number Samples	Uncapped Mean	C.V.	Capping Value	Number Capped	Capped Mean	Capped C.V.	Apparent Loss	% Capped
EX-C	730	12,930	19.2	3.92	125	474	12.89	2.19	32.9%	3.7%
EX-D	720	99	10.78	5.64	20	6	3.13	1.77	71.0%	6.1%
EX-G	740	257	10.93	4.3	35	14	6.11	1.58	44.1%	5.4%
EX-GNW	742	134	12.63	3.13	30	11	6.03	1.61	52.3%	8.2%
EX-B	760	511	17.28	4.08	60	32	8.49	1.93	50.9%	6.3%
X2-E1E	310	4,800	12.42	4.58	100	117	8.5	2.31	31.6%	2.4%
LC-E1E	410	2,464	11.78	3.68	75	96	7.96	2.24	32.4%	3.9%
LC-D	420	72	4.34	3.31	45	2	3.48	2.77	19.8%	2.8%
LC-C	430	102	8.23	7.74	60	1	2.47	2.84	70.0%	1.0%
IG-E1E	110	6,616	12.23	3.55	75	252	8.44	2.06	31.0%	3.8%
IG-D	120	3,930	12.65	3.77	75	135	8.08	2.04	36.1%	3.4%
IG-C	130	3,962	14.12	5.34	75	147	8.44	2.01	40.2%	3.7%
IG-E2	150	412	6.36	3.35	30	21	3.7	1.99	41.8%	5.1%
IG-D1	160	1,361	6.35	3.26	75	26	5.38	2.46	15.3%	1.9%
X1-E1E	210	3,239	14.96	3.53	75	130	9.53	1.87	36.3%	4.0%
GD-G2	620	264	81.9	4.09	75	42	18.8	1.52	77.0%	15.9%
GD-G3	630	264	18.32	3.95	75	18	8.13	2.46	55.6%	6.8%
GD-G6	660	215	37.11	5.31	75	18	13.54	1.72	63.5%	8.4%
All zones	Faces	41,632	15.21			1,542	9.74		36.0%	3.70%

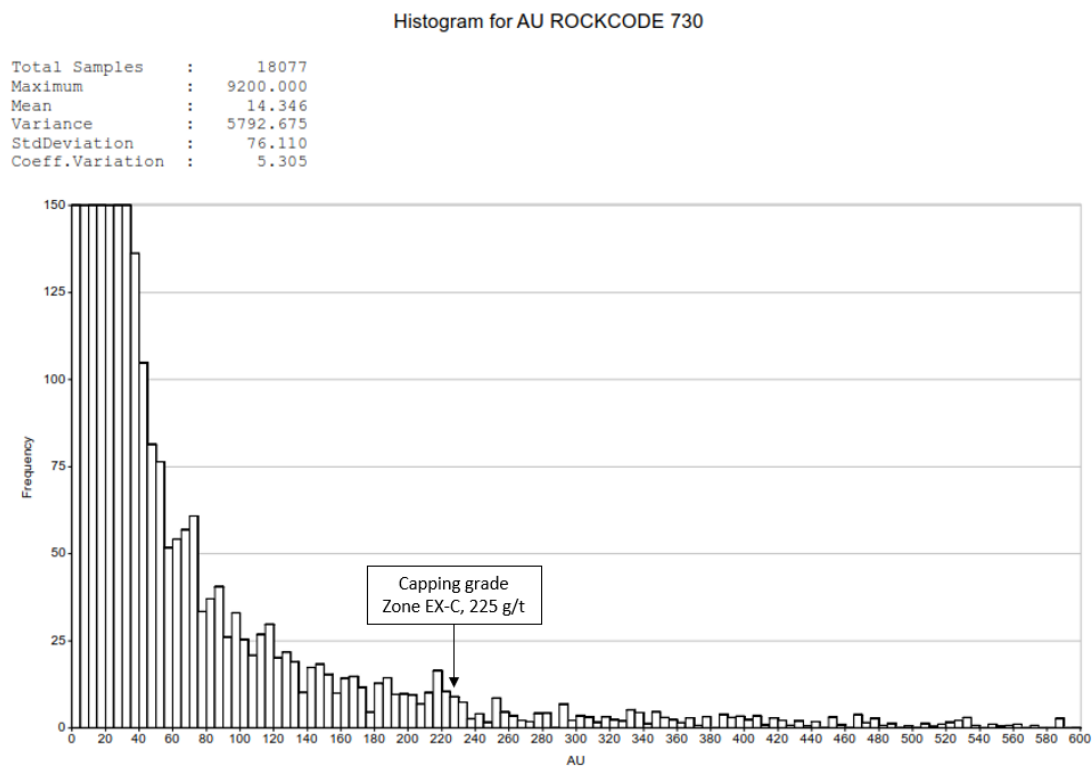


Figure 14-5 Histogram Plot Zone EX-C (Island Lower)

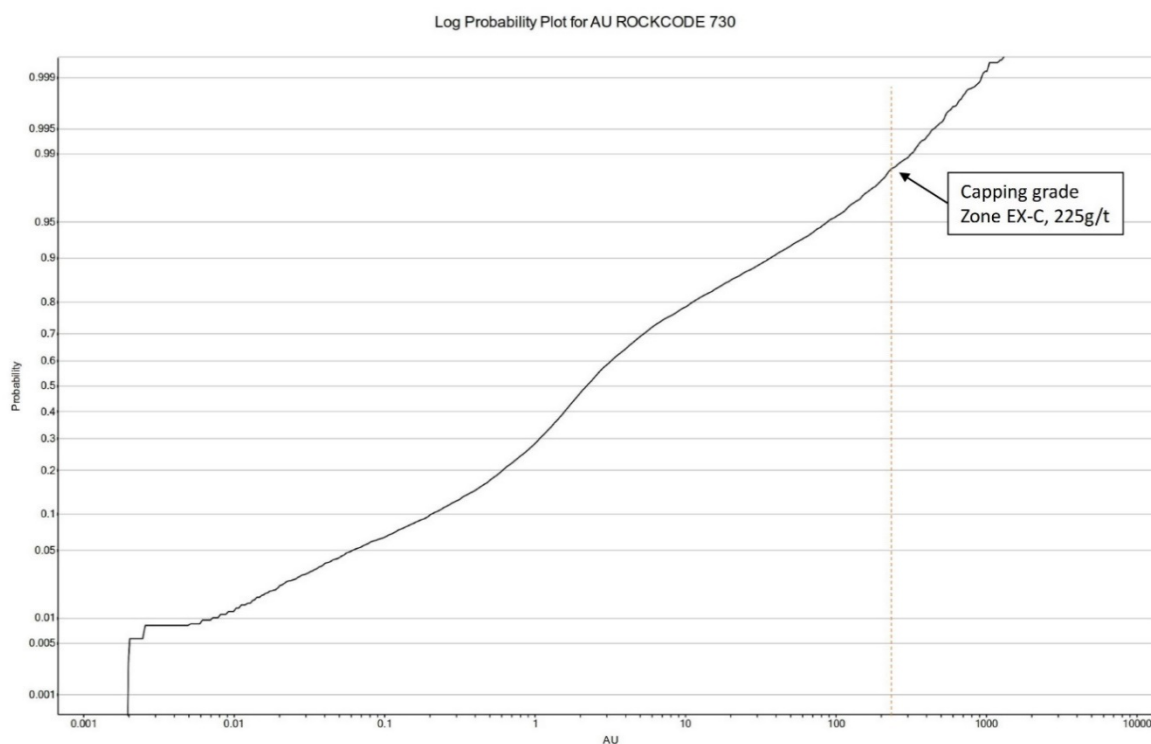


Figure 14-6 Probability Plot Zone EX-C (Island Lower)

14.7 Compositing

Intervals within each interpreted mineralized domain were coded by zone. Capped assays were composited over two-metres length, from drill hole collar to toe, within the interpreted zones. Factors influencing the selection of composite length included the relatively narrow nature of many of the mineralized domains (3 m to 8 m), the proposed block size (10 m x 10 m x 4 m in the Lower Island Domain and 10 m x 5 m x 2 m in the East Domain), and the original sample length (between 0.3 m to 1.5 m).

The composite length avoids de-compositing samples, which occurs when the sample length exceeds the composite length and provides a reasonable reconciliation to the raw data mean grade while sufficiently reducing the variation coefficient. All intervals within solids with no assays were given zero value during compositing. Composites of less than 0.75 m were discarded for grade interpolation.

Zones in the upper part of the mine, including the Lochalsh, Goudreau and Upper Island domains, were composited differently than the lower zones. The zones are typically thinner; therefore, a smaller block size is used for the estimation (5 m x 5 m x 2 m). As a result of these differences, the drill holes were composited over one-metre lengths, from drill hole collar to toe, within the interpreted zones. Composites of less than 0.3 m were discarded for grade interpolation.

Descriptive statistics of the coded composites used for the zones in Lower Island (EX) and East domains are shown in Table 14-5.

Table 14-5 Summary Statistic for 2 m Composites

Zone	Code	Number of Composites	Mean Au Grade g/t	Standard Deviation	Coefficient of Variation
EX-C	730 DDH	6,594	11.65	22.76	1.95
EX-D1	725 DDH	1,252	1.97	4.24	2.15
EX-D	720 DDH	1,472	1.72	3.85	2.24
EX-G	740 DDH	1,611	2.84	6.18	2.17
EX-G1	745 DDH	919	2	4.29	2.15
EX-GNW	742 DDH	260	5.26	8.08	1.54
EX-B	760 DDH	2,634	4.09	8.78	2.15
EX-SHT	750 DDH	44	4.29	4.75	1.11
X2-E1D	305 DDH	189	2.66	6.76	2.54
X2-E1E	310 DDH	4,214	8.3	16.59	2
EX-C	730 Faces	4,846	12.87	19.28	1.5
EX-G	740 Faces	97	5.53	5.89	1.06
EX-D1	725 Faces	6	8.23	6.44	0.78
EX-GNW	742 Faces	50	6.16	6.98	1.13
EX-B	760 Faces	205	7.92	9.63	1.21
X2-E1E	310 Faces	1,666	7.54	10.45	1.39

14.8 Density

The density determination of different zones in the lower part of the mine was conducted in 2016. Results confirm figures that have been used in the past, with density for ore zones established at 2.78 t/m³. This density has been used for December 2021 Mineral Resource estimates.

Table 14-6 Specific Gravity Measurements

Number of measurements	Mean	Median	Standard Deviation	Coefficient of Variation
4,828	2.78	2.77	0.078	0.028

14.9 Variography

Studio RM software is used to model the spatial continuity of the Island Gold deposit. Experimental variograms are generated from the composites within the zones.

The variography models the nugget effect and the spherical structures represent the larger scale spatial variability of the datasets. The modelled variograms for domains 730 and 310 (EX-C and X2 zones) are summarized in Table 14-7. The resulting orientations were visualized in Studio RM to determine if the directions of the axes were consistent with the solid orientations. The orientations usually fit well with the general orientations of the interpreted zones. Note that some reported rotations have been adjusted based on interpreted geological constraints. Search ellipse orientations are also shown overlain on a long section in Figure 14-7.

The nugget effect, which is determined by variography, is high, around 50% of the total variance for EX-C and X2-E1E. Variography provided information relevant to the estimation parameters used for ordinary kriging, as well as a guide for Mineral Resource classification.

Table 14-7 Island Gold Deposit Variography Study

Sector – Zone	Nugget	Ranges		Search Anisotropy		
		1st Structure	2nd Structure	Principal Azimuth	Principal Dip	Second Azimuth
East Domain E1E Zone Above 4500/4600 Elevation	113	X: 16	X: 50	0	-67	50
		Y: 17	Y: 45			
		Z: 18	Z: 8			
		Sill: 63	Sill: 42			
East Domain E1E Zone East of 15700E Below 4500/4600 Elevation	180	X: 21	X: 49	10	-65	50
		Y: 25	Y: 57			
		Z: 15	Z: 7			
		Sill: 84	Sill: 93			
East Domain E1E Zone West of 15700E Below 4500/4600 Elevation	375	X: 38		-175	74	45
		Y: 29				
		Z: 17				
		Sill: 371				
Lower Island Domain EX-C (730)	375	X: 38		-175	74	45
		Y: 29				
		Z: 17				
		Sill: 371				

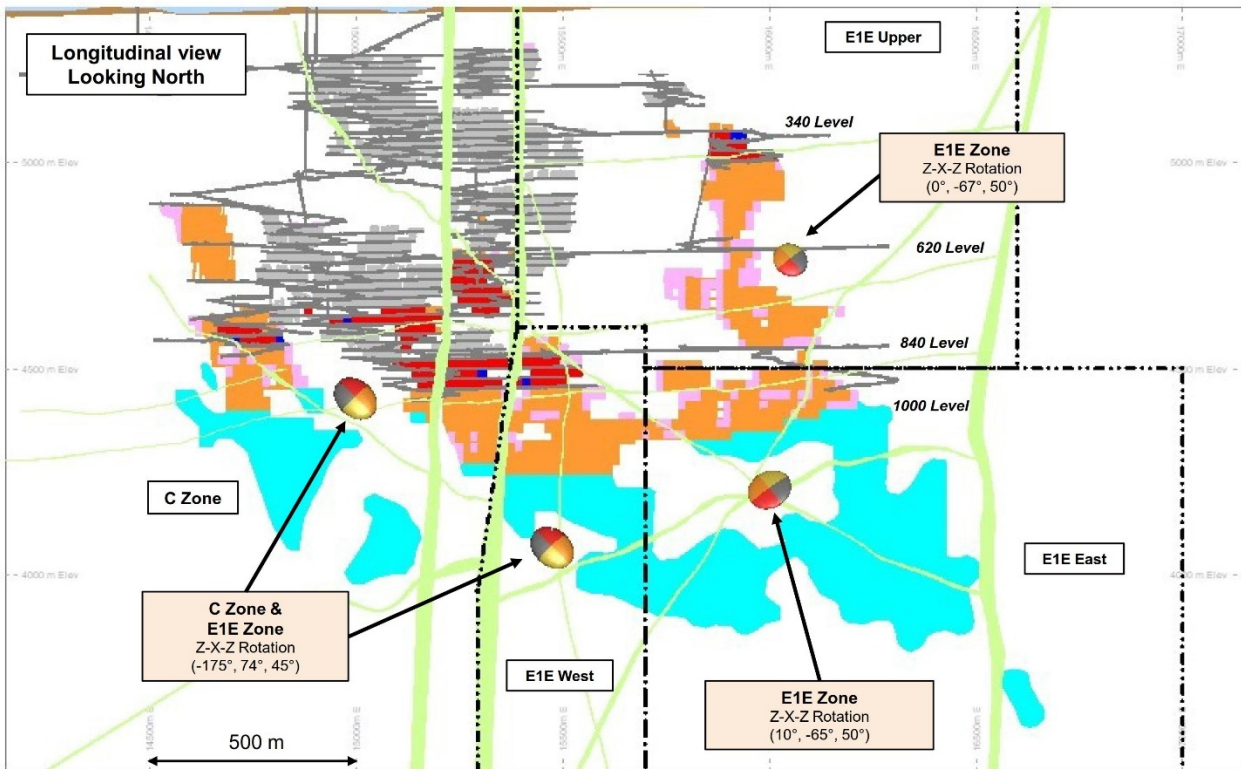


Figure 14-7 Search Ellipse Orientation Corresponding to Variography

14.10 Block Modelling

14.10.1 Upper Island Gold Mine

Block modelling consists of a 3D array of cells (blocks) with specified dimensions, for which each block's grade is determined by an interpolation method (algorithm). Two different models were produced for the upper Island mine, between the surface and a 400 m depth. The first one is for the Lochalsh and Upper Island domains and the second one is for the Goudreau Domain. The block dimensions for these sectors are 5 m along strike (east-west), 5 m in elevation, and 2 m across strike (north-south). The blocks are tagged to a specific rock code according to which mineralized zone they fall within. A percentage inside the solid is given to each block.

Block grade interpolation was carried out using inverse distance squared (ID2) for the Goudreau Domain and ordinary kriging for the two other domains. The following methodology is used to interpolate the grade of each block; (a) a minimum of 9 composites used for the first pass; (b) a minimum of 5 for the second pass; (c) a minimum of 1 for the third pass; and (d) a maximum of 16 composites for all passes. The search ellipse dimensions used for grade interpolation are different for each sector.

14.10.2 Lower Island Gold Mine

Two block models are used to estimate Mineral Resources for the lower mine. One for the west side in the Lower Island Domain for the C zone and parallel smaller structures (EX model) and the second for the E1E and E1D zone in the East Domain (X2 model).

Mineral Resources for the Lower Island Domain (EX model) were estimated by block modelling with 10 m block dimensions along strike, 10 m in elevation, and 4 m across-strike in the Studio RM software. To better model boundaries within the model space, the blocks have been sub-divided into smaller cuboid sizes (sub-cells) while keeping the storage and computational efficiency of the standard block model. Using 10 sub-cells in X and Z and seam filling in Y to allow the sub cell to be as large as necessary to fill the wireframe in Y direction, gives a minimum sub-cell size at 1 m x variable x 1 m in X, Y, Z.

Composites lengths are two metres and composites shorter than 0.75 m are discarded. Block model parameters for the EX model are given in Table 14-8.

Table 14-8 Block Model Parameters, EX Model

	East (X)	North (Y)	Elevation (Z)
Origin	14,000	3,880	5,500
Block Size	10	4	10
Number of Blocks	260	190	172
Model Rotation	0°		

For the East Domain (X2 model), block dimensions were changed to 10 m along strike, 5 m in elevation, and 2 m across-strike to reflect the thinner mineralized zone, and the shallower dip of approximately 65 degrees. Those dimensions fit better with the solids' orientation and dimensions. Sub-dividing of 10 sub-cells in X and Z and seam filling in Y provides a minimum sub-cell size at 1 m x variable x 0.5 m in X, Y, Z.

Block model parameters for the X2 model are given in Table 14-9.

Table 14-9 Block Model Parameters, X2 Model

	East (X)	North (Y)	Elevation (Z)
Origin	15,250	3,400	5,550
Block size	10	2	5
Number of blocks	180	700	450
Model rotation	0°		

14.10.3 Grade Interpolation Methodology for Island Lower Zones

The grade interpolation for the Island Gold Lower Domain block models was completed using ordinary kriging method. Anisotropic search ellipsoids were selected for the grade interpolation process based on the analysis of the spatial continuity of capped composites using variography (Table 14-10). Minimum and maximum numbers of composites are set for interpolation, and restrictions are placed on the number of composites used from each drill hole.

The grade estimates were generated using the 2 m composites. The blocks included in a zone are estimated only with the composites coded within this zone (hard boundary). The block models were estimated using the following parameters for both block models and all zones, except for some minor differences in the C zone estimate.

In addition to the capping of high-grade outliers in the EX-C zone, the spatial influence of high-grade assays was restricted in the third and fourth pass in the grade interpolation process.

Composites with grades higher than 125 g/t Au had their influence limited to a search ellipse of 10 m x 10 m x 10 m. The same restriction has been applied for all the parallel zones but only during the last pass, except for the GNW zone which does not have any restriction. This procedure is judged prudent to limit excessive smearing of high grade samples during interpolation of blocks in areas where drill spacing is wider for example, exploration holes beneath the mine.

Search parameters are as follows:

- First Pass: minimum of 9 and maximum of 16 composites found within a search ellipse that has the longest axis close to the range of the first structure identified by variography. A maximum of four composites per drill hole from a minimum of three different drill holes, can be used for any block estimate. Samples from development are used for this pass, therefore, only blocks within 10 m of development are estimated. Those blocks are classified as Measured Mineral Resources.
- Second Pass: minimum of 5 and maximum of 16 composites within a search ellipse with the longest axis that corresponds to the range of the second structure identified by variography. A maximum of four composites per drill hole from a minimum of two different drill holes, can be used for block estimates. Samples from development are used for this pass, therefore, only blocks within 10 m of development are estimated (corresponds to Measured Mineral Resources).
- Third Pass: minimum of 5 and maximum of 16 composites within a search ellipse with the longest axis that corresponds to the range of the second structure identified by variography. A maximum of four composites per drill hole from a minimum of two different drill holes, can be used for block estimates. Samples from development are not used for this pass, therefore, all blocks that have not been estimated in the previous interpolation passes can be estimated. Grades higher than 125 g/t have a restricted ellipse (C zone only and for zones without any development).
- Fourth Pass: minimum of 1 and maximum of 16 composites within a search ellipse with the longest axis that corresponds to 1.5X the range of the second structure identified by variography. A maximum of four composites per drill hole from a minimum of one drill hole, can be used for block estimates. Samples from development are not used in this pass, therefore, all blocks that have not been estimated in the previous interpolation passes can be estimated. Grades higher than 125 g/t have a restricted ellipse (except for GNW zone).

Table 14-10 Summary of Island Lower Zones Estimation Parameters

Zone	Pass	Method	Search Anisotropy			Sample Search			Sample			Data type	Restricted Ellipse
			Primary Azimuth	Prim. Dip	Int. Azmth	X	Y	Z	Min	Max	Max per Hole		
EX-C	1	OK	-175	74	45	25	20	10	9	16	4	DDH + Faces	No
EX-C	2	OK	-175	74	45	40	30	10	5	16	4	DDH + Faces	No
EX-C	3	OK	-175	74	45	40	30	10	5	16	4	DDH only	Yes
EX-C	4	OK	-175	74	45	60	45	20	1	16	4	DDH only	Yes
X2-E1E UP	1	OK	0	-67	50	30	25	10	9	16	4	DDH + Faces	No
X2-E1E UP	2	OK	0	-67	50	45	40	10	5	16	4	DDH + Faces	No
X2-E1E UP	3	OK	0	-67	50	45	40	10	5	16	4	DDH only	No
X2-E1E UP	4	OK	0	-67	50	75	70	20	1	16	4	DDH only	No
X2-E1E WEST	1	OK	-175	74	45	25	20	10	9	16	4	DDH + Faces	No
X2-E1E WEST	2	OK	-175	74	45	40	30	10	5	16	4	DDH + Faces	No
X2-E1E WEST	3	OK	-175	74	45	40	30	10	5	16	4	DDH only	No
X2-E1E WEST	4	OK	-175	74	45	60	45	20	1	16	4	DDH only	No
X2-E1E EAST	1	OK	10	-65	50	30	35	10	9	16	4	DDH only	No
X2-E1E EAST	2	OK	10	-65	50	50	55	10	5	16	4	DDH only	No
X2-E1E EAST	3	OK	10	-65	50	70	75	20	1	16	4	DDH only	No
EX-B	1	OK	-175	74	45	25	20	10	9	16	4	DDH + Faces	No
EX-B	2	OK	-175	74	45	40	30	10	5	16	4	DDH + Faces	No
EX-B	3	OK	-175	74	45	40	30	10	5	16	4	DDH only	No
EX-B	4	OK	-175	74	45	60	45	20	1	16	4	DDH only	Yes
EX-D	1	OK	-175	74	45	25	20	10	9	16	4	DDH + Faces	No
EX-D	2	OK	-175	74	45	40	30	10	5	16	4	DDH + Faces	No
EX-D	3	OK	-175	74	45	40	30	10	5	16	4	DDH only	No
EX-D	4	OK	-175	74	45	60	45	20	1	16	4	DDH only	Yes
EX-D1	1	OK	-175	74	45	25	20	10	9	16	4	DDH only	No
EX-D1	2	OK	-175	74	45	40	30	10	5	16	4	DDH only	No
EX-D1	3	OK	-175	74	45	60	45	20	1	16	4	DDH only	Yes
EX-G	1	OK	-175	74	45	25	20	10	9	16	4	DDH + Faces	No
EX-G	2	OK	-175	74	45	40	30	10	5	16	4	DDH + Faces	No
EX-G	3	OK	-175	74	45	40	30	10	5	16	4	DDH only	No
EX-G	4	OK	-175	74	45	60	45	20	1	16	4	DDH only	Yes
EX-GNW	1	OK	14	-69	59	25	20	10	9	16	4	DDH + Faces	No
EX-GNW	2	OK	14	-69	59	40	30	10	5	16	4	DDH + Faces	No
EX-GNW	3	OK	14	-69	59	40	30	10	5	16	4	DDH only	No
EX-GNW	4	OK	14	69	59	60	45	20	1	16	4	DDH only	No
EX-G1	1	OK	-175	74	45	25	20	10	9	16	4	DDH only	No
EX-G1	2	OK	-175	74	45	40	30	10	5	16	4	DDH only	No
EX-G1	3	OK	-175	74	45	60	45	15	1	16	4	DDH only	Yes
EX-STH	1	OK	-175	74	45	25	20	10	9	16	4	DDH only	No
EX-STH	2	OK	-175	74	45	40	30	10	5	16	4	DDH only	No
EX-STH	3	OK	-175	74	45	60	45	20	1	16	4	DDH only	Yes

Note:

- Restricted ellipse of 10 m x 10 m x 10 m is applied for composites greater than 125 g/t

14.11 Block Model Validation

Validation of the interpolated model was undertaken to confirm estimation parameters, to verify that the model reflects the input data on both local and global scales and particularly to verify that

the estimate is not biased. The validation was performed using a combination of different techniques, as follows:

- Inspection of block grades in plans and sections and visual comparison with drill hole grades;
- Statistical validation of sample means versus block estimates by zones; and
- Mean sample grade within a block versus interpolated grade

14.11.1 Visual Validation

Visual validation is a verification of the interpolated block model on a local block scale versus the composite grades. A visual inspection has been carried out on cross-sections and bench/level plans. Comparison between local block estimates and nearby composites is generally good and indicates that grade smoothing is not excessive in the block model.

14.11.2 Statistical Validation

With Studio RM software, statistics have been generated during the different steps of the block model estimation to validate the process. The process is as follows:

- Create a volume comparison table to evaluate the percent difference between the wireframe and the block model to make sure the block size is optimal (Table 14-11);
- Create a statistics table after interpolate pass with faces and/or DDH; and
- Create statistic tables for the entire block model showing the number of blocks estimated during passes associated with the minimum, maximum and mean grades (Table 14-12).

Table 14-11 Volume Comparison of the EX Model between Wireframes and Block Model

Zone Rock Code	Model Volume (m3)	Wireframe Volume (m3)	Percent Difference (%)
D-720	181,293	181,307	0.01
D1-725	64,102	64,295	0.30
C-730	3,953,883	3,953,853	0.00
D1N-732	57,349	57,310	-0.07
G-740	1,118,912	1,119,171	0.02
GNW-742	42,706	42,715	0.02
G1-745	1,451,797	1,451,764	0.00
STH-750	79,274	79,290	0.02
B-760	740,749	740,773	0.00

Table 14-12 Block Model Statistics for EX model

Zone Rock Code	Pass	Number of blocks	Min. block grade (g/t)	Max. block grade (g/t)	Mean grade (g/t)
C-730	-	538,472	0	0	0
C-730	1	251,300	0.17	92.49	11.70
C-730	2	1,697	0.80	15.26	6.57
C-730	3	402,855	0.00	76.64	9.02
C-730	4	731,853	0.00	78.91	4.96
B-760	1	14,486	1.23	34.83	8.47
B-760	3	329,741	0	51.50	3.50
B-760	4	62,375	0	15.36	2.57
D-720	-	24	0	0	0
D-720	1	8,549	0.10	13.07	4.54
D-720	2	491	0.53	2.91	1.75
D-720	3	76,634	0	15.32	2.11
D-720	4	24,033	0	8.87	1.84
D1-725	-	8,508	0	0	0
D1-725	1	1,506	4.78	6.20	5.34
D1-725	2	8,906	0.01	13.22	4.09
D1-725	3	17,335	0.01	14.87	8.16
D1N-732	1	19,993	1.27	56.08	12.98
D1N-732	2	12	2.41	2.41	2.41
D1N-732	3	9,160	2.10	45.77	11.27
D1N-732	4	31	6.20	6.20	6.20
G-740	-	39,694	0	0	0
G-740	1	7,561	1.28	13.40	6.12
G-740	3	329,612	0	42.50	2.64
G-740	4	277,116	0	24.30	2.16
GNW-742	1	3,749	1.96	13.16	6.52
GNW-742	3	18,831	0.63	13.46	4.42
GNW-742	4	104	1.64	2.55	2.14
G1-745	-	47,178	0	0	0
G1-745	1	30,452	0.01	11.08	2.09
G1-745	2	289,473	0	13.80	2.03
G1-745	3	442,044	0	18.77	1.51
STH-750	-	189	0	0	0
STH-750	2	16,756	0.78	11.74	4.90
STH-750	3	25,626	0.77	10.05	3.54

14.11.3 Mean Composite Grade within Blocks vs Interpolated Grade

A separate block model was done for the lower zones which interpolated only blocks that had a composite within it. The block was assigned the grade of that composite. The block grade from the new model is then compared to the original interpolated grade of the block. A successful grade interpolation protocol should result in block grade estimates that demonstrate a minimum amount of bias.

A total of 11,923 blocks with composites within the mineralized solids were identified in the EX block model (West side of lower Island Domain). The average gold grade of the composites within those blocks is 8.89 g/t Au while the average interpolated grade is 8.34 g/t Au. The comparison shows that there is no bias between the mean grade of the composites and the estimated grade. The analysis indicates that the Mineral Resource model provides a reasonable estimate of the Island Gold Mine deposit (Table 14-13).

Table 14-13 Comp. - Mean Sample Grade Within Block and Interp. Grade for Same Block

Zone	Number of Blocks with Composites	Avg Grade of Composites within Blocks (g/t Au)	Interpolated Grade OK (g/t Au)	Difference OK/CMP
EX-C	7146	12.33	11.53	-6.5%
EX-D1	24	6.17	4.66	-24.5%
EX-D	478	2.54	2.6	2.4%
EX-G	1303	3.22	3.03	-5.9%
EX-G1	760	2.06	2.04	-1.0%
EX-GNW	262	5.33	5.33	0.0%
EX-B	1917	4.82	4.48	-7.1%
EX-SHT	33	4.5	4.83	7.3%

14.12 Classification

Block model tonnage and grade estimates for the Island Gold deposit were classified according to the CIM Definition Standards for Mineral Resources and Mineral Reserves (May 2014).

A Mineral Resource classification is typically a subjective concept and industry best practices suggest that Mineral Resource classifications should consider the confidence in the geological continuity of the mineralized structures, the quality and quantity of exploration data supporting the estimates and the geostatistical confidence in the tonnage and grade estimates. Appropriate classification criteria should aim at integrating these concepts to delineate regular areas with similar Mineral Resource classifications.

The authors are satisfied that the geological modelling honours the current geological information and knowledge. The location of the samples and the assay data are sufficiently reliable to support Mineral Resource estimation.

The Mineral Resources at Island Gold Mine are classified into Measured, Indicated and Inferred Mineral Resource categories. The Mineral Resources were classified in two successive stages:

automated classification followed by manual editing of the final Mineral Resource categories on longitudinal sections for each individual zone.

Mineral Resources are classified based on drilling density which are as follows:

- **Measured Mineral Resources:** A maximum drill hole spacing of about 20 m and associated mining development completed. Mineralization must be exposed and continuity visually confirmed. Corresponds mainly to blocks estimated during the first pass.
- **Indicated Mineral Resources:** A maximum drill hole spacing of 20 m to 25 m between drill holes if no lateral development is above or below the Mineral Resource outline. All Mineral Resources classified as Indicated Mineral Resources are within the blocks estimated in the second and third passes.
- **Inferred Mineral Resources:** These blocks are represented mainly by areas where drill holes are spaced by more than 30 m (up to 75 m) where the mineralization is interpreted to be the extension of known mineralized zones. Extension is limited to a maximum of 30 m from the last drill holes. Blocks classified as Inferred Mineral Resources are estimated mainly during the fourth pass and partly during the third pass.

Additional infill drilling is required to support classification from Inferred to Indicated Mineral Resources and from Indicated to Measured Mineral Resources. It cannot be assumed that all or any part of an Inferred Mineral Resource will be upgraded to an Indicated or Measured Mineral Resource because of additional drilling.

Figure 14-8 is an illustration of the blocks estimated by different passes for Zone EX-C and E1E (EXT2). This information is used as a guide for classification. Measured Resources (small pink squares and red polygons) are limited to the blocks estimated in the first pass. Indicated Mineral Resources (light green squares and dark green polygons) are located well within the blocks estimated in the second and third passes. Inferred Mineral Resources (light blue squares and dark blue polygons) are estimated partly in the third pass or fourth pass.

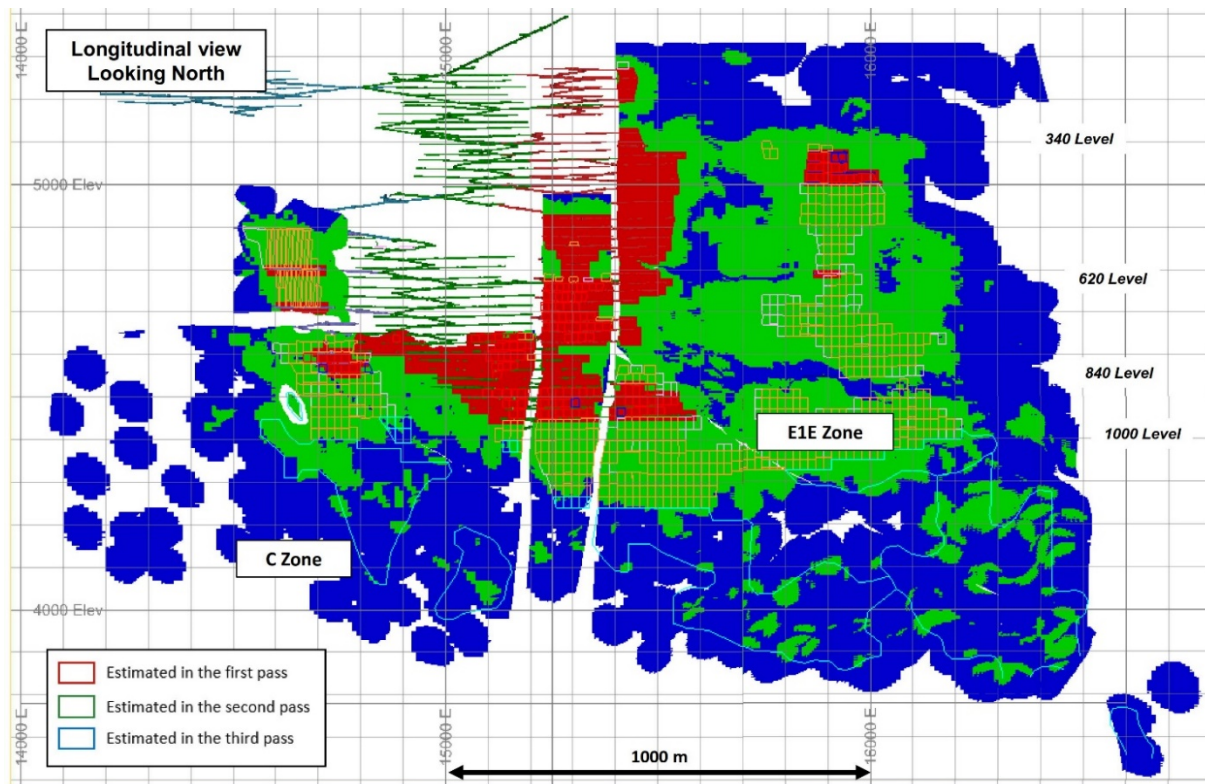


Figure 14-8 Longitudinal Showing Mineral Resource Classification

14.13 Mineral Resource Statement

CIM Definition Standards for Mineral Resources and Mineral Reserves (May 2014) defines a Mineral Resource as:

“[A] concentration or occurrence of diamonds, natural solid inorganic material, or natural solid fossilized organic material including base and precious metals, coal, and industrial minerals in or on the Earth’s crust in such form and quantity and of such a grade or quality that it has reasonable prospects for economic extraction. The location, quantity, grade, geological characteristics, and continuity of a Mineral Resource are known, estimated, or interpreted from specific geological evidence and knowledge.”

The “reasonable prospects for economic extraction” requirement generally implies that the quantity and grade estimates meet certain economic thresholds and that the Mineral Resources are reported at an appropriate cut-off grade considering extraction scenarios and processing recoveries.

Mineral Resources were estimated using a general undiluted cut-off grade of 3.74 g/t Au. This cut-off is based on a gold price of \$1,400 per ounce, an exchange rate of 0.75 and a minimum mining width of 2.0m. Depending on the zones, the capping values for the high-grade samples varies from 45 g/t Au (EX-D1 zone) to 225 g/t Au (EX-C).

Once interpolation has been carried out, the Mineral Resource areas were outlined on the vertical longitudinal section of the zone to a maximum lateral and vertical distance of approximately 30 m from drill hole intercepts for the C and E1E zones while a maximum lateral and vertical distance of 20 m was applied to other zones. All blocks within the outlined Mineral Resource shapes are

included in the estimation. Cut-off is applied to the overall Mineral Resource shapes and not to individual estimated blocks.

Island Gold Mine reports Mineral Resources exclusive of Mineral Reserves. The majority of Measured and Indicated Mineral Resources are at a higher grade than the economic cut-off and have been converted to Mineral Reserves (see Section 15). Therefore, the vast majority of Mineral Resources at Island Gold Mine are classified as Inferred Mineral Resources. The economic viability of Inferred Mineral Resources has not been demonstrated.

The estimation of Mineral Resources is a complex and subjective process and the accuracy of any such estimate is a function of the quantity and quality of available data and of the assumptions made and judgments used in geological interpretation.

14.13.1 Island Gold Mine - Mineral Resources

Island Gold Mine's Measured and Indicated Mineral Resources are estimated at 1,096,280 tonnes at 8.12 g/t Au for 286,300 ounces. As a matter of comparison, the Measured and Indicated Mineral Resources as of December 31, 2020, totalled 718,000 tonnes at an average grade of 7.18g/t Au for 165,854 ounces (Table 14-14 and Table 14-15).

The Inferred Mineral Resources are estimated at 7,905,600 tonnes at 13.59 g/t Au for 3,453,800 ounces. As a matter of comparison, the Inferred Mineral Resources as of December 31, 2020, totalled 6,915,250 tonnes at an average grade of 14.43 g/t Au for 3,208,300 ounces (Table 14-16).

Mineral Resources are presented exclusive of Mineral Reserves. The difference in grade between Measured and Indicated Mineral Resources and Inferred Mineral Resources is explained by the fact that the higher grade Measured and Indicated Mineral Resources have been converted to Mineral Reserves (after applying modifying factors). The increase in Inferred Mineral Resources (more than 245,000 oz) is due to the new drilling at depth in 2021 in the Extension 2 Domain and the additional zone E1D.

Table 14-17 summarizes the Mineral Resource by category, presented undiluted and in-situ and notes for all four tables follow Table 14-17.

A long section showing changes to the Mineral Resources in 2021 is presented in Figure 14-9.

A year over year reconciliation of Inferred Mineral Resources is presented in Figure 14-10.

Table 14-14 Island Gold Measured Mineral Resource Estimates as of Dec 31, 2021

Zone	Tonnes	Grade (g/t Au)	Ounces
Upper Island Domain – Upper Mine (IGU, LC and GD)	6,300	5.04	1,000
Lower Island Domain	14,000	4.87	2,200
Total Measured Resources	20,300	4.92	3,200

Table 14-15 Island Gold Indicated Mineral Resource Estimates as of Dec 31, 2021

Location	Zone	Tonnes	Grade (Au g/t)	Ounces
Upper Island Gold Mine	Lochalsh Domain - E2, E1E, D and C Zone	112,950	7.14	25,950
	Goudreau Domain - GD2, GD3, GD6, GD7 GD9, GP2 and GP5 Zone	68,850	9.51	21,050
	Upper Island Domain - C, E1E, D1, D, E1D and E2 Zone	69,200	6.39	14,200
Lower Island Gold Mine	Lower Island Domain - C Zone	129,900	8.16	34,050
	Lower Island Domain - Other Zones	245,500	7.84	61,900
	East Domain - E1E Zone	449,550	8.71	125,950
Total Indicated Resources		1,075,950	8.18	283,100

Table 14-16 Island Gold Inferred Mineral Resource Estimate as of Dec 31, 2021

Location	Zone	Tonnes	Grade (Au g/t)	Ounces
Upper Island Gold Mine	Lochalsh Domain - E2, E1E, D and C Zone	81,600	6.5	17,050
	Goudreau Domain - GD2, GD3, GD6, GD7 GD9, GP2 and GP5 Zone	23,750	11.81	9,050
	Upper Island Domain - C, E1E, D1, D, E1D and E2 Zones	30,900	8.81	8,750
Lower Island Gold Mine	Island Domain - C Zone	976,800	14.84	466,000
	Island Domain - Other Zones	383,350	8.62	106,200
	East Domain - E1E and E1D Zone	6,409,200	13.82	2,846,750
Total Inferred Resources		7,905,600	13.59	3,453,800

Table 14-17 Island Gold Mineral Resource Estimate Summary as of Dec 31, 2021

Mineral Resource	Tonnes	Grade (g/t Au)	Ounces
Measured Resources	20,350	4.92	3,200
Indicated Resources	1,075,950	8.18	283,100
Total Measured and Indicated Resources	1,096,300	8.12	286,300
Inferred Resources	7,905,600	13.59	3,453,800

Notes:

- CIM definitions of Mineral Resources were followed.
- Mineral Resources are estimated at an undiluted cut-off grade of 3.74 g/t Au.
- High-grade samples were capped at 75 g/t Au for most of the Upper Island Gold zones except IG-E1E and IG-C capped at 100 g/t Au, most of the Goudreau zones except for G2 and G6 capped at 100 g/t Au.

- *High-grade samples in Lochalsh were capped at 75 g/t Au for E1E, 45 g/t Au for D, 60 g/t Au for C and 55 g/t Au for E2.*
- *In the Lower mine high-grade samples were capped at 225 g/t for C, 90 g/t Au for B, 70 g/t Au for G and GNW, 45 g/t for G1, 50 g/t Au for D and STH, 45 g/t Au for D1, 80 g/t for ED and 185 g/t Au for E1E zones.*
- *Mineral Resources are estimated using a long-term gold price of \$1,400 per ounce.*
- *A minimum mining width of 2.00 m was used.*
- *A specific gravity value of 2.78 t/m³ was used for all zones except the Lochalsh at 2.82 t/m³*
- *Mineral Resources, which are not Mineral Reserves, do not have demonstrated economic viability.*
- *Totals may not match due to rounding.*

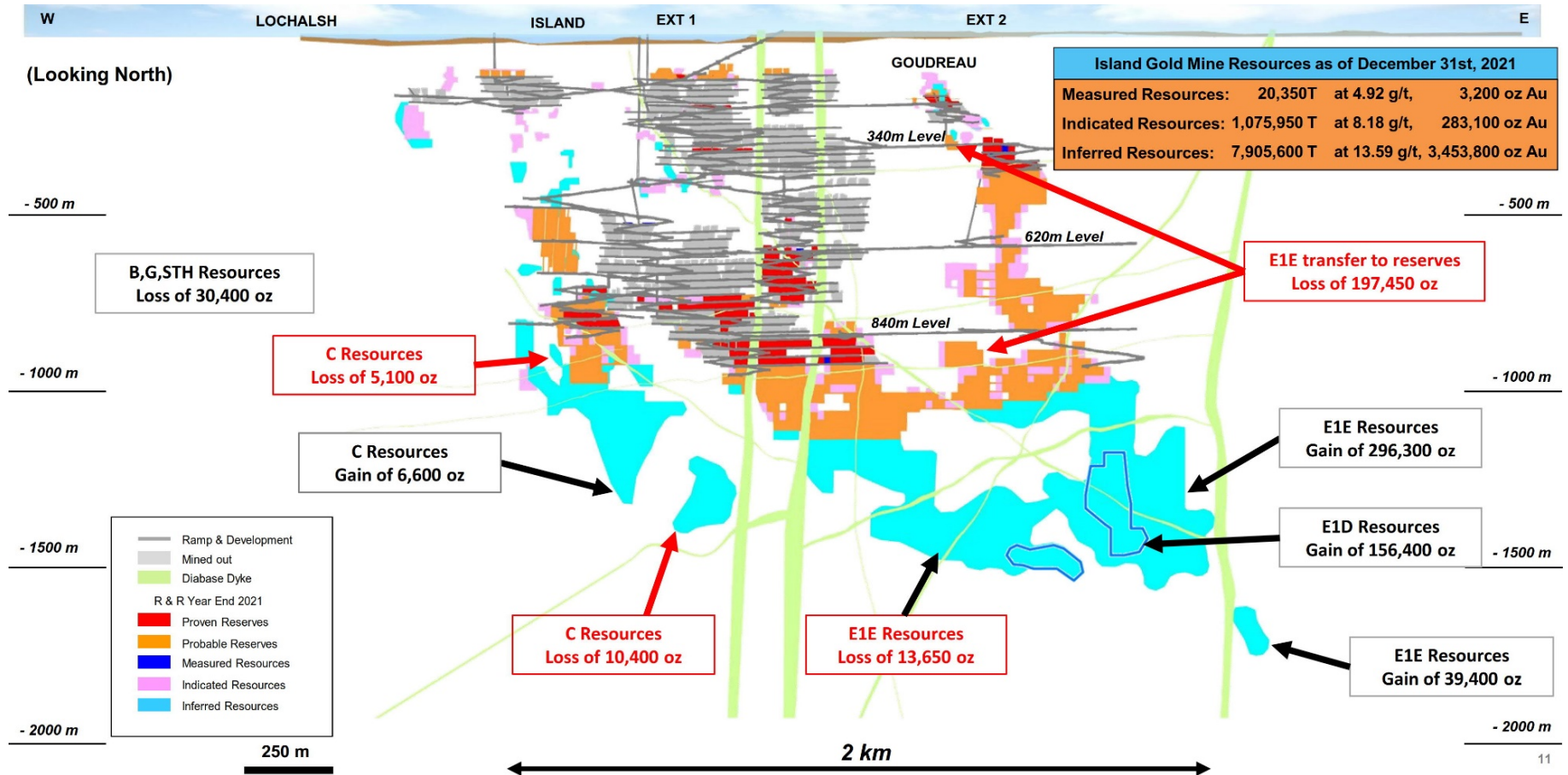


Figure 14-9 Change in Mineral Resources as of Dec 31st, 2021

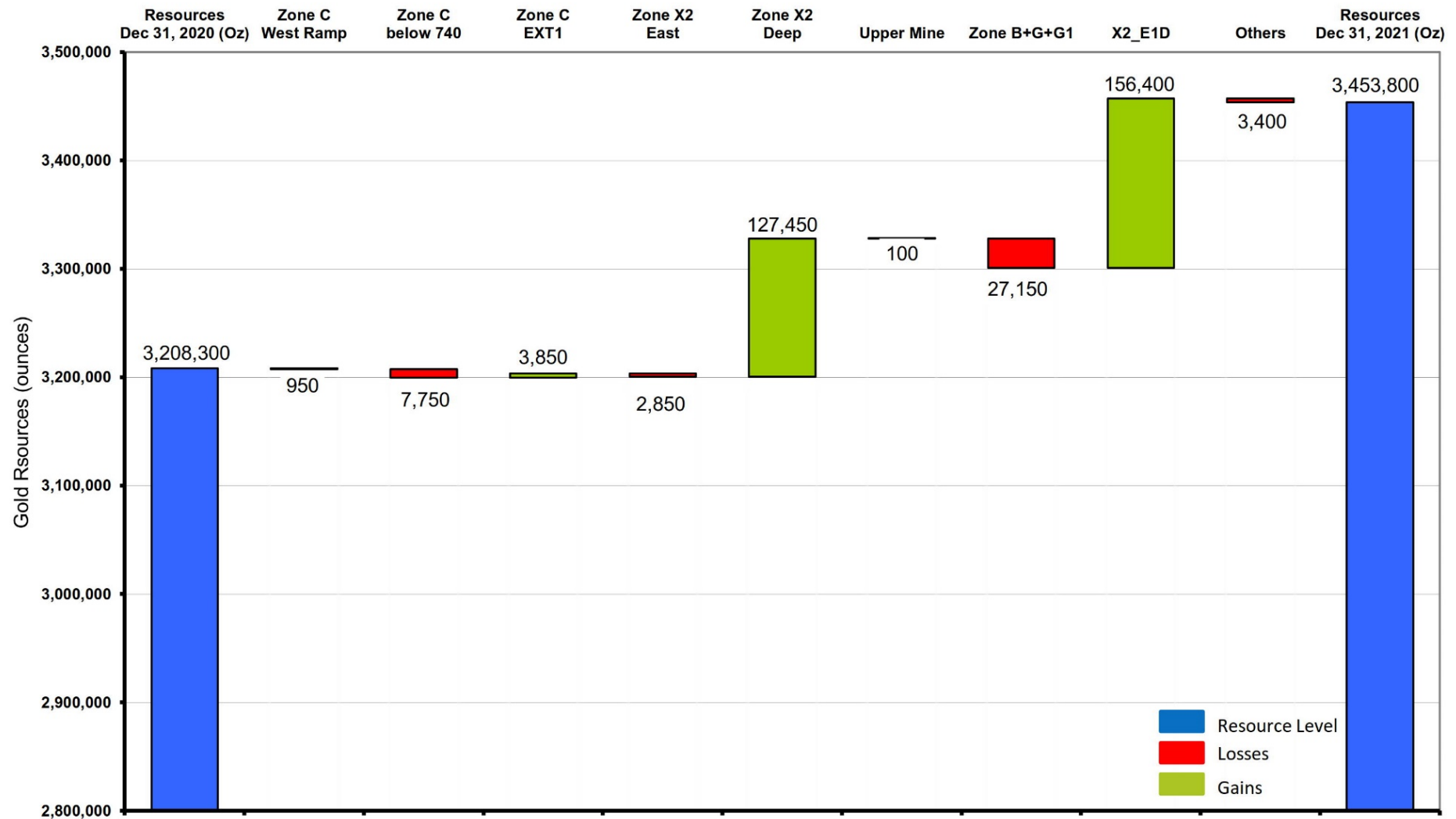


Figure 14-10 Inferred Mineral Resource Waterfall Graph (Dec 31st, 2020, vs Dec 31st, 2021)

15 MINERAL RESERVE ESTIMATES

Mineral Reserve calculations estimate the volume and grade of ore which can be mined and processed at a potential profit. The global Mineral Resource was reviewed by the Island Gold engineering department, with assistance from the geological staff, to define the Mineral Reserve blocks that could be economically extracted with a mining plan. The conversion of Mineral Resources into Mineral Reserves is based on the economic parameters detailed in Table 15-1. Only Mineral Resources that are classified as Measured or Indicated Mineral Resource categories were used in the economic calculations to estimate Mineral Reserves as of December 31, 2021.

Table 15-1 Mineral Reserve Estimation Parameters

Mineral Reserves Parameter	Value
Gold Price (USD)	\$1,250
Exchange Rate (USD/CAD)	0.75
Stope Cut-off Grade (g/t Au)	3.74
Development/Marginal Cut-off Grade (g/t Au)	3.06
Stope Dilution ¹ (%)	20%-50%
Development ¹ Dilution (%)	20%-30%
Dilution Grade (g/t Au)	0.50
Mining Recovery ¹ (%)	67%-95%
Process Recovery (%)	96.5%
Ore Specific Gravity ² (t/m ³)	2.78
Minimum Mining Width (m)	2.0
Mining, Processing and G&A Cost (CAD \$/t)	193

Notes:

1. Dependant on sector and mining method
2. 2.82 t/m³ for Lochalsh Domain

Mining costs and cut-off grades may vary depending on the mining method used and whether the ore block is already developed or not. The 3.74 g/t cut-off and \$193/tonne operating cost are for undeveloped zones utilizing the long hole mining method. Mining recovery also depends on the mining method and the sector.

The economic viability of the Mineral Resources converted into Mineral Reserves was determined by Island Gold's engineering department. Dilution, recovery rates and mining costs used in the Mineral Resource and Mineral Reserve calculations represent Island Gold's best estimates as of December 31, 2021. These factors and parameters are revised each year to take into consideration actual or realized factors.

The following definitions detail the nomenclature of the Mineral Reserve estimates as of December 31, 2021:

Proven Mineral Reserves: Ore development has been completed above and below the mining block (or only below in case of an upper stope). A minimum drill spacing of 20 m is necessary to confirm vein continuity. Only Measured Mineral Resources can be transformed into Proven Mineral Reserves. Economic feasibility was estimated by Island Gold Mine's engineering department to validate the block as Mineral Reserves.

Probable Mineral Reserves: No development was done, or development was done only above or below the mining block. Since the information from the ore development is lacking, a maximum drill hole spacing of 20 m to 25 m is necessary to validate vein continuity inside the mining block.

Only Measured and Indicated Mineral Resources can be transformed into Probable Mineral Reserves.

15.1 Island Gold Mine – Total Mineral Reserves

Total Island Gold Proven and Probable Mineral Reserves as of December 31, 2021, stand at 4,111,800 tonnes at a grade of 10.12 g/t Au for 1,338,500 contained ounces. As a matter of comparison, the Proven and Probable Mineral Reserves as of December 31, 2020, totaled 4,197,350 tonnes at an average grade of 9.71 g/t Au for 1,310,000 contained ounces. Total underground production in 2021 was 438,734 tonnes at an average grade of 10.27 g/t Au for 144,900 contained ounces. Table 15-2 and Table 15-3 summarize the Proven and Probable Mineral Reserve estimates. A combined Mineral Reserve estimate is presented in Table 15-4.

Mineral Reserves include 15% to 50% dilution at a grade of 0.5 g/t Au with an estimated mining recovery of 67% to 95%. Mineral Reserves are reported before mill recovery.

Table 15-2 Island Gold – Proven Mineral Reserve Estimates as of Dec 31, 2021

Location	Zone	Tonnes	Grade (Au g/t)	Ounces
Upper Mine	Upper Island Domain	28,550	6.96	6,400
	Goudreau Domain	29,950	12.58	12,100
Lower Mine	Broken Tonnes (UG and Surface Inventory)	41,300	5.12	6,800
	Island Domain - C Zone	503,850	10.28	166,550
	Island Domain - D1, CD1, and B Zone	76,700	12.46	30,750
	East Domain - E1E Zone	153,750	5.85	27,550
Total Proven Mineral Reserves		834,100	9.33	250,150

Table 15-3 Island Gold – Probable Mineral Reserve Estimates as of Dec 31, 2021

Location	Zone	Tonnes	Grade (Au g/t)	Ounces
Upper Mine	Upper Island Domain	66,400	6.17	13,150
	Goudreau Domain	45,850	10.95	16,100
	Lochalsh Domain	17,950	7.52	4,350
Lower Mine	Island Domain - C Zone	957,950	11.50	354,350
	Island Domain – B, CD1, D1, G, and GNW Zone	96,250	8.94	27,650
	East Domain - E1E Zone	2,093,350	9.99	672,700
TOTAL Probable Mineral Reserves		3,277,750	10.33	1,088,300

Table 15-4 Island Gold – Combined Mineral Reserve Estimate as of Dec 31, 2021

Mineral Reserve	Tonnes	Grade (g/t Au)	Ounces
Proven	834,100	9.33	250,150
Probable	3,277,750	10.33	1,088,300
Total Proven and Probable	4,111,850	10.12	1,338,450

Notes:

- CIM definitions of Mineral Reserves were followed.
- Mineral Reserves are estimated at cut-off grade of 3.06 g/t Au for developed areas and 3.74 g/t Au for undeveloped areas.
- Mineral Reserves are estimated using a long-term gold price of \$1,250 per ounce.
- A minimum mining width of 2.00 m was used.
- A specific gravity value of 2.78 t/m³ was used for all zones except the Lochalsh at 2.82 t/m³
- Totals may not match due to rounding.

Mineral Reserves presented herein are in large part estimates and production of the anticipated tonnages and grades may not be achieved or the indicated level of recovery may not be realized. The estimation of Mineral Reserves is a complex and subjective process and the accuracy of any such estimate is a function of the quantity and quality of available data and of the assumptions made and judgments used in engineering and geological interpretation. Mineral Reserve estimates may require revision based on various factors such as actual production experience, exploration results, fluctuations in the market price of gold, results of drilling, metallurgical testing, production costs or recovery rates. These factors may render the Proven and Probable Mineral Reserves unprofitable to develop. Also, the grade of ore mined may differ from that indicated by drilling results and this variation may have an adverse impact on production results.

15.2 Reconciliation of the 2021 Production with Mineral Reserve Models

The 2021 production tonnage and grades (stopes and development) have been reconciled to the December 31, 2020, and December 31, 2021, Mineral Reserves statements.

Total production reconciled to the mill was 438,734 tonnes at 10.27 g/t Au for 144,900 contained ounces of gold. Corresponding mining from Mineral Reserves was 382,226 tonnes at 11.37 g/t Au for 139,726 contained ounces for the December 31, 2020, model and 385,333 tonnes at 11.70 g/t Au for 145,004 contained ounces for the new December 31, 2021, model.

For the December 31, 2020, model, total reconciled production is 14% higher in tonnes, 9% lower in grade and 3% higher in gold ounces than Mineral Reserves (144,900 vs. 139,726 contained ounces).

For the December 31, 2021, model (current model), total reconciled production is 13% higher in tonnes, 11% lower in grade and 1% lower in gold ounces than Mineral Reserves (144,900 vs. 145,004 contained ounces).

Table 15-5 summarizes the comparison between 2021 production and the corresponding Mineral Reserves as of December 31st, 2020, and 2021.

Table 15-5 2021 Production Reconciliation to Mineral Reserves

Sector	Production 2021			Model December 31 st , 2020			Model December 31 st , 2021		
	Reconciled			Reserve			Reserve		
	Tonnes	Grade	Ounces	Tonnes	Grade	Ounces	Tonnes	Grade	Ounces
B	5,466	12.48	2,193	5,125	8.69	1,431	5,125	7.92	1,304
C-760	190,172	13.93	85,178	179,656	15.84	91,520	189,484	16.09	98,042
West Ramp	30,681	11.47	11,316	35,436	11.27	12,844	30,721	12.93	12,767
X2-East	42,523	5.66	7,741	32,613	7.50	7,869	27,953	5.70	5,119
G	23,061	6.19	4,592	17,843	5.22	2,997	17,287	5.06	2,810
X2>750	48,411	5.69	8,854	40,037	5.44	7,004	39,243	6.03	7,609
C-X1≤750	12,925	12.09	5,022	10,008	8.94	2,878	9,279	8.57	2,556
X2<750	44,346	9.30	13,258	36,583	7.37	8,666	39,055	8.27	10,382
C-X1>750	29,007	5.90	5,500	24,925	5.64	4,518	27,187	5.05	4,415
Stockpile Reclamation	12,143	3.19	1,246						
2021 Total	438,734	10.27	144,900	382,226	11.37	139,726	385,333	11.70	145,004
Production Vs Dec 31st, 2020, Model	114%	91%	103%						
Production Vs Dec 31st, 2021, Model	113%	89%	99%						

Based on this reconciliation, the new December 31, 2021, Mineral Resource and Mineral Reserve model is a good representation of what was mined in 2020 and an improvement compared to December 31, 2020, model. Comparison between actual production and the Mineral Reserve model will continue to be undertaken monthly going forward to confirm that geological interpretation, block model parameters and mining factors continue give a good estimation of future production.

15.3 Mineral Reserve Reconciliation 2020 vs 2021

Combined Proven and Probable Mineral Reserve estimates for December 31, 2021, are 1,338,450 contained ounces while the December 31, 2020, totaled 1,310,000 contained ounces. This represents an increase of 28,450 ounces. The gains are mainly attributed to definition drilling (transfer from Inferred Mineral Resources) in the Eastern area, partially offset by production depletion.

Table 15-6 shows reconciliation of Proven and Probable Reserve estimates between the December 31, 2020, and the December 31, 2021, estimations, and the gains and losses in different areas with their impact on the Mineral Reserve estimates. Figure 7-1 shows the same information but in a graphic form. The table and graph include the 2021 production to show the gain in Mineral Reserves, after depletion between, the two years.

Table 15-6 Mineral Reserve Reconciliation 2020 vs 2021 – Gains (losses) from Different Sectors

Sector	Dec 31 st , 2020, Reserves			2021 Production			Dec 31 st , 2021, Reserves			Gain/loss	Description
	Tonnes	Grade	Ounces	Tonnes	Grade	Ounces	Tonnes	Grade	Ounces		
Lochalsh (Upper Mine)	19,758	7.41	4,709	-	-	-	17,930	7.52	4,334	-375	BM Gemcom VS Studio
Island Gold (Upper Mine)	80,180	6.26	16,145	-	-	-	83,595	6.18	16,607	461	BM Gemcom VS Studio
Goudreau (Upper Mine)	85,539	10.64	29,258	-	-	-	75,788	11.59	28,251	- 1,007	GD7 turned into Resources
Alimak sector	162,574	6.80	35,521	-	-	-	162,610	6.80	35,551	30	No changes
B Zone	40,006	7.03	9,038	5,466	12.48	2,193	33,058	6.39	6,795	- 50	New drilling, reinterpretation
C>740	1,907	7.50	460	-	-	-	-	-	-	- 460	2 blocks (535) turned into Resources (ENG request)
C≤760	504,408	14.49	234,968	190,172	13.93	85,178	313,979	13.63	137,568	- 12,222	Lower grade
C-X1>750	238,219	5.09	39,001	29,007	5.90	5,500	196,519	4.93	31,129	- 2,372	New drilling, less tonnes at lower grade
C-X1≤750	432,187	12.79	177,651	12,925	12.09	5,022	406,437	13.37	174,685	2,056	Better grade
W-Ramp	532,622	10.45	178,970	30,681	11.47	11,316	492,632	11.84	187,545	19,892	New drilling, Reinterpretation
G + GNW Zone	48,616	6.23	9,736	23,061	6.19	4,592	29,527	6.33	6,007	863	G Reinterpretation
X1E1E	11,559	7.45	2,768	-	-	-	11,371	8.09	2,956	188	BM Gemcom VS Studio
X2>750	57,690	5.50	10,198	48,411	5.69	8,854	-	-	-	- 1,344	Less production tonnes because of merge stopes
X2<750	941,255	8.60	260,238	44,346	9.30	13,258	794,747	9.57	244,561	- 2,419	Transfer into Resources
X2-East	1,003,009	9.14	294,625	42,523	5.66	7,741	1,452,356	9.76	455,692	168,808	New drilling, transfer from Resources
Other	-	-	-	12,143	3.19	1,246	-	-	-	1,246	Stockpile reclamation
Inventory	37,841	5.51	6,704	-	-	-	41,278	5.12	6,800	96	Depletion of the stockpile
Total	4,197,369	9.71	1,309,991	438,734	10.27	144,900	4,111,824	10.12	1,338,482		

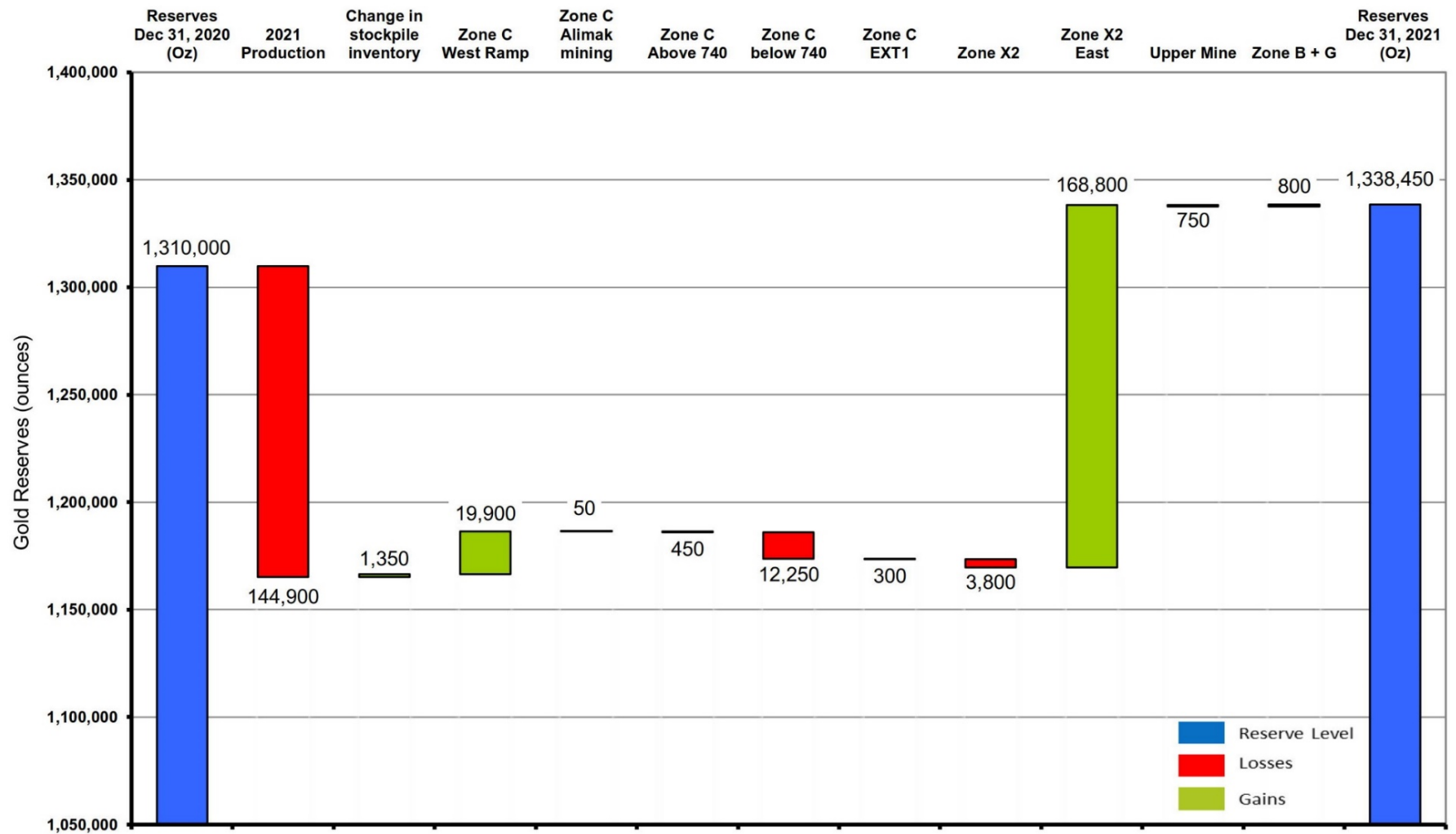


Figure 15-1 Mineral Reserve Reconciliation Waterfall Graph Dec. 31, 2020, vs Dec. 31, 2021

16 MINING METHODS

16.1 Overview

There are currently four active mining areas at the Island Gold Mine: the West Zone, the Main IG Zone, the Extension Zone, and the East Zone. The overall mine configuration is shown in Figure 16-1 below, which outlines the various mining zones as well as future stopes and development included in the life of mine plan (LOM).

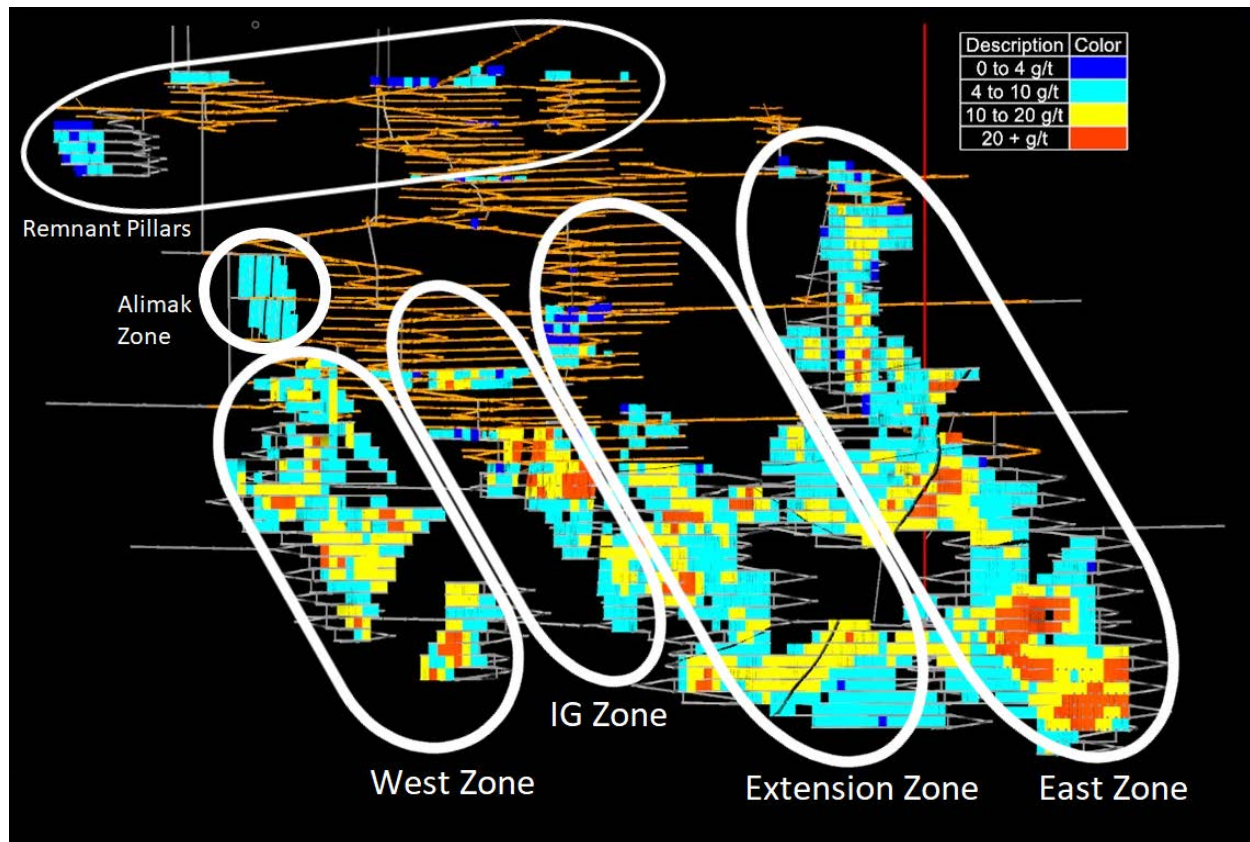


Figure 16-1 Life of Mine Design Looking North

16.2 Mine Access and Development

The Island Gold deposit is accessed via a single decline from surface down to the 425 Level, at which point multiple ramps are utilized to access the main IG, West, Extension and East zones. These ramps are also connected at numerous points throughout the mine allowing for easy travel between mining zones.

The life of mine plan includes the addition of a mine shaft which will be constructed between 2022 and 2026. Once commissioned, the shaft will be utilized to hoist ore and waste from the 1350 Level Loading Pocket to surface. Additionally, the shaft will be used to transport personnel and materials to any of the three shaft stations. From the shaft collar location ore and waste will be trucked to either the mill or the surface waste stockpile.

Level accesses are typically developed at 25 m intervals (floor to floor) and are designed south of internal ramps providing access to the footwall of the deposit. Once the ore is reached, sills

are developed along the ore contact, with their direction controlled by geology. Sill development is used as a drilling, mucking and backfilling platform for stope extraction. On some levels, additional footwall and cross-cut development is required when the width of the mineralization exceeds 10 metres. Standard drift dimensions are shown in Table 16-1.

Table 16-1 Standard Excavation Dimensions

Excavation Type	Dimensions
Lateral Development	
Jumbo sill	4.5 m W x 4.5 m H (6.5 m W max on variable width) (Arch)
Jumbo ramp	4.75 m W x 4.75 m H (Arch)
Level Sump	4.5 m W x 4.0 m H x 13.5 m L
Secondary access sump	4.5 m W x 4.0 m H x 7.0 m L
Electrical sub station	6.5 m W x 4.5 m H x 16.5 m L
Electrical bay	6.5 m W x 4.5 m H x 6.0 m L
Gear Bay	6.5 m W x 4.5 m H x 13.5 m L
Remuck in level	5.0 m W x 6.5 m H x 13.5 m L
Remuck in ramp	5.0 m W x 5.0 m H x 13.5 m L
Secondary remuck	5.0 m W x 4.5 m H x 13.5 m L
Vent Access #1	4.5 m W x 4.5 m H
Vent Access #2 (to raise)	5.5 m W x 4.5 m H
Safety bay	1.8 m W x 2.0 m H x 1.8 m L
DDH bay	6.0 m W x 5.0 m H x 13.5 m L
DDH bay used as turn-out	5.5 m W x 5.0 m H x 13.5 m L
Minimum turning radius for jumbo corner	3.5 m
Minimum turning radius for truck corner	5.0 m
Turning radius in ramp	20 m
Turning radius in sill	10 m / 5 m
Level intersection in ramp	5.5 m W x 5.5 m H x 42.5 m L
Truck load out	5.0 m W x 6.5 m H
Scoop load-out	5.0 m W x 5.5 m H
Truck turn-out	5.0 m W x 4.5 m H
Level entrance	5.5 m W x 5.5 m H
Vertical Development	
Alimak Raises	3.0 m W x 3.0 m L
Drop Raise (Ventilation Raises)	4.0 m W x 4.0 m L

A total of 124 km of lateral and vertical development are planned as part of the life of mine plan. Of this total approximately 28% is operating development, 66% is capital development and 6% is planned to support exploration activities.

Presently, level accesses are designed towards the center of the ore vein and stopes are mined longitudinally from sill extremities towards the level intersection. As mining progresses deeper level accesses are designed to access the extents of the deposit with stopes being mined from the center towards the extremities to support improved mining stress management.

A standard level configuration is shown in Figure 16-2.

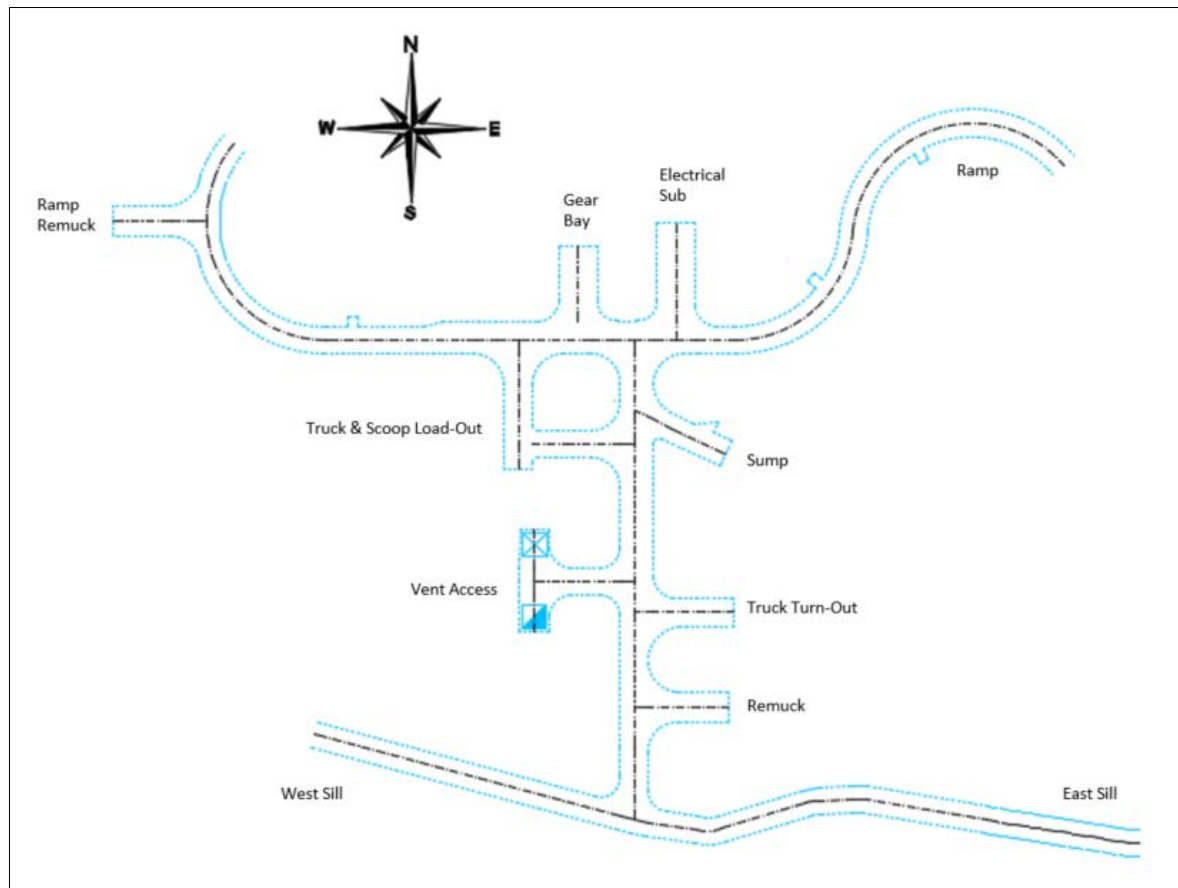


Figure 16-2 Typical Level Design

16.3 General Design Considerations

Island Gold has various tools and systems in place to gather and analyze geotechnical data for each new design. Underground geologists and ground control engineers perform regular mapping campaigns to keep the structural data up to date as mining advances deeper into the orebody. Face mapping and sampling is performed in all ore drives and major structures, quartz veining and shear zones are identified. This data is then digitized and shared with the planning team for future development and stoping designs. Each design is optimized by considering and mitigating risk with instability drivers such as faults, dykes, discontinuity families and changes in lithology.

In addition to mapping, Island Gold has also evaluated intact rock properties through several laboratory testing campaigns and has analyzed results from core samples at depth. In general, it was determined that the rock mass is considered blocky in nature and can be rated as “Good” to “Very Good” (RMR = 75-83), with a Q’ range (at the 50th percentile) of 20 to 40 (MDEng, 2018a, 2018b).

As the mine progresses deeper and new mining horizons are met, new testing campaigns are planned to ensure designs are created with accurate geotechnical information. This ensures that the design shape or sequence of excavation is optimized without compromising safety or the integrity of future excavations.

16.4 Stope Dimensions

Several different empirical assessments are performed on a regular basis at Island Gold. Empirical assessments allow the ground control department to quickly reference industry standards and published literature to ensure the initial design is in line with past experiences. The Matthews Stability Graph is the main empirical tool used to determine optimal stope dimensions that will yield a stable shape.

Geotechnical parameters (Q' , N') are calculated by using specific underground details of rock mass ratings or major geological intrusions. The N' parameter is then plotted on the stability graph, where an approximation of the ideal hydraulic radius is obtained for each face of the excavation. From there, the strike length of the stope can be calculated by using the fixed dimensions of the shape (level spacing and ore width). Stopes at Island Gold are typically designed with an approximate strike length of 18 m. This stope size has been proven to be successful in various zones throughout the mine, however this analysis is regularly performed to account for any new structures and geological features.

16.5 Stope Design

Mineral Resource shapes are created by geology then submitted to the production engineering team for design. A mineable stope shape is produced by engineering and is optimized by maximizing ore recovery and minimizing planned dilution. Recovery and external dilution factors are applied to the stope shape and its economic viability is evaluated. Stope shapes that have proven to be economically feasible are converted to Mineral Reserves and are mined using the most favorable mining method.

16.6 Mining Methods

The mining method for a particular stope is selected based on a variety of factors such as overall geometry of the mineralization, width of the ore zone, local stresses, mapping and geotechnical data, spatial location of the stope, and existing nearby development and infrastructure. Other factors considered include equipment size and limitations as well as available fill type. Ultimately, each stope is evaluated individually, and a stope package is produced to include detailed drilling plans, blast letters, ventilation, and gas check instructions, mucking plans as well as backfilling directives.

16.6.1 Longhole Open Stoping

The predominant mining method used at the Island Gold mine is longhole open stoping. This mining method is conducive to tabular, steeply dipping orebodies and is highly productive with low mining costs. The average dip of the orebody at Island Gold ranges from 75 to 85 degrees, making this a favorable mining method for ore extraction.

Longhole mining consists of drilling a series of sub-vertical down holes between two mining platforms, also known as the overcut and undercut sills. These holes are drilled with electric-hydraulic drill rigs. The top sill is typically used as the drilling and backfilling horizon, and the bottom sill is used as the mucking horizon. In some cases where there is no top sill development, up-holes are drilled from the bottom sill which acts as both the drilling and mucking platform. Once the stope is drilled, the ore is blasted in vertical slices towards an open void and retrieved from the bottom sill using remote LHDs. The material from the stope is trucked to surface, at which point the stope is then backfilled with unconsolidated rock fill (UCF), cemented rock fill (CRF) or paste fill. The backfilling process will be further discussed in Section 16.8.

There are two types of longhole stopping methods utilized at Island Gold: longitudinal open stopping and transverse open stopping. These two methods employ the same mining principles mentioned above; however, they differ by the stope's mining direction. Longitudinal stopes are mined along the strike of the ore vein and follow either a modified Avoca technique or a traditional blast hole stopping technique, whereas transverse stopes are mined perpendicular to the vein.

16.6.1.1 Longitudinal Open Stopping

Stopes are typically mined longitudinally when the ore width is narrow (usually under 10 m). For every first stope on a horizontal sublevel, a primary slot raise is drilled at the extremity of the ore contact. This raise is drilled using an in-the-hole (ITH) drill with a large reaming head that produces a large diameter hole. This large hole is used as a free face for the first blast. The stope is then fired towards the open void in several blasts (2-3 typically) which is achieved by retreating longitudinally towards the main level access (retreat is done in east/west direction). The broken ore is extracted after each blast is taken to ensure ample void for the following blast. Once the stope is empty, UCF, CRF, or paste fill is placed in the void to fill the opened excavation. This is shown in Figure 16-3, Figure 16-4, and Figure 16-5 below.

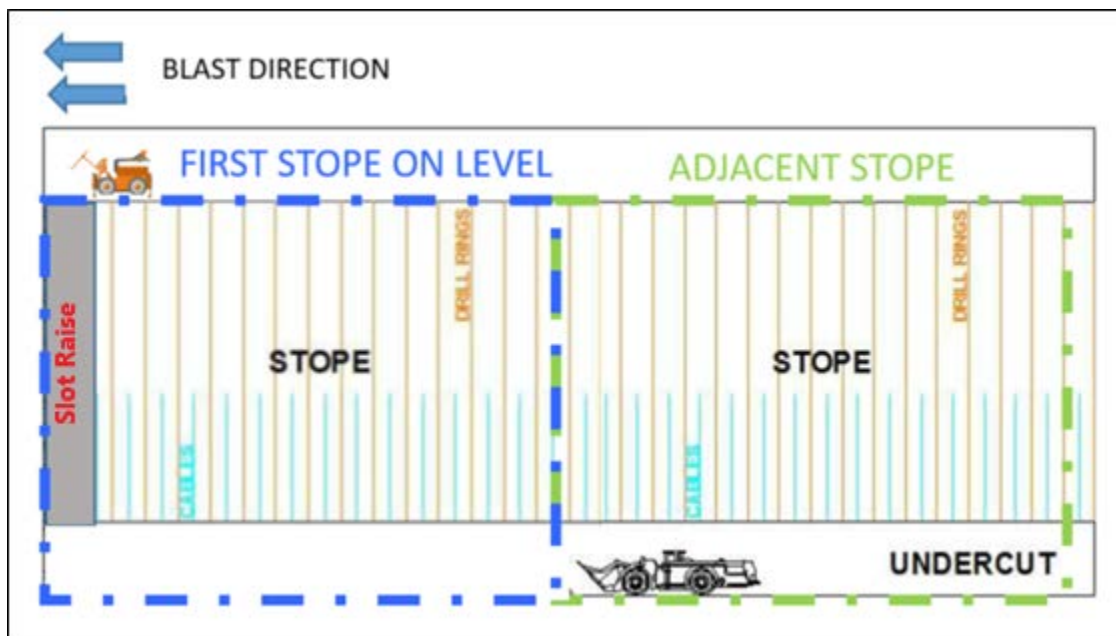


Figure 16-3 Drilling - Longhole Stope, Longitudinally Drilled and Mucked

To mine the adjacent stope, the UCF material from the first stope on the level is removed until an open brow is established and angle of repose of approximately 48 degrees is achieved. This creates a primary free face for blasting of the second stope, also known as a “pull void”. The second stope is then blasted, mucked, and backfilled. The process is repeated until the entire sublevel is mined out. Figure 16-6 illustrates the concept of this mining method. This method is commonly known as Modified Avoca Mining. Cavity monitoring surveys (CMS) are performed regularly to distinguish between the ore and backfill material. The cavity surveys along with the judgement of underground beat geologists help control grade dilution while mucking.

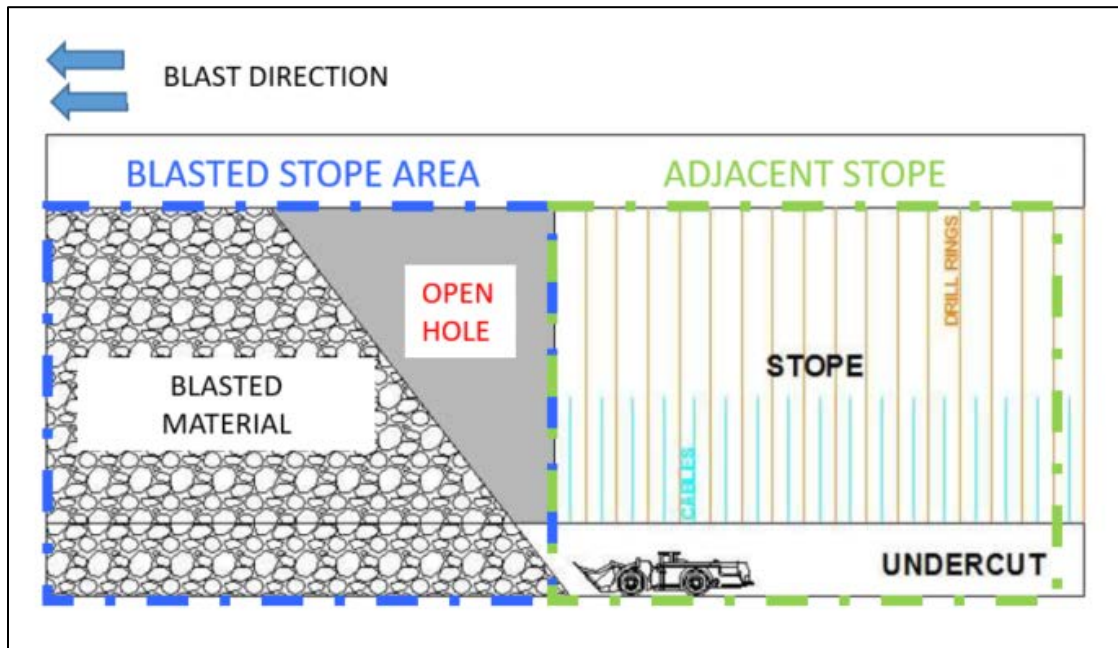


Figure 16-4 Blasting and Mucking - Longhole Stope, Longitudinally Drilled and Mucked

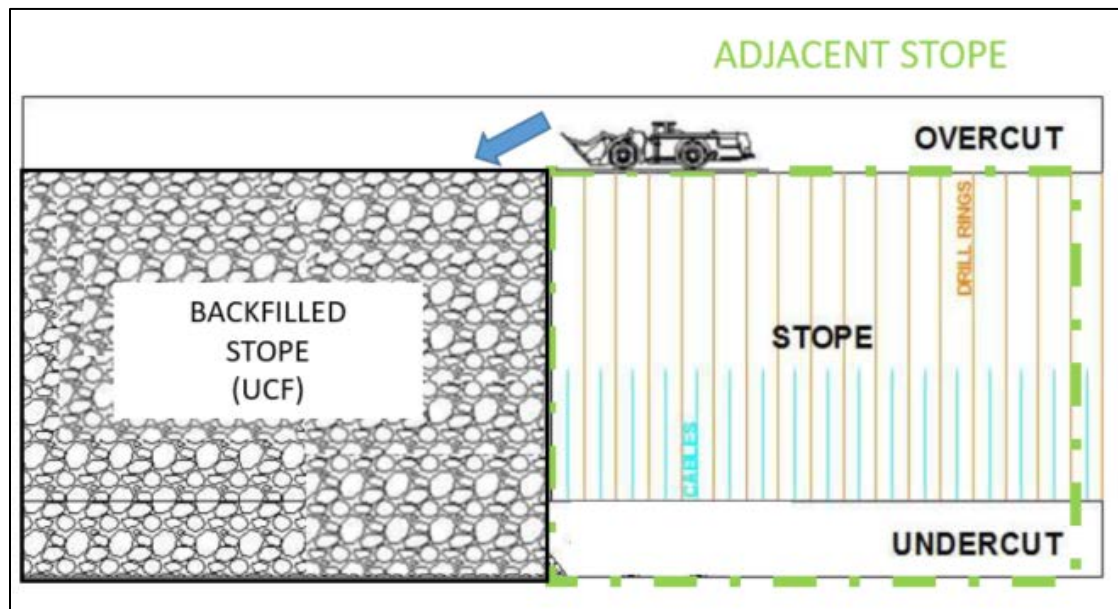


Figure 16-5 UCF Backfilling - Longhole Stope, Longitudinally Drilled and Mucked

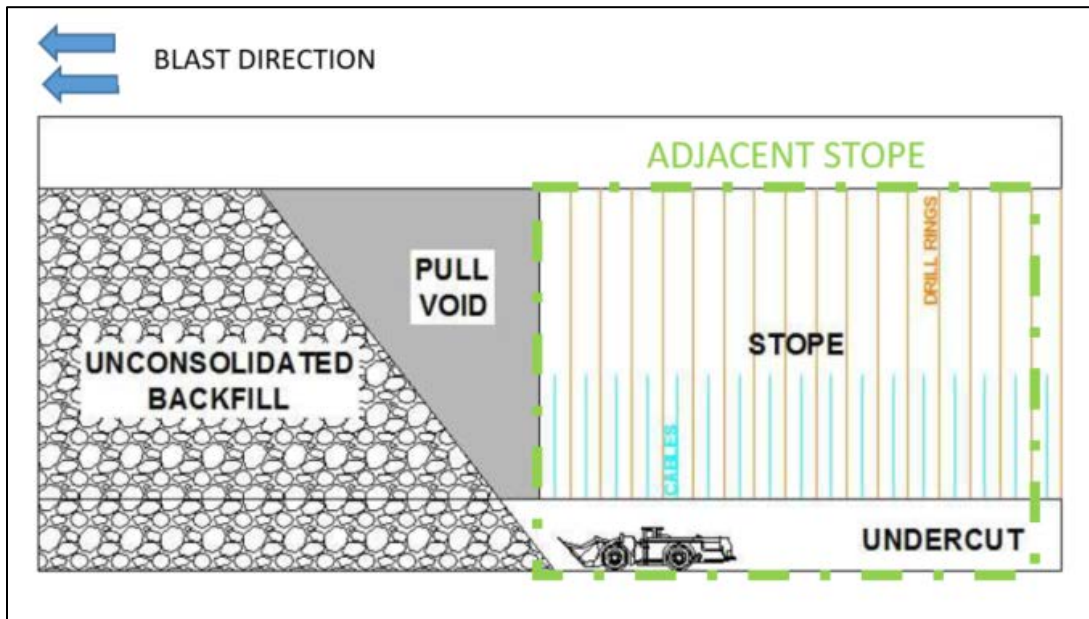


Figure 16-6 Pull Void - Longhole Stope, Longitudinally Drilled and Mucked

When CRF or paste fill material is used as fill, the backfill material solidifies due to the cement which acts as a binding agent. Therefore, the adjacent stope requires a slot raise for blasting. In this case, the stope must be mined in the same fashion as the first stope on the level, and ultimately follows the same process until the entire level is mined out. This is shown in Figure 16-7.

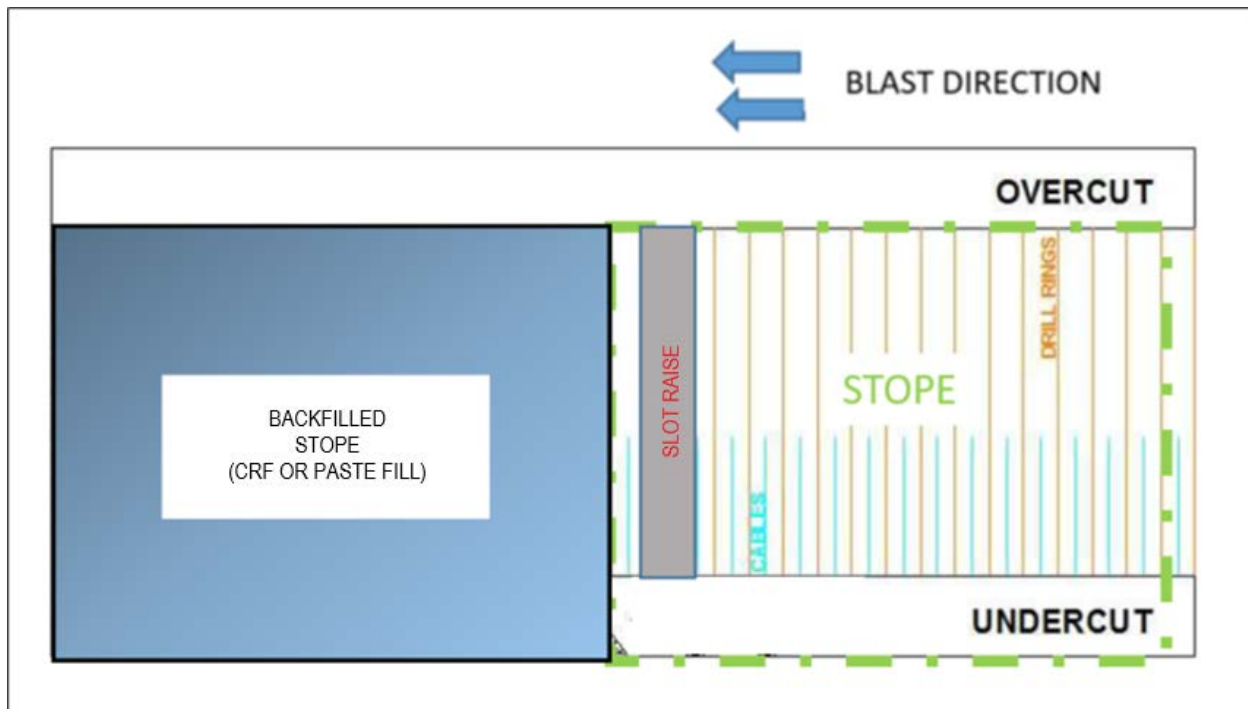


Figure 16-7 CRF or Paste Fill - Longhole Stope, Longitudinally Drilled and Mucked

16.6.1.2 Transverse Open Stopping

Stopes are typically mined transversely when the ore width is too wide to mine efficiently and safely using a longitudinal retreat method. Island Gold employs transverse mining where the mining direction runs perpendicular to the strike of the orebody (North/South). Stope mucking is done via multiple draw points that allow for line-of-sight mucking from the remote stand which optimizes mucking productivity. Furthermore, this method allows for production holes to be drilled parallel to the hanging wall and footwall and only requires fanning into the vertical stope ends which are inherently more stable. Each block is split into different panels employing a true primary/secondary sequence. Figure 16-8 displays how each block is accessed by its own drawpoint or access. This mining method requires the use of a raise as an initial void for blasting. This raise is typically designed in line with the drawpoint, which facilitates mucking as the material is blasted towards the drawpoint. Once the stope has been blasted and emptied, the void must be backfilled with UCF, CRF or paste fill. The primary stopes require a consolidated fill that creates solid end walls to withstand the blast energy while extracting adjacent stopes. The secondary stopes can be filled with UCF if there is no remnant mineralization nearby. One advantage of this method is it allows concurrent activities to take place on a single level improving mining cycle times and productivities.



Figure 16-8 Transverse Mining Access

16.6.2 Alimak Stopping

Alimak mining is planned to be utilized in a portion of the Island Gold West Zone. This method consists of using an Alimak climber as a means of development and production drilling instead of conventional horizontal development.

The process starts by driving a raise along the height of the stope that will serve for secondary support and production drilling access. The raise dimensions will typically be 3 m x 3 m.

Once the raise is completely driven, the raise screening is installed. Cable bolts are then drilled and installed as secondary support.

Production drilling is conducted on a horizontal axis on both sides of the raise. The blasting sequence consists of taking horizontal slices followed by void mucking only. Once the entire stope is blasted, continuous mucking can begin. Maintaining the stope filled with blasted material allows for better dilution control. Once mucking is completed, the Alimak stope will be backfilled with CRF. Figure 16-9 shows the Alimak mining sequence.

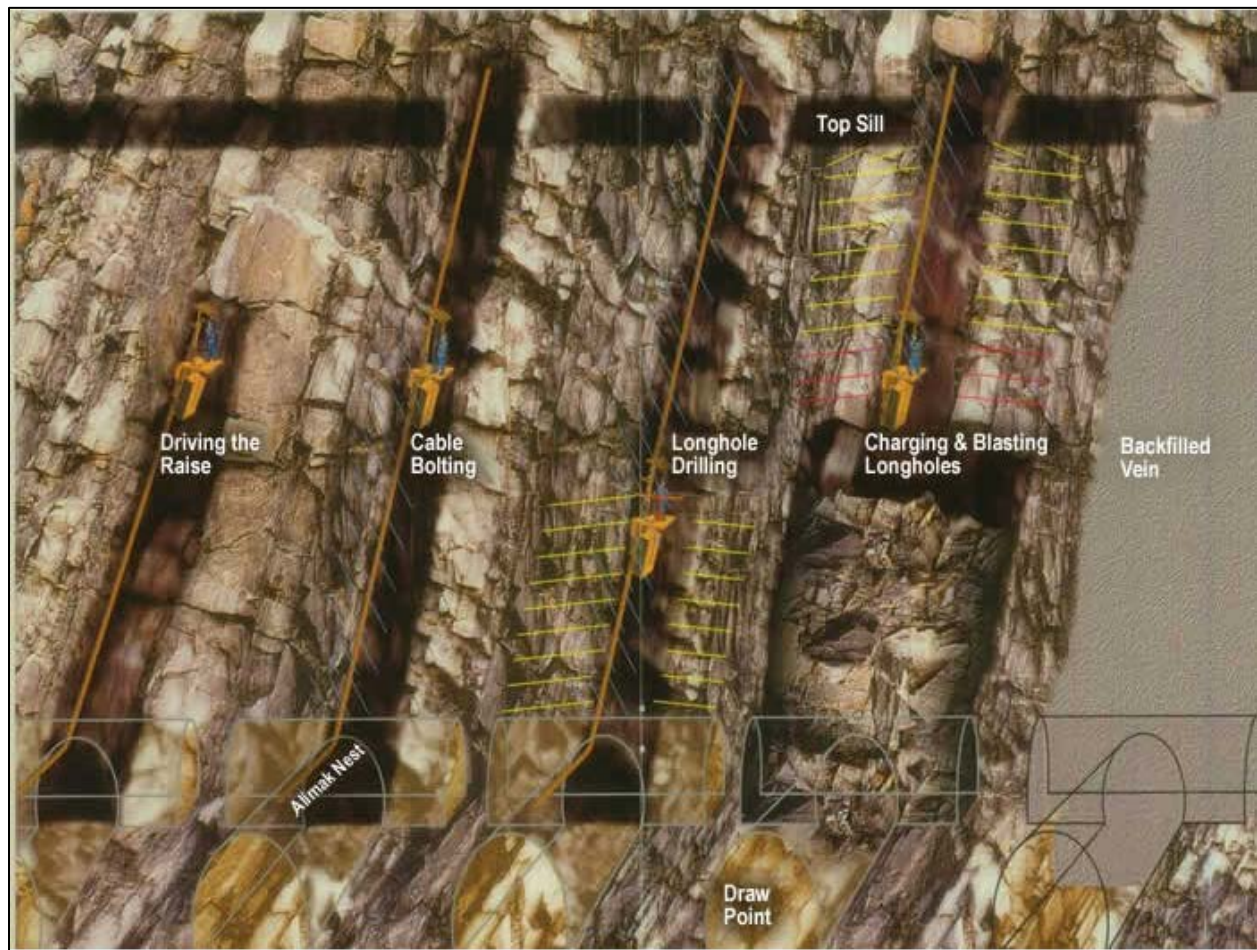


Figure 16-9 Alimak Stopping

Alimak mining has not yet been employed at the Island Gold, however some Alimak accesses and raise nests have been developed. This mining method is part of the life of mine design and production using this method is scheduled to begin in 2023.

16.7 Ground Control

A geomechanical sampling campaign was completed in 2019 and results were used to build a stress model. An update to this model was completed in 2022 to incorporate updated geotechnical structural features and mine designs. Regular third party geomechanical studies are completed involving the collection of new core samples underground, implementation of seismic data, as well the development and review of a new geomechanical models for the updated life of mine plan and proposed underground infrastructure (shaft and ore/waste handling

systems). Computerized numerical modeling programs are being utilized as a tool to assist in determining ground control practices and risk management of possible ground failures.

An update of the ground support standard was implemented in 2021 to address the increased stress levels with depth and to add robustness to the ground support standards. Ground support standards are regularly reviewed and updated based on observations and communications between engineering and mine operations.

A microseismic system was implemented in 2017 and 2018 with full commissioning of the system in 2019. The microseismic system covers every zone of the mine with regular expansion programs occurring as the mining front progresses deeper and laterally. A stope re-entry protocol was also implemented based on the micro seismic activity that stopes create after being blasted. The re-entry protocol limits access to areas deemed higher risk once a stope blast is taken (usually levels near the stope or sill levels), until the micro seismic activity returns to background activity. At this point, the “all clear” is given for workers to re-enter the barricaded levels to begin regular operations.

Stope ground support is used to control dilution. Dilution may come from a local structural failure or from inadequate drilling and blasting practices. Cable bolts are used to limit stope wall dilution and this method has provided good results. Cable bolts are installed along the undercut and overcut. Cables being used range from 6 metres to 12 metres in length.

Level access and stoping sequence methodologies were investigated below the 920 Level with a planned transition to a center out stope sequence (Figure 16-11) from the current practice of outside in (Figure 16-10). This is expected to improve mining stress management, as mining moves deeper, by shedding stress outward.



Figure 16-10 Current Outside-In Mining Sequence

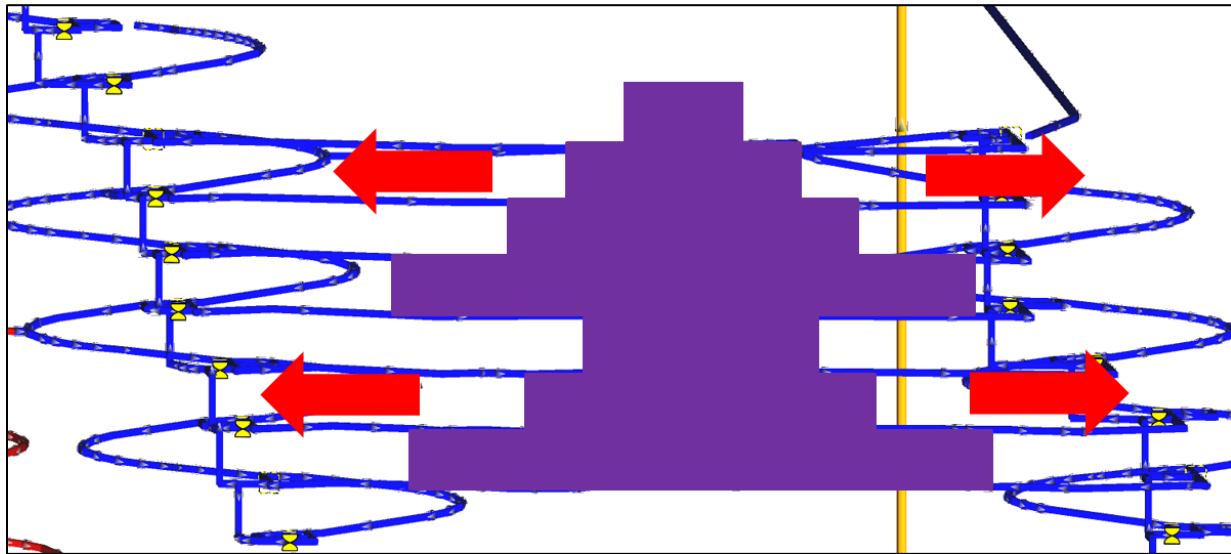


Figure 16-11 Proposed Centre-Out Sequence

16.8 Backfill

Island Gold is currently using two types of backfill methods: unconsolidated rock fill (UCF) and cemented rock fill (CRF) with the addition of paste fill planned as part of the Phase 3+ project. Both UCF and CRF fill methods are dumped from the top cut of a stope by an LHD. In the event of a hanging wall failure, a Rammer Jammer is employed to “ram” (push) muck tighter against the hanging wall to fill the voids.

16.8.1 Unconsolidated Fill (UCF)

Using UCF to fill empty stopes helps with mine waste management and minimizes stope wall failures by stabilizing them. UCF is not screened and comes directly from development faces. The angle of repose for UCF ranges between 45-54°, however 54° is usually used when modelling or estimating fill angles to remain conservative.

16.8.2 Consolidated Rock Fill (CRF)

CRF was implemented at Island in 2019, with placement focused on transverse mining zones, sill pillar recovery and problematic stopes that require CRF instead of UCF. Waste is hauled from development faces to the underground cement plant, where cement slurry is poured directly in the box of the haul trucks. The trucks then haul the CRF over to the top cut of the stope being backfilled, where it is dumped in a remuck. The LHD then picks up the CRF and dumps it into the stope.

It has been observed that the driving cycle from the cement plant to when the LHD picks up the muck and dumps is enough to mix the waste rock and slurry. BASF's Masteroc MF 701 is being added into the slurry mixture to help stabilize the cement's reactivity. This increases strength and increases workability time with the slurry to 4-6 hours.

Like the UCF, the CRF is still not considered an engineered product as there is limited quality control on the aggregates being used and no gradation curve can be obtained. However, samples of the cement slurry are sent out for analysis.

16.8.3 Paste Fill

Island Gold does not currently utilize paste fill onsite. The addition of a paste plant and an underground paste distribution system is included in the life of mine plan which will be utilized for filling future stopes and improving sill pillar recovery.

Filtered tailings from the process plant will be used as aggregate for paste backfill. The paste plant is incorporated into the process tailings dewatering circuit and includes the thickening, filtration, mixing, and batching equipment. The delivery of paste will be gravity-fed to the stopes via the underground distribution system (UDS) as shown in Figure 16-12.

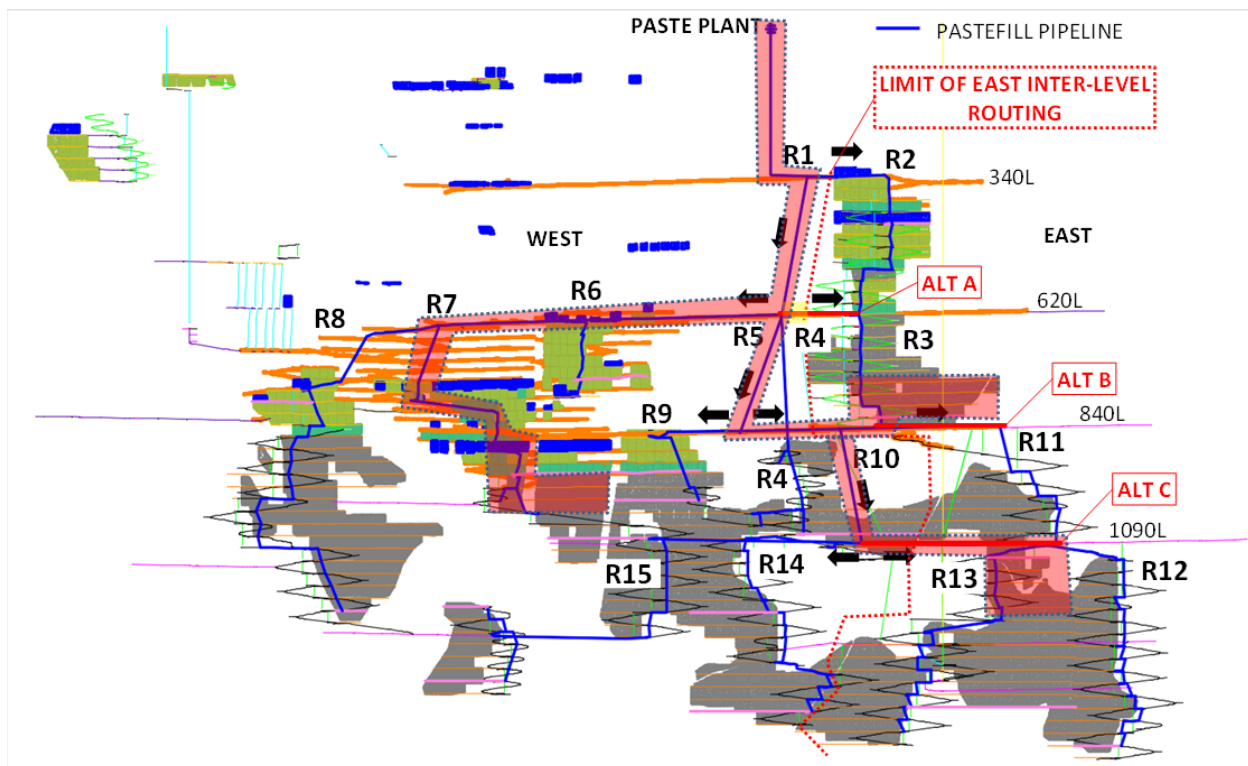


Figure 16-12 Paste Backfill Underground Distribution System

16.9 Stope Sequencing

The Island Gold Mine has multiple active stoping horizons in the Main Island Gold Zone, the Extension Zone, West Zone, and East Zone. This allows for flexibility in the production schedule as each mining horizon follows its own sequence and is independent of mining activities in other zones or horizons.

In areas where longitudinal retreat with UCF is employed, stopes are blasted, mucked, and backfilled starting from the eastern or western most point of the sill on the bottom horizon. This process repeats itself until the last intersection stope (at the level access and sill intersection) is excavated. Figure 16-13 show a typical longitudinal retreat mining sequence at Island Gold.

Once three stopes are taken on the bottom level of a horizon, production can begin on the level above, by retreating towards the level access. This process is repeated until the top horizon of the zone reached.

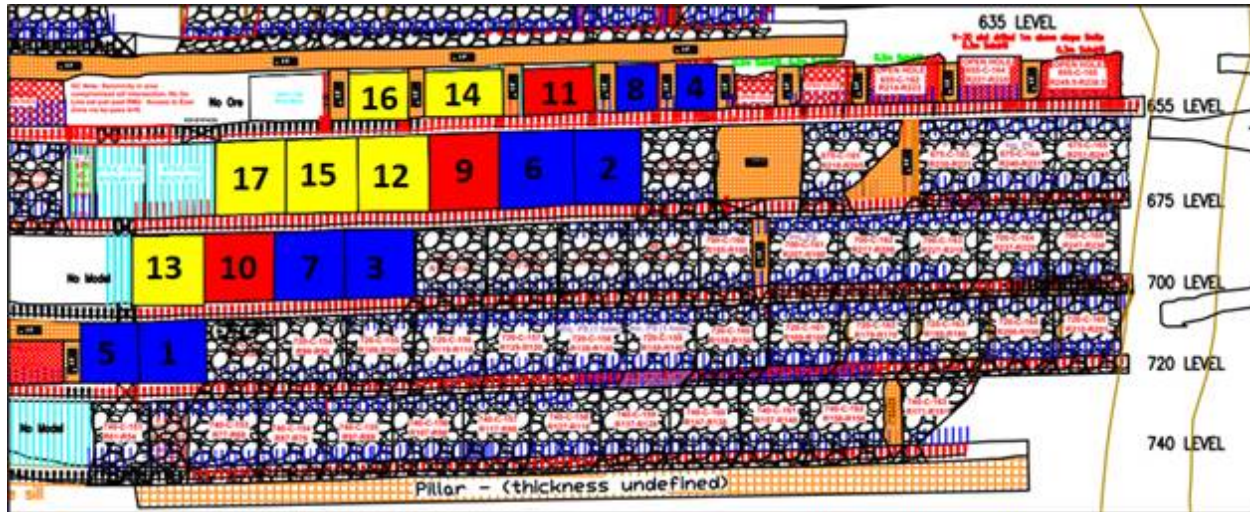


Figure 16-13 Example of Longitudinal Retreat Mining Sequence

Uphole stopes are taken on the top horizon where sill and rib pillars are left behind for stability purposes. With the addition of paste fill as a backfill method, future sill and rib pillars will be recovered by mining full uphole stopes to the paste fill. The addition of paste fill will allow for accelerated fill times, improved pillar recovery, and concurrent activities (filling and drilling) resulting in improved cycle times.

Transverse mining zones typically include a mix of transverse and longitudinally accessed mining blocks. These zones follow a traditional primary-secondary mining sequence with primary stopes being filled with a consolidated fill such as CRF or Paste Fill and secondary stopes being filled with either unconsolidated or consolidated fill as required.

16.9.1 Void Management Plan

Due to the nature of IGM's mining sequence and use of UCF in many areas, backfill subsidence and backfill run of muck is possible. To mitigate this risk, a Void Management Plan has been developed. This plan helps to track and manage voids as well as defines a series of steps to be taken when an unfillable or unexpected void occurs. A backfill action plan is developed and implemented, and a series of steps are taken afterwards to ensure the void is logged and considered during future designs.

16.10 Material Movement & Equipment

Island Gold currently utilizes an internal ramp system to haul ore and waste to surface. The ore is brought to the surface by a combination of Epiroc MT-42 and Caterpillar AD-30 haul trucks using a ramp system. A detailed list of primary equipment was developed for each scenario and the equipment list for the S2400 scenario can be found in Table 16-2. Capital cost for equipment was developed based on supplier quotes and includes provisions for rebuilds and replacements throughout the life of mine plan.

Once on the surface, ore is hauled by surface trucks to the mill located approximately 0.8 km from the portal of the ramp. The Phase 3+ project includes the addition of a mine shaft which will be utilized to hoist ore and waste from the 1350L Loading Pocket to surface. From the shaft collar location, ore and waste will be trucked to either the mill or the surface waste stockpile

Table 16-2 Island Gold Current and Phase 3+ Underground Equipment Fleet

Equipment Type	Application	Current Fleet	Average Post-Project
Jumbo	Development	4	3
Bolter	Ground Support	0	4
Scissor Lift	Ground Support / Services	17	8
UG Haulage Truck	Ore/Waste Transport	17	6
LHD 6 Yard	Production / Development	12	18
LHD 3.5 Yard	U/G Maintenance	4	4
Grader	Ramp Maintenance	2	2
Lube Truck/Water Truck	Maintenance	3	3
Boom Truck	Material Logistics	6	4
U/G Personnel Vehicles	Personnel Transport	31	35
Tractor	Maintenance	4	6
Excavator	Maintenance	1	1
Loader	Surface Works	4	2

16.11 Phase 3+ Expansion Study Mineral Inventory

In the past several years Island Gold has significantly increased its Mineral Reserve and Mineral Resources. For the purposes of the Study, it was assumed that the Mineral Reserves and a significant portion of the Mineral Resource would ultimately be available for mining. The bulk of Island Gold's Mineral Resources are in the Inferred Mineral Resource category and require infill drilling to upgrade them to Indicated Mineral Resources. Infill drilling is usually undertaken from exploration drives driven eastward and westward as the ramp system develops at depth. Generally, this infill drilling has been successful, and Island has had a historical conversion rate of over 80% over the last number of years.

Prior to the mine planning and sequencing process, Island Gold's geologists and engineers evaluated the December 31, 2021, Mineral Resource inventory (left hand side of Table 16-3) on a block by block basis and determined which blocks would support being included into a potential stope. Stopes were evaluated for mining shape, continuity of mineralization, and whether they would support the development required to access them. Stope blocks were created using the Deswik Stope Optimizer module using historical Island Gold mining, recovery and costing parameters and interrogated against the block model. Historical Island Gold mining dilution and recovery factors (Table 16-4 and Table 16-5), which vary by zone and stope type, were then applied. If, after applying dilution, the stoping shape made the cut-off grade (Table 16-6), it was then deemed eligible for inclusion with the mine plan. These Mineral Resources appear on the right hand side of Table 16-3. Total Mineral Reserves and Mineral Resources included within the mine plans for the Study are 13.5 Mt at a gold grade of 10.59 g/t. A reconciliation of the mineable resource gold ounces to the Dec 31, 2021, Mineral Reserve and Mineral Resource gold ounces can be found in Figure 16-14. A long section depicting the mineable resource stoping blocks categorized by grade is presented in Figure 16-15.

Table 16-3 Phase 3+ Expansion Study Mineable Resource

	December 31 2021			Undiluted Resource Used in P3+ Study			Diluted and Recovered Resource Used in P3+ Study		
	Tonnes (x,1000)	Grade (g/t)	Ounces (x,1000)	Tonnes (x,1000)	Grade (g/t)	Ounces (x,1000)	Tonnes (x,1000)	Grade (g/t)	Ounces (x,1000)
Mineral Reserves									
Proven	834	9.33	250				834	9.33	250
Probable	3,278	10.33	1,088				3,278	10.33	1,088
Total Mineral Reserves	4,112	10.12	1,338				4,112	10.12	1,338
Mineral Resources									
Measured	20	4.92	3	19	4.92	3	21	4.11	3
Indicated	1,076	8.18	283	991	8.18	261	1,128	6.83	248
Total Measured and Indicated	1,096	8.12	286	1,010	8.12	264	1,149	6.78	251
Inferred	7,906	13.59	3,454	7,283	13.59	3,182	8,289	11.34	3,023
P3+ Mine Plan							13,550	10.59	4,612

Table 16-4 Extension 1/2 Zone Dilution and Factors

Extension 1/2 Zone	
Stope Type	Dilution
Upper (>10 in height)	40%
Upper (<10 in height)	20%
Pillar Stopes	40%
Down	35%

Table 16-5 C and D1 Zone Dilution and Factors

C/D1 Zone	
Stope Type	Dilution
Upper (>10 in height)	45%
Upper (<10 in height)	25%
Pillar Stopes	40%
Transverse	20%
Down	25%

Table 16-6 Island Gold Cut-off Grades

Areas of Application	Reserve Cut-off
Undeveloped Areas	3.53
Developed Stopes	2.89
Alimak Stopes	4.22

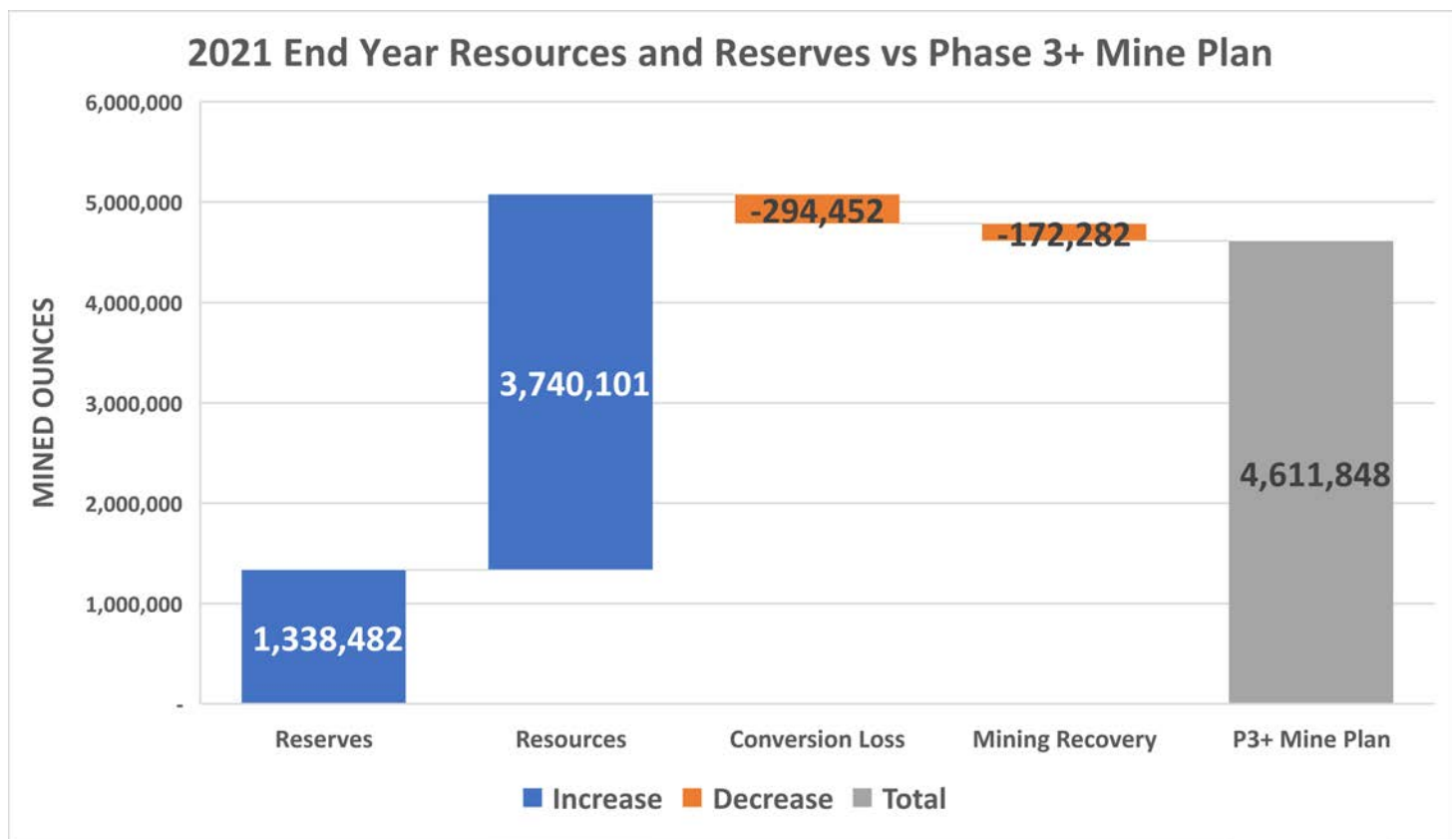


Figure 16-14 Reconciliation of Mineable Resource to Dec 31, 2021, Mineral Reserve and Resources

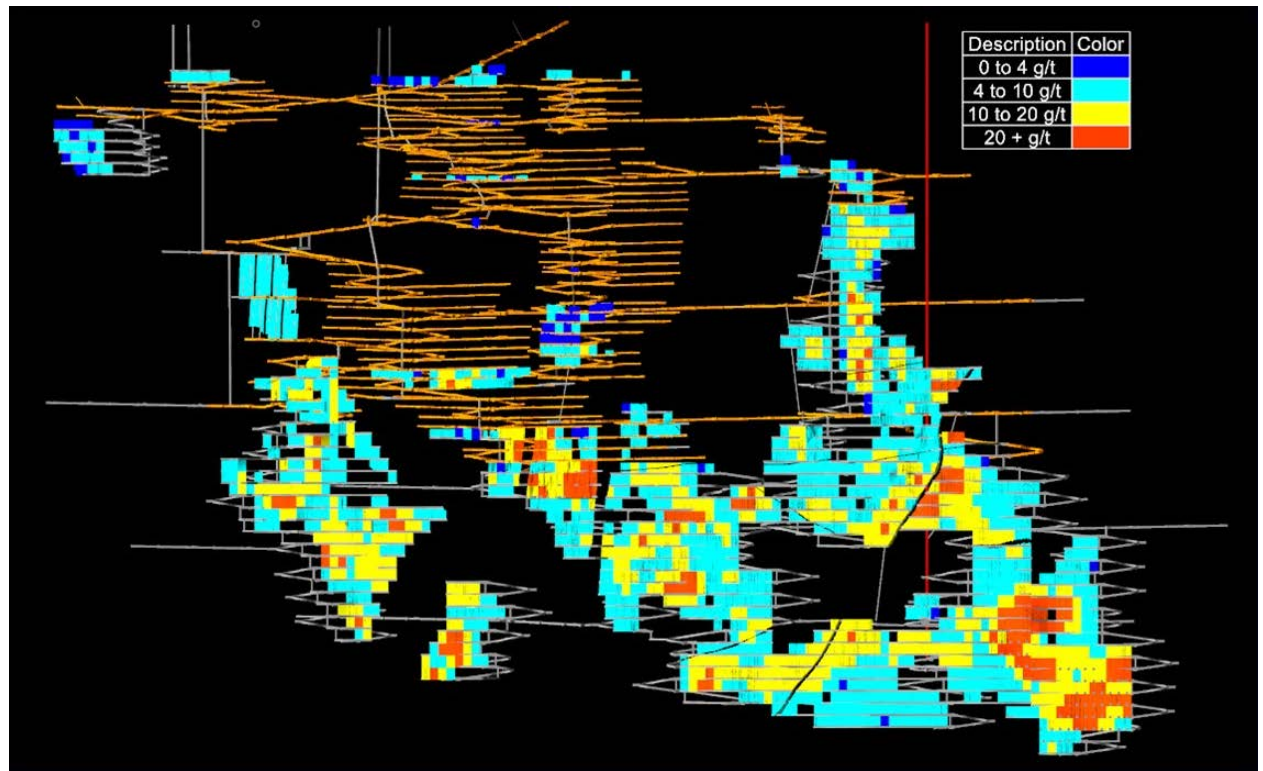


Figure 16-15 Mineable Resource Stopping Blocks Categorized by Grade

16.12 Phase 3+ Expansion Study Scenarios Examined

Two different mine capacity rates were examined, 2,000 tpd, and 2,400 tpd. The 2020 Phase III study laid out a scenario where the mill capacity of Island Gold would be expanded to 2,000 tpd. A study was undertaken to determine the maximum sustainable throughput of the underground mine and indicated that a 2,400 tpd mining capacity was viable with the Mineral Resource as defined in the December 31, 2021, Mineral Resource update. This increase in throughput is supported by the increase of mineable resources in the eastern extents of the deposit proximal the shaft infrastructure.

Multiple iterations of individual mine designs and sequences were undertaken for each scenario examined. Mine designs and resource levelled sequencing were developed with Deswik mine planning and scheduling software. Development and stope sequencing were resource driven and generated the annual physical quantities and equipment requirements.

Mine costing was from first principles and was guided by actual site cost and productivities experience at Island Gold.

16.12.1 Shaft at 2,000 TPD (S2000)

As defined in the 2020 Phase III Study the S2000 scenario anticipates sinking a 1,380 m deep shaft and increasing the mining rate to 2,000 tpd in 2027. Upon commissioning of the shaft and its associated infrastructure all ore and waste transport would be via the shaft. In addition, the shaft would be used to transport personnel and materials to any of the three shaft station levels thereby decreasing travel time significantly. The scenario has several significant capital investments including:

- The sinking of a 1,380 m deep, 5 m diameter, concrete lined shaft;
- Construction of a hosting plant and associated infrastructure. The design selected would be capable of hoisting 4,500 tpd of ore and waste from the 1,350 Level loading pocket and 3,500 tpd of ore and waste from the 2,000 m elevation should the shaft be required to be deepened in the future to access new Mineral Resources. Details of the shaft and associated infrastructure are fully described in Section 18;
- Construction of an ore and waste handling system underground, consisting of ore and waste passes, a grizzly, coarse ore bins, and a loading pocket;
- A paste fill plant, and the associated underground distribution system, capable of delivering 2,000 tpd of paste; and
- The mill capacity would be expanded to 2,000 tpd. This would require upgrades or additions to several of the areas of the process plant such as the crushing circuit, fine ore stockpile capacity, primary grinding circuit, pre-leach thickening, leaching, CIP, acid wash, process water management, and other equipment modifications. This is fully discussed in Section 17.

Total ventilation requirements for S2000 scenario are 850,000 cfm driven by geographical constraints of the ore body and not equipment operating constraints. A detailed life of mine ventilation design was undertaken supported by a third party consultant, which identified an opportunity to convert the existing portal to a fresh air source by converting the ventilation from a push system to a push-pull system. This change in ventilation design allows for a simplification of ventilation infrastructure around the shaft infrastructure, greatly reducing blast clearing timelines and will allow the switch to a 12 hour working shift further improving underground productivities. The change to a push-pull system will require the twinning of the west return air raise to surface with required fan installations, the conversion of the existing FAR#2 raise to an exhaust raise, and the relocation of existing mine air heaters at FAR#2 to the mine portal.

As most of the ore will be moved on the level to the ore passes, the truck requirements are significantly reduced from current (2022) requirement of eleven 42 tonne trucks. An average of five 42 tonne trucks will be required when the shaft infrastructure is in place.

Mine life in the S2000 scenario would go to 2042 and the annual extraction sequence is shown in Figure 16-16.

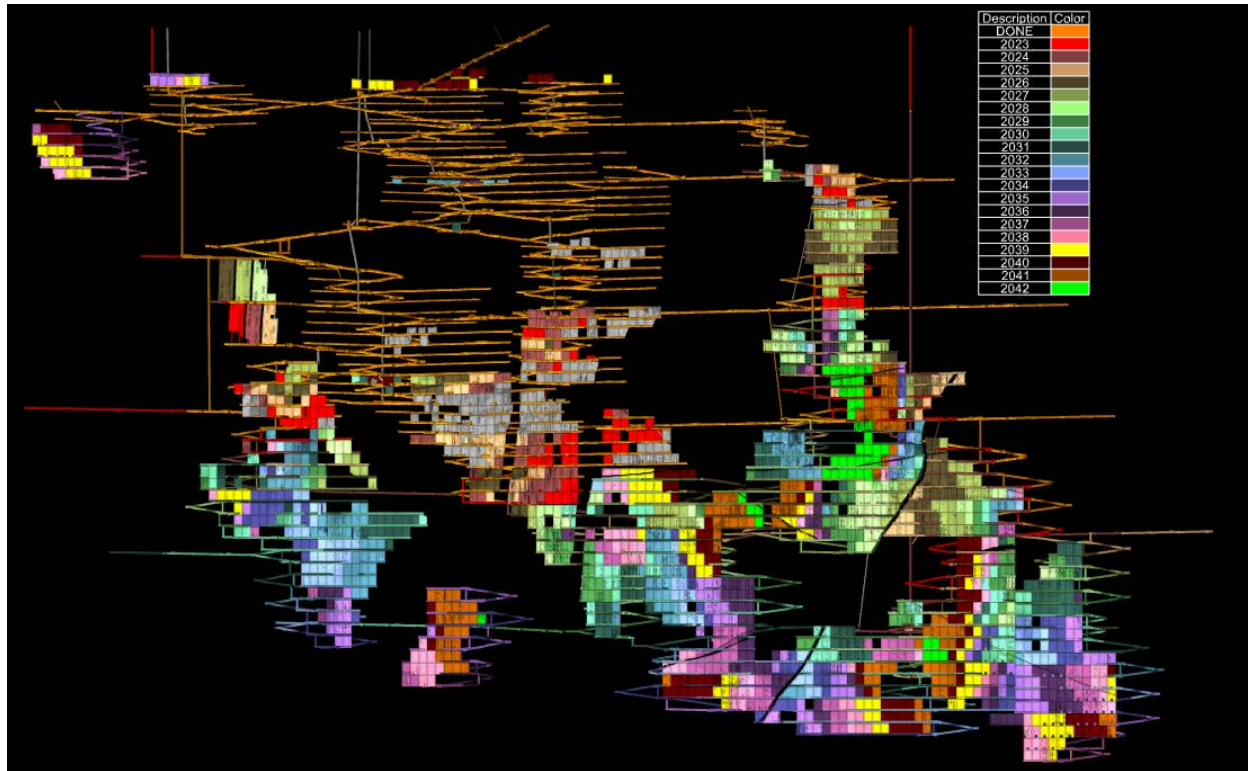


Figure 16-16 S2000 Annual Extraction Sequence

16.12.2 Shaft at 2,400 TPD (S2400)

A study was undertaken which determined the maximum sustainable throughput from the underground mine to be 2,400 tpd. The S2400 scenario will see an increase of the mining rate to 2,400 tpd in 2026. Upon commissioning of the shaft and its associated infrastructure all ore and waste transport would be via the shaft. In addition, the shaft would be used to transport personnel and materials to any of the three shaft station levels thereby decreasing travel time significantly. The scenario has several significant capital investments including:

- The sinking of a 1,380 m deep, 5 m diameter, concrete lined shaft;
- Construction of a hosting plant and associated infrastructure. The design selected would be capable of hoisting 4,500 tpd of ore and waste from the 1,350 Level loading pocket and 3,500 tpd of ore and waste from the 2,000 m elevation should the shaft be required to be deepened in the future to access new Mineral Resources. Details of the shaft and associated infrastructure are fully described in Section 18;
- Construction of an ore and waste handling system underground, consisting of ore and waste passes, a grizzly, coarse ore bins, and a loading pocket;
- A paste fill plant, and the associated underground distribution system, capable of delivering 2,400 tpd of paste; and
- The mill capacity would be expanded to 2,400 tpd. This would require upgrades or additions to several of the areas of the process plant such as the crushing circuit, fine ore stockpile capacity, primary grinding circuit, pre-leach thickening, leaching, CIP, acid wash, process water management, and other equipment modifications. This is fully discussed in Section 17.

Total ventilation requirements for S2400 scenario are 850,000 cfm driven by geographical constraints of the ore body and not equipment operating constraints. A detailed life of mine ventilation design was undertaken supported by a third party consultant, which identified an opportunity to convert the existing portal to a fresh air source by converting the ventilation from a push system to a push-pull system. This change in ventilation design allows for a simplification of ventilation infrastructure around the shaft infrastructure, greatly reducing blast clearing timelines and will allow the switch to a 12 hour working shift further improving underground productivities. The change to a push-pull system will require the twinning of the west return air raise to surface with required fan installations, the conversion of the existing FAR#2 raise to an exhaust raise, and the relocation of existing mine air heaters at FAR#2 to the mine portal.

As most of the ore will be moved on the level to the ore passes, the truck requirements are significantly reduced from current (2022) requirement of eleven 42 tonne trucks. An average of six 42 tonne trucks will be required when the shaft infrastructure is in place.

Mine life in the S2400 scenario would go to 2039 and the annual extraction sequence is shown in Figure 16-17.

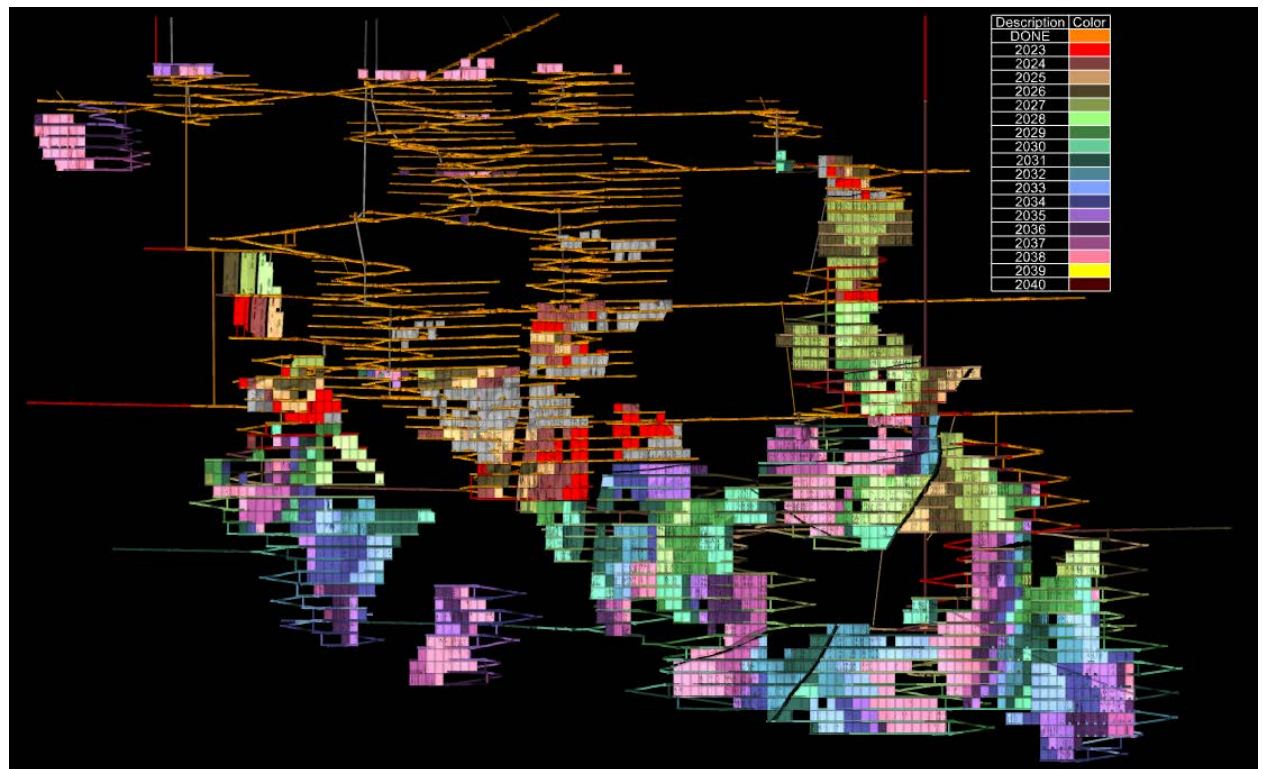


Figure 16-17 S2400 Annual Extraction Sequence

16.13 Phase 3+ Expansion Study Observations

Detailed capital and operating costing models were developed for each scenarios studied. Combining these models with the physicals, cash flow models were constructed that allowed for sensitivity analysis of costing and productivity input parameters. In comparing the two scenarios several observations were made that will focus on the S2000 and S2400 scenarios.

16.13.1 Capital Requirements

Capital costs increase moving from the S2000 scenario to the S2400 scenario. The increase in capital from the S2000 to the S2400 is attributable to the increase in daily mining capacity and addition to the cost associated with a further expansion in milling capacity and a larger paste plant. Capital requirements are discussed in detail in Section 21.

16.13.2 Productivity

Productivity levels between the S2000 and S2400 scenarios are very similar and dramatically increase with the use of the shaft to transport personnel underground. Using 2022 as a baseline, effective time at the face increases by 24% with the shaft. With additional time at the face, development metres per employee and stoping tonnes per employee increase. Increased productivity results in less personnel required to attain the same tonnage or alternatively the same number of personnel to achieve higher tonnage rates as is the case with the mine staffing requirements.

16.13.3 Operating Costs

Mine unit operating costs decrease moving from the S2000 scenario to the S2400 scenario driven by improved economies of scale, namely the sharing of fixed costs such as supervision, engineering, and geology, ventilation, pumping etc. over more tonnes. Operating costs are discussed in more detail in Section 21.

16.13.4 Economics

With the R1200 scenario considered as the Basecase, after-tax net present value (NPV) and IRR comparisons were undertaken on the S2000 and S2400 scenarios. It was determined that the S2400 scenario had the highest NPV. IRR's were undertaken on the deltas of the annual after-tax cash flow between the S2000 and S2400 scenarios and the Basecase. The S2000 scenarios has a lower, but still acceptable IRR.

Although the S2400 has higher initial or growth capital, this is more than offset by lower sustaining capital and operating costs than the S2000 scenario. In addition, the impact of higher annual revenue from the increase in annual ounce production positively impacts the NPV and IRR. The economics are discussed in more detail in Section 22.

16.13.5 Other Considerations

Due to the lower number of trucks operated, and the reduced life of mine ventilation requirements the S2400 option has the lowest greenhouse gas emissions both, on an annual basis (Figure 16-18) and a life of mine basis (Figure 16-19). It is expected that carbon emissions will decrease by 35% as compared to current operations. Battery electric vehicles (BEV) were not evaluated as part of this study and represent a further opportunity to decrease carbon emissions. It is expected that as BEV technology and reliability improve further evaluations will be completed to determine the feasibility of their application in the life of mine equipment fleet.

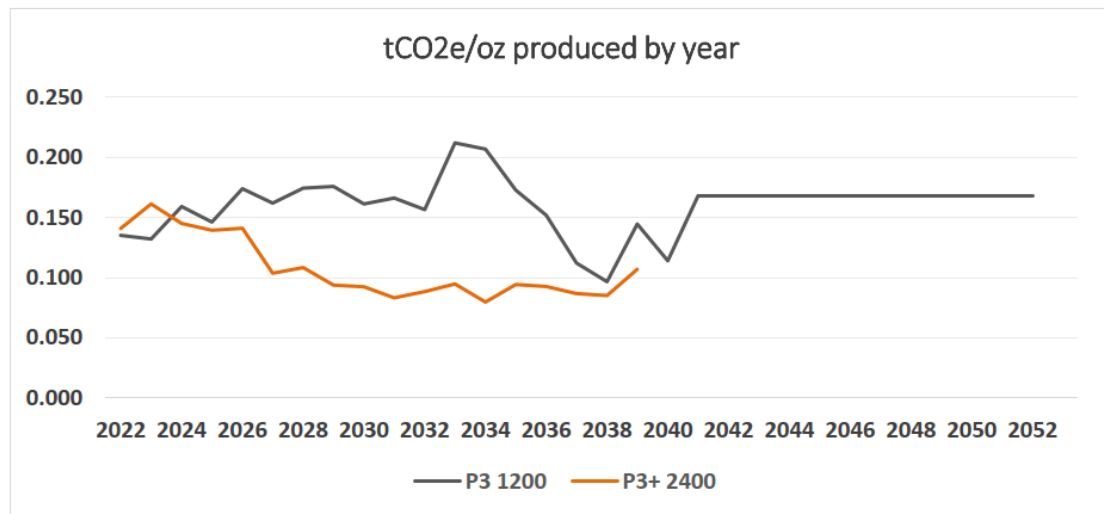


Figure 16-18 Annual CO2 Emissions Intensity

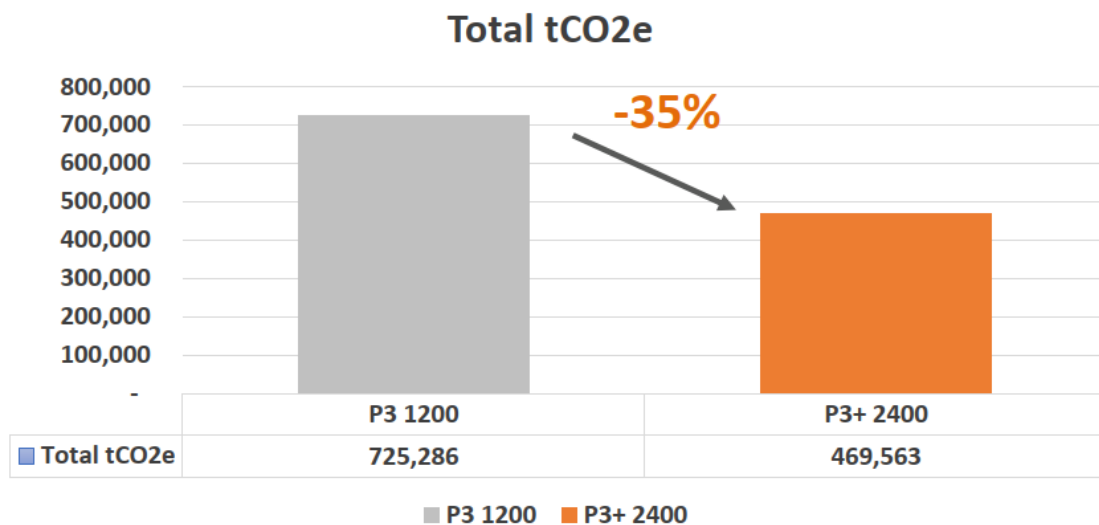


Figure 16-19 Life of Mine CO2 Emissions

In undertaking the mine planning exercise to generate the S2400 scenario it was demonstrated that a 2,400 tpd mine capacity is very viable. With the growth of the deposit in the lower eastern extents of the deposit, specifically the high grade (+20 g/t) domains in proximity to the shaft infrastructure, average annual gold production will meet or exceed 275k oz per year from 2027 through 2035.

Should additional Mineral Resources be discovered below the 1,500 metre elevation, the shaft is well positioned to access those potential Mineral Resources. Further study will be required to determine the optimum extraction methodology. In the scenario that haulage from below 1,500 m to the shaft infrastructure is not available, the planned shaft can be deepened, or a winze can be developed depending on the nature and orientation of the deposit at depth.

16.14 Phase 3+ Expansion Study Conclusions

The primary conclusions from the Study were:

- In expanding the Island Gold Mine, the shaft at a 2,400 tpd mining rate (S2400) scenario should be adopted given that it:
 - o Provides the strongest economics (free cash flow, after-tax NPV, and IRR) of all the scenarios studied;
 - o Provides the highest annual gold production;
 - o Has the lowest operating costs, as well as the lowest cash costs and mine-site all-in sustaining costs per ounce;
 - o Has the lowest combined operating and capital costs;
 - o Further de-risks the lower mine operation; and
 - o Provides a significantly reduced carbon footprint and reduced exposure to diesel price and carbon tax increases.

Based upon these conclusions Island Gold is proceeding with the permitting and construction of the paste plant, the shaft complex and mining at a rate of 2,400 tpd.

16.15 Island Gold Life of Mine Plan

With the commitment to implement the S2400 scenario, the mine plan associated with that scenario has been adopted as the life of mine plan for Island Gold and is presented in Table 16-7.

Table 16-7 Life of Mine Production Physicals

	Units	LOM	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039
Stope Tonnes	(kt)	11,363	339	345	316	344	498	620	625	619	658	696	752	772	847	830	843	844	874	542
Stope Grade	(g/t)	11.09	10.44	9.32	12.36	13.39	11.80	11.52	10.82	13.12	13.06	13.98	12.49	10.29	12.14	10.19	9.81	9.43	8.67	7.27
Development Tonnes	(kt)	2,187	99	93	124	135	201	256	253	257	218	180	126	104	29	46	36	32	0	0
Development Grade	(g/t)	8.00	7.21	9.32	8.28	9.83	6.85	7.69	8.19	7.93	7.74	8.00	6.83	9.73	4.79	8.69	7.67	4.60	0.00	0.00
Total Ore Tonnes	(kt)	13,550	438	438	439	479	699	876	878	876	876	876	878	876	876	876	878	875	874	542
Total Grade	(g/t)	10.59	9.73	9.40	11.29	12.47	10.45	10.46	10.12	11.66	11.79	12.81	11.73	10.27	11.94	10.16	9.78	9.32	8.73	7.34
Waste Tonnes	(kt)	6,182	493	579	647	676	537	393	393	370	427	412	376	227	219	229	122	81	0	0
Operating Development	(m)	34,980	1,530	1,580	2,067	2,255	3,151	4,049	3,977	4,006	3,439	2,850	2,092	1,655	471	739	582	538	0	0
Capital Development	(m)	81,075	7,328	7,470	7,143	9,309	6,790	4,637	5,053	5,334	6,063	5,622	4,469	2,733	3,110	3,235	1,679	1,099	0	0
Exploration Development	(m)	7,802	964	627	2,155	152	670	985	566	0	0	230	857	482	80	33	0	0	0	0
Total Development	(m)	123,856	9,822	9,677	11,365	11,715	10,611	9,672	9,596	9,340	9,502	8,703	7,419	4,869	3,661	4,007	2,261	1,637	0	0
Alimak Raising	(m)	1,561	0	138	0	458	965	0	0	0	0	0	0	0	0	0	0	0	0	0
Conventional Vent Raising	(m)	2,943	292	243	191	227	227	188	204	169	280	247	158	146	64	126	123	58	0	0
Raisebore Vent Raising	(m)	891	529	362	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

17 RECOVERY METHODS

17.1 Overview

The existing process plant at Island Gold treats ore via a conventional cyanidation process. Run-of-mine (ROM) ore is processed using a conventional two-stage crushing circuit followed by a two-stage ball mill grinding circuit, cyanide leach, partial carbon-in-leach (CIL), carbon-in-pulp (CIP) circuit and associated gold recovery and carbon handling circuits to produce gold doré.

Plant performance over the last three years is summarized in Table 17-1 below.

Table 17-1 Plant Performance (2019-2021)

	2019	2020	2021
Ore Processed, t/y	401,276	386,591	435,297
Ore Processed, t/d	1,099	1,056	1,193
Feed Grade, g/t Au	11.85	11.62	10.35
Average Recovery, %	97%	97%	96%

The process plant was upgraded in 2018 to treat 1,200 t/d ore and will be further expanded to treat 2,400 t/d ore.

The feed grade for the plant expansion is 11.4 g/t Au with a design grade of 22 g/t Au.

The following section describes the existing and future recovery methods.

The existing Island Gold process plant consists of the following unit operations:

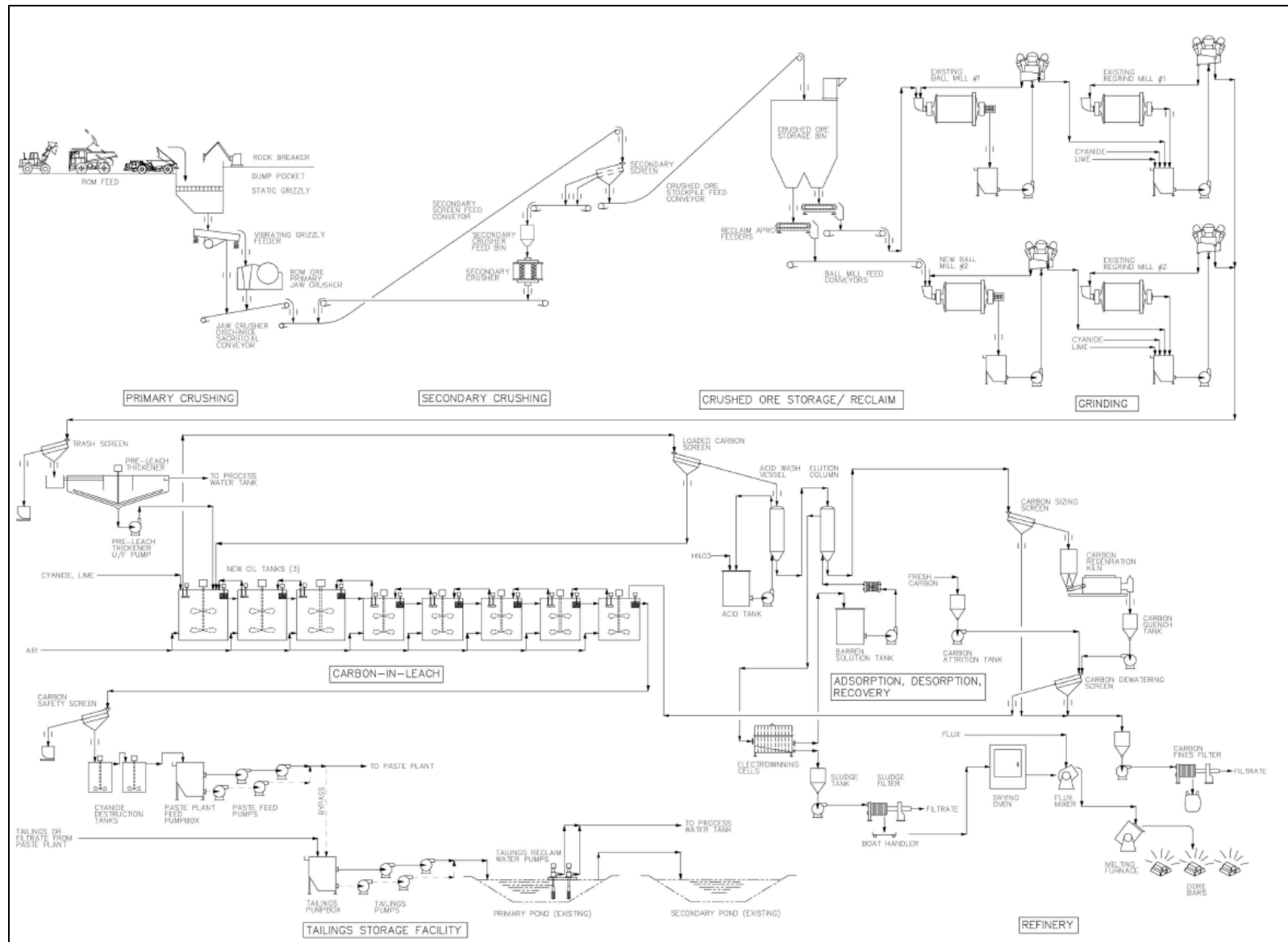
- Primary and secondary crushing circuit and associated material handling equipment;
- Crushed ore storage bin and associated reclaim systems;
- Two stage ball mill grinding circuit, cyclone classification and associated pumping and material handling systems. The existing circuit consists of a primary ball mill and two smaller secondary ball mills one of which is kept on standby) to produce a primary grind size of P80 of 65 µm;
- Pre-leach thickening;
- Leach (4 tanks) and CIL (1 tank) circuit providing approximately 20 hours residence time;
- CIP circuit;
- Acid wash and elution, and carbon reactivation (1.7 tonne ADR plant);
- Gold electrowinning and smelting (gold room); and
- Tailings pumping to the primary tailings storage pond.

The existing process plant will be expanded as follows:

- Throughput increased from 1,200 t/d to 2,400 t/d;
- Replace the existing crushing plant with a new primary and secondary crushing circuit to process the higher ore tonnage. Crushing plant availability will be 75%;
- Install a new crushed ore storage day bin with a new associated reclaim system;
- Install a new primary ball mill and convert the existing grinding circuit to a twin train grinding circuit to produce a primary grind size of P80 of 65 µm. Mill availability will be 92% (similar to existing grinding operations);
- Upgrade the feed well, pumps and pipes at the existing pre-leach thickener;
- Convert the existing five leach/CIL tanks to CIL and add three new larger CIL tanks providing approximately 20 hours residence time;
- Decommission the existing CIP circuit;
- Replace the existing 1.7 tonne ADR plant with a new packaged 5 tonne ADR plant;
- Replace the existing gold room with a new gold room for the increased doré production;
- Install a cyanide destruction circuit; and,
- Upgrade tailings pumping system for the increased throughput and pumping to a new paste backfill plant.

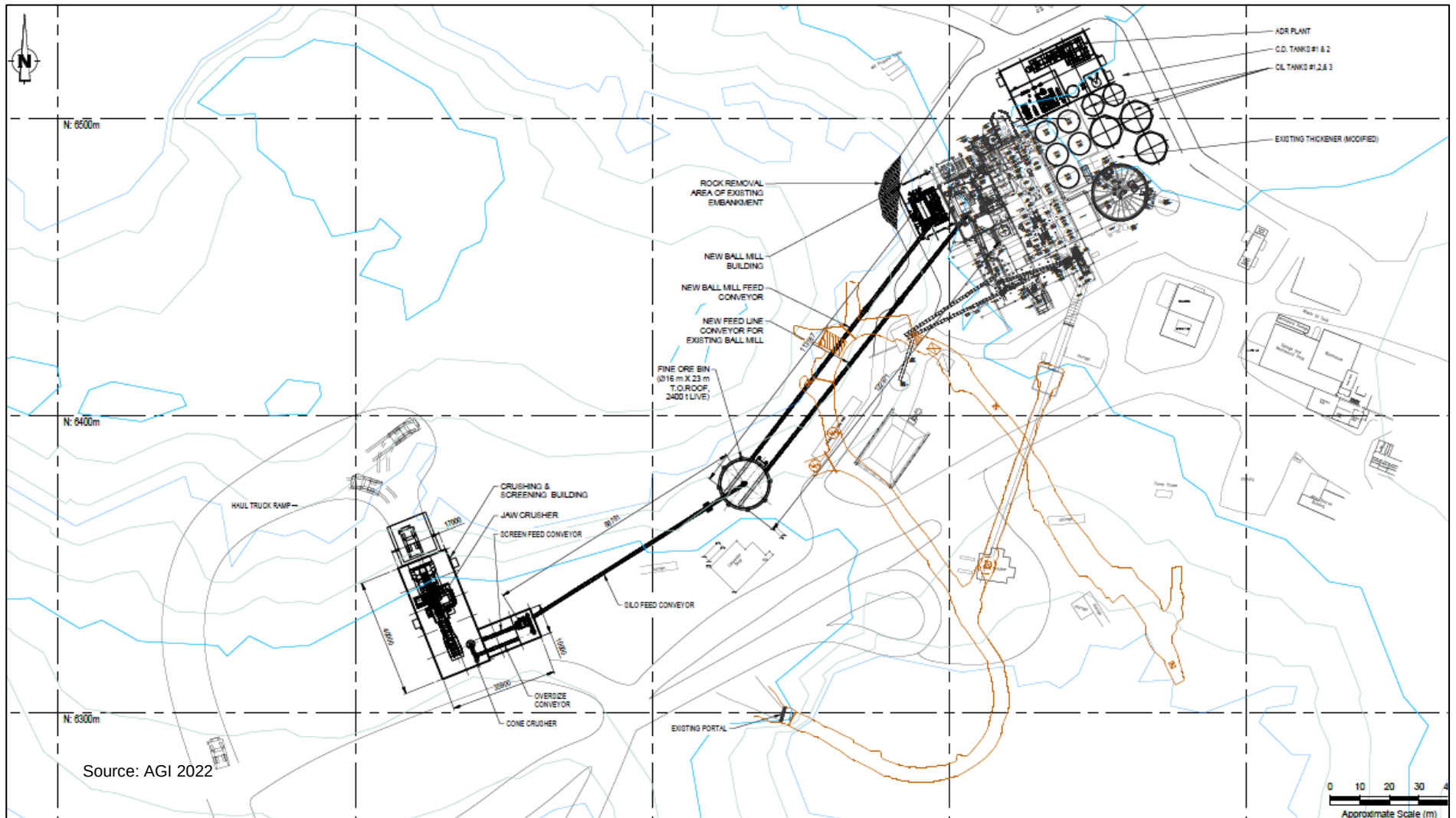
A simplified process flowsheet of the proposed expanded Island Gold process plant is shown in Source: AGI 2022

Figure 17-1 and a conceptual layout for the expansion is shown in Figure 17-2



Source: AGI 2022

Figure 17-1 Plant Process Flowsheet (Upgrade for 2,400 t/d)



Source: AGI 2022

Figure 17-2 Conceptual Plant Process Layout (Upgrade for 2,400 t/d)

17.2 Process Description

17.2.1 Crushing Circuit and Crushed Ore Storage

The existing crushing circuit consists of a primary crusher, sizing screen, secondary cone crusher and associated material handling equipment. The existing crushing plant is undersized to process 2,400 t/d and will be replaced with a new complete crushing circuit with 75% availability as follows. The crushing circuit will produce a crushed ore size of approximately P80 of 13 mm.

ROM from the underground mine, at a maximum lump size of approximately 300 mm (after the Shaft project has been completed), will be transported from the shaft bin house to the plant by 40-tonne capacity dump haul trucks. The trucks will tip directly into the ROM bin.

Ore will be withdrawn from the ROM bin and feed directly to a vibrating grizzly. Oversize from the grizzly will report directly to a new jaw crusher, which will operate in open circuit. Crushed ore from the jaw crusher will discharge, together with undersize from the vibrating grizzly, on to the primary crusher discharge sacrificial conveyor that will feed a new double deck sizing screen via a secondary screen feed conveyor.

Oversize ore from the top and middle deck of the sizing screen will feed a secondary crusher feed bin that will feed a new secondary cone crusher. Discharge from the new cone crusher will be conveyed to the secondary screen feed conveyor. Undersize from the sizing screen will be conveyed to a new day ore bin with 2,400 tonnes live capacity via the crushed ore feed conveyor.

The ore bin will have two outlets to reclaim ore via belt or apron feeders to a twin grinding circuit.

17.2.2 Grinding

The existing grinding circuit consists of a primary ball mill (3 m diameter by 5.79 m EGL, 8,95kW motor) and two secondary ball mills (2.74 m diameter by 3.25 m EGL, 373kW each). Currently, only one of the two secondary ball mills is operated, with second ball mill kept on standby. The grinding circuit will be upgraded and reconfigured to process 2,400 t/d and produce a primary grind size of P80 of 65 µm as follows.

The grinding circuit will be reconfigured from a single train to a twin train circuit. A new primary ball mill of similar size to the existing primary ball mill will be installed. Each circuit will consist of a primary ball mill in closed circuit with cyclones and an existing secondary ball mill in closed circuit with cyclones. Reclaimed ore from the new crushed ore bin will feed each train with its own feed conveyor. A belt scale on each of the ball mill feed conveyors will monitor the feed rate. Process water will be added to each primary ball mill to maintain a 75% slurry discharge density. Mill discharge for each of the primary and secondary mills will pass through its own trommel screen to remove grinding media scats. Each train will have the capacity to process 1,200 t/d of ore, similar to the existing operations.

The existing secondary regrind mills will be refurbished to improve circuit reliability. The existing trash screen will be upgraded for the higher capacity.

Sodium cyanide and lime slurry will be added to the primary ball mill pump boxes.

Relevant pumps, pipelines, chutes, and pump boxes will be upgraded for the new grinding configuration and higher throughput. The current pipeline to the pre-leach thickener will be replaced with a larger diameter pipeline.

17.2.3 Pre-Leach Thickening

Product from the grinding circuit flows to an existing 14 m diameter pre-leach thickener. For the plant expansion, the pre-leach thickener will be upgraded as follows.

The thickener rake and rake drive unit will be refurbished, and the feed well will be modified to a high rate, auto-diluting design. Thickener underflow slurry will continue to be drawn out at 54-55% solids.

Flocculant addition rates will increase at the higher throughput rate and additional dosage of flocculant can be addressed by mixing solutions more frequently, and by dosing a higher strength solution to the thickener.

The existing thickener underflow pumps and pipeline will be changed for the higher throughput. A second pipeline for the thickener overflow will be installed for the increased flowrate and a second process water tank will be installed for the additional process water.

17.2.4 Leach and CIL

Pre-leach thickener underflow currently pumps slurry to a leach-CIP circuit. The existing leach circuit consists of five tanks and is configured as four leach tanks and one CIL tank. The CIP circuit is a traditional CIP circuit consisting of five tanks with launder screens.

For the plant expansion, the existing leach-CIP circuit will be converted to a full 8-stage CIL circuit as follows.

The existing four leach tanks will be converted to CIL tanks and three larger CIL tanks will be installed to provide an overall circuit residence time of 20 hours for the higher throughput and the tanks will be sparged with air. The CIL tanks will be equipped with interstage screens and pumps to advance the loaded carbon upwards to the next CIL tank. Activated carbon will be added into the seventh and eighth tanks and loaded carbon will leave the CIL circuit from the first and second tanks. Activated carbon concentration will vary between 10 and 20 g/L slurry within the CIL circuit.

Sodium cyanide will be added to the CIL circuit to dissolve gold and lime slurry will be added to maintain a slurry pH of approximately 11-11.5.

The loaded carbon will be transferred to the carbon stripping circuit, while the leach residue will be sent to an upgraded carbon safety screen, to recovery any carbon fines. Undersize from the carbon safety screen will flow to a new cyanide destruction circuit.

17.2.5 Acid Wash and Elution

The existing acid wash and elution circuit has a strip capacity of 1.7 tonnes and is undersized. For the plant expansion a new 5 tonne Pressure Zadra vendor packaged circuit will be installed and the existing plant will be decommissioned. The plant will nominally complete one strip per day and has the capacity to double the strips per day when mill head grades increase to 22 g/t Au of gold.

Loaded carbon from the CIL circuit will be pumped and screened to the new acid wash column where it will be treated with hydrochloric acid to remove inorganic foulants. The carbon will first be rinsed with fresh water. Acid will then be pumped from the acid wash circulation tank to the acid wash column and then pumped upward through the acid wash vessel and overflow back to

the acid wash circulation tank. The carbon will then be rinsed with fresh water to remove the acid and any mineral impurities. A recessed impeller pump will then transfer the carbon from the acid wash vessel into the elution vessel using recycled carbon transfer water.

The carbon stripping (elution) cycle will utilize barren solution to strip gold rich carbon to create a pregnant solution. During the strip cycle, solution containing approximately 2% sodium hydroxide and 0.2% sodium cyanide, at a temperature of 150°C and 500 kPa will be circulated through the strip vessel. Solution exiting the top of the elution vessel will be cooled below its boiling point by the heat recovery heat exchanger. Heat from the outgoing solution will be transferred to the incoming cold solution.

17.2.6 Carbon Regeneration Circuit

The existing carbon regeneration circuit will be replaced with a larger circuit.

Once stripped of gold, transport water will transfer the carbon from the elution column to the carbon dewatering screen. Oversize carbon from the screen will discharge by gravity to the carbon regeneration kiln feed hopper. Screen undersize will drain by gravity into the carbon fines tank. An electrically fired kiln will treat 5 tonnes of carbon per day. The regeneration kiln discharge will be transferred to the carbon quench tank by gravity, cooled by process water and stored in the regenerated sized carbon tank prior to pumping back to the CIL circuit. Fresh carbon will be periodically added, to make up carbon losses due to attrition, to the carbon pre-attrition tank along with fresh water to mix and activate the carbon. The fresh carbon will then drain into the regenerated carbon tank.

17.2.7 Electrowinning and Gold Room

The existing electrowinning circuit and gold room will be replaced with a newer and larger circuit.

The pregnant solution generated from the elution column will be pumped to two electrowinning cells. These cells will operate on a single-pass basis to produce gold sludge. The barren solution will be collected in the barren solution pump box where it will be pumped to the barren solution tank.

The electrowinning cathodes will be washed with a high-pressure washer to dislodge gold sludge from the cathodes. The sludge will be filtered by a filter press and the filter cake will be dried in a drying oven. The resulting filtrate will be pumped back to the barren solution pump box within the refinery. Dried filter cake will then be transferred manually into the induction smelting furnace with flux materials where it will be batch smelted into gold doré bars and stored in a secure vault.

17.2.8 Cyanide Destruction

The process plant currently does not have a cyanide destruction circuit. For the plant expansion, a new cyanide destruction circuit will be installed and will consist of two mechanically agitated tanks. The conventional SO₂/air process will be used for cyanide destruction. Treated slurry will flow by gravity to the tailings pump box for pumping to either the new paste plant or the existing primary pond.

The cyanide destruction circuit will treat leach residue from the CIL circuit, process spills from various contained areas and process bleed streams (cold cyanide barren solution effluent, acid wash effluent).

Air will be sparged into the cyanide destruction tanks. Lime will be added to maintain a pH of 8.5 and copper sulphate added as a catalyst. SO₂ gas will be sparged into the tanks. The process will reduce weak acid dissociable (WAD) cyanide levels to 10 mg/L.

17.2.9 Reagents

The process plant consists of the existing reagents and reagent make up system that will be modified as noted for the plant expansion:

- Quick Lime: Delivered by tanker truck, and is discharged into a silo, and fed by screw conveyor to the mix tank to be slaked into a milk of lime slurry. The milk of lime overflows into a lime distribution tank that is pumped to the primary ball mill discharge pump box via a piping loop and is used to maintain pH at the appropriate level to avoid HCN gas generation. Unused lime returns to the distribution tank. The existing system is suitable and distribution pumps will be sped up and piping routed for the additional users;
- Sodium Cyanide – Delivered as a liquid and distributed to primary ball mill pump box and barren solution tank. The existing system is suitable and distribution pumps will be sped up and piping routed for the additional requirements;
- Sodium Hydroxide (also known as caustic) - the existing system is suitable, and piping will need to be rerouted to new equipment;
- Flocculant – Delivered in 25kg bags and the solution is made up for the pre-leach thickener. A new dosing pump will be required for the additional requirements;
- Copper Sulphate – a new system will be installed and will consist of receiving copper sulphate in 1 tonne bags and preparing a copper sulphate solution for use in the cyanide destruction circuit; and
- Sulphur Dioxide (SO₂) – a new system will be installed and will consist of receiving liquid SO₂ and vaporizing the liquid to produce gaseous SO₂ for use in the cyanide destruction circuit.

17.2.10 Air

Blowers currently supply low pressure process air for the process plant. Additional air blowers will be required for the CIL and cyanide destruction circuits.

Air compressors currently supply plant and instrument air for the process plant. This system will be upgraded and piping rerouted for the additional equipment.

17.2.11 Water

The following water networks exist at the process plant:

- Process water – sourced from the pre-leach thickener overflow;
- Fresh water – sourced from the Maskinonge Lake and the Kremzar Mine;
- Reclaim water – sourced from the existing primary pond; and
- Gland seal water – reclaim water from the existing primary pond.

The water networks will be upgraded, and the piping rerouted for new and modified areas

17.3 Metallurgical Accounting

Several samplers are provided throughout the plant to generate composite shift samples from key process streams. Two types of sampling are performed; metallurgical and process control sampling. Metallurgical sampling will be upgraded for the plant expansion.

Metallurgical samplers are used to generate shift composite samples that will be assayed for plant metallurgical accounting. The following process streams will be equipped with metallurgical samplers:

- Primary cyclone overflow for each grinding train; and
- CIL Tailings.

The metallurgical samplers sample feed and tailings product which will allow an accurate metal balance for the plant to be completed.

Process control samplers generate samples used to monitor unit processes in the plant. The process control samplers are used to generate shift composite samples on process streams that will provide plant operation performance data.

The following process streams will be equipped with process control samplers:

- CIL feed;
- CIL tailings;
- Final tailings;
- Pregnant solution to electrowinning; and
- Barren solution after electrowinning.

A weightometer on the crushed ore storage bin feed conveyor will measure primary crushed ore tonnage, and a weightometer on the each of the primary ball mill feed conveyors will determine mill feed tonnage.

A manual belt cut sampling point on each of the primary ball mill feed conveyors will allow for the collection of a mill feed head grade sample for cross-checking with the calculated head grade. This sample will also be utilized to establish the moisture content of the mill feed.

Regular surveys of the gold in circuit will continue to allow a reconciliation of precious metals in the feed compared to doré production.

Water supplied and used in the various areas will continue to be continuously monitored.

Reconciliation of the reagents used over relatively long periods will be achieved by delivery receipts and stock takes. On an instantaneous basis, reagent usage rates to unit operations will be measured and accumulated using flowmeters.

17.4 Plant Consumption

17.4.1 Water

A preliminary water balance for the plant expansion was completed. Since reclaim water from the primary pond is the main source of the fresh water for the plant, no additional fresh water will be required for the plant expansion.

17.4.2 Energy

The power demand for the upgraded process plant, along with the rest of the project, will be provided by grid power. The additional average power demand for the process plant will be approximately 4 MVA.

17.4.3 Reagents

Existing reagent storage, mixing and pumping facilities will be provided for all reagents for the process plant. Reagents usage for the upgraded plant is summarized in Table 17-2.

Table 17-2 Reagent Consumption (post expansion)

	Unit Consumption	Nominal Daily Usage
Quick Lime	0.93 kg/t ore	2.2 t/d
Sodium Cyanide	0.56 kg/t ore	1.3 t/d
Activated Carbon	0.037 kg/t ore	89 kg/d
Sodium Hydroxide	0.12 kg/t ore	288 kg/d
Flocculant	10 g/t ore	24 kg/d
Copper Sulphate	0.15 kg/t ore	360 kg/d
SO ₂	0.31 kg/t ore	740 kg/d

18 PROJECT INFRASTRUCTURE

18.1 Local Resources and Services

The Island Gold Mine is an established producer in a mining district that has been historically active since the early 1980s with the former Kremzar, Magino, Edwards and Cline mines. The nearest town is Dubreuilville, Ontario, approximately 17 km from the mine, accessed via Goudreau Road. Dubreuilville is located at the end of Highway 519, heading east off the Trans-Canada Highway, 72 km northwest of Wawa.

The town of Dubreuilville has an estimated population of 635 inhabitants and the main available services consist of an elementary school, a secondary school, a health centre, a motel, a restaurant and bar, a grocery store, a Canada Post outlet, the town hall, a hardware store, a gas station, a municipal library and an arena.

Over 50% of the mine's employees are local from the surrounding region (Dubreuilville, Wawa, White River, etc.). Out of town workers live in camp accommodations (newly installed 2019/2020) near Dubreuilville, where they have individual rooms, recreational services, and a cafeteria. Chartered flights are provided from Toronto, Sudbury, and Rouyn-Noranda and which fly into the Wawa airport daily.

18.2 Current Site Infrastructure

18.2.1 Surface Site Infrastructure

The Island Gold Mine is accessed via a singular portal (Main Portal) and decline. The ore stockpile pad and waste storage are located directly adjacent to the mine portal. The mill feed is hauled approximately 1 km from the stockpile to the mill complex. The maintenance facility, surface office and dry complex are located adjacent to the mill. The Kremzar Portal accessing old mine workings and detached from current underground mine, is utilized to access the bottom of the surface ore bins which are used to feed the crusher section of the mill.

The primary tailings pond is located approximately 500 m west of the mill and the secondary pond is located just west of the portal area. Mine ventilation is via two surface fresh air fans and raises located adjacent to Goudreau Road, approximately 2.5 km away from the main site along with one exhaust raise located across from the fresh air fans. Figure 18-1 displays the relative location of the surface infrastructure.



Figure 18-1 Island Gold Mine Site - Surface General Arrangement

(Alamos 2022)

18.2.2 Power Supply

Site power is supplied by Algoma Power Inc. (API) via one 44kV transmission line. Two 44kV / 4160V transformers (7.5MVA each) are located at the Main Portal and mill complex, respectively. One 44kV / 13.8kV (10MVA) transformer is located off the main site not far from the surface fans.

The Main Portal transformer (44kV / 4160V, 7.5MVA) feeds the compressor room, fresh air raise (FAR) #1 fans and two underground feeders. The two underground feeders are 3C 4/0 AWG, 5kV and feed the upper portion of the mine down to the 620 Level and the new east ramp

descending from 340 Level. The compressed air is backed with a diesel compressor up in the event of a power loss.

The mill transformer (44kV / 4160V, 7.5MVA) feeds the mill, surface shops, warehouse, water treatment plant, dry complexes, fire hall, tailings pond, core shack and office complex. A portion of the mill (critical equipment) and the surface buildings are backed up with a 600V, 1MW diesel generator.

The 13.8kV skid transformer (44kV / 13.8kV, 10MVA) feeds the FAR #2 fans and one 3C 350MCM, 15kV underground feeder that descends down a borehole to the 620 Level where it is distributed to the bottom portion of the mine (620 Level downwards)

18.2.3 Underground Ventilation

The primary ventilation system currently consists of a push type system served with two fresh air raises. FAR #1 currently supplies approximately 140,000 cfm while FAR #2 supplies approximately 350,000 cfm, totalling 490,000 cfm. Return Air Raise (RAR) #2 and the Main Portal serve as the main exhaust routes for the return air. Table 18-1 provides a summary of the surface fan arrangement at Island Gold.

Table 18-1 Key Surface Ventilation Fans Data

Item	FAR #2	FAR #1
Fan Type	Centrifugal	Axial
Fan Arrangement	Parallel	Parallel
Total Horsepower	2,400 HP	400 HP
Variable Frequency Drive	Yes	Yes

FAR #2 serves to ventilate the Island Deep and West Zones, while FAR #1 serves in ventilating the Extension and East Zones of the deposit. FAR #1 is supplemented with an underground booster fan located at the 390/400 Level. An underground booster fan arrangement is also located on 660W Level to ventilate the West Zone. This fan arrangement serves to pull air from FAR #2 into the West Zone to ventilate during development. A fresh air transfer drift is planned to be established between the Island Deep and West Zone to improve air quality in the West Zone and provide enough ventilation for production when the zone's production front comes online.

Propane powered mine air heaters are located on both surface fresh air fans. FAR #1 has a 30 MBTU heater while FAR #2 has 20 MBTU which serve to heat the air to 7°C during the winter months.

18.2.4 Mine Process & Domestic Water

Surface process and potable water is supplied via Maskinonge Lake. Water is pumped from the lake and distributed to the existing site and mill. The raw water is filtered via a 4-stage treatment process: multi-media filtration, reverse osmosis membrane filtration, chlorine injection and ultraviolet rays.

Underground process water is collected underground in a naturally filling cavity (via groundwater and diamond drill holes) near the 190 Level and the 735 Extension Ramp. Process water is distributed via pipeline down the ramp to working horizons via gravity, employing pressure reducing valves (PRV) to control pressure. A process water reserve is employed in the 315 Level and 630 Level underground maintenance facility fire suppression systems.

18.2.5 Mine Dewatering

The Island Gold Mine employs a clean water dewatering system. Water cascades down level sumps, via boreholes or pumps, until it reaches an intermediate dewatering pump station, at which point all water is pumped via the ramp or boreholes to the main dewatering pump stations, located on the 425 Level and the 125 Level. The main pump stations pump the water directly to either the primary or secondary pond on surface depending on pond status and water quality. On average, Island Gold dewateres approximately 1,200 m³/day from the underground workings.

Island Gold is currently designing and implementing a new main dewatering station in the deep portion of the mine to sustain its operations at depth. This system will employ a series of clarification and pump stations to allow pumping of clear water to surface and disposal of residual suspended solids underground.

18.2.6 Mine Compressed Air

Island Gold's underground mine operations are serviced via one surface compressed air plant. The plant is located adjacent to the portal and is composed of a series of 200, 300 and 600 hp compressors totalling 2,100 hp and produces approximately 9,920 cfm. One backup diesel unit provides 1,600 cfm in case of emergency. Underground workings are fed via one 8" air line down the ramp, stepping down to 6" lines to deliver air to the mine's auxiliary levels.

18.2.7 Miscellaneous Underground Infrastructure

There are currently two underground workshops, one located on the 315 Level, and one located on the 620 Level. The 315 Level workshop provides basic mechanical services to Island Gold's primary production fleet as well as secondary vehicles. Construction of a new underground workshop located on the 620 Level was completed in 2021. The new shop is substantially larger and allows Island to perform major mechanical work underground, not having to bring equipment to surface or offsite. The new shop is composed of two mechanical bays, a welding bay, a warehouse, a waste bin and hose bay, tire bay as well as a washbay. Construction of a third workshop is planned in the vicinity of the 1305L Shaft Station to allow equipment to be serviced closer to the active mine workings as mining progresses deeper. It is expected that this shop will be constructed after the shaft is fully commissioned.

An underground cement plant is located on the 720 Level which provides cement produce cemented rockfill. The 720 Level was designed to facilitate the movement of cement and truck traffic on the level. Trucks pull onto the level loaded with waste and get backfilled with slurry, to then haul the cemented rock fill to the backfill site.

18.2.8 Automation, Communication and Controls

The primary means of communication underground is via a 16 channel leaky feeder system. Femco lines are also installed in every refuge station as a secondary communication system. Island Gold is currently in the process of installing LTE underground to increase wireless data capacity and coverage. On surface, leaky feeder, phone lines and Voice over Internet Protocol (VoIP) are utilized.

Island has a fibre-optic network installed underground (and on surface) servicing the main ramps, underground infrastructure and microseismic system. This system also serves as the backbone for the ventilation on demand (VOD) network, recently implemented in 2019. The blasting system is also routed via the fiber network with the leaky feeder network as backup.

Each truck operating underground is also configured with a remote system with the ability to track payload and cycle times allowing the proper management of the underground fleet. The system is planned to be deployed on all prime movers in the fleet in the near future.

Island Gold also has a comprehensive microseismic system installed throughout its underground workings. This allows for continuous monitoring of microseismic activity underground and the implementation of re-entry protocols based on seismicity intensity underground following blasts.

A central control room (CCR) was implemented in 2019, where all communication and automation reports to. The CCR is used as the heart of the operation through which all information flows. This allows for the optimization of traffic flow, monitoring of microseismic activity, monitoring the VOD system as well as to relay key information to underground workers as required.

18.2.9 Tailings Management

The tailings management facility (TMF) represents the main water retention structures on the mine site. It consists of two ponds, the Primary Pond and the Secondary Pond which acts as a polishing pond in addition to water transfer systems via a siphon system (see aerial view of site plan in Figure 18-1). The Primary Pond (built in the former Miller Lake basin) occupies an area of 109 ha. The Secondary Pond has an area of 22 ha. The TMF is operated in accordance with ECA No. 5444-BNPL46 issued April 30, 2020, by the MECP.

Tailings slurry is conveyed by a pressurized pipeline from the mill and spigotted around the inside perimeter of the Primary Pond. The surface of the tailings forms a sloped beach allowing for a pond to form at the lowest part. Water is reclaimed (pumped) from the Primary Pond to the mill. Both the primary tailings and reclaim pipes are placed in an engineered ditch, with drainage to an emergency catchment section (with an area of 0.8 ha) at its lowest points and reinforced by construction of earthen berms. The TMF also includes seepage collection and pump back systems at dykes Nos. 1 and 2 at the Primary Pond; these were built to prevent any migration of seepage to Maskinonge Lake.

In 2020 and 2021 an Offline-Dyke Barrier (ODB) was constructed downstream of Dyke 1 and 4 ensuring that the natural water features are not touching the toe of the TMF dykes. By constructing this dyke barrier Island Gold's tailings facility is considered offline by the Ministry of Energy, Northern Development and Mines (MENDM) and the approval process will go through MENDM for any future dam raises and not through the Ministry of Natural Resources and Forestry (MNRF), as was previously the case.

The initial TMF was constructed in 1988 under the direct supervision of engineers of Gibson and Associates Inc. The Primary Pond capacity was increased in 2011 by raising, expanding, and adding additional dams or dykes under the supervision of AMEC Environmental. The capacity of the Primary Pond of the TMF was further augmented in November 2015 to provide an extra six years of tailings deposition by increasing the height of all existing dykes to an elevation of 424 masl, as well as by adding a new dyke, to accommodate an additional storage capacity of up to 2.5 Mm³. A 3 m tertiary dam raise was undertaken in 2020 to bring the dam to elevation to 427 masl. All 5 dykes were raised 3 metres by the modified downstream method. The spillway was also raised to an elevation of 425.8 masl.

The maximum operating water level (OWL) is 425.5 masl, with the emergency spillway located 0.3 m metres above at an elevation of 425.8 masl. The minimum OWL is 417 masl to ensure enough water is present in the pond for reclaiming continuously throughout winter.

The Secondary Pond was initially constructed in 1988 alongside the Primary Pond and with Dam A built to capture tailings solution transferred from the Primary Pond and discharge as effluent

to the environment pending passing water quality, A subsequent raise was done on the Secondary Pond in 2011 with Dam B and Dam C constructed to an elevation of 399 masl. A spillway was constructed at an elevation of 398.8 masl to convey any water during upset conditions. The minimum OWL is 396.4 masl while the maximum OWL is 398.8 masl.

The dams at the Primary Pond have been designed and constructed using the downstream construction method. The body of the dam consists of engineered granular fill, placed in controlled lifts, and compacted. The embankment consists of a partially zoned construction, consisting of an exterior shell, an upstream membrane, a cut-off below grade, and filter systems.

Annual inspections have been conducted by geotechnical specialists (WOOD group formerly AMEC Foster Wheeler), confirming overall good performance of the dykes. In 2019 a comprehensive Dam Safety Review (DSR) was completed by Golder Associates in conjunction with the annual Dam Safety Inspection (DSI) to review all phases of the construction, operations, and maintenance of the tailings facility. Nothing deficient was found of the tailings facility, and recommendations for minor improvements were completed with the 2020 dam raise.

18.2.10 Water Treatment

Water treatment is managed through natural degradation in the Primary Pond and Secondary Pond. Natural degradation is primarily active during ice-free periods, and batch discharge cycles are timed to accommodate the natural processes. Cycle duration is typically 40 days:

- 10 days transfer from the Primary Pond to the Secondary Pond
- 20 days of final polishing at the Secondary Pond, and
- 10 days discharge of treated water from the Secondary Pond to the receiving environment in a series of streams, wetlands, and ponds, eventually discharging into the central part of Goudreau Lake.

This process can be expedited with the introduction of coagulant and flocculent in the transfer process to reduce total suspended solids (TSS) and metals levels. Total cyanide and ammonia naturally break down via sunlight which is enhanced by adding sulphuric acid to control pH levels and thus increasing microbial activity. Based on these processes, the water treatment is known to reduce site-wide water inventory all while meeting water quality limits prior to discharge.

Water quality is routinely monitored in the Primary Pond and Secondary Pond, and in Goudreau Lake at the discharge point and downstream. A comprehensive water monitoring program has been implemented for the site, and includes twelve compliance sampling locations, and effluent limits as mandated by the MECP. Limits have been established for TSS, total cyanide, copper, nickel, lead, zinc, un-ionized ammonia, oil, and grease, arsenic, and pH. Effluent objectives have also been established for iron, phosphorus, total ammonia, and oil and grease (daily). Notwithstanding, Island Gold Mine also conducts sampling and analyses for other parameters of concern, for example: metals, anions, hydrocarbons.

Water discharge and takings are recorded and kept in such a manner as to maintain compliance with the applicable regulations. Flow measurement devices have been installed to monitor the discharge of treated water from the Secondary Pond.

Under the previous 2012 ECA, discharge of treated water was seasonal, with the annual treatment and release window being from May 15 to December 31. However, with the amended ECA issued by MECP in November 2016, Island Gold can discharge treated water continuously, which allows increased operational flexibility. This operational flexibility is still maintained with the newly revised April 2020 ECA.

An Operations, Monitoring and Surveillance Manual (OMS Manual) has been prepared for the TMF, which includes operating procedures, inspection programs, repair and maintenance programs, contingency plans and procedures for dealing with potential spills, bypasses and any other abnormal situations and for notifying the MECP, and complaint procedures for receiving and responding to public complaints.

18.3 Phase 3+ Infrastructure Expansion

Island Gold recently undertook engineering studies on a possible Phase 3+ Expansion to increase underground mined tonnage rates and implement associated infrastructure upgrades and additions as required. The studies involved the evaluation of several scenarios, which demonstrated that the Shaft Expansion at an increased production rate of 2,400 tpd is the most economic, most efficient, and productive alternative. This also best positions Island Gold to capitalize on further growth in Mineral Reserve and Mineral Resources. The Phase 3+ Expansion will result in the construction of a new shaft, amongst many other infrastructure upgrades and additions, as listed below:

- Development of a new production/service shaft to 1,380 m in depth (initial depth);
- Development/implementation of a new ore and waste handling system underground;
- Upgrade to the site grid power supply;
- Construction of a paste plant and underground distribution system;
- Upgrade of the existing mill from 1,200 tpd to 2,400 tpd;
- Upgrade to the Tailings Management Facility to suit higher and longer life of mine production; and
- Upgrade to mine dewatering and effluent treatment system.

The location of the new planned surface infrastructure is shown in Figure 18-2.

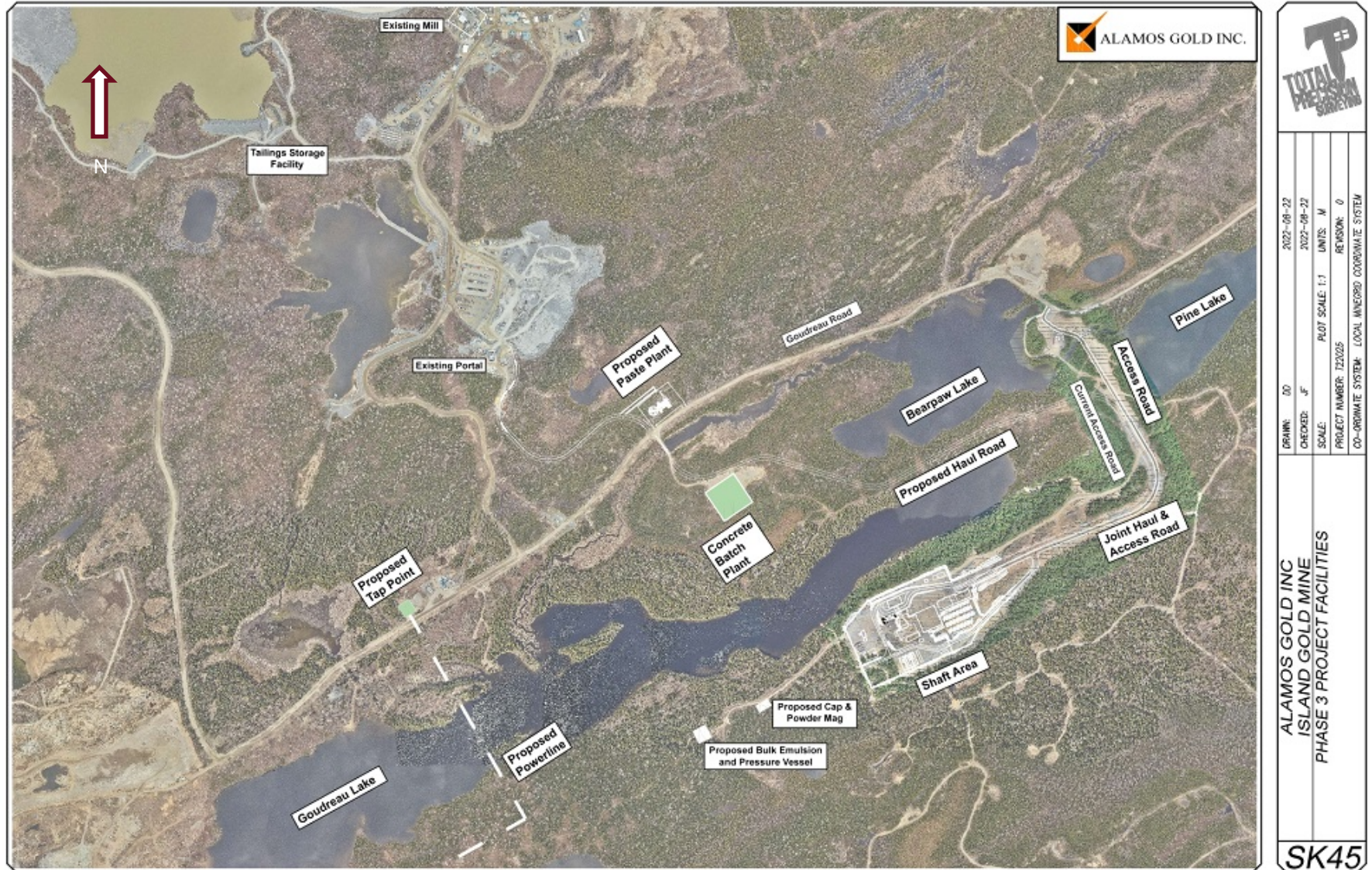


Figure 18-2 Phase 3+ Expansion - Surface Infrastructure Location

(Alamos 2022)

The new shaft and hoists are the foundation for accessing the orebodies at depth. This facility is designed for future capability of a higher production rate and an increase in ultimate depth from the current planned depth of 1,380 m to 2,000 m. Figure 18-3 displays the new mine design and underground infrastructure required to support the planned expansion (orange is existing development, grey is planned development).

The shaft location was chosen based on surface topography/bedrock profile and optimal placement relative to the long-term underground workings. The shaft will cut across the orebody near the 620 Level exploration drift, within an area that has been delineated as an uneconomic zone. It is also positioned east of the centroid of the orebody once the shaft comes online, positioning in a prime location based on the eastward trend of the deposit for any future Mineral Resources and Mineral Reserves that are added into the life of mine plan from exploration results.

18.3.1 Shaft Site

The shaft site is located approximately 2 km from the existing mill complex and will be accessed via an approximate 5 km access road (being a combination of existing roads and new access road). The site will house the required infrastructure to support operations and maintenance of the shaft, an ore and waste handling system as well as basic infrastructure to support a dry and administrative office complex. Figure 18-4 displays the planned infrastructure to be located on the shaft site.

The shaft site is composed of the shaft complex (hoist house, hoist drive cooling building, headframe with bin house, collar house) as well as a new substation and electrical supply, ventilation plenum, and office/maintenance complex.

18.3.1.1 Shaft Complex

The shaft complex is composed of the headframe, hoist house and collar house. The hoist house will be composed of an insulated pre engineered building and will house the two hoists (one production and one service hoist). An E-House south of the Hoist House will not only feed the shaft complex, but also be the main underground feed for the deep portion of the mine, complementing the already existing electrical system underground. The production hoist will be centered square to the shaft whilst the service hoist will be flanked. Figure 18-4 shows the shaft site general arrangement, which also outlines location of the hoist house with respect to the shaft and collar house. Figure 18-5 shows the production and service hoist arrangements in the hoist house. Room has been allocated in the west end of the hoist house for temporary sinking compressors. The hoisting/conveyance system will be designed for 2000 m in depth operating at 3,500 tonnes per day (ore and waste). At the loading pocket level of 1350 m, it will have a 4,500 tpd operating capacity, more than enough to withstand peak mining rates. The service hoist will allow all underground mine personnel to access the underground workings in under 30 minutes, greatly increasing underground productivity when compared to the current ramp access system.

Table 18-2 presents key data on the hoisting plant.

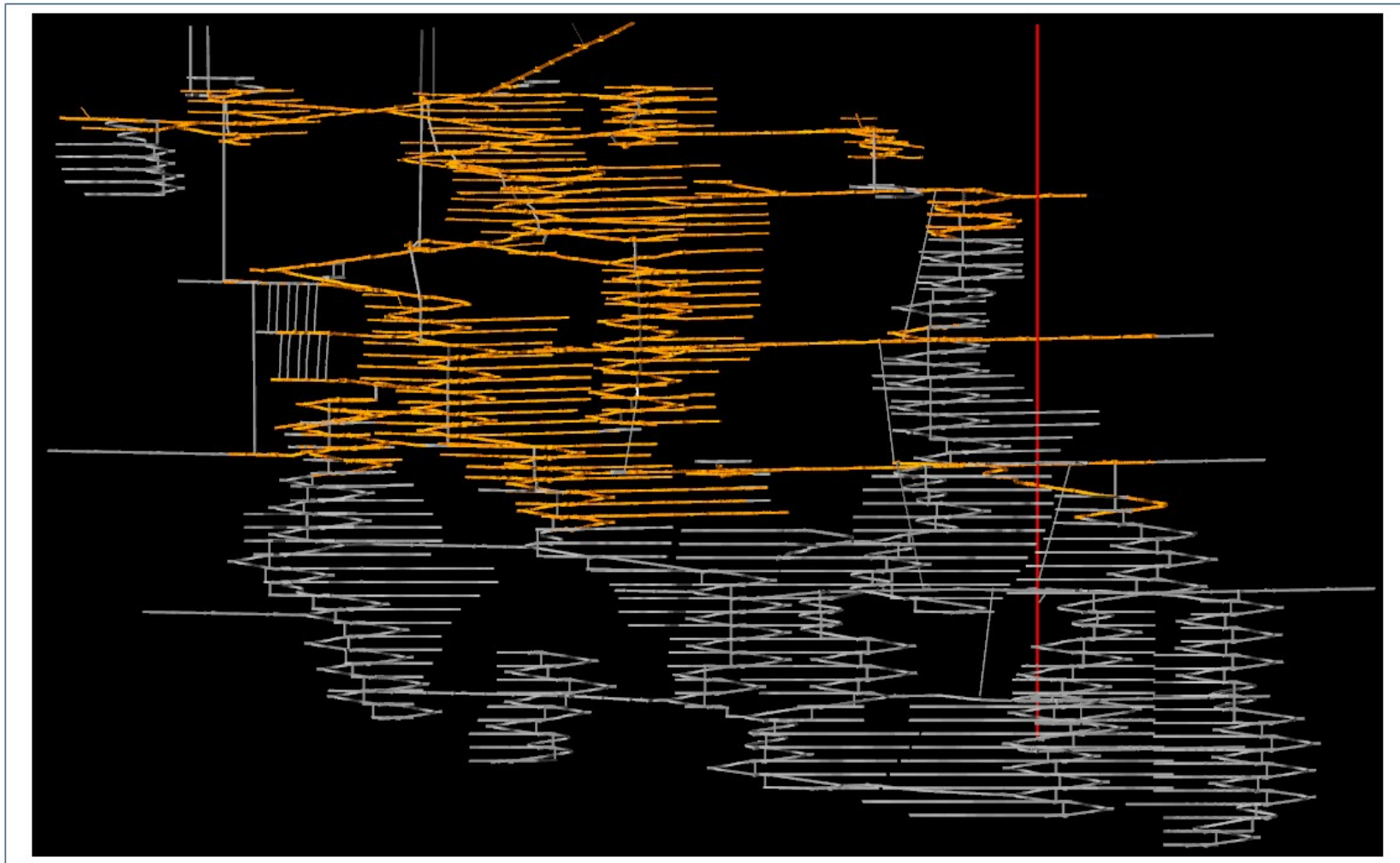


Figure 18-3 Phase 3+ Expansion – Underground Development and Infrastructure

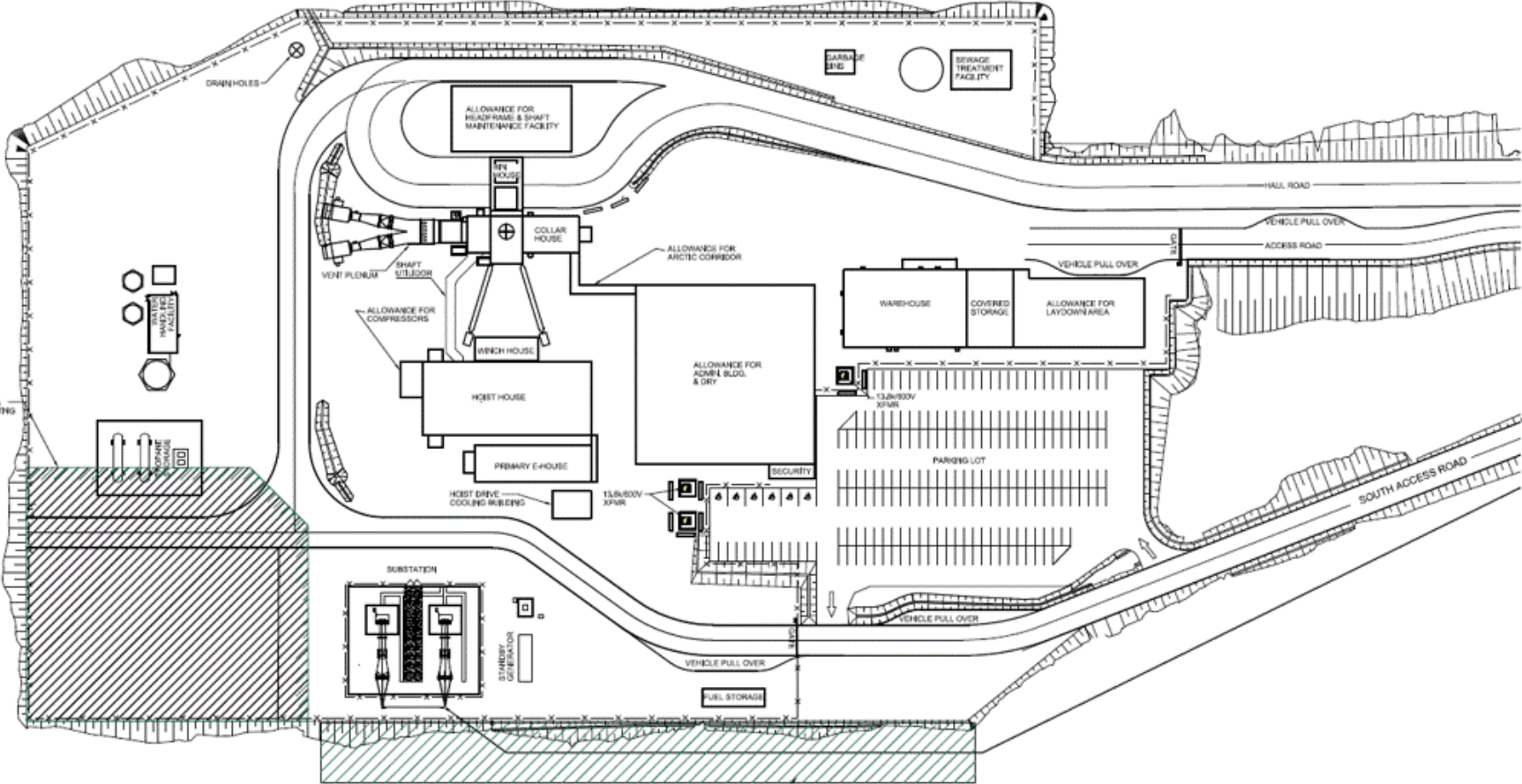


Figure 18-4 Shaft Site General Arrangement

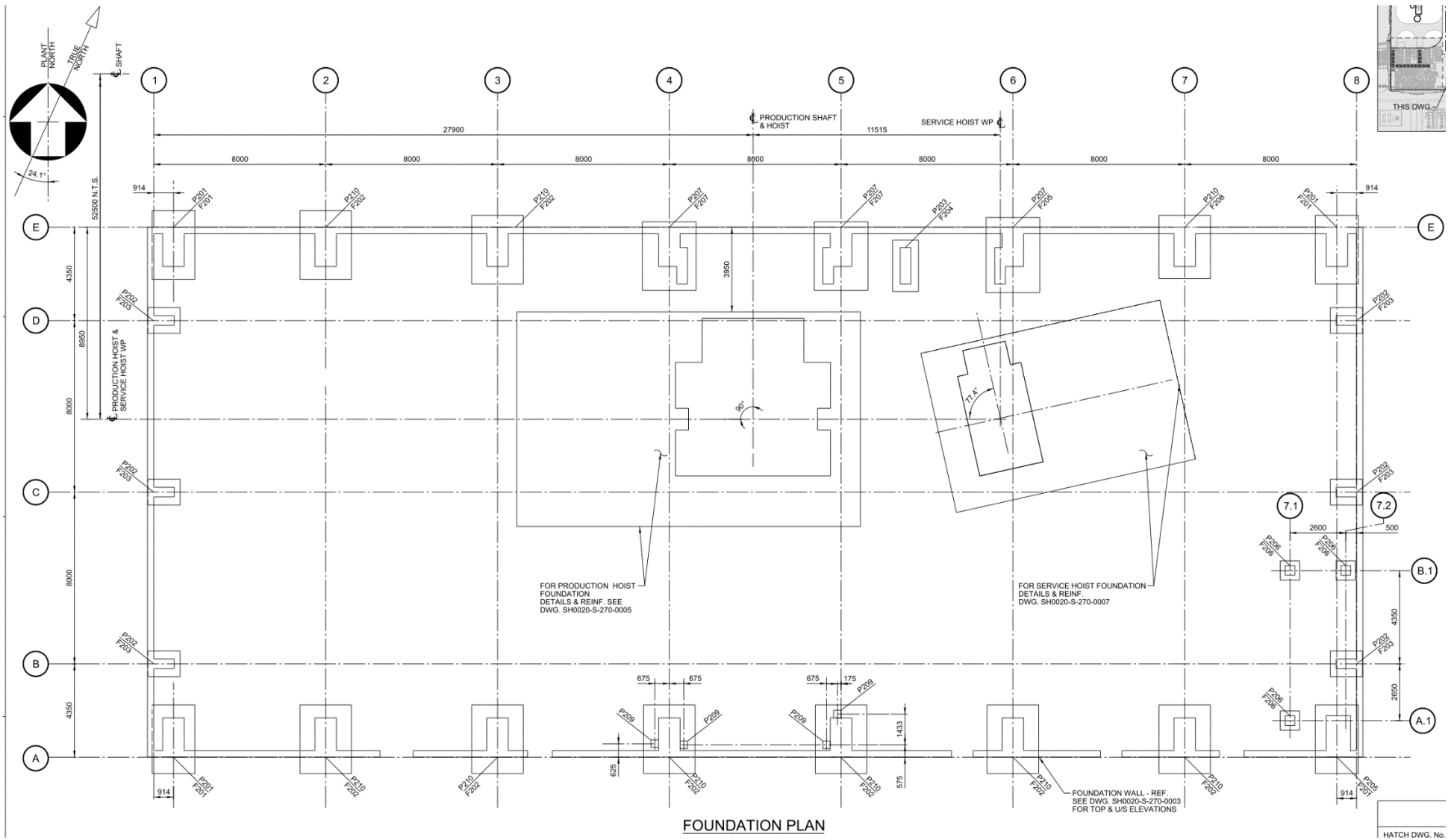


Figure 18-5 Production and Service Hoist Arrangement in Hoist House

Table 18-2 Hoisting Plant Key Data (Production and Service Hoists/Plants)

Hoisting Plant Key Data	
Production Hoist	
Maximum Hoisting Depth	2,000 m
Maximum Hoisting Capacity at Depth:	
Ore	2,400 tonnes/day
Waste	1,100 tonnes/day
Total	3,500 tonnes/day
Hoist Availability	Mechanical availability 22 hours per day, average utilization 16.5 h/d, 345 d/yr
Production Hoist	5 m x 2 m double-drum, double clutch, 3,969 kW RMS.
Skips	2 x 12 tonnes capacity
Hoisting Speed	15 m/s
Hoisting Cycle and Capacity from 2,000 m	198s, 3,500 tonnes/day
Hoisting Cycle and Capacity from 1505 Loading Pocket	154s, 4,500 tonnes/day
Skip Chairing	Fixed chairs with buffers
Applicable Safety Factors	Ontario standard (SANS exception not applied)
Service Hoist	
Maximum Hoisting Depth	2,000 m
Maximum Payload at Depth	6 tonnes
Hoist Availability	20 h/day. Bratticed shaft to allow concurrent production and service hoisting
Service Hoist Layout	4.5 m x 1.8 m double-drum, single clutch, 1,925 kW RMS.
Cage	1 x 6 tonnes double deck. 33 persons/deck
Hoisting Speed	10 m/s
Cage Charing	Cage-mounted guide clamps
Applicable Safety Factors	Ontario standard (SANS exception not applied)

The shaft will house two 12 tonne skips in dedicated compartments for ore and waste movement, along with a double deck service cage for the transport of personnel and materials.

The steel headframe will be erected at 65 m in height and will be insulated. It will have two sheave deck levels to serve the production and service hoists, as well as the sinking winches with one set of rear structural supports. The service hoist sheave deck will be located at 46 m in elevation while the production hoist sheave deck will be located at 57 m in elevation. Figure 18-6 displays an isometric view of the headframe and hoist.

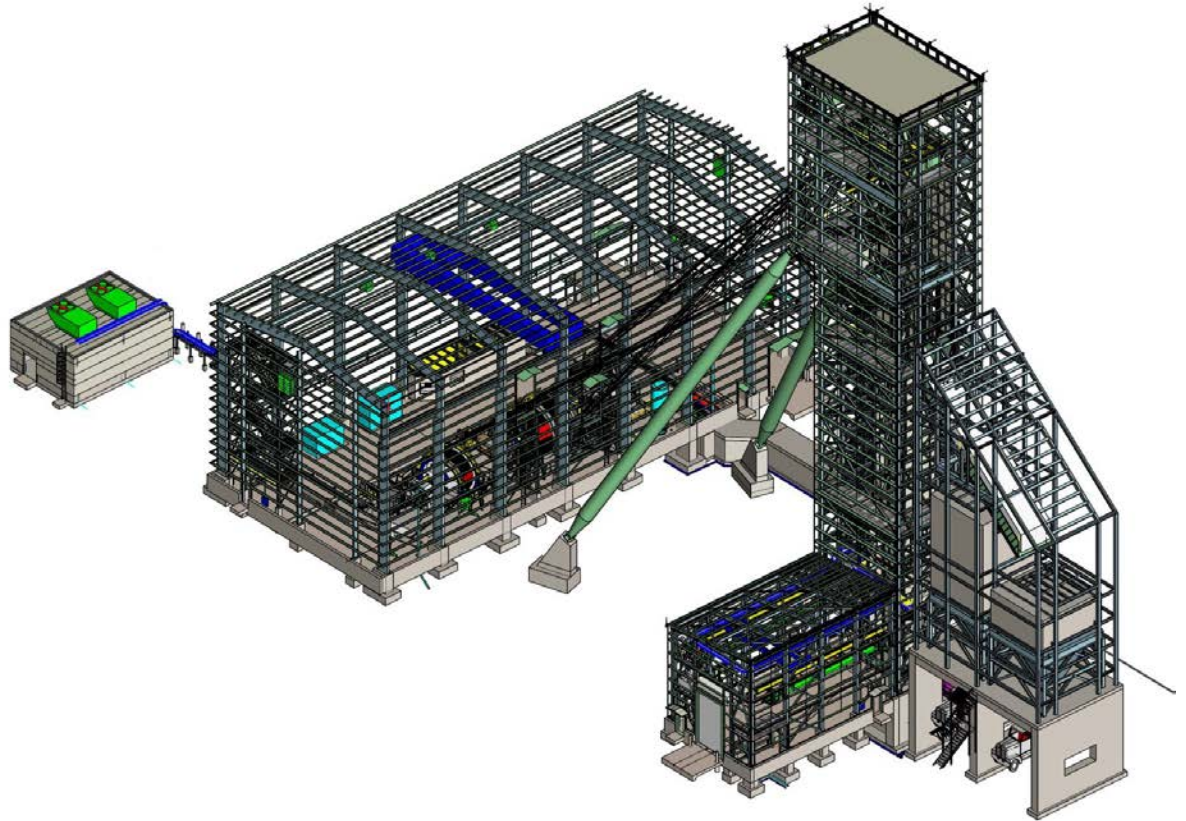


Figure 18-6 Headframe and Hoisting Plant Isometric

The concrete shaft sub collar will be primarily used for routing services in and out of the shaft, as well as for the ventilation plenum. The collar house will be constructed with metal cladding and will be insulated. It will be located on the cage side of the shaft and will be the primary access point for personnel and material.

18.3.1.2 Shaft and Shaft Stations

The shaft layout is shown in Figure 18-7. The shaft will be concrete lined, measuring 5 m internal finished diameter. The shaft size has been determined based on several parameters including ore and waste production requirements, intake ventilation, installed permanent services, service hoisting and constructability during shaft sinking.

The shaft will have three shaft stations (840 Level, 1095 Level, and 1305 Level) as well as a loading pocket on the 1350 Level and ramp access to the shaft bottom on 1380 Level. Each shaft station will be utilized to access the main mine horizons and coincide with the main exploration drifts underground. Figure 18-8 displays a shaft riser diagram.

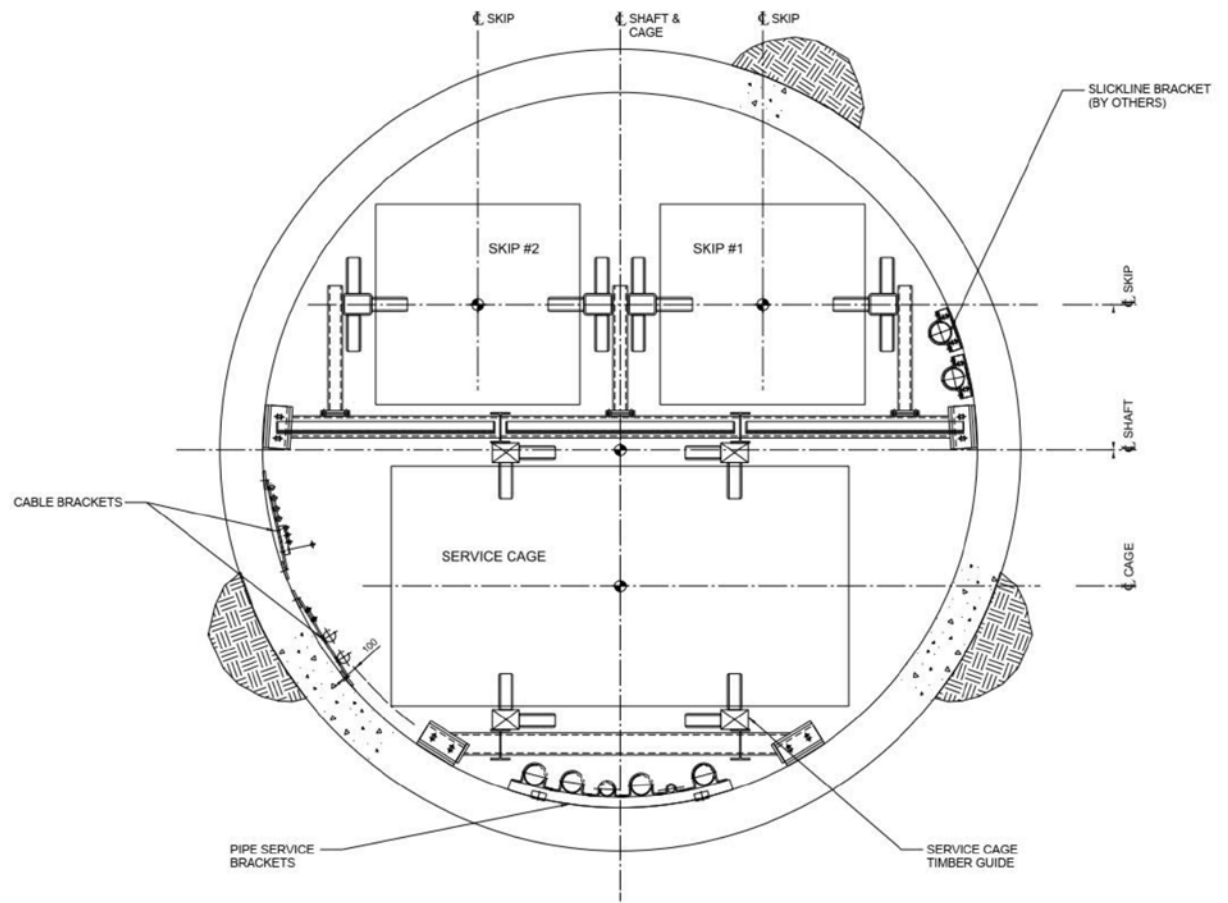


Figure 18-7 Production Configuration Shaft Cross Section

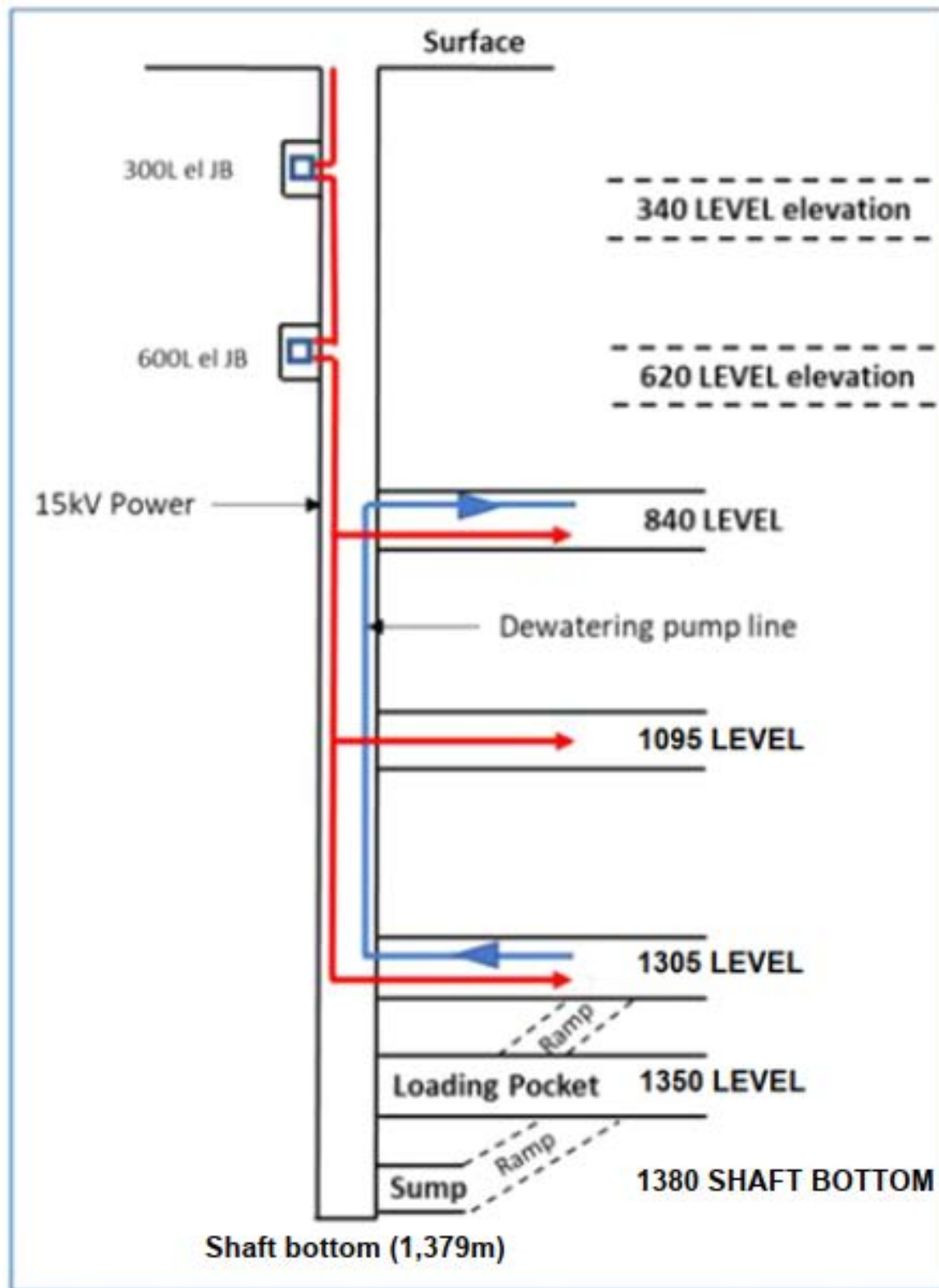


Figure 18-8 Shaft Riser Diagram

Note:

- (Note: Shaft Bottom 1379 m and 1380 Level are Same Elevation)

All shaft stations will have an electrical substation as well as other main mine services. The shaft will have a single loading pocket located on the 1350 Level and the shaft bottom will be accessed via a ramp.

The initial sinking depth will be set at 1,380 m from the collar, with the loading pocket set at the 1350 Level. It is important to note that Island Gold is not limited to this initial sinking depth and has the option of extending the shaft bottom to as deep as 2,000 m, with the possibility of repositioning the loading pocket/bin arrangement to a deeper level, should the orebody prove out at depth (beyond the 1,500 m elevation).

18.3.1.3 Ore & Waste Handling System

The underground ore and waste handling system is designed to accommodate 2,400 tpd of ore and an additional 1,500 tpd of waste for any future mine expansion. This system is designed to transport the mine ore and waste rock from all underground mining zones to one common rock breaking station. In general, the material will be moved through the respective passes with dump access and finger raises (intermediate station) to the rock breaker station.

The underground ore and waste handling system will be composed of one loading pocket with two underground bins (ore and waste) and three primary dump points in ore and waste passes. The ore and waste passes will be excavated between the main exploration levels. Figure 18-9 depicts the process flow diagram for the system.

The ore and waste handling system will be sized for hoisting at the 1305 Level rock breaker station. This will be fed directly by the ore pass system above the level and will be located above the bin. The bin collar will be on the 1305 Level and the bin bottom on the 1350 Level. The rock breaker station will consist of a recessed grizzly, scalping bars, rock breaker and picking arm, with a feed chute complete with press frame and control chains (Figure 18-10).

The material sized through the rock breaker station will be dropped into the underground bins to then discharge onto the 1350 Level apron feeder. This will then discharge onto the loadout conveyor and subsequently to the loading pocket. The grizzly will be sized to 400 mm x 400 mm.

The loading pocket will be a conventional design, like those in use at Alamos' other Ontario operation Young-Davidson. The conveyor will discharge to a transfer car to alternatively divert material to one of two measuring flasks. Each measuring box discharges into the parked skip in the shaft via dedicated discharge gates and arc gate controls.

Both the ore/waste bin and the loading pockets will have load cells and volumetric measurements to ensure optimal loading of skips that will be monitored remotely and locally.

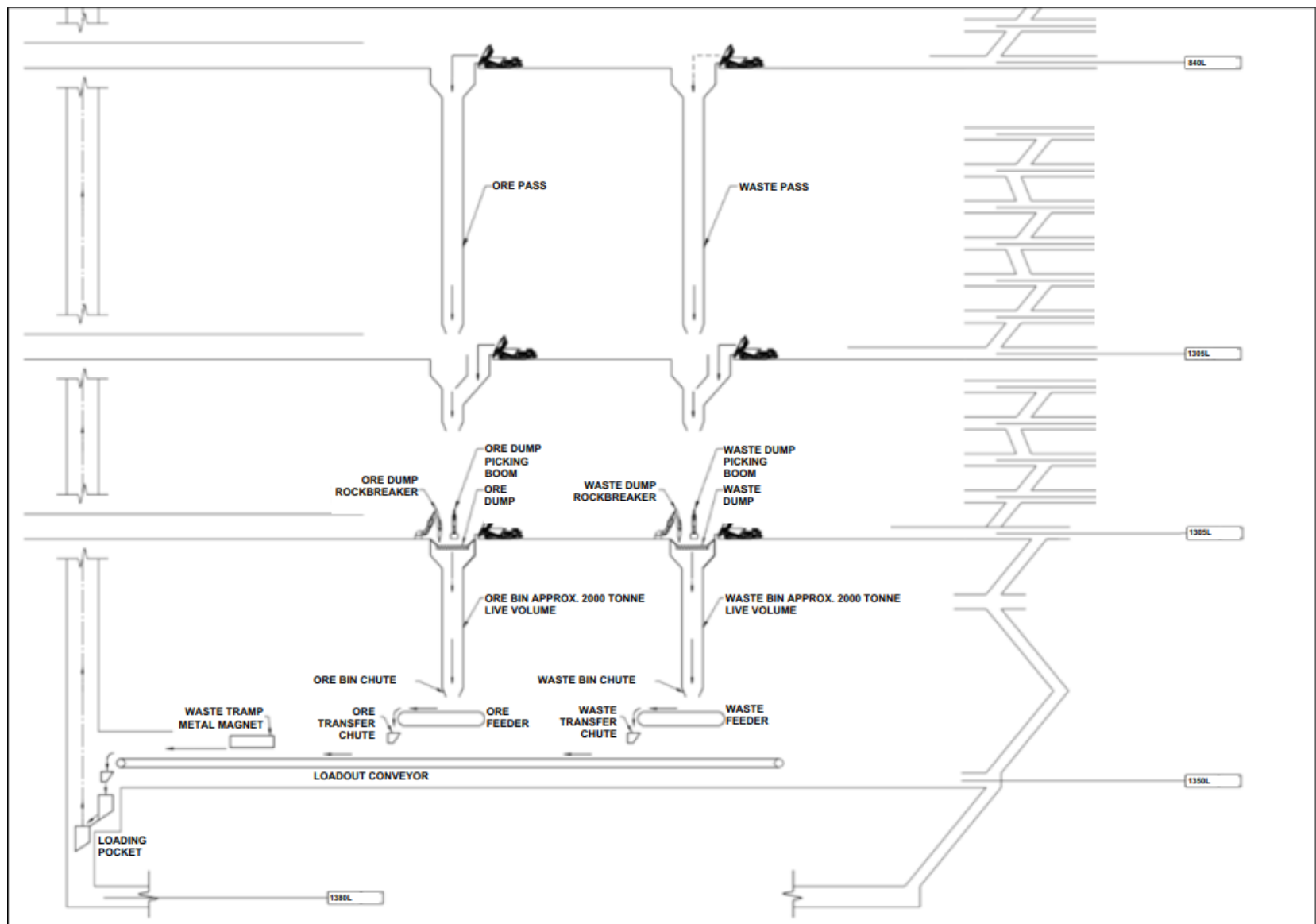


Figure 18-9 Underground Ore and Waste Handling System Process Flow Diagram

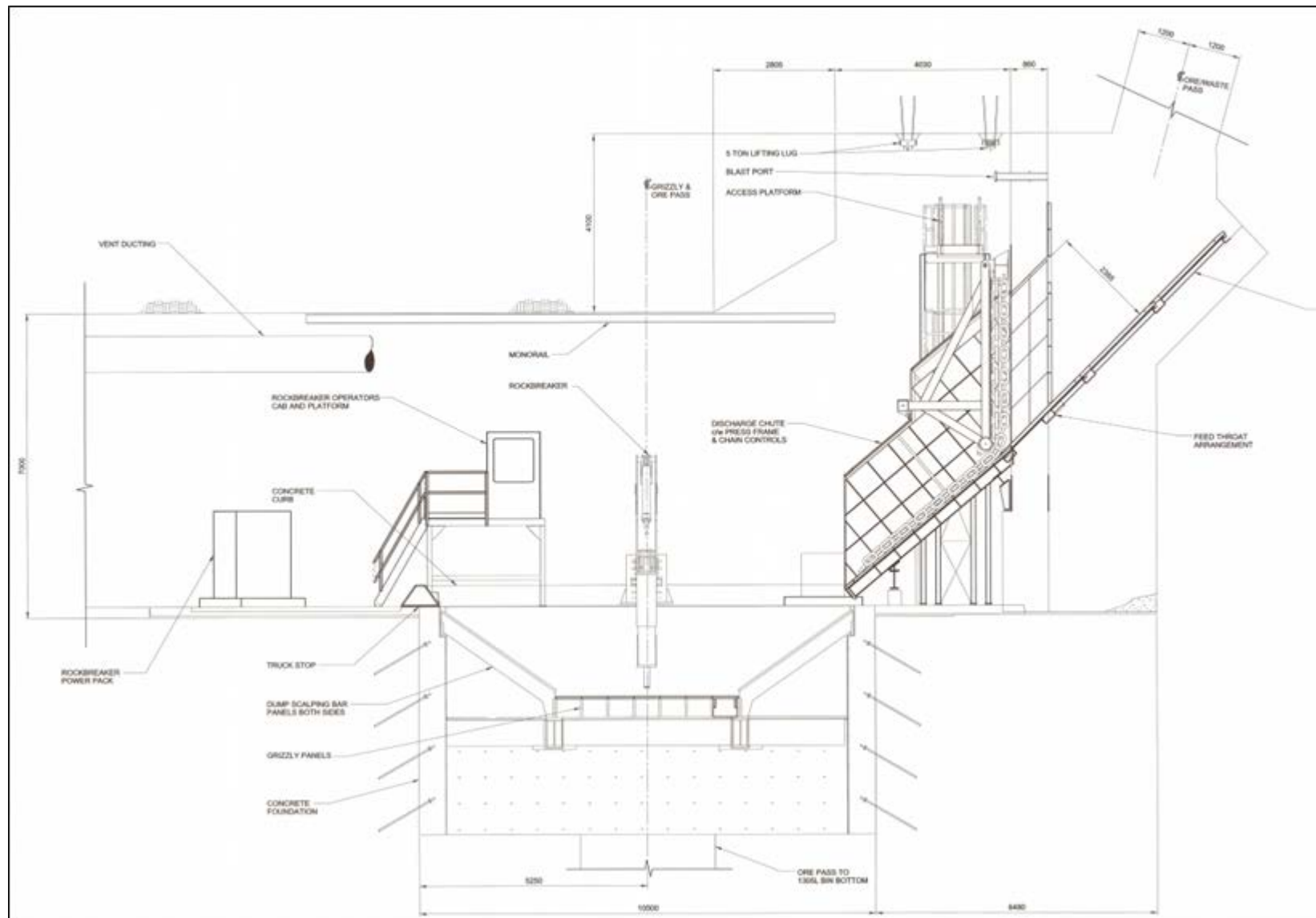


Figure 18-10 1305 Level Rock Breaker Station – Section View

18.3.1.4 *Electrical Upgrades*

A new 15 kV substation, as well as a pole line tying into the Algoma Power supply, will be installed and commissioned on the new shaft site. The substation will provide power to the surface infrastructure on the shaft site, as well as provide power underground to the deeper portion of the mine. The substation will be designed in such a way as to not only provide power to the deep portions of the mine, but also to support existing operations if the existing substation fails, by back feeding the underground circuit. This provides redundancy to the underground 15 kV supply, thereby de-risking the underground operation substantially.

An emergency diesel generator will also be installed to supply power to the service hoist for emergency back up power, should the mine site lose power and need to power up the hoist to bring the cage to surface.

18.3.1.5 *Ventilation Upgrades*

The shaft will also be a new source of fresh air for the deep portion of the mine. A plenum will be driven from surface to tie into the shaft just beneath the sub collar. Two axial fans in parallel will be used to supply approximately 400,000 cfm to the bottom of the shaft. A series of airlock doors and booster fans will be installed at each shaft station to direct the fresh air from the shaft to active mine workings. Direct fired propane burners will also be installed on the intake portion of the fans to heat the air during winter months.

A new series of internal return/fresh air raises will be driven to accommodate the new influx of fresh air from the shaft. These raises will tie together the main exploration drifts, reducing the air velocities in the ramp and mitigating dust related issues in the ramp. As part of this expansion additional exhaust fans will be installed on the west return air raises and the existing Fresh Air Raise #2 will be converted to an exhaust raise. This change facilitates the conversion of the ventilation system from a push system to a push-pull system, the main benefits of which will be the conversion of the existing portal from exhaust to a fresh air source allowing for more rapid clearing of blast gases and longer working shifts to be utilized. This change also drastically simplifies required ventilation controls in the proximity of the shaft greatly de-risking movement of personnel and material through the shaft infrastructure. Figure 18-11 **Error! Reference source not found.** displays the primary ventilation distribution system in its final state at the end of the S2400 mine life.

18.3.2 *Site Power Transmission Upgrade*

Island Gold's electrical power supply is currently provided by API's 44 kV powerline running from Hawk Junction to site. This line currently supplies power to Dubreuilville, as well as a limited amount to back country residential properties.

Island Gold is currently allocated approximately 12 MW and utilizes 9.5 MW on average per year. With the Phase 3+ Expansion, Island is expecting to require a peak 33.5 MW when production reaches 2,400 tpd. The existing distribution line does not have sufficient capacity to supply the additional electrical power required for the expanded project.

Island Gold has provided API with its forecasted load demand (peak/connected and average) and is in the process of discussing the power upgrade alternatives and execution schedule to meet expected power demand at site.

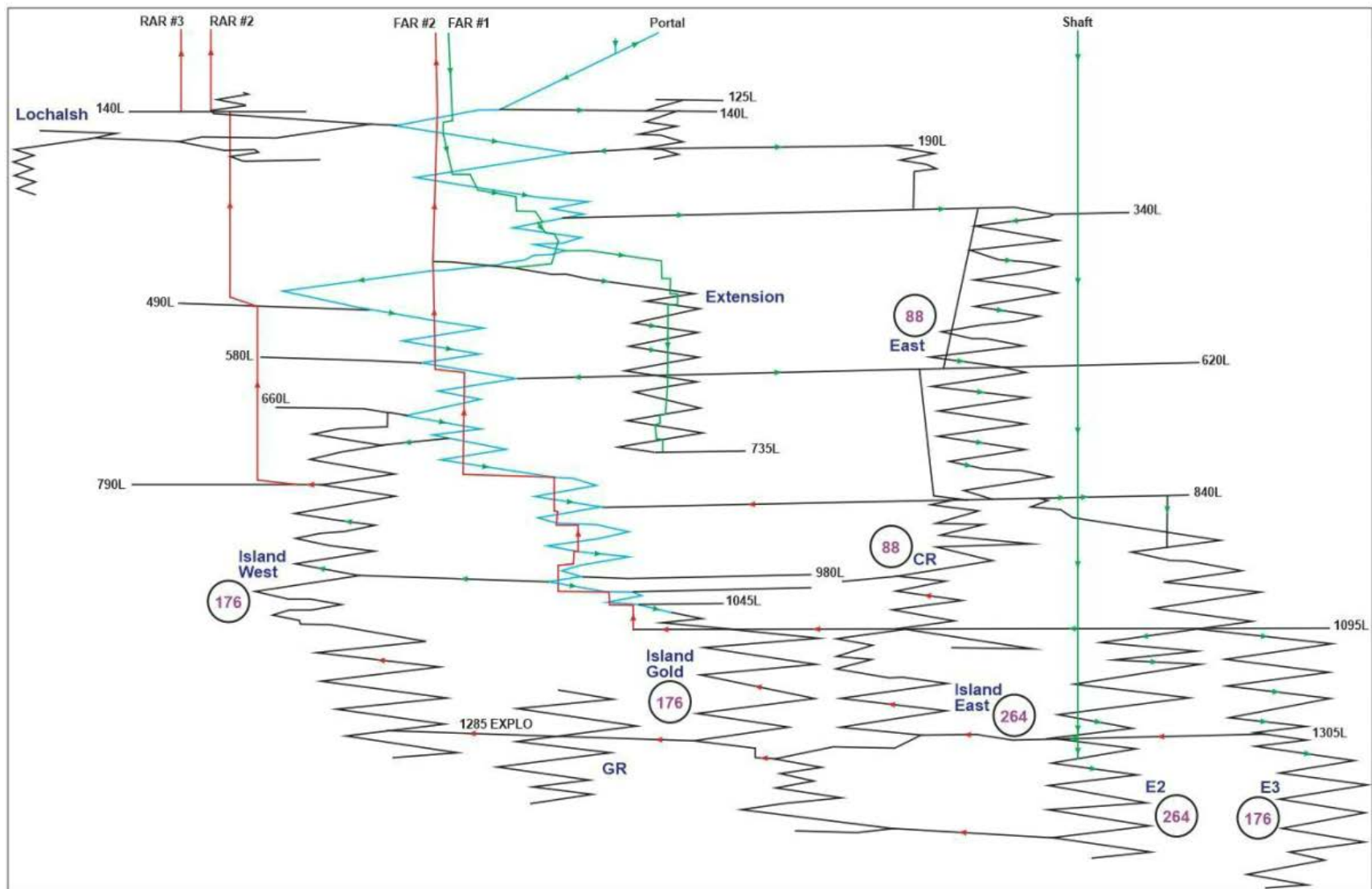


Figure 18-11 Schematic of LOM Flow Distribution and Flow Requirements per Mining Zone

18.3.3 Paste Backfill

Paste backfill is required for the mine expansion and increased mining rate. CIL tailings from the process plant will be dewatered and mixed with cement and binders to produce a paste in a new paste plant that will be located adjacent to the mine. Paste will be pumped to the stopes via a new underground distribution system (UDS).

18.3.3.1 Paste Test Work

Significant amount of paste test work has been completed on Island Gold ore by Golder and Paterson and Cooke. The most recent test work was completed by Paterson and Cooke, and pertinent results are summarized below and have been used as the basis for the design of the paste plant.

During 2021, three tailings samples representing typical ore blends that are treated at the existing process plant were tested at Paterson and Cooke's laboratory in Sudbury, Ontario.

Test work included:

- Material Characterisation: pH, Conductivity, Particle Size Distribution (PSD), Solids Density, Zeta Potential, Mineralogy, Water Analysis;
- Tailings Desliming Tests;
- Dewatering: Static Settling (flocculant screening and thickener feed dilution optimisation), High-Rate Batch Dynamic Thickening, Dynamic Batch Thickener Bed Consolidation, Vacuum Filtration;
- Rheology: Static Vane, Dynamic Rotational; and
- Unconfined Compressive Strength (UCS).

The material characteristics of the three samples are summarized in Table 18-3.

Table 18-3 Material Characteristics

Test	Material				
	Sample 1 (Hi/Lo Blend)	Sample 2 (Cooked Ore)	Sample 3 (Regular Blend)	50-50 Deslimed Regular Blend	75-25 Deslimed Regular Blend
pH	11.5	10.9	10.6	-	-
Conductivity (mS/cm)	1.1	1.0	3.4	-	-
Density (kg/m ³)	2,730	2,724	2,647	-	-
D ₉₀ (µm)	81	69	56	88	72
D ₅₀ (µm)	22	18	17	40	25
% <20 µm	47%	53%	56%	13%	42%
Zeta Potential (mV)	-6.3	-	-5.7	-	-
Mineralogy	Silicates: ~69% Clays: ~22%	Silicates: ~68% Clays: ~25%	Silicates: ~68% Clays: ~24%	-	-

The Particle size distributions (PSD) of the materials were determined for each sample by laser diffractions, using a Malvern Mastersizer 3000.

Figure 18-12 shows the PSD curves.

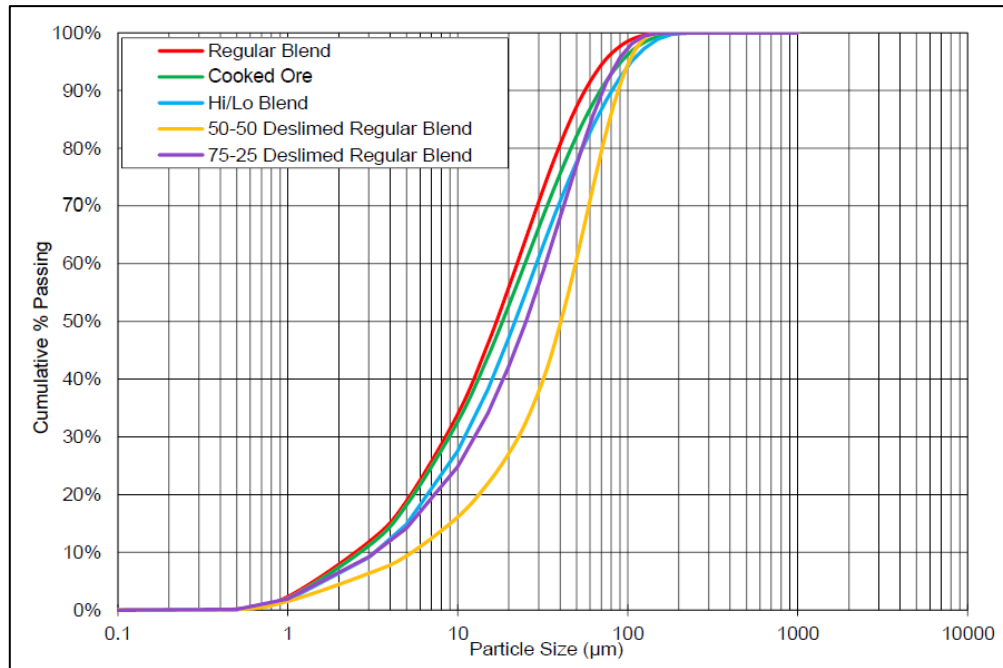


Figure 18-12 Particle Size Distribution Curves

Flocculant screening was completed using a suite of Nalco flocculants as Nalco's Optimer 83949 flocculant is already in use at the process plant. For all materials tested, the Optimer 83949 resulted in the fastest settling rates, while maintaining good underflow densities.

Following the static cylinder flocculant screening and optimal thickener feed concentration tests, batch dynamic thickening tests were performed. This test is a two-step approach with step one focused on the optimal flocculant dosage rate and step two focused on the optimal thickener solids loading rate. The flocculant dosage results showed an optimal dosage rate of between 25g/t and 30g/t for all tested materials. Optimal thickener solids loading rate tests were conducted using the optimal flocculant dosage rate.

Batch consolidation tests (24 hours without raking and then an additional 24 hours with raking) demonstrated that the thickener underflow mass concentration can be increased with between 3 and 5% points in a four-hour period.

Vacuum filtration test work was conducted on the Hi/Lo Blend, Regular Blend, 50-50 Deslimed Regular Blend and 75-25 Deslimed Regular Blend at feed solids concentration varying between 55% and 65% by weight and varying simulated filter cycle times. Table 18-4 shows the results obtained for the 90 second cycle time that is typically used for filter sizing calculations.

Table 18-4 Filtration Data used for Filter Sizing

Material	Feed @ ~55% m and 90 s Cycle Time		Feed @ ~60% m and 90 s Cycle Time		Feed @ ~65% m and 90 s Cycle Time	
	Effective Filtration Rate	Cake Thickness	Effective Filtration Rate	Cake Thickness	Effective Filtration Rate	Cake Thickness
	(kg/h)/ m^2	mm	(kg/h)/ m^2	mm	(kg/h)/ m^2	mm
Hi/Lo Blend	1,416	17.8	1,566	21.5	2,154	26.0
Regular Blend	1,134	16.9	1,246	18.6	1,806	26.9
50-50 Deslimed Regular Blend	2,137	30.5	2,470	34.9	3,442	46.5
75-25 Deslimed Regular Blend	1,149	18.0	1,484	40.4	2,767	38.4

Figure 18-13 shows the moisture content in the vacuum filter cake for the different materials at different feed concentrations.

Figure 18-13 Vacuum Filtration Cake Moistures

Static vane rheology test work was conducted for all materials with and without cement across a range of solids concentrations.

The rheological properties of the slurries were determined using rotational viscometer laminar flow data. The data was analysed by applying the Bingham plastic model which is a two-parameter model describing the slurry rheology. The two parameters are the Bingham plastic viscosity (KBP) and Bingham yield stress (τ_B).

The UCS test program was developed to evaluate the affect that varying binder concentrations have on the strength of the paste backfill over a range of cure times. The binder addition was varied to provide various water-to-binder (W:B) addition ranges across all mixes to enable interpolation of results. Figure 18-14 provides an overview of all the UCS results obtained during the test campaign.

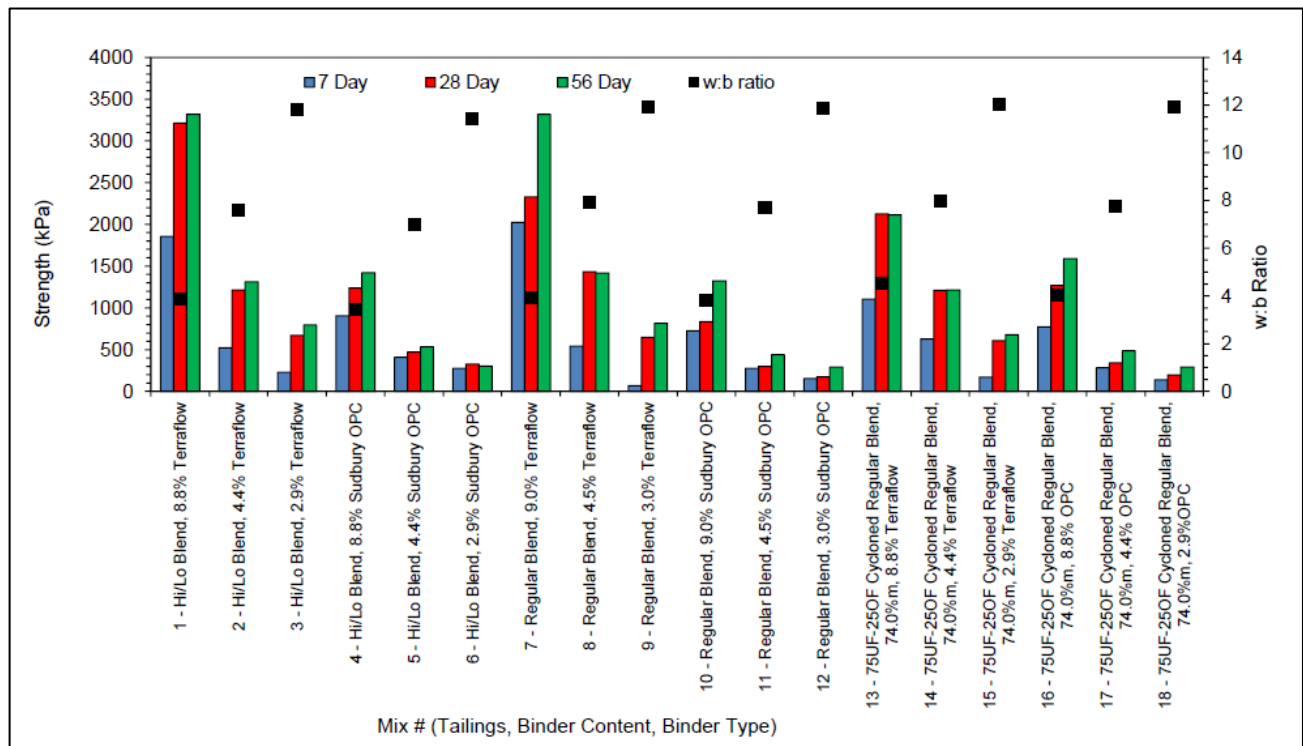


Figure 18-14 Static Yield Stress vs Mass Concentration

18.3.3.2 Paste Plant Process Description

The paste plant has been designed to utilize the full instantaneous nominal tailings available at 2,400 t/d with a maximum instantaneous throughput of 2,600 t/d. This equates to a flow rate range of 70 to 90 m³/h using whole tailings after year 2026. The design availability of the paste plant is 92% and is expected to have a maximum utilization of 60%.

Treated CIL slurry tailings from the new paste plant feed pump box located near the new cyanide destruction circuit at the process plant will be pumped to an agitated paste plant feed tank located at the new paste plant via a new overland pipeline. The slurry tailings will then be pumped to the thickener feed box before thickening in an 18 m diameter high-rate thickener.

Thickener underflow will be pumped to the filter feed tank at approximately 60% solids and then pumped to the conventional vacuum disc filters for further dewatering. Thickener overflow will flow by gravity to a process water tank at the paste plant.

There will be two vacuum disc filters at the paste plant; one duty and one standby. The vacuum disc filters will further dewater the thickened slurry tailings to produce a filter cake with a maximum moisture content of 25.5% moisture by weight. The filter cake will then be conveyed to a continuous paste mixer and the filtrate will be pumped to the tailings thickener feed box.

Cement and binder from the binder silo will be added together with process water and the filter cake in the paste mixer at predetermined recipes based on requirements for the underground mine. Paste is then discharged to the paste hopper. A positive displacement paste pump is provided to pump the paste underground via the underground distribution system (UDS). A gravity system is provided as a backup to the paste pump.

A dust collector will be provided at the paste mixer to collect any dust that will be reused in the process.

The paste plant will include a 600-tonne silo for the binder and a flocculant mix and storage facility.

Excess process water and any used slurry tailings from the paste plant will be collected in an agitated return tank and then pumped to the primary tailings pond via an overland pipeline.

Process water, gland seal water, potable water and fire water systems will be provided at the new paste plant. A dedicated air compressor system will be provided for plant air users and instrument air.

To ensure paste quality and strengths are consistent and meet the required strength targets, quality assurance and control functions will include periodic slump, filter cake moisture content and UCS cylinders testing at set intervals.

A general arrangement drawing of the new paste plant is provided in Figure 18-15 for reference.

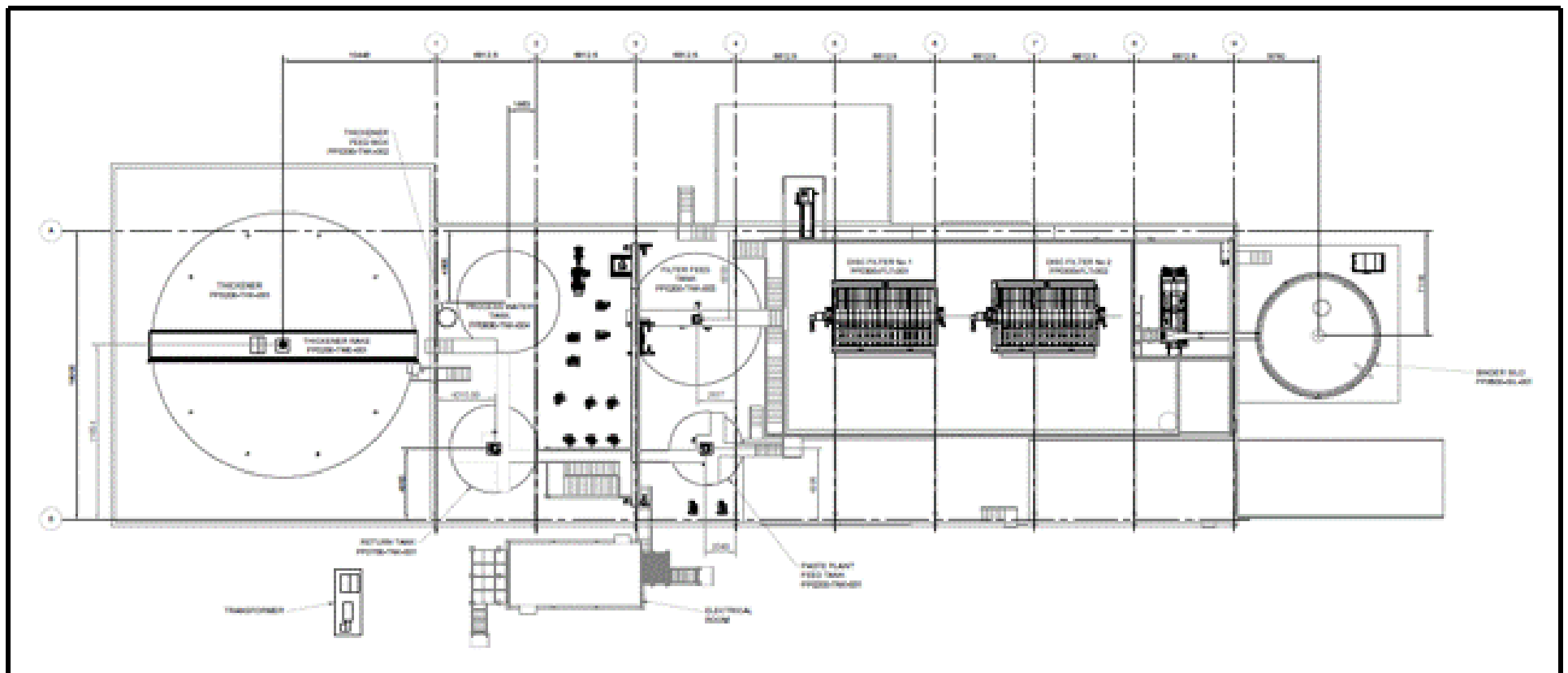


Figure 18-15 Paste Plant General Arrangement Overview

18.3.3.3 Paste Plant Location

The new paste plant will be located near the northwest corner of the new haul road intersection at Goudreau Road as shown in Figure 18-16.

The key elements for selecting this site included the proximity to an existing electrical supply, the surface paste boreholes will be directly adjacent to the paste plant, ease of access for cement and binder supply trucks, and suitable ground conditions (geotechnical) for construction.

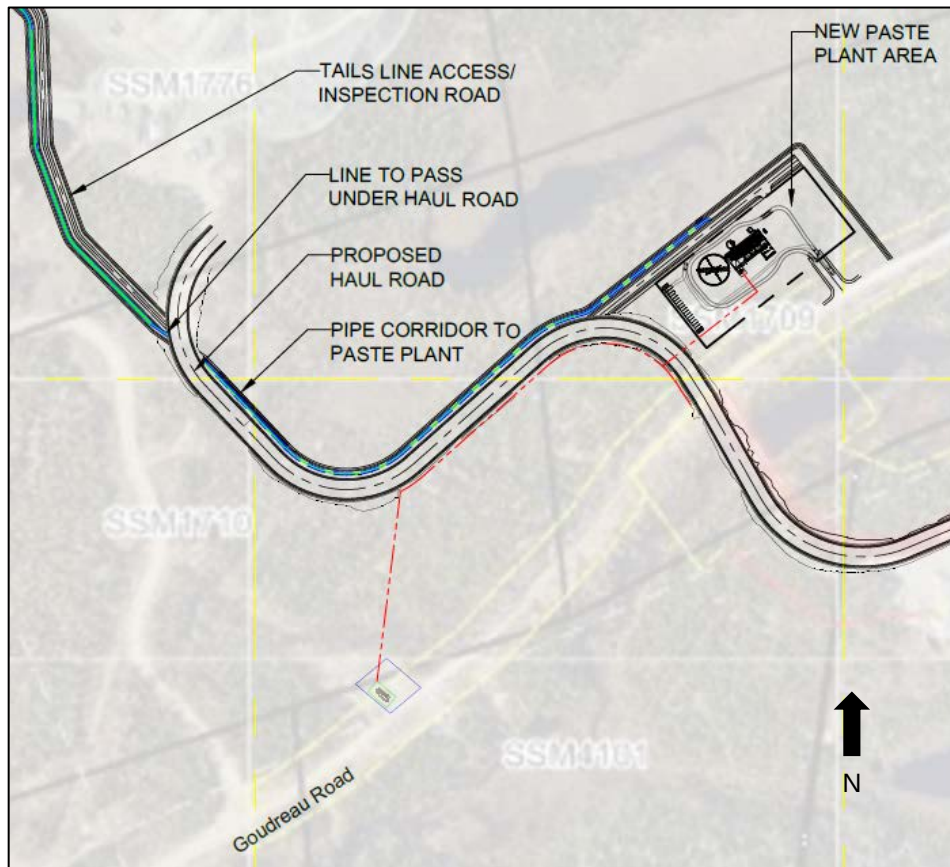


Figure 18-16 Paste Plant Location

(Alamos 2022)

18.3.3.4 Underground Distribution System (UDS)

A preliminary UDS has been designed to accommodate future mining needs.

The paste will flow through the underground distribution piping network at a flow rate dependent on the paste plant production rate. The UDS design will include pressure instrumentation throughout the pipeline to allow for real-time monitoring and observation of the pour behaviour and for troubleshooting in the case of upsets. Cameras will be provided for monitoring the pour discharge at the stope and other key locations. Emergency dump clean-out tees will be provided at critical locations, which allow for remote pressure release of the pipeline in the case of blockage. Manual valves near the discharge to the stopes will be used to divert water during a flush for small stopes. Rupture disks will be provided at the base of every borehole to prevent over pressurizing the pipeline.

Paste backfill will be pumped underground from boreholes drilled adjacent the paste plant with an angle of approximately 60° from horizontal. Provision have been made for a total of four surface boreholes (initial one duty and one stand-by with provision for an additional two if required in future). Two holes will be drilled and cased with unlined carbon steel pipe, one for redundancy to allow for paste fill operations to continue in the event one of the surface boreholes becomes blocked. Additionally, if the paste pump fails, the system can resume operations in a limited capacity as a gravity-fed system only as an emergency back-up while the paste pump is being serviced.

The surface boreholes will breakthrough into a dedicated paste fill cut-out within a new development of approximately 430 m from the existing underground drift on the 340 Level. From there, paste fill piping will be routed into the new mining areas, with multiple boreholes that will be drilled with approximately 60 to 70° angle from horizontal. This angular range allows the paste to slide down the footwall of the hole and absorbs some of the energy while avoiding a free fall drop that may be common with vertical holes. Free falling 'batches' of paste tend to act as a piston and compress the air ahead which creates a pressurized local area that can lead to blowback, drying out, and potentially segregation of the paste.

The boreholes will branch off main levels (340, 620, 840 and 1090 Levels) to the required filling areas. Mining and paste filling will advance from multiple ramps, resulting in several borehole systems running through different areas of the same levels.

Underground, distribution of the paste to the various working areas will be via inter-level boreholes and manual switchovers from the main trunk lines to the stope piping. Automated diverter valves will be added in areas where frequent pipe route changes are required, to reduce labour and time requirements.

The pipeline routes and deposition points analyzed are shown in

Figure 18-17

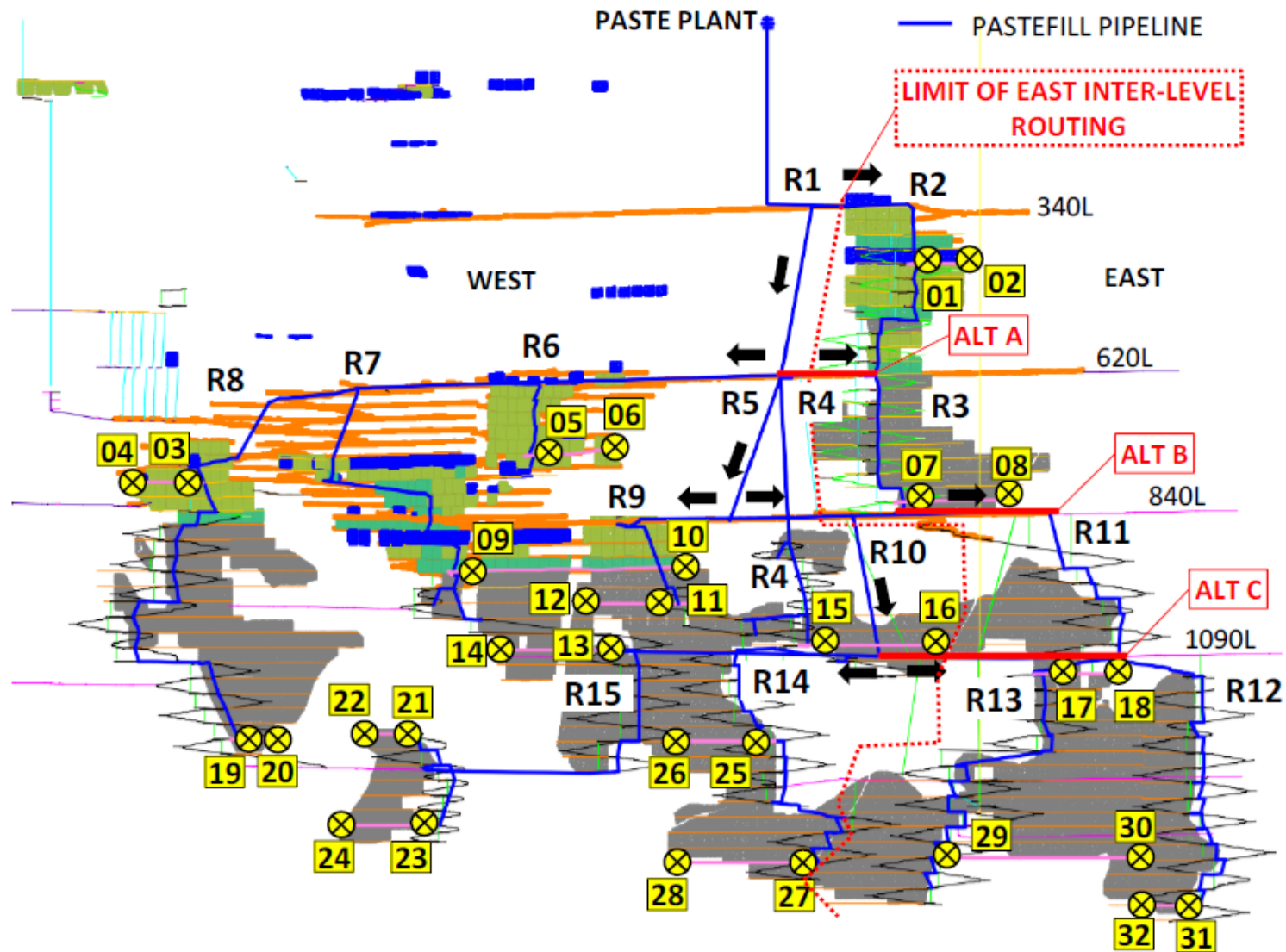


Figure 18-17 Underground Distribution System (Long Section View)

18.3.4 Tailings Management under Phase 3+ Expansion

For the purposes of the study, it was assumed that the earliest the cemented paste backfill plant would start operation would be at the beginning of 2026. Therefore, with respect to Primary Pond construction, it was assumed that dyke raises to elevation 430 m will be required in 2024, a dyke raise to elevation 432 m will be required in 2029, and a dyke raise to elevation 433 would be required in 2034.

No changes to the Secondary Pond capacity and general water management strategy were considered with respect to its use as a secondary clarifier of tailings pond water prior to discharge to the environment.

A summary of the dyke raising requirements within the Primary Pond are provided in Table 18-5.

For the conventional disposal with cemented paste backfill option, only three additional dam raises above elevation 427 m would be required (Stages 5, 6a, and 6b) to an ultimate crest elevation of 433 m to store the projected life of mine tailings. Figure 18-18 provides the ultimate configuration of the Primary Pond TSF. Dykes 1, 3 and 4 would be raised via the downstream method of construction, while Dykes 2 and 5 would be raised via the centre-line method. In general, it was determined that this option would be the simplest of the technologies to implement and requires the least intensive effort from an operational oversight perspective. Raising of Dykes 2 and 5 by the centre-line method would also save on some earthworks quantities as compared with downstream raising, while satisfying stability requirements.

The current dam footprint is sufficient for construction of additional lifts beyond the 6b lift should additional Mineral Reserves and Resources be delineated and brought into the mine plan.

Table 18-5 TSF Dam Construction Timelines and Storage Capacity Summary

Construction Stage	Maximum Dam Crest Elevation (m)	Storage Capacity for Tailings above Elev. 424 m (M-m ³)	Dyke Raise Timeline
Stage 4	427	1.47	Completed in 2020
Stage 5	430	3.12	2024
Stage 6a	432	4.12	2029
Stage 6b	433	4.62	2034
Stage 7a	435	5.62	N/A
Stage 7b	436.5	6.32	N/A

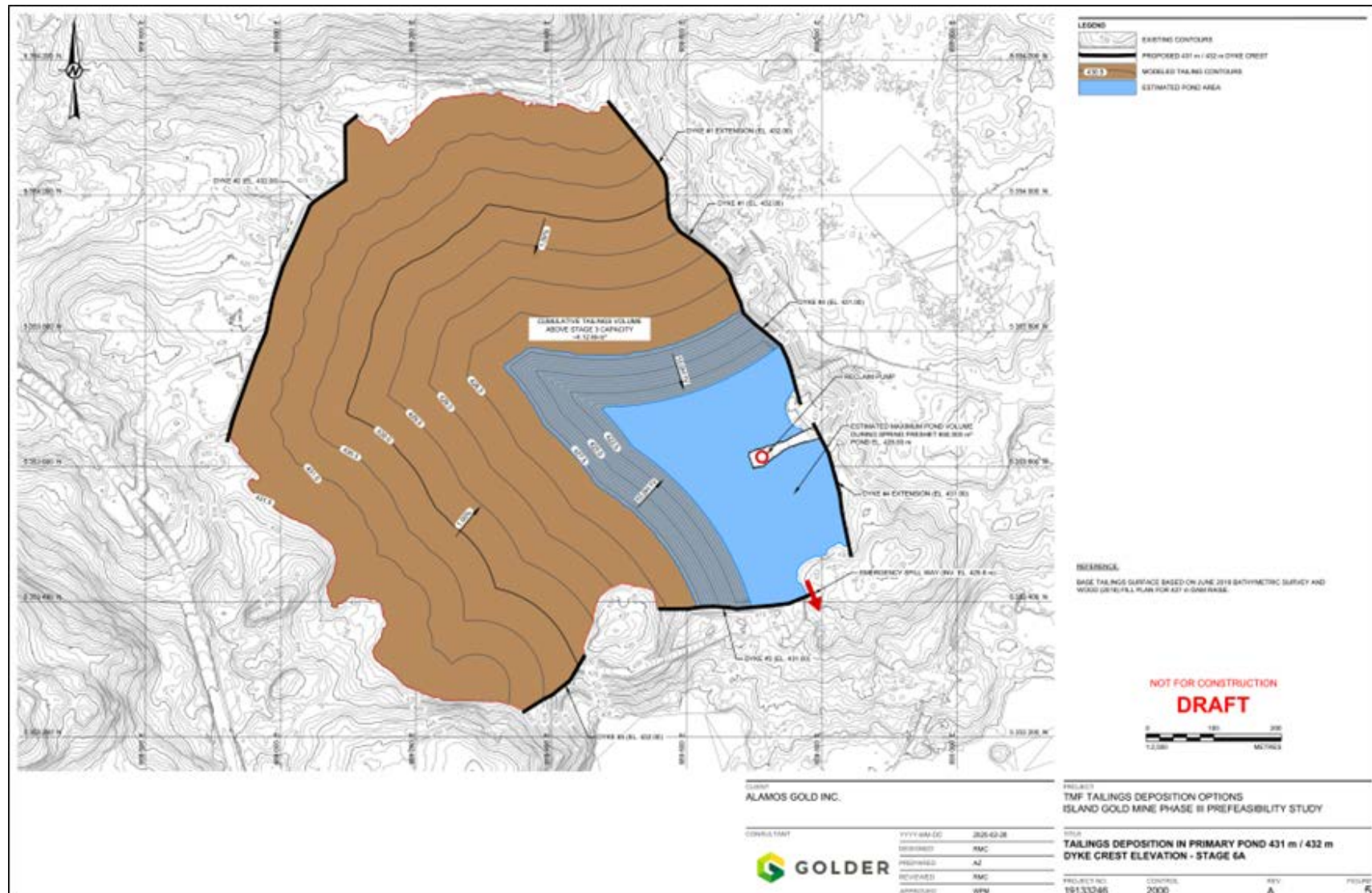


Figure 18-18 Ultimate Configuration of the Primary Pond TSF
(Golder 2020)

18.3.5 Water Treatment

The Island Gold Mine has limited options for storing mine contact water upstream of the water treatment plant. The treatment plant needs to have an availability of 95%.

There are two sources of water requiring treatment:

- Underground mine water; and
- Excess water from the Primary Pond.

Underground mine water will be pumped to the Primary Pond. The flow rate from the underground mine was set to 2,200 m³/day over the mine life by Alamos and Golder. The Primary Pond also gains water from precipitation and run on.

Water from the Primary Pond will be treated year round as required. The treatment system will be sized to manage the annual excess water volume accumulating in the Primary Pond during a 1 in 25 wet year. The design influent rate from Primary Pond is 3,800 m³/day. The nominal flow rate from Primary Pond is 3,300 m³/day. The flow sheet is depicted in Figure 18-19.

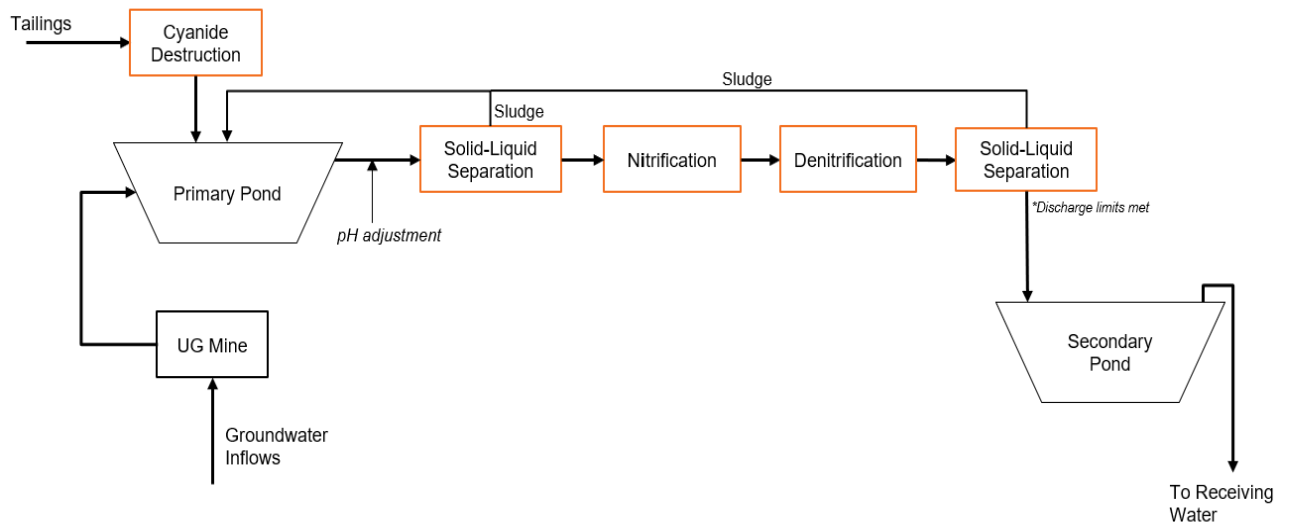


Figure 18-19 Treatment Plant Conceptual Process Flow Diagram

18.3.5.1 Effluent Design Basis

Island Gold will be amending its Environmental Compliance Approval permit. Treated water will be discharged from the Water Treatment Plant to the Secondary Pond before discharge to the environment. An Assimilative Capacity Study as well as the test work that will be conducted will inform the revised effluent limits.

19 MARKET STUDIES AND CONTRACTS

19.1 Market studies

No market studies were conducted by Alamos. Gold is a freely traded commodity on the world market for which there is a steady demand from numerous buyers.

19.2 Metal Pricing

- US \$1,800/oz gold, US \$22.00/oz silver and a \$0.79 USD/CAD were used for revenue for 2022;
- US \$1,650/oz gold, US \$18.00/oz silver and a \$0.78 USD/CAD were used for revenue for 2023 onwards;

20 ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

From exploration to operations to closure, one of the goals at the Island Gold Mine is to safeguard the environment, educate its employees and the communities about the mine's environmental programs and commitments, and apply best management practices to prevent or mitigate any potential environmental impacts. The operations at Island Gold use a range of materials and consumables that includes explosives, chemicals, and fuels.

This section will provide a description of the environmental, permitting, social and community, and sustainability components relating to the Island Gold Mine operations, and will also cover mine closure and reclamation at Island Gold.

20.1 Existing Conditions

20.1.1 Past and Current Land Use

The Goudreau area has a history of mining dating back to the discovery of gold in the early 1900s. Several small gold mines and open pit pyrite mines have been active in the area in the past. Between the late 1920s and early 1940s, the area was also subjected to intensive prospecting for gold.

Current land use in the Island Gold Mine area consists primarily of forestry operations, mining and exploration activities, tourism, and recreation. The project area is occupied and surrounded by historically and periodically harvested forest lands.

Exploration programs have periodically occurred throughout the region's past to support potential future mining operations.

Recreational activity consists primarily of fishing, hunting, and snowmobiling. A hunting/fishing camp, for summer cottagers, is in the Lochalsh town site, approximately 15 km northeast of the project site. Summer cottage homes also exist approximately 8 km west of the mine site, near the old town of Goudreau.

20.1.2 Topography and Soils

The site is located within a physiographical region described as a bedrock-drift complex. Regional topography is bedrock-controlled and is characterized by a sequence of east-northeast trending rounded hills and ridges. The valleys and low-lying areas between ridges are generally characterized by the presence of interconnected wetlands, streams, and lakes (such as Maskinonge Lake and Goudreau Lake). Site relief is generally low, with low to moderate surface slopes and elevation differentials typically in the range of 5 m to 10 m. The site's highest elevations are encountered north of the primary pond area (approximately 470 masl) and the site's lowest elevation is at Goudreau Lake (381 masl).

Overburden soils are relatively uniform with types generally being topsoil, sand, and till. These overlay bedrock at greater depths.

20.1.3 Climate

The Wawa region climate is humid continental. Temperature extremes are moderated, and precipitation patterns are altered by its proximity to Lake Superior. In 2019 the average annual temperature was 1.1°C with a monthly average minimum temperature of -16.5°C recorded in

January and a monthly average maximum temperature of 15.2°C recorded in July, respectively. Daily extreme temperatures of -40.8°C and 27.4°C were recorded at Environment Canada's Wawa A station (ID 6059413).

Average total annual precipitation is estimated around 980 mm, of which approximately 70% falls as rain and 30% as snow. Total average evaporation is projected to be approximately 520 mm. The precipitation and evaporation vary year-over-year and can fluctuate based on numerous factors. Nearly half of the annual snowfall occurs in December and January, while maximum rainfall occurs from June to October. Snow is generally present on the ground from November to April. Ice on the Primary and Secondary Ponds do not completely dissipate until Mid-May.

20.1.4 Water

The Island Gold Mine is located within the Maskinonge Lake and Goudreau Lake sub-watersheds (total area of 48.2 km²), approximately 40 km south of the Arctic drainage divide. Both sub-watersheds are part of the Michipicoten-Maggie watershed and Lake Superior Drainage Basin. Surface water drainage at the site is bedrock-controlled, generally flowing from northeast to southwest within the valleys between the elongated hills and ridges.

The Maskinonge Lake catchment covers the northeastern part of the mine site. Drainage from the northeast part of the mine site reports to this catchment, including drainage north of the mill and the wetland area to the northeast of the primary pond. The water flows in a southerly direction through the upper stretches of Goudreau Creek via meandering stream and wetland, eventually reporting to Bearpaw Lake and, ultimately, to Goudreau Lake. The Goudreau Lake catchment covers the entire site, including both the tailings management facility and Maskinonge Lake catchments, and drainage areas to the east of the site. Goudreau Lake outflows to a creek in a southerly direction towards the Michipicoten river and ultimately discharges to Lake Superior.

Water depths in Goudreau Lake vary substantially. A 2019 bathymetry was conducted on Goudreau Lake. Goudreau Lake can be divided into two distinct sections based on its bathymetry; an upstream basin, where depths between 14 m and 23 m are observed in deeper pool sections, and a second, much shallower basin, where depths do not exceed 2 m. The second basin is connected to the lake discharge and to the first deeper basin by a narrow and shallow corridor.

Water quality studies have been conducted since 1985 and they show some alteration in water quality in Goudreau Lake because of the impact of two historical mining operations in the area. Historically, sampling at stations upstream of Goudreau Lake (i.e., Maskinonge Lake, Miller Lake and Goudreau Creek) showed lower background levels of pH, conductivity, and alkalinity similar to the Goudreau lake downstream stations. Additionally, the former Magino Gold mine discharged tailings directly to the west of the upper basin of Goudreau Lake during the middle to late 1930s. Sediments in this portion of the lake are composed of natural sediments and historic tailings. Despite these increased loadings, data has indicated that the water quality in the upper basin of Goudreau Lake was quite good. Historic tailings have been also deposited into Pine Lake from the operations of the Edwards Mine. Drainage from historic abandoned iron mines which flows from Teare lake into Goudreau lake impacts the downstream portion of Goudreau Lake.

The mine is required to have an on-going water balance model which tracks all inputs and outputs out of the system monthly.

Final treated effluent from the mine flows into the upper portion of Goudreau Lake via Goudreau Creek. Fresh water is taken from Maskinonge Lake, which is then treated via a domestic water

treatment plant (WTP) to provide water for the mine and makeup water for processing. Water for mill use is reclaimed from the Primary Pond of the Tailings Management Facility. Water in the mine is currently reused, any excess is pumped, via a multi-stage pumping system (comprised of sumps/pumps at various levels in the mine) to the Primary Pond on the surface. Underground water from the Lochalsh workings is pumped to either the Primary or Secondary Pond. A visual representation of the site's hydrologic system site is shown in Figure 20-1.

Tailings, water management and final effluent monitoring and quality requirements are regulated under an amended ECA (No. 544-BNPL46) which was issued in April 2020. This ECA also allows for a mill production rate of up to 38,480 tonnes per month.

Additional monthly surface water quality monitoring is conducted by Island Gold at three locations in Goudreau Lake (the receiving water body), one on Maskinonge Lake and one on Pine Lake. Both Maskinonge Lake and the upper basins of Goudreau Lake would be characterized as meeting provincial objectives. The majority of metal concentrations were below their respective Provincial Water Quality Objectives (PWQO). Annual results have been comparable from 2007 to 2022.

In addition to the monitoring completed in conjunction with Environmental Compliance Approval (ECA) requirements, the site is subject to the Federal *Metal and Diamond Mining Effluent Regulations (MDMER)*. As required under MDMER, Environmental Effects Monitoring (EEM), studies started in 2005 and have continued since then, with the most recent field program planned to occur for the Cycle 6 EEM Study in the fall of 2022 (final report is expected in late 2023).

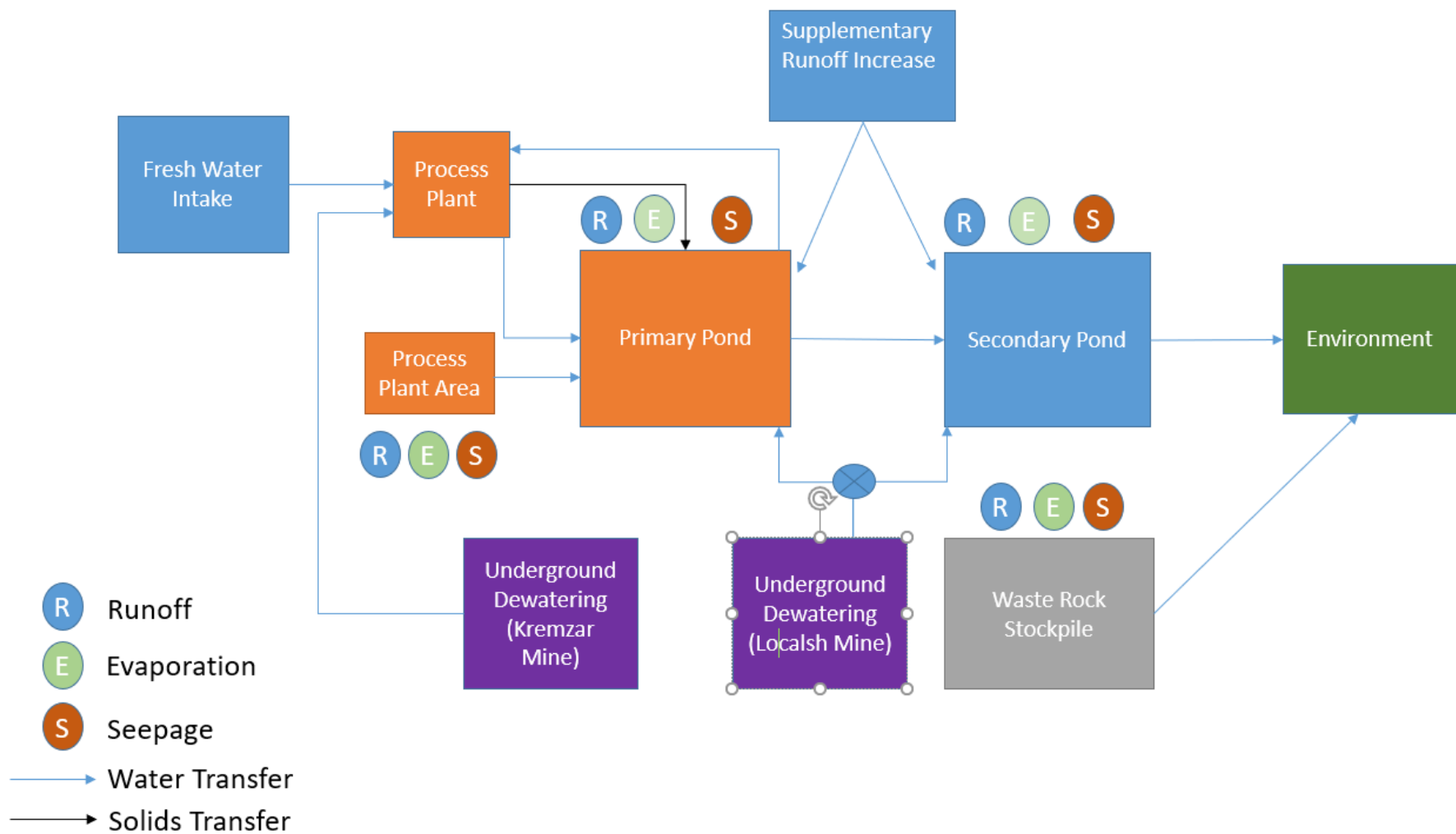


Figure 20-1 Island Gold Hydrologic System

20.1.5 Groundwater

The site is located within a bedrock-drift complex physiographical region characterized by thin overburden or exposed bedrock in the highland areas and waterlogged wetlands in low-lying areas. There are no known or potential groundwater users within several kilometres of the site. There is no current usage of groundwater resources on site.

A hydrogeological study of the Island Gold Mine site was conducted by Exp Services Inc. in 2013 with additional monitoring wells drilled in 2017. This included installation of ten groundwater monitoring wells across the site. Depth to groundwater ranged from 1.32 m to 11.7 m below ground surface. Regional groundwater flow direction in both overburden and bedrock is southward toward Goudreau Lake.

A groundwater monitoring program has been in place since 2013 with regular monitoring of groundwater levels and quality. Additional wells were installed in 2022. Samples from the groundwater wells have been tested for various parameters including metals, cyanide, hydrocarbons, and anions, with no exceedances of the Ontario drinking water quality guidelines.

20.1.6 Air

Air and noise discharges are regulated under an amended ECA, No. 1821 BAWLAC which was issued in May 2019 to the Island Gold Mine, allowing for an annual ore processing rate of 461,760 tonnes per year.

The ECA requires that the Island Gold Mine be in compliance with Ontario Regulation 419/05, applicable MECP Guidelines for Air and Noise, and other performance requirements as specified in their conditions. It allows modifications such as process changes, de-bottlenecking, or addition of new equipment subject to limits on operational flexibility.

Emission Summary and Dispersion Modelling (ESDM) reports have been prepared in accordance with Section 26 of Ontario Reg. 419/05 in 2009 by Blue Heron Environmental ("Blue Heron") in 2014 and annually thereafter by WSP Golder Associates Ltd ("Golder"). These ESDM reports were prepared to support applications for the Air ECA and to demonstrate ongoing compliance. ESDM considers the potential contaminants from the various air emission sources generated at site, modelled downstream effects along the Island Gold property boundary, and is updated annually to include additions/deletions of equipment across the site. The potential contaminants included ammonia, carbon monoxide (CO), copper, lead, nickel, nitrogen oxides (NO_x), sulphur dioxides (SO_x) and total suspended particulates (TSP). All modelled potential contaminants were compared against MECP criteria, with all below their respective limits.

The Province of Ontario and Federal government have released different GHG management programs; Ontario with the Emission Performance Standards and the Federal government with the Output Based Pricing system (OBPS) with the goal to encourage industries to transition from high-intensity to low-intensity GHG emissions generation. Alamos has registered with both the OBPS and EPS systems, and in 2021 Island Gold mine generated 13,083.6 tonnes CO_{2e}. Alamos as a corporation intends to reduce GHG Emissions by 30% by 2030.

20.1.7 Terrestrial Plant and Animal Life

The area is largely composed of trembling aspen, white birch, balsam poplar, black spruce, white spruce, balsam fir, and jack pine.

Wildlife populations in the area are regionally typical with the noted presence of moose, wolves, foxes, black bears, beavers, otters, muskrats, mink, snowshoe hares and red squirrels.

Habitats are generally favourable to moose because of past and ongoing forestry operations. Local moose populations are subject to considerable hunting pressure because of easy road access. Black bears and moose are prominent in the mine area and are sighted regularly. The beaver, otter and mink that inhabit the local area are the focus of trapping activity. Other organisms like owls have been reported.

Tree clearing was conducted in 2020 and 2022 for the Phase 3+ Mine Expansion. Blue Heron and Golder completed the Species at Risk (SAR) screening survey in June 2015, with a desktop review to compile data from the area to assess the potential for SAR to utilize the habitat. Based on the desktop review, there was a potential for thirty-four designated species to occur within the Wawa District. Seven of these species have a moderate potential to inhabit the study area and twenty-seven species have a low potential to inhabit the site. Three of the species with moderate potential to occur on site have been listed as either endangered or threatened under both the Species at Risk Act (SARA) and the Endangered Species Act (ESA). These species are Whip-Poor-Will, Northern Myotis and Little Brown Myotis bats, with field surveys conducted by Golder and Blue Heron at the Island Gold Mine site in June and July 2015. Northern Myotis and Little Brown Myotis were introduced onto the SAR list both federally and provincially in 2014.

An expanded desktop study was conducted by Golder in early 2020 as part of the Phase III Expansion Study. Golder's study area covers approximately 3,800 ha. During this study and from a review of historical information for neighbouring mines, potential habitat areas for Species at Risk were identified. The potential SARS are found in Table 20-1.

Table 20-1 Identified Species at Risk (SAR)

Species at Risk Observed	Ontario ESA Status	Federal SARA Status
Bald eagle	Special concern	Not listed
Canada warbler	Special concern	Threatened
Chimney swift	Threatened	Threatened
Common nighthawk	Special concern	Special concern
Eastern whip-poor-will	Threatened	Threatened
Little brown bat	Endangered	Endangered
Northern long-eared bat	Endangered	Endangered
Olive-sided flycatcher	Special concern	Special concern

Further fieldwork in 2020 confirmed Northern Myotis and Little Brown Myotis were around the Phase III Expansion Area. To offset the potential loss in bat habitat, 150 bat houses were installed to provide alternate habitat. Another 75 are planned to be installed later in 2022.

20.1.8 Aquatic Life

Lake trout are restricted to Mountain Lake, where a self-sustaining population exists. Walleye has been introduced into both Pine and Goudreau Lakes. Goudreau Lake, which constitutes the receiving water body for the treated effluent discharge from the Secondary Pond, supports northern pike, white suckers, and various minnow species. Perch occur in Pine Lake. Maskinonge Lake supports a northern pike population.

Fish surveys have been conducted as part of on-going EEM studies in Goudreau Lake, Maskinonge Lake, and Pine Lake. Based on the field program conducted in 2019 the most abundant species found in Goudreau Lake was the common shiner and secondly was the yellow perch. Occasional white sucker, walleye and pike were collected. In Maskinonge Lake, yellow perch was the most abundant fish species and second was the common shiner. Occasional white suckers, pike and walleye were collected.

20.1.9 Waste Management

Alamos Gold's strategy is to reduce consumption, reuse any waste generated, and dispose final waste in a safe and responsible manner. A Waste Management Procedure (WMP) has been developed and implemented for the site; it provides guidance to site and non-site personnel on the handling, processing, and disposal of waste, including hazardous waste and domestic materials generated during the normal operations of the facility.

The WMP is consistent with the requirements of Reg. 347 (Waste Management), Reg. 207/96 (burning of domestic waste), and Dubreuilville By-Law No. 2012-44 (domestic waste produced by the mine site and the camp).

20.1.10 Geochemistry of Waste Rock and Tailings

Excess underground waste rock is transported to the surface and stockpiled for use as future backfill and for constructing site roads and ongoing/future dam raise projects for the tailings facility.

In 2019, Golder was subcontracted to conduct an assessment on the geochemistry of Island Gold tailings and waste rock. Historical documentation, from Wood PLC, (formerly AMEC) was also reviewed. Golder determined that the waste rock did not generate any ARD nor metal leaching and recommended reducing the sample analyses to monthly.

Weekly analyses were conducted for tailings and ore for metal leaching and acid-rock drainage (ARD). The tailings did not generate any mobile metals, but results show an unknown potential to generate acid due to static Neutralization Potential Ratio (NPR) testing. The NPR is a ratio of the Neutralizing Potential (NP) to Acid Potential (AP) driven by the concentration of sulphides. Acid generation criteria are discussed in Table 20-2.

Table 20-2 Acid Generation Criteria

Acid Generation Potential	Criteria	Comments
Potentially Acid Generating (PAG)	NPR<1	Potentially acid generating, unless sulphide minerals are non-reactive
Uncertain	1<NPR<2	Possibly acid generating, if NP is insufficiently reactive or is depleted at a rate faster than sulphides
Non-Potentially Acid Generating (non-PAG)	NPR>2	Not expected to generate acidity.

From the 75 results analyzed, 54% of the samples are classified as non-PAG while 46% are classified as uncertain or PAG. Therefore, Golder recommended to continue weekly analysis of tailings static geochemistry and initiate four 60-week humidity cells, two if NPR is less than 1 and two when NPR is between 1 and 2. Three humidity cell tests were completed (2 at week 60 and another to week 80). No net-acid generation was observed. Weekly analyses will continue for the tailings with future decisions being made on the need for humidity cell tests.

20.2 Anticipated Permitting Activities

Relevant regulatory agencies for the anticipated Phase 3+ permitting needs include the provincial Ministry of the Environment, Conservation and Parks (MECP), Ministry of Natural Resources and Forestry (MNRF), Ministry of Energy and Ministry of Mines (Mines). There may also be permitting requirements from the Federal Department of Fisheries and Oceans (DFO),

All permitting activities will cover modifications and/or additions to the site including but not limited to:

- increased production rates;
- updated water management and effluent discharge strategies;
- new air and noise discharges;
- infrastructure additions/modifications related to the paste fill plant and new shaft area;
- new access roads;
- aggregate sources; and
- potential impacts to terrestrial habitats and natural water bodies including related fisheries resources.

To facilitate the Phase 3+ expansion, Alamos would be required to amend some existing operational permits and acquire a series of new permits and/or authorizations for both future operational requirements and Phase 3+ construction related activities.

Currently, Island Gold is fully permitted to be operated at a production rate of 461,760 tonnes per year (not to exceed 38,480 tonnes per month) of gold-bearing ore. An amended ECA for Air & Noise (No. 1821 BAWLAC) was issued in May 2019 allowing for an annual ore processing rate of 461,760 tonnes per year. The ECA requires that the Island Gold Mine be in compliance with Ontario Regulation 419/05, applicable MECP Guidelines for Air and Noise, and other performance requirements as specified in permit conditions. It also allows modifications such as process changes, de-bottlenecking, or addition of new equipment subject to limits on operational flexibility.

An amended ECA for Industrial Sewage Works (No. 9118-B9CM3R) was also issued in May 2019 to allow for a mill production rate of up to 38,480 tonnes per month and amended in April 2020 to include raising the tailings facility to 427 masl. This permit includes all components for site water management, tailings management and domestic sewage treatment.

In addition to ECA's the site is permitted for water taking activities under various Permits to Take Water (PTTW). Table 20-3 lists the PTTWs for the various locations.

Table 20-3 List of Permits to Take Water (PTTWs)

Permit Id	Location	Maximum Allowable Water Taking (litres/day)	Expiry
8571-8PENN8	Lochalsh	10,000,000	September 8, 2027
6138-9ABJ9Z	Kremzar	1,500,000	August 7, 2023
4231-A8BNM7	Maskinonge Lake	434,500	April 5, 2026
P-300-187434989	Exploration Drills	360,000	April 21, 2027

Within Ontario, both ECA's and PTTW's fall under the regulatory mandate of the Ministry of Environment, Conservation and Parks (MECP). Environmental Compliance Approvals (ECAs) are issued under the *Environmental Protection Act*. Permits to Take Water (PTTW) are issued under the *Ontario Water Resources Act*.

Proponents in Ontario are required to file and maintain an updated Closure Plan under the authority of the *Mining Act*, which falls under the mandate of the Ministry of Mines. The latest Closure Plan Amendment (CPA) was filed in March 2022 which provided authorization to start construction activities for the Phase 3+ Mine Expansion.

The current Closure Plan Amendment details the decommissioning strategy for the Island Gold Mine. It reflects the current and expected site conditions and defines a program which ensures the long-term chemical and physical stability of the site. The goal of the Closure Plan is to ensure that chemical and physical impacts to the site are minimized during operations and that the site is returned as closely as possible to pre-development conditions at close-out. The Closure Plan has been developed using data collected during physical, chemical, and biological studies of the site (treated effluent, surface water, ground water, ore/waste rock characterization, etc.) and the surrounding environment during the production phases. As detailed in the Closure Plan the total cost estimate for remediation of the Island Gold Mine in its current state stands at \$26,345,369 CAD.

As part of the ongoing Phase 3+ expansion, Alamos will be required to amend both Island Gold ECAs and complete another Closure Plan Amendment (CPA).

In addition to amending these operational permits, Alamos will likely be required to acquire new permits or authorizations for future operations and to support construction activities outside the scope of operational permits.

Alamos would be required to obtain a new Permit to Take Water for any domestic water supply needed to service the shaft surface facilities. In addition, given the potential for new disturbance associated with the shaft area, access roads, and aggregate sources there may be permits required under the Ontario *Endangered Species Act*. These permitting activities fall under the regulatory authority of the Ministry of Environment, Conservation and Parks (MECP).

Additional permits or authorizations would also need to be acquired through the different legislative requirements that fall under the mandate of the Ministry of Natural Resources and Forestry (MNRF). Included within the MNRF mandate would be aggregate permits for till for tailings dam lifts. These types of materials would typically be reserved to the Crown (i.e., Ontario) and their use is regulated under the *Aggregate Resources Act*. Other approvals include Forest Resource Licenses (tree clearing activities) issued under the *Crown Forest Sustainability Act*, Work Permits issued under the *Public Lands Act* for activities such as culvert installations or constructing portions of the new access road where it crosses public lands. There may also be a requirement for Land Use Permits under the *Public Lands Act* to allow temporary occupation of Crown Land for site development activities such as powerlines.

Alamos may also require an authorization issued under Section 35 the Federal *Fisheries Act* from the Federal Department of Fisheries and Oceans (DFO). It is not confirmed at this time whether this authorization is needed for Phase 3+ as design details for project development activities that may have impacts to fisheries resources have not been advanced sufficiently. Submission of a formal “request for review” to DFO will initiate this process.

The remaining Phase 3+ expansion, as proposed, is expected to be fully permitted within an 18-month timeframe once formally initiated.

20.3 Environmental Emergency Response

Island Gold’s environmental programs are designed with the goal of preventing all environmental incidents. However, in the event of unplanned incidents, the mine maintains a high degree of emergency preparedness with appropriate plans, resources, and training to minimize the impact

on workers, operations, the environment, and the community should an unplanned incident occur. A Spill Prevention and Control Plan (SPCP) is mandated under the regulatory requirements of Ontario Reg. 224/07 Spill Prevention and Contingency Plans, the primary purpose of which is to prevent and reduce the risk of spills of pollutants, and to prevent, eliminate or ameliorate any adverse effects that result from spills of pollutants that is updated annually.

A SPCP, as part of the environmental emergency response program, is in place for the overall Island Gold site. It outlines/mandates response to a leak or spill, to limit effects on employees, the community, and the environment.

It also includes roles and responsibilities of all employees, containment procedures, reporting aspects (both to internal management and external agencies), and follow-up/close-out procedures. Additional steps are taken to complete remediation and clean-up of a spill once the emergency containment has been completed.

20.4 Social and Community Considerations

20.4.1 Communities

The two most significant communities in the vicinity of the Island Gold Mine are the Town of Dubreuilville (closest to the mine) and the Municipality of Wawa, both in the Algoma District. Two other communities situated in the Algoma District include the Township of White River, which is 93 km north of Wawa, and the small community of Hawk Junction, which is approximately 30 km northeast of Wawa. Seasonal cottagers use the area as well.

The Mine is located 17 km from the Town of Dubreuilville which has a population of approximately 635 permanent residents. The Island Gold site is accessible from Dubreuilville by an all-weather road from Highway 519. The town contains accommodations for some mine personnel. Dubreuilville is accessible by car or train.

Historically, forestry and to a lesser degree mining, have been major contributors to Dubreuilville's economy. In November 2007, Dubreuil Lumber Inc. ("Dubreuil") filed for bankruptcy protection and ceased its logging operations. In 2008, Dubreuil was reduced to four employees. The collapse of the forestry industry dramatically impacted the town, leaving hundreds without work.

Statistics Canada data shows that Dubreuilville's population steadily decreased from 990 people in 1996 to 613 people in 2011 (a decline of 38%). The median age of the total population in 2016 was 42 years. The majority of the Town's citizens are bilingual and speak French as their primary language.

Educational facilities include a Catholic elementary school and a public high school, both of which are francophone and have small class sizes. Students must travel to Wawa for English education. Daycare services are also offered. Residents have access to Contact North, which offers access to university and college courses through distance learning and online education.

The Dubreuilville Health Centre has two full-time registered nurses and receives six physician visits per month. The community also offers homecare, tele-health video consultations and mental health referrals. Dubreuilville provides community support services such as a food bank. The nearest hospital is the Lady Dunn Health Centre, approximately 75 km away by road in Wawa.

20.4.2 Industry

The Island Gold Mine is the primary mining operation in the area. An adjacent open pit mine is under construction (owned by Argonaut Gold). Other junior exploration programs are on-going. Previous operations include the Kremzar, the Magino, the Edwards, the Cline Lake Gold mines, and small bulk samples such as Markes, Vega, and Morrison Mines.

The Island Gold Mine is in the southeast corner of the timber management area controlled by Dubreuil Forest Products, of Dubreuilville. No forestry operations are taking place in the mine area at present, and none are expected in the near term as the area has been extensively harvested. There is a local sawmill located in Dubreuilville which has been closed since 2008.

20.4.3 Recreation

Within 100 km of the Town of Dubreuilville there are numerous lakes that provide for recreational boating and fishing opportunities. Locally, fishing has been restricted on Goudreau Lake in recent years following the stocking of the lake with walleye by local conservation organizations, limiting fishing in the immediate mine area to Pine Lake, which has reasonable public access. Maskinonge Lake is restricted by access through the mine site. All three lakes support northern pike, white sucker, common shiner, and a variety of minnow species. Yellow perch also occur in Pine Lake, as well as walleye. Relatively easy access has resulted in intense fishing pressure by residents, as well as the presence of some non-resident anglers who return to the area annually.

Moose and partridge are the primary game animals and hunting pressure is considerable by both residents of Dubreuilville and non-residents. There are other game animals like marten, mink, and beaver. A grouse population also attracts hunters. Black bear hunting is popular in the area with a local outfitter operating out of Dubreuilville.

The project area is adjacent to a provincial snowmobile trail route, with lodging in Dubreuilville, resulting in increased traffic in the area during winter.

20.4.4 Community and Benefits

Island Gold's philosophy is to maximize local hiring of employees from the labour pool in the surrounding communities. This has increased the economic stability of the local communities of Dubreuilville, Wawa and White River who have been hit hard by the downturn of the forestry industry.

As of end of July 2022 there were 456 employees and approximately 442 contractors employed by Island Gold. This increase in contractors has largely been due to supporting contractors for the Phase 3+ Shaft construction. Island Gold's employees have significantly augmented the local economy by living locally or supporting the local businesses when residing at the mine workforce accommodation facilities in Dubreuilville for approximately six months out of the year.

Island Gold supports the local businesses and various non-profit organizations through its substantial local donations, purchase of goods, services, and materials, use of area motels and many home and apartment rentals for workforce accommodations. Support is also reflected through company programs such as the health and wellness program which provides yearly funds to encourage employees to join a fitness centre along with rental of local facilities such as the arena and school gym for employee activities or events. The recreational committee's various activities with the local communities (curling & golfing tournaments) aid in developing relations with the town and supporting their economic developments which is of important need to reconnect post the COVID-19 pandemic.

Island Gold encourages employees to relocate to the local. Island Gold helps to supplement the local health care system by securing the services of a registered nurse on-site to provide health care services and a health and wellness program onsite for our employees to promote a healthy lifestyle. The services include health care, referrals to local doctors, awareness training and a vaccination program (that includes hepatitis, COVID-19, twinRix and flu shots).

The mine has made donations to various initiatives in the communities, most recently with support for meals-on wheels and funding and providing supplies during the COVID-19 pandemic.

Public consultation activities are ongoing. Several information sessions are held in Dubreuilville, to provide updates of mine activities and to outline any proposed changes to the mine. Feedback garnered from consultation activities have been incorporated into the decision-making processes. Primary feedback has been related to employment opportunities at the mine for residents of Dubreuilville.

There have also been regular meetings with the Dubreuilville Town Council to discuss common interests such as the Town's landfill and incentives to Island Gold's employees to buy a house in Dubreuilville.

The construction of a 98-unit bunkhouse complex within the community was undertaken in the fall of 2015. A further expansion in bunkhouse access is partially complete and it includes adding four 44 bed-bunkhouses and one 38 bed bunkhouse with a new kitchen and recreational facility. Phase I camp expansion was completed in December 2019 which included commissioning one half of the bunkhouse capacity and a fully operational kitchen. The Phase II camp expansion was completed in 2021. Construction is on-going for additional temporary bunkhouses north of the existing camp to support the Phase 3+ Project.

20.5 Indigenous Engagement

Indigenous engagement initiatives for Island Gold were initiated in December 2003 by Patricia and continued with Richmond Mines. Alamos has increased Indigenous engagement efforts since acquiring the mine in 2017. The corporation's site and executive management team is actively engaged with all Indigenous engagement initiatives.

To date the following Indigenous groups have been identified as having varying degrees of interest around the Island Gold Mine: Michipicoten First Nation (MFN), Missanabie Cree First Nation (MCFN), Batchewana First Nation (BFN), the Garden River First Nation (GRFN).

There have also been engagement activities to outline any proposed significant changes to the project. Discussions have centered on the opportunities for employment, contracting, training, environmental effects of the project, and community development.

Alamos is committed to working with all affected Indigenous groups as Island Gold operations progress with the objective of arriving at mutually beneficial arrangements with appropriate Indigenous communities. A Community Benefits Agreement (CBA) was signed with the Missanabie Cree First Nation in March 2017. Alamos signed a Community Consultation and Benefit Agreement with Michipicoten First Nations in April 2022. Alamos is actively working with other communities on formal agreements.

21 CAPITAL AND OPERATING COSTS

21.1 Capital Cost Estimate Input

The capital costs for the Study have been estimated based on the scope of work defined in the sections below. The parties below have contributed to the preparation of the capital cost estimate in specific areas, as listed.

- Hatch:
 - o Shaft headframe and hoist house; and
 - o Underground ore and waste handling system.
- Redpath
 - o Shaft sinking, support, and lining.
- Halyard and DRA
 - o Process plant expansion.
- Golder
 - o Tailings management facility expansion; and
 - o Mine water treatment plant.
- Paterson and Cooke
 - o Paste fill plant.
- Alamos
 - o Administration building, warehouse, and mine dry;
 - o Camp and kitchen;
 - o Powerline upgrade;
 - o Delineation drilling;
 - o Underground capital development.
 - o Underground infrastructure, including dewatering and electrical; and
 - o Mobile equipment requirements;

21.2 Capital Cost Estimate Summary

The capital estimates conform, unless otherwise noted, to AACE Class 3 guidelines for a PFS Estimate with a -10% to +30% accuracy. The estimate is based on a contractor engineering and owner procurement and construction management execution approach.

The following parameters and qualifications were considered:

- The estimate was based on Q2 2022 pricing;
- All mining equipment would be purchased;
- A \$0.78 USD/CAD exchange rate was assumed, and no allowance has been made for exchange rate fluctuations; and
- There is no escalation added to the estimate.

Data for the estimates have been obtained from numerous sources, including:

- Pre-feasibility level engineering design for;
 - o Process plant expansion; and
 - o Paste fill plant
- Detailed engineering for:
 - o Shaft headframe and hoist house;
- Construction contracts in place:
 - o Warehouse;
 - o Underground capital development;
 - o Site clearing; and
 - o Shaft sinking, support, and lining.
- Mine schedules;
- Topographical information obtained from site survey;
- Budgetary equipment quotes;
- Budgetary unit costs from local contractors for civil, concrete, steel, electrical and mechanical works; and
- Data from similar recently completed studies and projects.

Major cost categories (permanent equipment, material purchase, installation, subcontracts, indirect costs, and Owner's costs) were identified and analyzed. To each of these categories on a line item basis, a contingency was allocated based on the accuracy of the data, and an overall contingency amount was derived in this fashion.

21.3 Capital Expenditures

As this report covers an expansion scenario at Island Gold, capital expenditures are divided into two distinct categories: growth capital and sustaining capital.

Growth capital expenses are defined as expenditures that allow the mine to expand from the current throughput of 1,200 tpd to 2,400 tpd. Once the Shaft Expansion is completed and production has sustainably reached the targeted tonnage for a period of three months, all further capital expenditures are classified as sustaining capital.

Sustaining capital expenditures are expenditures related to sustaining the existing production and operating plan and allow Island Gold to mine its current Mineral Reserves, and a portion of its Mineral Resources, during, and post, the project period. The project period is defined as 2022 to 2026 and the post project period is defined as 2027 to the end of the life of the mine.

As the life of mine plan detailed in this study incorporates mineable resources, which are, in terms of tonnes, 272% larger than the current Mineral Reserve, the mine life has grown significantly from the current 8 year Mineral Reserves life to an 18 year mine life with a substantially deeper mine. As a result, Island Gold will be required to invest in certain key areas to allow sustained production through 2039.

21.3.1 Sustaining Capital Investment

The sustaining capital requirements reported in the Study for the Shaft Expansion total \$995 million and include \$15 million of reclamation costs and \$41 million of delineation drilling. In the Shaft Expansion mine development will require the largest sustaining capital investment, totaling \$559 million from 2022 to 2039. These sustaining capital expenditures have increased over the previous Phase III S2000 case because the mineable resource increased by 55%, requiring more capital development and underground infrastructure. Table 21-1 presents the life of mine sustaining capital expenditures for the S2400 scenario.

Table 21-1 Sustaining Capital Investment (2022 – 2039)

Sustaining Capital Expenditure	C\$ million	US\$ million
Tailings Storage Facility	\$41	\$32
Surface Facilities	\$17	\$13
Paste Distribution System	\$30	\$23
General Underground	\$27	\$21
Underground Dewatering	\$8	\$6
Underground Electrical	\$103	\$80
Ventilation	\$12	\$9
Mobile Equipment	\$143	\$111
Delineation Drilling	\$41	\$32
Capital Development	\$559	\$436
Total Sustaining Capital	\$980	\$765
Reclamation	\$15	\$12
Total (including Reclamation)	\$995	\$777

Annual sustaining capital is detailed in Table 21-2.

Table 21-2 Annual Sustaining Capital

Sustaining Capital Expenditure	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039
Tailings Storage Facility	0.0	2.7	9.4	0.0	0.0	0.0	0.0	13.3	0.0	0.0	0.0	0.0	15.0	0.0	0.0	0.0	0.0	0.0
Surface Facilities	7.7	0.0	0.0	9.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Paste Distribution System	0.0	0.0	0.0	0.0	5.8	4.6	6.3	3.2	3.8	1.5	0.2	1.4	1.5	1.2	0.3	0.0	0.0	0.0
General Underground	2.5	1.5	1.4	0.7	0.8	9.1	3.4	1.4	1.2	1.0	0.9	0.6	0.1	1.2	0.9	0.5	0.1	0.0
Underground Dewatering	0.5	2.6	1.9	0.3	0.1	0.5	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.4	0.1	0.1	0.0	0.0
Underground Electrical	2.5	8.1	6.5	7.7	6.5	8.1	8.5	10.4	8.3	7.7	5.6	5.2	5.6	4.7	3.9	2.6	0.9	0.0
Ventilation	2.0	0.8	0.4	0.2	4.0	0.5	0.8	0.4	0.6	0.9	0.4	0.4	0.2	0.6	0.3	0.2	0.0	0.0
Mobile Equipment	8.8	5.8	10.5	10.1	7.3	11.2	8.5	7.9	11.9	7.8	5.8	9.0	6.7	5.5	12.3	9.0	4.4	0.0
Delineation Drilling	5.1	5.0	5.0	2.5	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9	0.0	0.0
Capital Development	19.5	41.4	38.9	45.1	42.4	43.8	43.9	41.8	46.8	45.8	42.3	27.4	26.6	27.6	15.3	10.2	0.0	0.0
Total Sustaining Capital	48.5	67.9	74.1	75.7	68.7	79.7	73.5	80.6	74.9	67.1	57.4	45.9	57.9	43.1	34.9	24.4	5.4	0.0
Reclamation	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.5
Total (including Reclamation, C\$ millions)	48.5	67.9	74.1	75.7	68.7	79.7	73.5	80.6	74.9	67.1	57.4	45.9	57.9	43.1	34.9	24.4	5.4	15.5
Total (including Reclamation, US\$ millions)	38.3	53.0	57.8	59.1	53.6	62.1	57.4	62.9	58.4	52.3	44.7	35.8	45.1	33.6	27.2	19.1	4.2	12.1

21.3.2 Growth Capital Investment

The Island Gold growth capital expenditures for the Shaft Expansion are estimated to be \$966 million, including; \$507 million of direct costs, \$82 million of indirect costs (contractor indirects and owner's costs), \$70 million of contingency, and \$207 million of capital development (Table 21-3). This is expected to be spent between 2022 and 2026, with the bulk of this spending occurring between 2022 and 2025.

Within the \$966 million of growth capital there are \$307 million of expenditures that are accelerated spending in that they are expenses that would have been incurred without the Shaft Expansion, albeit at a later point in time. These accelerated growth capital expenditures are required to be spent during the project period to both advance the ramp system to the bottom of the shaft and to prepare mining areas in advance of expanding the mining rate to 2,400 tpd. These expenditures consist of lateral capital development and the establishment of underground infrastructure including ventilation and electrical expansions and are detailed in Table 21-4.

Specific surface infrastructure projects would be required even without the Shaft Expansion. These include:

- Addition of a paste plant (\$66M);
- Power line upgrade (\$30M); and
- Addition of an effluent treatment plant (\$21M).

Table 21-3 Annual Growth Capital Expenditure

Growth Capital Cost	2022	2023	2024	2025	2026	Total
Shaft Surface Complex	\$45.3	\$21.2	\$1.5	\$7.2	\$0.0	\$75.2
Shaft	\$67.4	\$61.7	\$64.9	\$22.2	\$1.1	\$217.3
Mill	\$1.5	\$16.5	\$40.9	\$38.1	\$0.4	\$97.5
Paste Plant	\$0.6	\$0.6	\$22.0	\$42.7	\$0.3	\$66.1
Power Upgrade	\$2.7	\$15.5	\$12.2	\$0.0	\$0.0	\$30.3
Effluent Treatment Plant	\$0.3	\$7.5	\$4.0	\$9.3	\$0.0	\$21.0
General Indirect Costs	\$21.3	\$17.3	\$18.7	\$19.0	\$6.0	\$82.4
Contingency	\$10.3	\$14.2	\$20.3	\$24.4	\$0.3	\$69.6
Total Growth Capital	\$149.3	\$154.5	\$184.4	\$162.9	\$8.1	\$659.1
Underground Equipment & Infrastructure (mostly accelerated)	\$29.1	\$16.5	\$16.1	\$13.1	\$25.9	\$100.6
Accelerated Capital Development	\$34.9	\$64.2	\$39.7	\$39.8	\$28.0	\$206.7
Total Growth Capital including Accelerated Spend (C\$, millions)	\$213.2	\$235.2	\$240.2	\$215.8	\$62.0	\$966.4
Total Growth Capital including Accelerated Spend (US\$, millions)	\$168.5	\$183.4	\$187.4	\$168.4	\$48.3	\$755.9

Table 21-4 Annual Accelerated Growth Capital Expenditure

Accelerated Growth Capital	2022	2023	2024	2025	2026	Total
Ore Waste Handling System	\$0.9	\$0.0	\$0.0	\$1.9	\$17.0	\$19.8
General Underground Infrastructure	\$0.0	\$0.3	\$0.4	\$0.3	\$0.0	\$1.0
Mobile Equipment	\$4.6	\$0.0	\$7.8	\$3.9	\$0.0	\$16.3
UG Ventilation Expansion	\$6.5	\$8.9	\$1.0	\$0.6	\$0.4	\$17.3
UG Dewatering Expansion	\$3.0	\$0.5	\$0.6	\$0.7	\$2.6	\$7.4
Surface Growth Projects	\$9.2	\$0.0	\$0.0	\$0.0	\$0.0	\$9.2
Underground Electrical	\$4.9	\$6.5	\$5.6	\$5.1	\$5.6	\$27.7
General Underground Infrastructure	\$0.0	\$0.3	\$0.7	\$0.5	\$0.3	\$1.8
Accelerated Capital Development	\$34.9	\$64.2	\$39.7	\$39.8	\$28.0	\$206.7
Total Accelerated Growth Capital (C\$, millions)	\$64.0	\$80.7	\$55.8	\$52.9	\$53.9	\$307.3
Total Accelerated Growth Capital (US\$, millions)	\$50.5	\$62.9	\$43.6	\$41.2	\$42.0	\$240.3

21.4 Operating Expenses

Operating expenses were calculated using the Island Gold's 2022 budget as a reference point, where applicable, and were developed from first principles when budgetary items were not available. The 2022 budget costs were adjusted to reflect increases related to mining at greater depths, increased operational efficiencies associated with shaft access, higher underground throughput, and recent inflationary impacts on inputs. Fixed and variable components of cost centers were considered. Costs were adjusted to reflect the total volume of material moved (waste and ore) per year. Costs were also adjusted to reflect the reduction in labour related to the completion of capital development activities.

The summarized cost structure consists of the following categories and sub-categories:

- Mining
 - o Administration and Haulage;
 - o Ore Development;
 - o Stopping;
 - o Geology;
 - o Engineering; and
 - o Maintenance.
- Milling
- G&A
 - o Administration;
 - o Human Resources;
 - o IT;

- o Environment;
- o Health and Safety;
- o Lodging and Transportation;
- o Warehouse and Purchasing; and
- o Surface Maintenance.
- Royalties

21.4.1 Mining Costs

Mining costs were generated on a departmental basis. Table 21-5 summarizes the underground departmental mining costs the S2400 scenario in project and post project periods. Overhead, haulage, and maintenance costs decrease after the shaft is commissioned due to the combined effects of increased tonnage and a decrease in the absolute number of trucks required for material movement. Stopping costs increase after the post project period due to the introduction of paste backfill, the cost of which is offset by higher mining recoveries.

Some of the fixed costs, on a unit cost basis, increase toward the end of the mine as less capital development is undertaken. Island Gold uses an allocation system to allocate a portion of overheads to capital costs, and as capital development decreases, more of these costs stay with operating costs.

Table 21-5 S2400 Unit Mining Costs

Department	2022-2026 \$/t	2027-2039 \$/t	LOM \$/t	Total LOM C\$M
Overhead and Haulage	25.47	21.80	22.47	305
Development	13.42	4.63	6.25	85
Stopping	30.07	34.69	33.84	458
Geology	3.55	4.71	4.50	61
Engineering	3.46	3.72	3.67	50
Maintenance	33.08	30.05	30.60	415
Total	109.05	99.59	101.33	1,373

21.4.2 Processing Cost

The mill expansion from 1,200 tpd to 2,400 tpd in the Shaft Expansion mainly benefits the labour unit costs as labour is predominantly a fixed cost. Most consumables are utilized at approximately the same rate post and pre project completion. Table 21-6 presents the S2400 processing costs.

Table 21-6 S2400 Unit Processing Costs

	2022-2026 \$/t	2027-2039 \$/t	LOM \$/t	Total LOM C\$M
Electricity	2.43	3.19	3.05	41
Reagents	11.05	11.73	11.60	158
Propane	0.74	0.66	0.68	9
Water Treatment	0.48	2.76	2.34	32
Maintenance	3.56	2.07	2.34	32
Labour	14.19	11.73	12.19	166
Mill Haulage	4.17	1.56	2.05	28
Laboratory	0.12	0.40	0.35	5
Total (\$/t milled)	36.73	34.10	34.59	470

21.4.3 General and Administrative Costs

As many general and administrative (G&A) costs are fixed costs, the G&A unit costs tend to decrease with an increase in tonnage rates Table 21-7 presents the S2400 processing costs.

Table 21-7 S2400 General and Administrative Cost

Department	2022-2026 \$/t	2027-2039 \$/t	LOM \$/t	Total LOM C\$M
Administration	10.13	7.06	7.64	104
Human Resources	3.51	1.78	2.11	29
IT	5.46	3.01	3.46	47
Health and Safety	4.90	4.66	4.70	64
Lodging and Transportation	12.01	11.01	11.19	152
Warehouse and Purchasing	1.91	2.02	2.00	27
Surface Maintenance	3.50	3.58	3.57	49
Environmental	2.28	2.67	2.60	35
Total (\$/t milled)	43.71	35.80	37.27	507

21.4.4 Total LOM Operating Costs

Total life of mine operating costs of the S2400 Shaft Expansion are \$2,412 million (Table 21-8). Total operating costs are expected to average \$174 per tonne of mill feed post completion of the shaft and mill expansion by 2027. This includes average mining costs of \$100 per tonne. Without the expansion, and with a ramp only scenario, mining costs would increase substantially on an annual basis as the mining horizons get deeper and haulage distances increase.

Table 21-8 S2400 Life of Mine Operating Costs

Operating Cost	2022-2026 \$/t	2027-2039 \$/t	LOM \$/t	Total LOM C\$M
Mining	109.05	99.59	101.33	1,373
Processing	36.73	34.10	34.59	470
General and Administrative	43.71	35.80	37.27	507
Royalties	6.49	5.74	5.88	80
Silver Credit	-1.34	-1.28	-1.29	-18
Total Operating Cost	194.63	173.94	177.78	2,412

Annual mining, process and G&A costs are presented in Table 21-9, Table 21-10, and Table 21-11.

Table 21-9 S2400 Scenario Annual Unit Mining Costs

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	LOM
Overhead and Haulage	22.64	22.81	22.94	26.58	29.73	19.93	20.18	20.56	19.62	21.04	21.40	23.24	23.25	23.20	24.75	24.46	19.32	22.83	22.47
Development	11.98	15.04	10.78	15.07	13.84	9.65	9.44	9.54	8.26	6.87	5.00	3.99	1.12	1.78	1.42	1.30	0.00	0.00	6.25
Stoping	31.96	32.60	27.32	29.89	29.14	29.22	29.96	28.13	30.52	32.52	35.03	35.84	39.21	38.56	38.32	38.51	39.09	36.89	33.84
Geology	3.84	3.52	3.30	3.21	3.79	4.27	4.25	4.35	4.16	4.23	4.34	4.93	4.95	4.92	5.46	5.70	4.52	5.43	4.50
Engineering	3.91	3.59	3.37	3.29	3.28	3.56	3.55	3.63	3.47	3.53	3.62	4.11	4.01	3.80	4.21	4.34	2.74	3.89	3.67
Maintenance	32.16	31.57	32.75	34.01	34.18	28.57	28.74	29.12	28.32	28.47	28.87	30.68	30.75	30.70	32.24	32.78	30.03	32.14	30.60
Total (\$/t mined)	106.49	109.13	100.46	112.05	113.95	95.20	96.13	95.33	94.34	96.65	98.26	102.79	103.27	102.96	106.42	107.09	95.69	101.17	101.33

Table 21-10 S2400 Annual Unit Processing Costs

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	LOM
Electricity	2.41	2.41	2.41	2.20	2.61	3.21	3.20	3.22	3.22	3.23	3.22	3.22	3.21	3.22	3.22	3.23	3.24	2.61	3.05
Reagents	10.64	10.59	10.92	11.06	11.65	11.68	11.60	11.88	11.90	12.10	11.92	11.68	11.89	11.66	11.64	11.54	11.47	11.32	11.60
Propane	0.73	0.73	0.73	0.67	0.79	0.67	0.66	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.54	0.68
Water Treatment	0.00	0.00	0.00	0.00	1.65	2.78	2.77	2.79	2.79	2.80	2.79	2.79	2.78	2.79	2.79	2.80	2.80	2.26	2.34
Maintenance	4.17	4.17	4.18	4.17	2.06	2.06	2.06	2.06	2.06	2.07	2.07	2.07	2.06	2.07	2.07	2.07	2.07	2.08	2.34
Labour	15.31	15.32	15.32	14.00	12.30	11.93	11.86	11.94	11.90	12.04	11.96	11.92	11.64	11.68	11.70	11.55	11.52	10.29	12.19
Mill Haulage	5.15	5.15	5.17	5.15	1.80	1.64	1.64	1.63	1.66	1.66	1.64	1.54	1.53	1.54	1.48	1.45	1.40	1.40	2.05
Laboratory	0.00	0.00	0.00	0.00	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.35
Total (\$/t milled)	38.41	38.38	38.73	37.24	33.26	34.38	34.18	34.58	34.60	34.98	34.66	34.29	34.17	34.03	33.97	33.71	33.58	30.90	34.59

Table 21-11 S2400 Annual Unit General and Administrative Costs

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	LOM
Administration	14.88	10.80	10.79	9.86	6.73	7.52	7.47	7.52	7.52	7.55	7.52	7.54	7.52	7.55	6.15	6.17	5.21	6.26	7.64
Human Resources	3.51	3.89	4.14	3.97	2.63	2.08	2.07	2.08	1.92	1.93	1.92	1.73	1.72	1.72	1.70	1.69	1.07	1.40	2.11
IT	6.13	6.16	6.33	5.90	3.84	3.15	3.14	3.15	3.15	3.17	3.16	3.14	3.12	3.13	3.12	3.12	1.75	2.62	3.46
Health and Safety	5.56	5.13	4.99	4.80	4.38	4.35	4.34	4.44	4.24	4.31	4.30	4.85	4.86	4.83	5.32	5.54	4.33	4.95	4.70
Lodging and Transport	13.50	12.29	11.75	11.59	11.39	10.49	10.06	10.28	9.83	9.98	10.22	11.45	11.49	11.42	12.19	12.66	11.80	11.33	11.19
Warehouse and Purchasing	2.20	2.01	1.89	1.81	1.75	1.88	1.83	1.87	1.79	1.82	1.87	2.12	2.13	2.12	2.35	2.46	1.83	2.36	2.00
Surface Maintenance	3.90	3.58	3.53	3.59	3.16	3.26	3.24	3.32	3.18	3.23	3.30	3.73	3.74	3.73	4.11	4.29	3.75	3.81	3.57
Environmental	2.51	2.29	2.15	2.27	2.23	2.39	2.38	2.44	2.33	2.37	2.43	2.76	2.77	2.76	3.06	3.20	3.00	2.95	2.60
Total (\$/t milled)	52.17	46.16	45.57	43.79	36.11	35.11	34.54	35.10	33.97	34.36	34.72	37.33	37.36	37.26	38.01	39.11	32.74	35.68	37.27

22 ECONOMIC ANALYSIS

An engineering economic model was developed to estimate annual cash flows and sensitivities for Island Gold. After-tax estimates were developed to approximate the true investment value.

Sensitivity analyses were performed for variation in metal prices, foreign exchange rate, operating costs, and capital costs to determine their relative importance as value drivers.

The estimates of capital and operating costs have been developed specifically for Island Gold and are summarized in Section 21 of this report. They are presented in Q2, 2022 Canadian dollars unless otherwise stated. The economic analysis has been run with no additional inflation (constant dollar basis).

22.1 Assumptions

All costs and economic results are reported as CAD\$, unless otherwise noted. Table 22-1 outlines the planned life of mine tonnage and grade estimates.

Table 22-1 Life of Mine Plan Summary

Parameters	Unit	Value
Mine Life ¹	Years	18
Total Mill Feed	Kt	13,550
Processing Rate ²	tpd	2,400
Average Au Head Grade	g/t	10.59
Total Au Production over Life of Mine	koz	4,460
Au Production (Years 2022 to 2039)	Average koz/a	255
Au Production (Years 2026 to 2039) ³	Average koz/a	287

Notes:

- 2022 to 2039
- 1,200 tpd from 2022 until shaft completion in 2026, after which the mill ramps up to 2,400 tpd by 2027
- Post-shaft completion in 2026
- Other economic factors and assumptions used in the economic analysis include the following:
- US \$1,800/oz gold, US \$22.00/oz silver and a \$0.79 USD/CAD were used in the cash flow model for 2022;
- US \$1,650/oz gold, US \$18.00/oz silver and a \$0.78 USD/CAD were used in the cash flow model for 2023 onwards;
- Discount rate of 5%;
- Closure cost of \$15.5M (US \$12.1M);
- No salvage assumed at the end of mine life; and
- Exclusion of all costs prior to 2022. However, 60% of Alamos' Canadian tax pools at December 31, 2021, are utilized in the tax calculations.

22.2 Revenue and Working Capital

Working capital assumptions were not included in the economic analysis as the mine is currently operating with adequate working capital.

Mine revenue is derived from the sale of gold doré into the international marketplace. The mine has contractual arrangements for refining. The parameters used in the economic analysis are consistent with current agreements, as shown in Table 22-2.

Figure 22-1 illustrates the annual recovered gold and cumulative recovered gold by project year.

Table 22-2 NSR Assumptions Used in the Economic Analysis

Assumptions	Unit	Value
Au Payable	%	99.96%
Au Refining and Transportation Charge	\$/oz	4.20

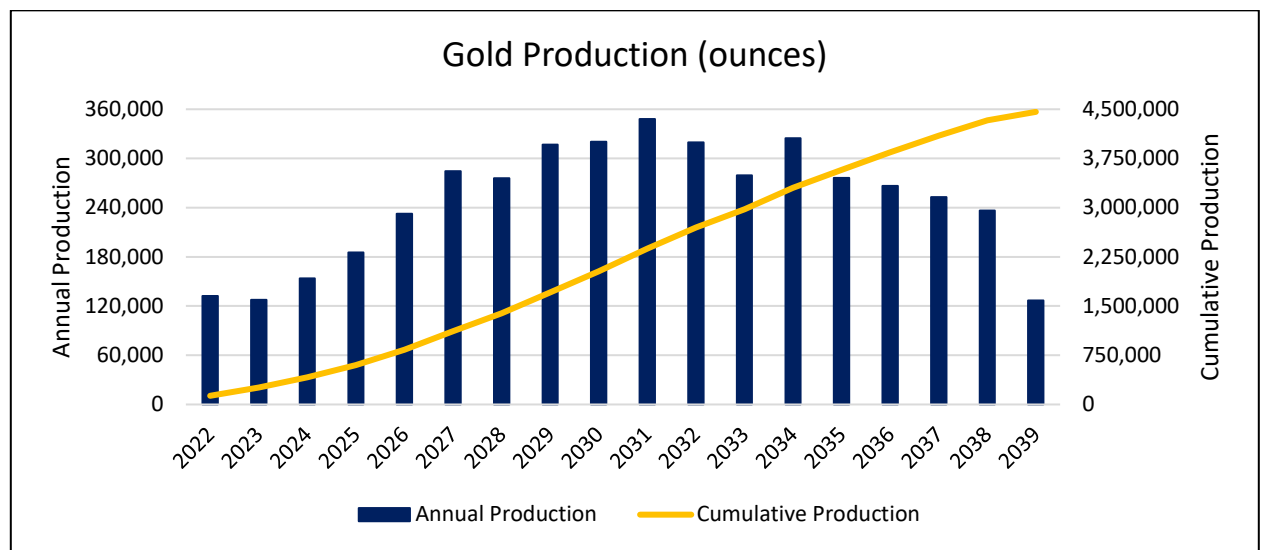


Figure 22-1 Annual and Cumulative Gold Production

22.3 Summary of Operating Costs

Total life of mine operating costs, as presented in Table 22-3, amount to \$2,412M (US \$1,882M), including silver by-product credits, royalties and refining and transportation charges. This translates into an average cost of \$178/t processed over the life of mine (\$174/t from 2027 to 2039 post shaft completion). A detailed analysis of the operating costs can be found in Section 21.4 of this report.

Table 22-3 Summary of Operating Costs

Operating Cost	C\$/t Processed	LOM C\$M	US\$/t Processed	LOM US\$M
Mining	\$101	\$1,373	\$79	\$1,071
Processing	\$35	\$470	\$27	\$367
G&A	\$37	\$507	\$29	\$395
Subtotal	\$173	\$2,350	\$135	\$1,834
Silver Credit	-\$1	-\$18	-\$1	-\$14
Royalties	\$6	\$80	\$5	\$62
TOTAL Operating Costs	\$178	\$2,412	\$139	\$1,882

22.4 Summary of Capital Costs

The capital costs used for the economic analysis are set out below. Table 22-4 summarizes the capital costs used in the economic analysis, and Table 22-5 and Table 22-6 show a breakdown by sustaining and growth capital. Detailed information can be found in Section 21 of this report.

Table 22-4 Total Capital Costs

Total Capital Cost	LOM C\$M	LOM US\$M
Sustaining Capital	\$995	\$777
Growth Capital	\$966	\$756
Total Capital Costs	\$1,961	\$1,533

Table 22-5 Sustaining Capital Costs

Sustaining Capital Cost	LOM C\$M	LOM US\$M
Tailings Storage Facility	\$41	\$32
Underground Infrastructure	\$197	\$154
Mobile Equipment	\$143	\$112
Delineation Drilling	\$41	\$32
Capital Development	\$559	\$436
Total Sustaining Capital	\$980	\$765
Reclamation	\$15	\$12
Total Sustaining Capital (including Reclamation)	\$995	\$777

Table 22-6 Growth Capital Costs

Growth Capital Cost	LOM C\$M	LOM US\$M
Shaft Surface Complex	\$75	\$59
Shaft	\$217	\$170
Mill	\$97	\$76
Paste Plant	\$66	\$52
Power Upgrade	\$30	\$24
Effluent Treatment Plant	\$21	\$16
General Indirect Costs	\$82	\$64
Contingency	\$70	\$55
Total Growth Capital	\$659	\$516
Underground Equipment & Infrastructure (mostly accelerated)	\$101	\$79
Accelerated Capital Development	\$207	\$162
Total Growth Capital (including Accelerated Spend)	\$966	\$756

22.5 Reclamation and Mine Closure Plan

The Closure Plan anticipates a cost of \$15.5M (US \$12.1M) for reclamation and closure. The bulk of the closure costs and reclamation activity will occur beyond 2039, after mining and processing have been completed at Island Gold.

22.6 Taxes

Island Gold will be subject to provincial, federal, and mining taxes as follows:

- Ontario Mining Tax: 10%;
- Ontario Provincial Income Tax: 10%; and
- Federal Income Tax: 15%.

The rates above are current as of the date of this report and are subject to change in the future. Based on these rates and the financial assumptions used in this report, Island Gold is expected to have payable income and mining taxes of \$1,301M (US\$1,015M) over its 18-year life. Alamos has various Canadian tax pools that could be applied against future income from its Canadian operations, and 60% of the tax pools as of December 31, 2021, were used in this study to reduce taxes payable at Island Gold in the economic analysis.

22.7 Royalties

Production from Island Gold is subject to third-party royalties that average approximately 2.5% over the life of mine based on ounces produced. However, approximately 90% of the royalties at Island Gold are paid in-kind (ounces) to a third-party. The accounting treatment requires that in-kind royalties be recorded at production cost, which lowers the royalty expense with an offsetting reduction in revenue given in-kind ounces transferred to royalty holders do not meet the definition of sales. As a result, the average royalty included in the economic analysis is approximately 0.9% of revenue over the life of mine. There is no net impact on gross margin

from the accounting for in-kind royalties compared to cash-paid royalties given the lower royalty expense is offset by a reduction in ounces sold. Total royalties included in this report are \$80M (US\$62M).

22.8 Economic Analysis

The expansion option is economically viable with an after-tax internal rate of return (IRR) of 23%. IRR which is calculated on the differential after-tax cash flow between the expansion scenario and the scenario of continuing to mine at 1,200 tpd with ramp-only access (R1200). After-tax net present value at 5% (NPV^{5%}) is \$2,093M (US \$1,632M).

Figure 22-2 shows the projected cash flows used in the economic analysis and based on the assumptions in Section 22.1. Table 22-7 shows the detailed results of this evaluation.

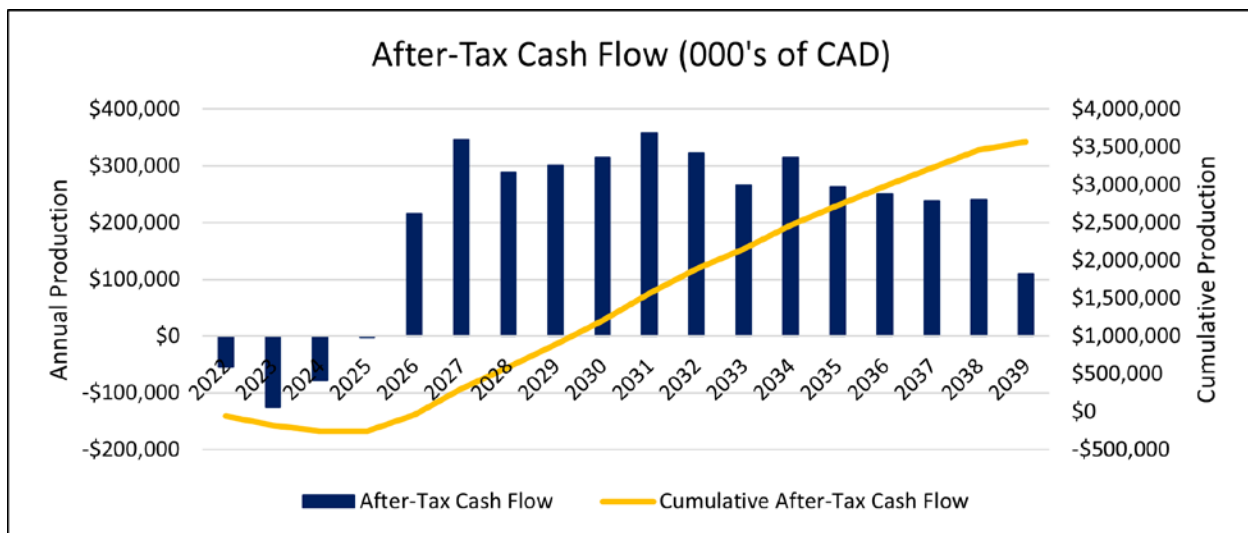


Figure 22-2 Annual and Cumulative After-Tax Cash Flow at USD \$1,650 Gold

Table 22-7 Summary of Economic Results

Category	Unit	Value (C\$)	Value (US\$)
Net Revenues	\$M	\$9,247	\$7,216
Operating Costs ¹	\$M	\$2,412	\$1,882
After-Tax Cash Flow from Operations ²	\$M	\$5,533	\$4,318
Total Capital & Closure Costs	\$M	\$1,961	\$1,533
Total Cash Cost (2022-2039)	\$/oz	\$553	\$432
Mine Site All-In Sustaining Cost (2022-2039)	\$/oz	\$781	\$610
Total Cash Cost (2026-2039) ³	\$/oz	\$545	\$425
Mine Site All-In Sustaining Cost (2026-2039) ³	\$/oz	\$739	\$576
Net After-Tax Cash Flow	\$M	\$3,572	\$2,786
After-Tax NPV5%	\$M	\$2,093	\$1,632
After-Tax IRR	%	23%	

Notes:

1. Operating Costs include mining, processing, G&A, royalties, transport & refining costs, and silver credit.
2. Cash Flow from Operations includes payable taxes.
3. Post-shaft completion in 2026.
4. IRR is calculated on the differential after-tax cash flow between the expansion scenario and the scenario of continuing to mine at 1,200 tpd with ramp-only access.

22.9 Sensitivities

A sensitivity analysis was performed to test value drivers on Island Gold's NPV using a 5% discount rate. The results of this analysis are demonstrated in Table 22-8 and Table 22-9 and illustrated in **Error! Reference source not found.** Island Gold proved to be most sensitive to changes in metal price followed by foreign exchange, capital costs and operating costs. A sensitivity analysis of the after-tax results was performed using various gold prices.

Figure 22-4 represents the annual after-tax cash flow and cumulative cash flow at a gold price of USD \$1,800 in 2022 and \$1,850 from 2023 onwards.

Table 22-8 After-Tax NPV^{5%} Sensitivity Results

(\$M of CAD)	-10%	-5%	100%	5%	10%
Gold Price	\$1,698	\$1,896	\$2,093	\$2,290	\$2,486
Canadian Dollar	\$2,531	\$2,300	\$2,093	\$1,905	\$1,733
Capital Costs	\$2,209	\$2,151	\$2,093	\$2,035	\$1,976
Operating Costs	\$2,200	\$2,147	\$2,093	\$2,039	\$1,986

(\$M of USD)	-10%	-5%	100%	5%	10%
Gold Price	\$1,324	\$1,479	\$1,632	\$1,785	\$1,939
Canadian Dollar	\$1,772	\$1,702	\$1,632	\$1,562	\$1,491
Capital Costs	\$1,723	\$1,678	\$1,632	\$1,587	\$1,541
Operating Costs	\$1,716	\$1,674	\$1,632	\$1,590	\$1,548

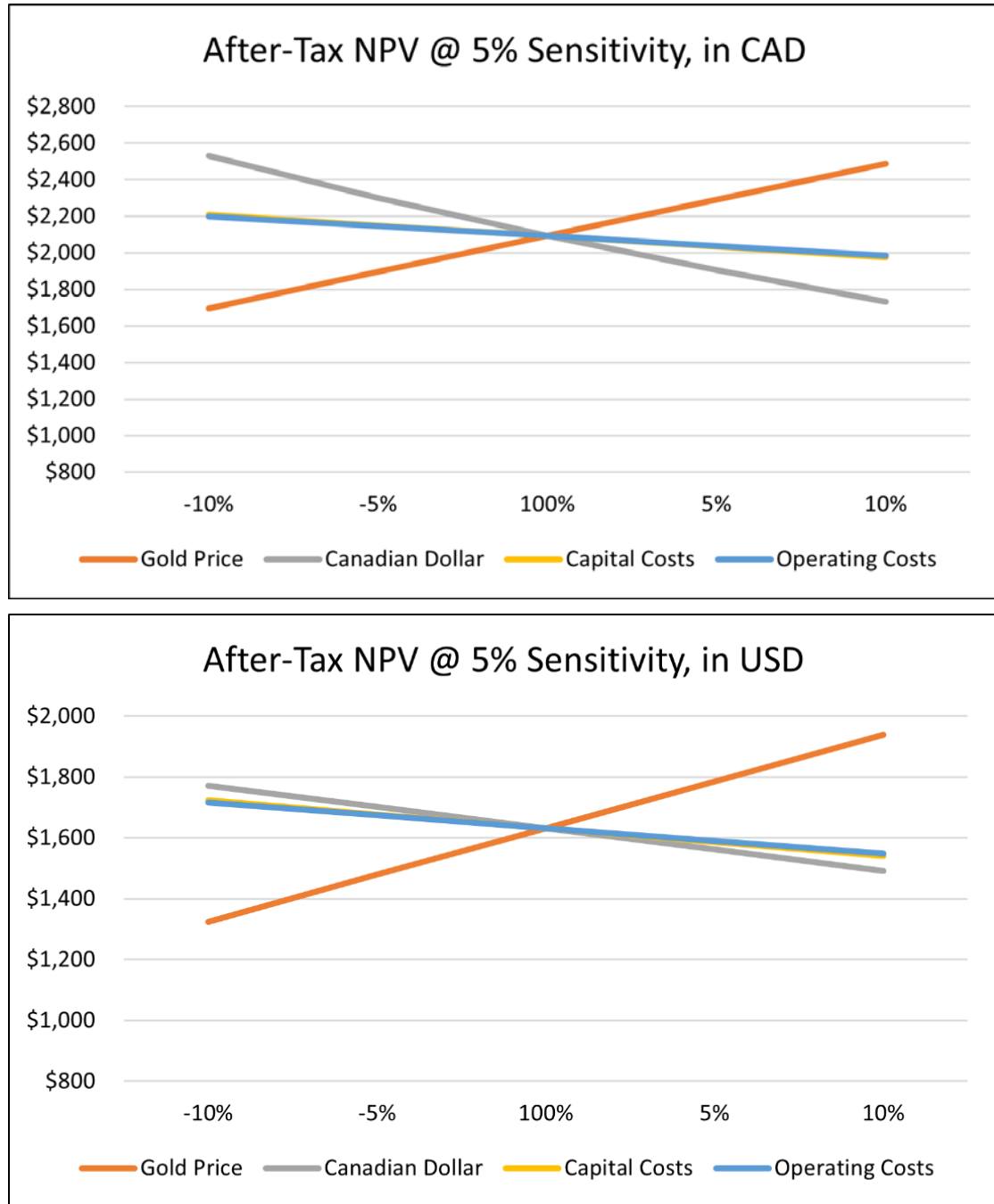


Figure 22-3 After-Tax NPV^{5%} Sensitivity Results

Table 22-9 Gold Price Sensitivity on NPV

Gold Price	After-Tax NPV (C\$M)	After-Tax NPV (US\$M)	After-Tax IRR ¹ (%)
\$1,450	\$1,614	\$1,258	20%
\$1,550	\$1,854	\$1,446	22%
\$1,650	\$2,093	\$1,632	23%
\$1,750	\$2,459	\$1,818	24%
\$1,850	\$2,570	\$2,004	25%
\$1,950	\$2,808	\$2,189	26%

Notes:

1. IRR is calculated on the differential after-tax cash flow between the expansion scenario and the scenario of continuing to mine at 1,200 tpd with ramp only access

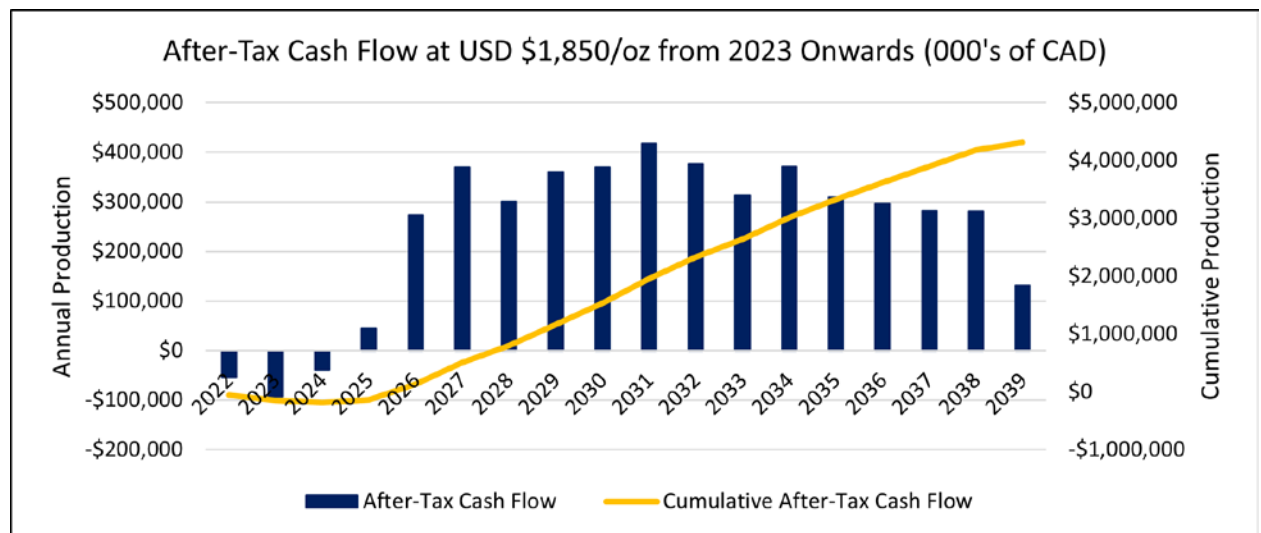


Figure 22-4 Annual and Cumulative After-Tax Cash Flow at USD \$1,850 Gold

Note:

- USD \$1,800 is used in 2022

A summary of the Island Gold financial model is shown in Table 22-10.

Table 22-10 Island Gold Financial Model Summary

	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	Total
Mill Feed Mined (tonnes)	438,000	437,997	439,200	478,969	698,697	875,991	878,399	875,996	875,673	875,994	878,400	876,000	876,000	875,930	878,400	875,095	873,668	541,520	13,549,929
Waste Mined (tonnes)	493,342	579,009	646,867	675,798	536,895	392,842	393,383	370,233	427,460	411,766	376,177	227,350	219,262	229,257	121,717	80,744	0	0	6,182,102
Total Tonnes mined	931,342	1,017,006	1,086,067	1,154,767	1,235,593	1,268,832	1,271,782	1,246,229	1,303,133	1,287,760	1,254,577	1,103,350	1,095,262	1,105,188	1,000,117	955,839	873,668	541,520	19,732,031
Grades (g/t Au)	9.73	9.40	11.29	12.47	10.45	10.46	10.12	11.66	11.79	12.81	11.73	10.27	11.94	10.16	9.78	9.32	8.73	7.34	10.59
Gold Production (oz)	132,414	127,790	153,828	185,287	232,577	284,401	275,792	317,015	320,326	348,148	319,683	279,249	324,615	276,227	266,537	253,014	236,529	126,766	4,460,201
Operating Costs																			
Unit Mining Costs (C\$/tonne)	\$106	\$109	\$100	\$112	\$114	\$95	\$96	\$95	\$94	\$97	\$98	\$103	\$103	\$103	\$106	\$107	\$96	\$101	\$101
Unit Milling Costs (C\$/tonne)	\$38	\$38	\$39	\$37	\$33	\$34	\$34	\$35	\$35	\$35	\$35	\$34	\$34	\$34	\$34	\$34	\$34	\$31	\$35
Unit G&A Costs (C\$/tonne)	\$52	\$46	\$46	\$44	\$36	\$35	\$35	\$35	\$34	\$34	\$35	\$37	\$37	\$37	\$38	\$39	\$33	\$36	\$37
Total Unit Operating Costs ¹ (C\$/tonne)	\$202	\$199	\$190	\$199	\$181	\$169	\$169	\$169	\$167	\$171	\$172	\$179	\$179	\$179	\$183	\$185	\$166	\$172	\$178
Total Cash Costs (US\$/oz)	\$539	\$542	\$431	\$408	\$461	\$416	\$431	\$374	\$365	\$342	\$378	\$448	\$387	\$452	\$480	\$508	\$489	\$584	\$432
Mine-site AISC (US\$/oz)	\$833	\$964	\$813	\$733	\$696	\$641	\$645	\$577	\$553	\$496	\$521	\$579	\$530	\$577	\$584	\$585	\$507	\$681	\$610
Capital Expenditures																			
Sustaining Capex (US\$ M)	\$38	\$53	\$58	\$59	\$54	\$62	\$57	\$63	\$58	\$52	\$45	\$36	\$45	\$34	\$27	\$19	\$4	\$12	\$777
Growth Capex (US\$ M)	\$168	\$183	\$187	\$168	\$48	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$756

Notes:

1. Total unit costs are inclusive of royalties and silver credits.

23 ADJACENT PROPERTIES

23.1 Magino Mine

The first discovery of gold in the area was in 1918 by J. W. Webb on what is now referred to as the Magino deposit, located about 1 km to the southwest of Island Gold's western property boundary. The deposit is reported to be hosted in the Webb Lake Stock, a quartz porphyritic granodiorite, which intrudes mafic volcanic rocks. Both lithologies occur within the GLDZ and have been highly altered because of deformation associated with this structure. Gold is present within subparallel grey quartz veins and in silicified wall rocks of east-west striking shear zones in granodiorite within the GLDZ. According to public records, the Magino mine operated intermittently between 1933 and 1939 during which time 105,792 tonnes of ore at a grade of 2.57 g/t Au were mined, producing approximately 8,700 ounces. The mine was closed from 1940 until 1988, when it was reopened by McNellen Resources Inc. and Muscocho Exploration Ltd. During the 1988 to 1992 period, a total of 696,413 tonnes of ore averaging 4.56 g/t Au was mined using bulk mining methods, producing 105,543 ounces of gold. Golden Goose Resources Inc. ("Golden Goose") acquired the Magino mine property in 1996. In April 2004, Golden Goose filed a Mineral Resource estimate to verify the potential for a large tonnage low grade gold mineralization amenable to open pit mining.

This property was later acquired by Prodigy Gold Inc. ("Prodigy Gold"), formed through a union between Golden Goose and Kodiak Exploration Ltd., which did further exploration and defined an open pit Mineral Resource of 223 million tonnes averaging 0.87 g/t Au for 6.25 million ounces of gold.

In 2012, Argonaut Gold Inc. ("Argonaut") acquired 100% of all issued and outstanding shares of Prodigy Gold. On December 2013, Argonaut announced pre-feasibility study results for the Magino project with after-tax IRR of 18% and total cash flow of US\$350 million.

In February 2016, Argonaut announced pre-feasibility study results for the Magino project with after-tax IRR of 22.9% (at a foreign exchange rate (US\$: CAD\$) of 0.78) and total cash flow of US\$715 million. As of February 22, 2016, the Probable Mineral Reserves were 105.4 million tonnes at 0.89 g/t Au for 3.019 million ounces of gold. (JDS Energy & Mining Inc., 2016).

In December 2017, Argonaut released the results of a feasibility study for the Magino Project with after-tax IRR of 19.5% (at a foreign exchange rate (US\$: CAD\$) of 0.78) and total cash flow of US\$540 million. As of December 21, 2017, the Total Proven and Probable Mineral Reserves were 59.0 million tonnes at 1.13 g/t Au for 2.137 million ounces of gold. (JDS Energy & Mining Inc., 2017).

In 2021, blasting in support of construction commenced, with current plans to reach first gold pour by Q2 2023, and full production in Q4 2023.

In February 2022, Argonaut published an updated Mineral Resource and Mineral Reserve estimate. As of February 14, 2022, the Total Proven and Probable Mineral Reserves were 65.6 million tonnes at 1.15 g/t Au for 2,427 million ounces of gold (Independent Mining Consultants Inc, 2022)

Argonaut's Magino Property consists of seven patented mining claims, four leases claims, and 69 unpatented mining claims totalling 2,204.5 hectares (Argonaut Gold, <https://www.argonautgold.com/English/assets/development/magino/default.aspx>).

23.2 Trillium Property: Edwards Mine and Cline Mines

The Edwards and Cline Mines are currently owned by Alamos Gold Inc. as part of a 5,700 hectare property to the east of the Island Gold Property. The property was purchased in December 2020, and remains a separate entity not yet incorporated into the Island Gold Mine property. Trillium Mining Corporation completed a 20 hole, 5,021 m drill program in 2020 testing the extensions of known occurrences on the Cline/Edwards property. This is the only exploration work completed since the historical work described below.

Since 2021, Regional exploration efforts have focused on large property-scale surveys such as airborne geophysics and glacial till surveys to built up the regional data. Historic data compilation and validation of the historic mines on the property is ongoing.

23.2.1 Edwards Mine

The Edwards mine property is located to the northeast of Island Gold's eastern boundary. The property was originally staked in 1924 by Peter Edwards. In 1933, Gold Lands Syndicate optioned the property and sunk an inclined shaft to a depth of 105 feet. In 1935, Edwards Gold Mines Ltd. acquired the property and deepened the shaft to 300 feet and erected a 75 ton per day mill. During this period 1,573 tons of ore were milled producing 435 ounces of gold (at a recovered grade of 0.31 ounces per ton). Between 1939 and 1960 the property laid dormant, until staked by A. Paquette followed by several other company options, until 1986 when it was acquired by Spirit Lake Explorations Ltd. (Spirit Lake). In late 1996, River Gold Mines Ltd. ("River Gold") agreed to purchase the two leasehold mining claims comprising the Edwards mine from VenCan Gold Corporation ("VenCan Gold" was formerly Spirit Lake). From 1996 to 2001, River Gold exploited three zones on which VenCan Gold had concentrated its drilling. River Gold ramped down to a depth of 300 m extracting 144,000 ounces of gold. The Edwards mine zones consist of a series of steeply dipping, sub-parallel, mineralized shoots hosted within deformed rocks of the GLDZ. The zones vary in width from 1 m to 5 m and are reported to extend to depth. The deposit was being mined as a low tonnage, high grade operation by River Gold. Ore was stockpiled on site and trucked to River Gold's mill located to the west of Wawa. In July 2001, River Gold closed the Edwards mine and put it on care and maintenance.

In July 2002, The Edwards mine was sold to Strike Minerals Inc. ("Strike"). Strike conducted more than 40,000 feet of drilling on the property that delineated several parallel auriferous quartz vein systems in addition to the vein systems mined in early 2000.

In 2012, Strike dewatered the mine and did some development in the upper portion of the deposit. The deposit has been dewatered to the 140 metre Level. Development of the crosscut on the 60 metre Level intersected the Edwards #1 and Edwards #5 zones and Strike planned to continue development on the 60 metre Level past the Rusty Weathered Zone to the Plowman #1 and #3 zones. On the 90 metre Level, Strike planned to develop the crosscut through the New North 2, New North 1, Edwards #1, Edwards #5, Rusty Weathered, Plowman #1 and Plowman #3 zones.

In March 2013, Strike announced sampling results from the first lift on the Edwards #1 zone above the 60 metre Level. The lift created approximately 225 tonnes of mineralized material. Muck samples at 8 foot intervals from the first lift returned an average grade of 38.98 g/t Au over a 1.5 m width for a length of 24 m. Chip samples taken across the back after removal of the first lift returned a weighted average grade of 15.39 g/t Au over 1.5 m for a length of 24 m. Initial back sampling of the Edwards #1 zone on the 60 metre Level returned a weighted average grade of 25.45 g/t Au over 1.5 m for a length of 21 m.

23.2.2 Cline Mine

The Cline mine is located approximately two kilometres northeast of Island Gold's eastern boundary and is northeast of the Edwards mine. The Cline mine zones comprise a series of steeply dipping quartz veins that are hosted by highly carbonated and silicified sheared granodiorite, felsic porphyry, and intermediate volcanic rocks. Deformation is related to splays developing off the of east-west trending Edwards-Cline shear. The gold bearing zone has been identified along a strike length of 150 m and to vertical depths exceeding 200 m. Gold bearing mineralization was discovered on the Cline property in 1918 and extensively explored during the 1920's via two openings. During this period, the No. 1 vertical shaft was sunk to a depth of approximately 45 m and the No. 2 inclined shaft to a depth of approximately 60 m. During the period from 1936 to 1942, additional works on the Cline deposit included development and mining from the No. 4 shaft, sunk to a depth of approximately 360 m. During 1965-1966, a shallower shaft, No. 3, was sunk to a depth of approximately 35 m to complete further exploration. Production from the property is reported at 63,328 ounces of gold and 10,598 ounces of silver from 301,000 tonnes of ore.

23.3 Goudreau Property

The Goudreau property is owned by Manitou Gold Inc., and consists of 22,500 hectares consolidated prospective gold properties (Manitou Gold, <https://www.manitougold.com/projects/goudreau-property/overview/>). The Goudreau Property is located to the east of Trillium Mining's property and extends approximately 37 km east towards the past producing Renabie Mine.

24 OTHER RELEVANT DATA AND INFORMATION

There is no other relevant data or information that is material to this report.

25 INTERPRETATIONS AND CONCLUSIONS

25.1 Summary

The completion of the Phase 3+ Shaft Study confirms the technical feasibility and economic viability of the 2,400 tpd Shaft Expansion.

25.2 Geology and Mineral Resource Estimate

Alamos personnel reviewed and audited the historical exploration data available for the Island Gold Mine as well as the exploration methodologies adopted to generate the data. Exploration work is professionally managed, and procedures are adopted that meet accepted industry best practices. The author is of the opinion that the exploration data is sufficiently reliable to interpret with confidence the boundaries of the gold mineralization and support evaluation and classification of Mineral Resources in accordance with generally accepted CIM Estimation of Mineral Resource and Mineral Reserve Best Practices Guidelines and CIM Definition Standards for Mineral Resources and Mineral Reserves.

The drilling database includes information from 6,907 drill holes (813 from surface and 6,094 from underground) comprising 1,365,397 m of drilling. The Mineral Resource statement, effective December 31, 2021, is provided in Table 25-1.

Table 25-1 Island Gold Mineral Resource Estimate Summary as of Dec 31, 2021

Mineral Resource	Tonnes	Grade (g/t Au)	Ounces
Measured	20,350	4.92	3,200
Indicated	1,075,950	8.18	283,100
Total Measured and Indicated	1,096,300	8.12	286,300
Inferred	7,905,600	13.59	3,453,800

Notes:

- CIM definitions of Mineral Resources were followed.
- Mineral Reserves are estimated at undiluted cut-off grade of 3.74 g/t Au
- Mineral Resources are estimated using a long-term gold price of \$1,400 per ounce.
- A minimum mining width of 2.00 m was used.
- Depending on the zone, the capping values for the high-grade samples varies from 45 g/t Au to 225 g/t Au.
- Mineral Resources, which are not Mineral Reserves, do not have demonstrated economic viability.
- Totals may not match due to rounding.

25.3 Mining Methods and Reserves

The mining methods used at Island Golf include longitudinal open stoping and transverse open stoping and are deemed suitable considering the geometry of the orebody. The summary of Mineral Reserves is contained in Table 25-2.

Table 25-2 Island Gold – Combined Mineral Reserve Estimate as of Dec 31, 2021

Mineral Reserve	Tonnes	Grade (g/t Au)	Ounces
Proven	834,100	9.33	250,150
Probable	3,277,750	10.33	1,088,300
Total Proven and Probable	4,111,850	10.12	1,338,450

Notes:

- CIM definitions of Mineral Reserves were followed.
- Mineral Reserves are estimated at cut-off grade of 3.06 g/t Au for developed areas and 3.74 g/t Au for undeveloped areas.
- Mineral Reserves are estimated using a long-term gold price of \$1,250 per ounce.
- A minimum mining width of 2.00 m was used.
- A specific gravity value of 2.78 t/m³ was used for all zones except the Lochalsh at 2.82 t/m³
- Totals may not match due to rounding.

Island Gold performs regular reconciliations between production and the reserve block model and results have generally been within industry acceptable ranges. Island Gold uses the reconciliation process to validate its Mineral Resource estimation parameters and procedures.

Island Gold has undertaken a detailed engineering and economic study of several scenarios to mine the Mineral Reserves and Mineral Resources. Island Gold has concluded that constructing a shaft to a depth of 1,380 m and expanding the mining and milling capacity to 2,400 tpd is the best way to proceed.

25.4 Processing

The current 1,200 tpd mill consistently achieves recoveries of greater than 96%. In undertaking the Shaft Expansion, the mill will be expanded to 2,400 tpd with the addition of another primary ball mill, a new crushing circuit and other upgrades, additions, and expansions within the circuit.

25.5 Infrastructure

As part of the Shaft Expansion and to support sustainable development of the mine going forward the following infrastructure upgrades will be undertaken:

- Development of a new production/service shaft down to 1,380 m in depth (initial depth);
- Development/implementation of a new ore and waste handling system underground;
- Upgrade to the main site power supply;
- Constructions of a paste plant and underground distribution system; and
- Upgrade to mine dewatering and water treatment system.

25.6 Environmental Considerations

The Island Gold Mine is operating within environmental compliance.

A number of operational permits will need to be amended to allow for the expansion up to 2,400 tpd. In addition, a number of other permits will need to be acquired for construction activities. Permitting is expected to take between 18 and 24 months.

The Island Gold Mine has been and will continue to be major contributor to the local economy. Alamos will continue to engage and work with area Indigenous communities and other communities of interest.

25.7 Capital and Operating Costs

Capital and operating costs have been undertaken to a pre-feasibility study level of detail. Life of mine capital costs for the Shaft Expansion are summarized in Table 25-3. Life of mine unit operating costs are summarized in Table 25-4.

Table 25-3 Total Capital Costs

Total Capital Cost	LOM C\$M	LOM US\$M
Sustaining Capital	\$995	\$777
Growth Capital	\$966	\$756
Total Capital Costs	\$1,961	\$1,533

Table 25-4 Summary of Operating Costs

Operating Cost	C\$/t Processed	LOM C\$M	US\$/t Processed	LOM US\$M
Mining	\$101	\$1,373	\$79	\$1,071
Processing	\$35	\$470	\$27	\$367
G&A	\$37	\$507	\$29	\$395
Subtotal	\$173	\$2,350	\$135	\$1,834
Silver Credit	-\$1	-\$18	-\$1	-\$14
Royalties	\$6	\$80	\$5	\$62
TOTAL Operating Costs	\$178	\$2,412	\$139	\$1,882

25.8 Economic Analysis

The Shaft Expansion option is economically viable with an after-tax internal rate of return (IRR) of 23% IRR, which is calculated on the differential after-tax cash flow between the 2,400 Shaft Expansion scenario and the Basecase scenario of continuing to mine at 1,200 tpd with ramp only access. After-tax net present value at 5% (NPV^{5%}) is \$2,093M (US \$1,632M).

26 RECOMMENDATIONS

At the conclusion of the Phase 3+ Expansion Study the following recommendations are being made:

- Continue to invest in the surface exploration drilling program to potentially add to the Mineral Resource base;
- Continue with the underground delineation drilling program to convert Inferred Mineral Resources to Indicated Mineral Resources;
- Utilize the geology model to aid in identifying additional targets on the Island Gold Property;
- Continue with the production to Mineral Reserve reconciliations to further refine Mineral Resource estimation parameters and methodologies;
- Proceed with the paste fill plant construction;
- Continue with the 2,400 tpd Shaft Expansion project;
- Continue with the environmental baseline program to support the permitting program; and
- Continue with the process of amending existing operational permits and acquire a series of new permits and/or authorizations for both future operational requirements and Phase 3+ construction related activities.

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28 UNITS OF MEASURE, ABBREVIATIONS, ACRONYMS

Abbreviations and Acronyms

AACE	Association for the Advancement of Cost Engineering International
Ag	Silver
Ai	Abrasion Index
ALS	ALS Global
AP	Acid Potential
API	Algoma Power Inc.
AQTK	Drill Core Size (35.5 mm diameter)
ARD	Acid Rock Drainage
ASL	Analytical Solutions Ltd.
Au	Gold
BQ	Drill Core Size (36.4 mm diameter)
BWI	Ball Mill Work Index
CAD	Canadian dollars
CAPEX	Capital and Sustaining Capital Expenditure
CCR	Central Control Room
CIL	Carbon in Leach
CIM	Canadian Institute of Mining, Metallurgy and Petroleum
CIP	Carbon-In-Pulp
CN	Cyanide
CO _{2e}	Carbon Dioxide Equivalent
CRF	Cemented Rock Fill
CRM	Certified Reference Materials
CSS	Close Size Setting
CV	Coefficient of Variation
DFO	Fisheries and Oceans Canada
ECA	Environmental Compliance Approval
EEM	Environmental Effects Monitoring
ENDM	Ministry of Energy, Northern Development and Min
EPCM	Engineering Procurement and Construction Management
ESA	Endangered Species Act
FAR	Fresh Air Raise
Fe	Iron
G&A	General and Administrative
GHG	Green House Gases
GLDZ	Goudreau Lake Deformation Zone
GPS	Global Positioning System
ICP	Inductively Coupled Plasma

ID2	Inverse Distance Weighting Method
IRR	Internal Rate of Return
ISO	International Standards Organization
LHD	Load Haul Dump
LiDAR	Light Detection and Ranging
LOM	Life of Mine
MDL	Method Detection Limit
MECP	Ministry of the Environment and Conservation and Parks
MENDM	Ministry of Energy, Northern Development and Mines
MGB	Michipicoten Greenstone Belt
MLF	Maskinonge Lake Fault
MNRF	Ministry of Natural Resources and Forestry
N'	Geotechnical Parameter
NaCN	Sodium Cyanide
Non-PAG	Not-Potentially Acid Generating
NP	Neutralizing Potential
NPI	Net Profit Interest
NPR	Neutralization Potential Ratios
NPV	Net Present Value
NQ	Drill Core Size (47.6 mm diameter)
NSR	Net Smelter Return
OK	Ordinary Kriging
OPEX	Operating Expenditure
OREAS	Ore Research and Exploration Pty Ltd Assay Standards
OWL	Operating Water Level
PAG	Potentially Acid Generating
PR	Provincial Road
PTTW	Permit to Take Water
PWQO	Provincial Water Quality Objectives
PWQO	Provincial Water Quality Objectives
Q'	Geotechnical Parameter
QA	Quality Assurance
QC	Quality Control
QP	Qualified Person
RAR	Return Air Raise
RMR	Rock Mass Rating
ROM	Run of Mine
RPA	Roscoe Postle Associates
RQD	Rock Quality Designation
SAB	SAG and Ball Mill
SAG	Semi-Autogenous Grinding (mills)

SAR	Species at Risk
SARA	Species at Risk Act
SD	Standard Deviation
SQL	Structured Query Language
TMF	Tailings Management Facility
TSP	Total Suspended Particulates
TSS	Total Suspended Solids
UCF	Unconsolidated Rock Fill
UCS	Uniaxial Compressive Strength
UDS	Underground Distribution System
URSTM	Unité de Recherche et de Service en Technologie Minérale
USD	United States Dollars
VLD	Very Low Frequency
VOD	Ventilation on Demand
vs.	Versus
WI	Work Index
WMP	Waste Management Procedure
WTP	Water Treatment Plant

Units of Measure

°	Degrees
°C	Degrees Celsius
cfm	Cubic feet per minute
µg/m ³	Microgram per cubic metre
µm	Micrometre (Micron)
g	gram
g/L	Grams per litre
g/t	Grams per tonne
h	Hours
ha	Hectare
hp	Horsepower
K	Thousand
kg	Kilogram
km	Kilometre
km ²	Square kilometre
kt	Thousand tonnes
kW	Kilowatt
KWh	Kilowatt hour
L	Litre

m	Metre
M	Million
m ³	Cubic metre
Ma	Millions of Years Before Present
masl	Metres above sea level
mg/L	Milligram per litre
min	Minute
mL	Millilitre
mm	Millimetre
MVA	Megavolt amperes
MW	Mega Watt
MWh	Megawatt-hours
oz	Troy ounce
oz	Ounce
Pa	Pascal
pH	Measure of a solution's acidity
ppb	Parts per billion
ppm	Parts per million
t	Tonne
t/m ³	Tonne per cubic metre
tpd	Tonne per day
V	Volt
w/w	Weight for weight
wt%	Weight percent
y	Year

29 CERTIFICATES OF QUALIFIED PERSONS

CERTIFICATE OF QUALIFIED PERSON

I, Raynald Vincent, P.Eng., M.G.P., as an author of this report entitled “NI 43-101 Technical Report for the Island Gold Mine, Dubreuilville, Ontario, Canada” prepared for Alamos Gold Inc. and with an effective date of June 28, 2022, do hereby certify that:

1. I am a Geological Engineer employed as Exploration Superintendent of the Island Gold Mine by Alamos Gold Inc., located at Goudreau Road, Dubreuilville, Ontario;
2. I received a bachelor in Geological Engineering from the University of Laval (Quebec, Canada) in 1983 and a Master's in Project Management from the University of Quebec in Abitibi-Témiscamingue (Quebec, Canada) in 2002;
3. I am a registered member of the Ordre des Ingénieurs du Québec (OIQ licence no 42761) and a member of the Professional Engineers of Ontario (PEO licence no 100210071). I have worked as an Engineer for more than 30 years since my graduation. I have worked mainly in exploration and production geology in the mining industry for different companies with increasing levels of responsibilities;
4. I have read the definition of “qualified person” set out in National Instrument 43-101 (“NI 43-101”) and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a “qualified person” for the purposes of NI 43-101;
5. I have worked at the Island Gold Mine for the last eight years;
6. I am the author of Sections 4 to 12, 14, and 23, and co-author of Sections 1, 2, 3, 15 and 24 to 27 of the NI 43-101 report entitled “NI 43-101 Technical Report for the Island Gold Mine, Dubreuilville, Ontario, Canada ” with an effective date of June 28, 2022;
7. I have no personal knowledge, as of the date of this certificate, of any material fact or change, which is not reflected in this report;
8. I have been an employee of Alamos Gold Inc. since January 2009, first as Senior Geologist, Exploration Division and later as Chief Geologist, Island Gold Mine, and currently as Exploration Superintendent, Island Gold Mine;
9. I am not independent of the issuer, as described in Section 1.5 of NI 43-101;
10. I have prepared this Technical Report in compliance with National Instrument 43-101 and in conformity with generally accepted Canadian mining industry practices. As of the date of this certificate, to the best of my knowledge, information and belief, the Technical Report contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading; and
11. I consent to the filing of the Technical Report with any stock exchange and other regulatory authority and any publication by them for regulatory purposes, including electronic publication in the public company files on their websites accessible by the public.

Dated this 29th day of August 2022

(Signed & Sealed) “Raynald Vincent”

(Original signed and sealed)

Raynald Vincent, P.Eng. M.G.P. (OIQ no. 42761, PEO no 100210071)

CERTIFICATE OF QUALIFIED PERSON

I, Nathan Eugene Gerard Bourgeault, M.Eng, P.Eng., PMP as an author of this report entitled "NI 43-101 Technical Report for the Island Gold Mine, Dubreuilville, Ontario, Canada" prepared for Alamos Gold Inc. and with an effective date of June 28, 2022, do hereby certify that:

1. I am a licensed Mining Engineer employed as Chief Engineer of the Island Gold Mine, located at Goudreau Road, Dubreuilville, Ontario;
2. I received a Bachelor of Engineering in Mining Engineering from Laurentian University (Ontario, Canada) in 2007 and a Master of Engineering with a specialization in Natural Resources Engineering from Laurentian University (Ontario, Canada) in 2014;
3. I am a registered member of the Professional Engineers of Ontario (PEO licence no 100149936). I have worked as an Engineer for more than 15 years since my graduation. I have worked mainly in project development and operations in the mining industry for different companies with increasing levels of responsibilities;
4. I have read the definition of "qualified person" set out in National Instrument 43-101 ("NI 43-101") and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101;
5. I have worked at the Island Gold Mine for the last five years;
6. I am the author of Sections 16, 19, 21 and 22, Section 18 except for Sections 18.3.3.1 to 18.3.3.3, and co-author of Sections 1, 2, 3, 15 and 24 to 27 of the NI 43-101 report entitled "NI 43-101 Technical Report for the Island Gold Mine, Dubreuilville, Ontario, Canada " with an effective date of June 28, 2022;
7. I have no personal knowledge, as of the date of this certificate, of any material fact or change, which is not reflected in this report;
8. I have been an employee of Alamos Gold Inc. since November 2017, first as Senior Production Engineer, Island Gold Mine and currently as Chief Engineer, Island Gold Mine;
9. I am not independent of the issuer, as described in Section 1.5 of NI 43-101;
10. I have prepared this Technical Report in compliance with National Instrument 43-101 and in conformity with generally accepted Canadian mining industry practices. As of the date of this certificate, to the best of my knowledge, information and belief, the Technical Report contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading; and
11. I consent to the filing of the Technical Report with any stock exchange and other regulatory authority and any publication by them for regulatory purposes, including electronic publication in the public company files on their websites accessible by the public.

Dated this 29th day of August 2022

(Signed & Sealed) "Nathan Eugene Gerard Bourgeault"

(Original signed and sealed)

Nathan Eugene Gerard Bourgeault, P.Eng. (PEO no 100149936)

CERTIFICATE OF QUALIFIED PERSON

I, Colin Webster, P. Eng., as an author of this report entitled "NI 43-101 Technical Report for the Island Gold Mine, Dubreuilville, Ontario, Canada" prepared for Alamos Gold Inc. and with an effective date of June 28, 2022, do hereby certify that:

1. I am an Environmental Engineer employed as Vice President, Sustainability & External Affairs for Alamos Gold Inc. located at 181 Bay Street, Suite 3910, Toronto, Ontario, M5J 2T3;
2. I received a Bachelor of Science in Mining Engineering from Queen's University (Kingston, Ontario) in 1990 and a diploma in Environmental Technology from Fanshawe College (London, Ontario) in 1994;
3. I am a registered member of the Professional Engineers of Ontario (PEO licence no 90498825). I have worked as an Engineer for more than 25 years since my graduation. I have worked mainly in environmental management and sustainability within the mining and consulting industries for different companies with increasing levels of responsibilities;
4. I have read the definition of "qualified person" set out in National Instrument 43-101 ("NI 43-101") and certify that by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I fulfill the requirements to be a "qualified person" for the purposes of NI 43-101;
5. I have worked at Alamos Gold Inc. for the last six years;
6. I am the author of Section 20 of the NI 43-101 report entitled "NI 43-101 Technical Report for the Island Gold Mine, Dubreuilville, Ontario, Canada " with an effective date of June 28, 2022;
7. I have no personal knowledge, as of the date of this certificate, of any material fact or change, which is not reflected in this report.
8. I have been an employee of Alamos Gold Inc. since January 2016 as Vice President, Sustainability & External Affairs;
9. I am not independent of the issuer, as described in Section 1.5 of NI 43-101;
10. I have prepared this Technical Report in compliance with National Instrument 43-101 and in conformity with generally accepted Canadian mining industry practices. As of the date of this certificate, to the best of my knowledge, information and belief, the Technical Report contains all scientific and technical information that is required to be disclosed to make the Technical Report not misleading; and
11. I consent to the filing of the Technical Report with any stock exchange and other regulatory authority and any publication by them for regulatory purposes, including electronic publication in the public company files on their websites accessible by the public.

Dated this 29th day of August 2022

(Signed & Sealed) "Colin Webster"

(Original signed and sealed)

Colin Webster, P.Eng. (PEO no 90498825)

CERTIFICATE OF QUALIFIED PERSON

I, Neil Lincoln, P.Eng, as an author of this report entitled “NI 43-101 Technical Report for the Island Gold Mine, Dubreuilville, Ontario, Canada” prepared for Alamos Gold Inc. and with an effective date of June 28, 2022, do hereby certify that:

- 1) I am an Independent Metallurgical Consultant of Lincoln Metallurgical Inc located at 1565 Lords Manor Lane, Ottawa, Ontario, K4M 1K3, Canada.
- 2) I graduated from the University of the Witwatersrand, South Africa, in 1994 with a Bachelor of Science in Metallurgy and Materials Engineering (Minerals Process Engineering) degree.
- 3) I am a professional engineer in good standing with the Professional Engineers of Ontario (PEO) in Canada (no. 100039153).
- 4) I have practiced my profession in the mining industry continuously since graduation. I have over 27 years experience as a metallurgist and study manager. I have sufficient relevant experience having worked on numerous projects ranging from scoping studies, prefeasibility and feasibility studies to project implementation related to mineral processing plants. My mineral processing commodity and unit operations experience includes precious metals, base metals and industrial minerals covering metallurgical test work to process plant design. As a result of my experience and qualifications, I am a Qualified Person as defined in NI 43-101.
- 5) I have read the definition of “qualified person” set out in the National Instrument 43-101 (“NI 43-101”) and certify that by reason of my education, affiliation with a professional association and past relevant work experience, I fulfil the requirements to be a qualified person for the purposes of NI 43-101.
- 6) I have participated in the preparation of the Technical Report and am responsible for the supervision or creation of the following sections and sub-sections: 13, 17, 18.3.3 to 18.3.3.3.
- 7) I have visited the site property that is the subject of this report several times; most recent visit was on August 9, 2022.
- 8) I have prepared this Technical Report in compliance with National Instrument 43-101 and in conformity with generally accepted Canadian mining industry practices. As of the date of this certificate, to the best of my knowledge, information and belief, the sections and sub-sections of the Technical Report listed in item 6 above contain all scientific and technical information that is required to be disclosed to make these sections and sub-sections of the Technical Report not misleading.
- 9) I have read and understand NI 43-101 and I am considered independent of the issuer as defined in Section 1.5 of NI 43-101.
- 10) I consent to the filing of the Technical Report with any stock exchange and other regulatory authority and any publication by them for regulatory purposes, including electronic publication in the public company files on their websites accessible by the public.

Dated this 29th day of August 2022

/signed and sealed/

Neil Lincoln, P.Eng.